END OF WELL REPORT

AZADEGAN OIL FIELD

END OF WELL REPORT

AZNS – 127 (PAD #54), TYPE: IIF

Drilling Department

DECEBER 2024

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TABLE OF CONTENTS

1. General Information 10
1.1 Field Introduction 10
1.2 AZNS – 127 Well Location Map 11
2. Well History 12
2.1 Well Informaion 12
2.2 AZNS – 127 Well History 13
2.3 AZNS – 127 ROP Chart 34
2.4 Bit Record Table 35
2.5 Bit Parameters Plot 36
3. Time Analysis Report 37
3.1 Time vs. Depth Chart 37
3.2 Total Time Break-Down of Operation 38
WorkOver Section 38
Side Track Section 39
Completion Section 40
3.3 NPT Break-Down 41
4. Geological & Logging Reports 47
4.1 Logging and Testing Reports 47
5. Bottom Hole Assembly Record 48
6. Casing Information 76
.7 Cement Job Report 77
7.1 7” Liner Cementing Operation Information 77
7.2 Engineering Data 77
7.3 Execution of Operation 78
7.4 Operation Ticket 79
7.5 Job Log & Summary 80
7.6 Job Graph 81
7.7 Operation Instructions 81
Pre – Job Preparation 81
Pre – Job Preparation 81
7.8 Pumping Schedule 82
7.9 Amount of Material & Used in Operation 82
7.10 4 – 1/2” Liner Cementing Operation Information 83
7.11 Engineering Data 83
7.12 Execution of Operation 84
7.13 Operation Ticket 85
7.14 Job Log & Summary 86
7.15 Job Graph 87
7.16 Operation Instructions 87
Pre – Job Preparation 87

Pre – Job Preparation 87
7.17 Pumping Schedule 88
7.18 Amount of Material & Used in Operation 88
8. Completion Report 89
8.1 Completion Tubing Information 89
8.2 Completion Tools Specification Checklist 89
8.3 Assembling Test Report 93
8.4 Completion Operation Checklist 111
8.5 Completion Sketch & Tally 115
8.6 Completion String Pressure Test 124
9. Wellhead Report 127
9.1 Final Well Status 127
9.2 Wellhead & X-mass tree Specification 128
9.3 Wellhead Schematic 130
9.4 X-mass Tree Specification (Section D) 131
9.5 X-mass Tree Pressure Test Report (Section D – befor install) 132
9.6 X-mass Tree Pressure Test Report (Section D – after install) 132
9.7 Wellhead Related Test Reports 134
10. Drilling Fluid Information 138
10.1 Summary 138
10.2 Interval Remark 139
WorkOver Section 139
8 – 1/2” Hole Section 141
6 – 1/8” Hole Section 142
10.3 Drilling Fluid type 144
10.4 Material Consumption Analysis 145
10.5 Volume Analysis 147
10.6 Daily volume Breakdown 148
10.7 Loss Distribution 151
10.8 Interval Fluid Properties Recap 154
10.9 Solid Control Disscusion 155
10.10 8 – 1/2” Hole Section Drilling Fluid Types 156
10.11 8 1/2” Interval Material Consumption Analysis 157
10.12 8 1/2” Interval Volume Analysis 158
10.13 8 1/2” Interval Daily Volume Breakdown 160
10.14 8 – 1/2” Interval Loss Distribution 163
10.15 8 – 1/2” Interval Fluid Properties Recap 165
10.16 Solid Control Disscusion 170
10.17 6 – 1/8” Hole Section Drilling Fluid Types 171
10.18 6-1/8” Interval Material Consumption Analysis 172
10.19 6-1/8” Interval Volume Analysis 174

10.20 6-1/8" Interval Daily Volume Breakdown 175
10.21 6-1/8" Interval Loss Distribution 178
10.22 6-1/8" Interval Fluid Properties Recap 180
10.23 6-1/8" Interval Solid Control Disscusion 186
10.24 Interval Completion Section 187
11. Waste Management Report 188
11.1 Well & Hole Section 188
11.2 De-Completion Operation's Recap 189
11.3 8 – 1/2" Hole Section 196
11.4 8 – 1/2" Operation's Recap 197
12. List of Contractors 204
13. Appendix 205
13.1 DDR Summary 205
14. Comments & Revision 220

LIST OF TABLES

TABLE 2-1: AZNS-85 WORKOVER WELL INFORMATION 12
TABLE 2-2: AZNS-127 CASING HISTORY 14
TABLE 2-3: AZNS-127 BIT RECORD TABLE 35
TABLE 3-1: AZNS-127 WORKOVER TIME BREAKDOWN 38
TABLE 3-2: AZNS-127 SIDE TRACK TIME BREAKDOWN 39
TABLE 3-3: AZNS-127 COMPLETION TIME BREAKDOWN 40
TABLE 3-4: AZNS-127 NPT TIME BREAKDOWN 46
TABLE 4-1: AZNS-127 LOGGING INFORMATION 47
TABLE 6-1: AZNS-127 FINAL CASING DATA 76
TABLE 7-1: AZNS-127 ENGINEERING DATA 77
TABLE 7-2: AZNS-127 EQUIPMENT USED 78
TABLE 7-3: AZNS-127 OPERATION CREW 78
TABLE 7-4: AZNS-127 PUMPING SCHEDULE 82
TABLE 7-5: AZNS-127 MATERIAL USED TABLE 82
TABLE 7-6: AZNS-127 ENGINEERING DATA 83
TABLE 7-7: AZNS-127 EQUIPMENT USED 84
TABLE 7-8: AZNS-127 OPERATION CREW 84
TABLE 7-9: AZNS-127 PUMPING SCHEDULE 88
TABLE 7-10: AZNS-127 MATERIAL USED TABLE 88
TABLE 8-1: AZNS-85 TUBING DATA 89
TABLE 8-2: AZNS-127 EQUIPMENT CHECKLIST 91
TABLE 8-3: AZNS-127 EQUIPMENT CHECKLIST 92
TABLE 8-4: AZNS-127 OPERATION CHECKLIST 114
TABLE 9-1: AZNS-127 FINAL WELL STATUS 127
TABLE 9-2: AZNS-127 EQUIPMENT SPECIFICATION 129
TABLE 9-3: AZNS-127 SECTION "D" SPECIFICATION 131
TABLE 9-4: AZNS-127 SECTION "D" TEST REPORT BEFOR INSTALLATION 132
TABLE 9-5: AZNS-127 SECTION "D" TEST REPORT AFTER INSTALLATION 133
TABLE 10-1: AZNS-127 SUMMARY OF FLUID 138
TABLE 10-2: AZNS-127 FLUID PROPERTIES RECAP TABLE 144
TABLE 10-3: AZNS-127 MATERIAL CONSUMPTION ANALYSIS TABLE 145
TABLE 10-4: AZNS-127 VOLUME ANALYSIS TABLE 147
TABLE 10-5: AZNS-127 DAILY VOLUME BREAK DOWN TABLE 149
TABLE 10-6: AZNS-127 LOSS DISTRIBUTION 152
TABLE 10-7: AZNS-127 FLUID PROPERTIES RECAP 154
TABLE 10-8: AZNS-127 SOLID CONTROL DESSCUSION 155
TABLE 10-9: AZNS-127 8-1/2" FLUID PROPERTIES RECAP TABLE 156
TABLE 10-10: AZNS-127 8-1/2" INTERVAL MATERIAL CONSUMPTION ANALYSIS 157
TABLE 10-11: AZNS-127 8-1/2" INTERVAL VOLUME ANALYSIS 158
TABLE 10-12: AZNS-127 8-1/2" INTERVAL DAILY VOLUME BREAKDOWN 160
TABLE 10-13: AZNS-127 8-1/2" INTERVAL LOSS DISTRIBUTION 163
TABLE 10-14: AZNS-127 8-1/2" INTERVAL FLUID PROPERTIES 165
TABLE 10-15: AZNS-127 8-1/2" SOLID CONTROL DESSCUSION 170
TABLE 10-16: AZNS-127 6-1/8" FLUID PROPERTIES RECAP TABLE 171
TABLE 10-17: AZNS-127 6-1/8" INTERVAL MATERIAL CONSUMPTION ANALYSIS 172
TABLE 10-18: AZNS-127 6-1/8" INTERVAL VOLUME ANALYSIS 174
TABLE 10-19: AZNS-127 6-1/8" INTERVAL DAILY VOLUME BREAKDOWN 175
TABLE 10-20: AZNS-127 6-1/8" INTERVAL LOSS DISTRIBUTION 178
TABLE 10-21: AZNS-127 6-1/8" INTERVAL FLUID PROPERTIES 180
TABLE 10-22: AZNS-127 6-1/8" SOLID CONTROL DESSCUSION 186

TABLE 10-23: AZNS-127 COMPLETION SECTION 187
TABLE 11-1: AZNS-127 FIXATION TABLE 190
TABLE 11-2: AZNS-127 FIXATION ANALYSIS 191
TABLE 11-3: AZNS-127 DE-WATERING RECAP 193
TABLE 11-4: AZNS-127 DE-WATERING ANALYSIS 194
TABLE 11-5: AZNS-127 CHEMICAL 195
TABLE 11-6: AZNS-127 FIXATION TABLE 197
TABLE 11-7: AZNS-127 FIXATION ANALYSIS 199
TABLE 11-7: AZNS-127 DE-WATERING RECAP 200
TABLE 11-7: AZNS-127 DE-WATERING ANALYSIS 201
TABLE 11-10: AZNS-127 CHEMICAL TABLE 203
TABLE 12-1: AZNS-127 CONTRACTORS INVOLVED 204
TABLE 13-1: AZNS-127 DAILY DRILLING REPORT SUMMARY 219

LIST OF FIGURES

FIGURE 1-1: AZNS OILFIELD LOCATION MAP 10
FIGURE 1-2: AZNS-127 WELL LOCATION MAP 11
FIGURE 2-1: AZNS-127 WELL SKETCH 33
FIGURE 2-2: AZNS-127 ROP CHART 34
FIGURE 2-3: AZNS-127 BIT PARAMETERS PLOT 36
FIGURE 3-1: AZNS-127 WORKOVER WELL TIME VS. DEPTH AFTER SIDE TRACK 37
FIGURE 3-2: AZNS-127 NPT BREAK-DOWN CHART 41
FIGURE 7-1: AZNS-127 7" LINER JOB GRAPH 81
FIGURE 7-2: AZNS-127 4 – 1/2" LINER JOB GRAPH 87
FIGURE 8-1: AZNS-127 TRSSSV TEST REPORT 93
FIGURE 8-2: AZNS-127 TRSSSV CONNECTIONS & BODY PRESSURE TEST GRAPH 95
FIGURE 8-3: AZNS-127 PERMANENT PACKER TEST REPORT 96
FIGURE 8-4: AZNS-127 ANCHOR TUBING SEAL TEST REPORT 97
FIGURE 8-5: AZNS-127 PERMANENT PACKER CONNECTIONS & BODY PRESSURE TEST GRAPH 98
FIGURE 8-6: AZNS-127 SSD TEST REPORT 99
FIGURE 8-7: AZNS-127 SSD CONNECTIONS & BODY PRESSURE TEST GRAPH 100
FIGURE 8-8: AZNS-127 EX JT TEST REPORT 101
FIGURE 8-9: AZNS-127 EX JT CONNECTIONS & BODY PRESSURE TEST GRAPH 102
FIGURE 8-10: AZNS-127 LANDING NIPPLE TEST REPORT 103
FIGURE 8-11: AZNS-127 LANDING NIPPLE CONNECTIONS & BODY PRESSURE TEST GRAPH 104
FIGURE 8-12: AZNS-127 FIRST CIV TEST REPORT 105
FIGURE 8-13: AZNS-127 FIRST CIM TEST REPORT 106
FIGURE 8-14: AZNS-127 FIRST CIM CONNECTIONS & BODY PRESSURE TEST GRAPH 107
FIGURE 8-15: AZNS-127 SECOND CIV TEST REPORT 108
FIGURE 8-16: AZNS-127 SECOND CIM TEST REPORT 109
FIGURE 8-17: AZNS-127 SECOND CIM CONNECTIONS & BODY PRESSURE TEST GRAPH 110
FIGURE 8-18: AZNS-127 COMPLETION SKETCH 115
FIGURE 8-19: AZNS-127 COMPLETION TALLY 123
FIGURE 8-20: AZNS-127 COMPLETION STRING & TRSSSV PRESSURE TEST GRAPH 125
FIGURE 8-21: AZNS-127 SETTING PACKER, ANNULUS & FINAL COMPLETION STRING PRESSURE TEST 126
FIGURE 9-1: AZNS-127 WELLHEAD SKETCH 130
FIGURE 9-2: AZNS-127 X-MASS TREE BODY PRESSURE TEST GRAPH BEFORE INSTALLATION 134
FIGURE 9-3: AZNS-127 X-MASS TREE BODY PRESSURE TEST GRAPH AFTER INSTALLATION 135
FIGURE 9-4: AZNS-127 WELLHEAD MOM BEFOR INSTALLATION 136
FIGURE 9-5: AZNS-127 WELLHEAD MOM AFTER INSTALLATION 137
FIGURE 10-1: AZNS-127 MUD CHEMICAL USAGE 146
FIGURE 10-2: AZNS-127 EFFECTIVE CONCENTRATION 146
FIGURE 10-3: AZNS-127 VOLUME ANALYSIS CHART 147
FIGURE 10-4: AZNS-127 DAILY VOLUME BREAK DOWN CHART 150
FIGURE 10-5: AZNS-127 LOSS ANALYSIS 153
FIGURE 10-6: AZNS-127 8 -1/2" EFFECTIVE CONCENTRATION 157
FIGURE 10-7: AZNS-127 8-1/2" MUD CHEMICAL USAGE 158
FIGURE 10-8: AZNS-127 8-1/2" INTERVAL VOLUME ANALYSIS 159
FIGURE 10-9: AZNS-127 8-1/2" INTERVAL DAILY VOLUME BREAKDOWN 161
FIGURE 10-10: AZNS-127 8-1/2" LOSSES ANALYSIS 162
FIGURE 10-11: AZNS-127 8-1/2" LOSSES ANALYSIS 164
FIGURE 10-12: AZNS-127 8-1/2" DENSITY GRAPH 166
FIGURE 10-13: AZNS-127 8-1/2" VISCOSITY GRAPH 166
FIGURE 10-14: AZNS-127 8-1/2" PV & YP GRAPH 167
FIGURE 10-15: AZNS-127 8-1/2" PH GRAPH 167
FIGURE 10-16: AZNS-127 8-1/2" CALCIUM GRAPH 168
FIGURE 10-17: AZNS-127 8-1/2" TOTAL HARDNESS GRAPH 168
FIGURE 10-18: AZNS-127 8-1/2" SOLID% GRAPH 169
FIGURE 10-19: AZNS-127 8-1/2" CHLORIDE GRAPH 169
FIGURE 10-20: AZNS-127 8-1/2" KCL % WT GRAPH 170

FIGURE 10-21: AZNS-127 6-1/8" EFFECTIVE CONCENTRATION 173
FIGURE 10-22: AZNS-127 6-1/8" MUD CHEMICAL USAGE 173
FIGURE 10-23: AZNS-127 6-1/8" INTERVAL VOLUME ANALYSIS 174
FIGURE 10-24: AZNS-127 6-1/8" INTERVAL DAILY VOLUME BREAKDOWN 176
FIGURE 10-25: AZNS-127 6-1/8" LOSSES ANALYSIS 177
FIGURE 10-26: AZNS-127 6-1/8" LOSSES ANALYSIS 179
FIGURE 10-27: AZNS-127 6-1/8" DENSITY GRAPH 181
FIGURE 10-28: AZNS-127 6-1/8" VISCOSITY GRAPH 181
FIGURE 10-29: AZNS-127 6-1/8" PV & YP GRAPH 182
FIGURE 10-30: AZNS-127 6-1/8" PH GRAPH 182
FIGURE 10-31: AZNS-127 6-1/8" CALCIUM GRAPH 183
FIGURE 10-32: AZNS-127 6-1/8" TOTAL HARDNESS GRAPH 183
FIGURE 10-33: AZNS-127 6-1/8" WATER LOSS% GRAPH 184
FIGURE 10-34: AZNS-127 6-1/8" CHLORIDE GRAPH 184
FIGURE 10-35: AZNS-127 6-1/8" KCL % WT GRAPH 185
FIGURE 11-1: AZNS-127 FIXATION GRAPH 190
FIGURE 11-1: AZNS-127 DE-WATERING GRAPH 194
FIGURE 11-3: AZNS-127 FIXATION GRAPH 198
FIGURE 11-3: AZNS-127 DE-WATERING GRAPH 202

1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun, Khuzestan province on a large long – axis anticline with N – S trending with approximate dimensions about 20^{km} width and 60^{km} length near to Iran – Iraq border, It is located in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45$^{°c}$ in summer and Highest Temperature is 52$^{°c}$. The Lowest Temperature is -4.4$^{°c}$ in winter. Maximum Rainfall is 78^{mm} in February, Winter Season.

The trap area of the south Azadegan is limited to 43.2x17.2km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

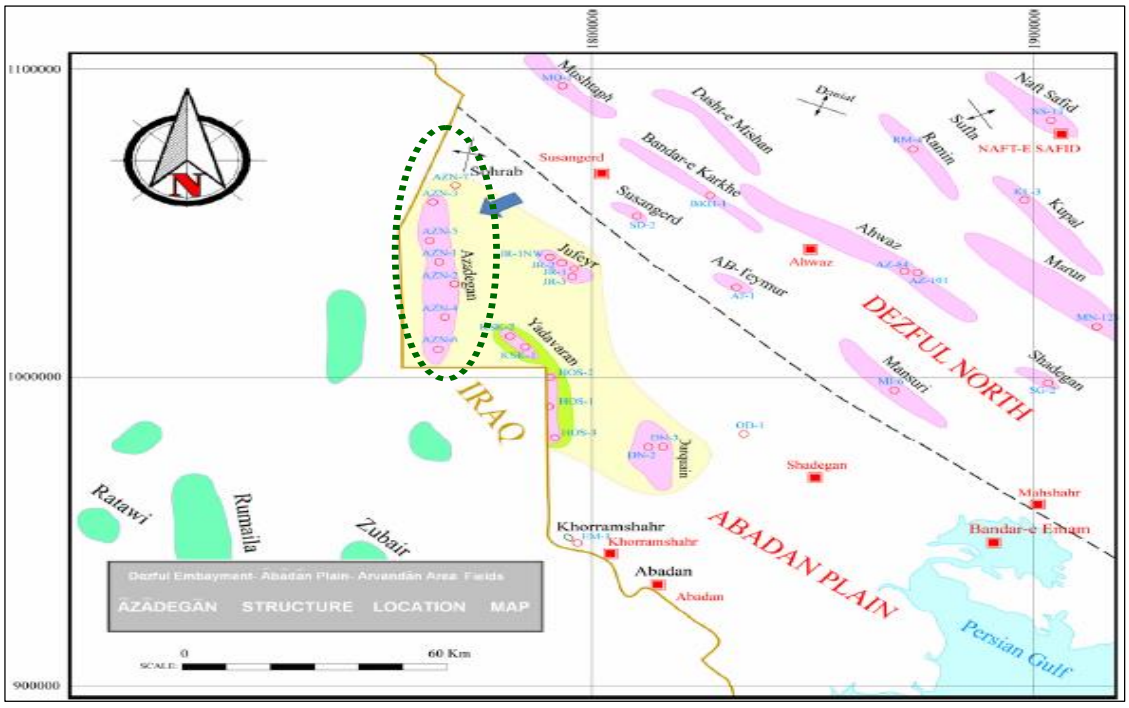


Figure 1-1: AZNS Oilfield Location Map

1.2 AZNS – 127 Well Location Map

This well is located on "Pad – 54" of South Azadegan Oil field in Block – II.

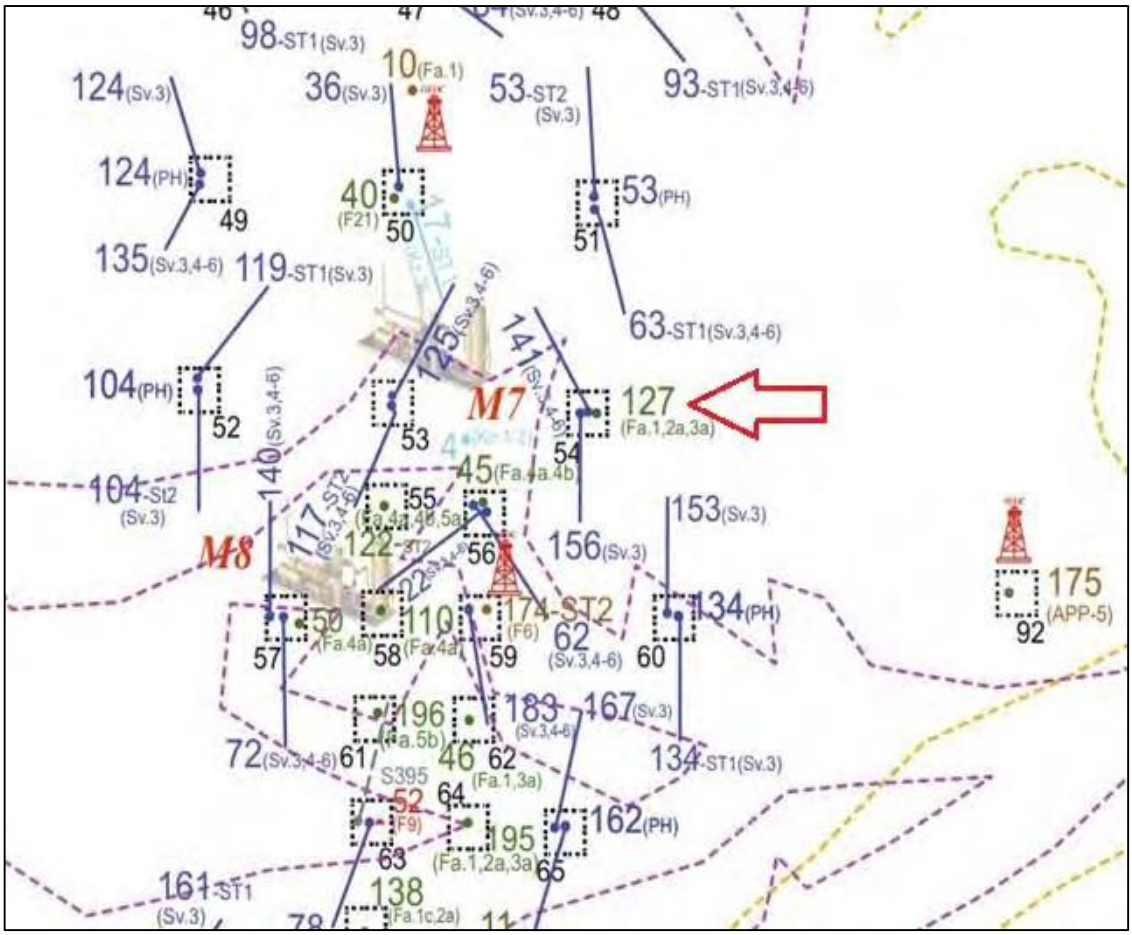


Figure 1-2: AZNS-127 Well Location Map

2. Well History

2.1 Well Informaion

Rig name	208	**Owner**	PEDEC/PPL
Rig type	Land Rig 2000HP	**Contractor**	MSA/OEOC
Well Name	AZNS – 127	**Management Contractor**	MSA
Well Type	Vertical IIF	**Field**	South Azadegan
Completion Type	4-1/2" * 3-1/2" 10K	**RTE**	13.45
Actual TD	4226^{m}	**GLE**	3.95
Surface Coordinate		N: 3,448,561 – E: 196,167	
Target Coordinate		N: 3,448,538.7 – E: 196,163	
Drilling & Suspension		**Completion**	
Spud Date	08/05/2024	**Spud Date**	16/09/2024
Released Date	25/09/2024	**Released Date**	25/09/2024
Rig Days Plan / Actual	50 (main plan) + 80 (side tracking & Completion plan) / 141		

Size (in)	Shoe Depth (mMD)	Specification
20	146	K – 55, 94^{ppf}, BTC
13 – 3/8	1441	N – 80, 68^{ppf}, BTC
9 – 5/8	3909	T – 95, 53.5^{PPF}, TSH BLUE T – 95, 53.5^{PPF}, TPCQ
7 (main hole)	3826 – 4226	C110, 29^{PPF}, TPCQ
7 tie-back	3777 – 3826	C110, 29^{PPF}, TPCQ
7 (side track)	3147 – 3891	110SS, 29^{PPF}, TPG2
4/5 (SY)	3780.41 – 4139.5	110SS, 13.5^{PPF}, TPG2

Table 2-1: AZNS-85 Workover Well Information

2.2 AZNS – 127 Well History

The AZNS – 127 well was drilled & completed by TDDC on date 1395.10.08 at 12:00 as describe below:

- 26" hole size drilled to 150^{m} w/ Mw 66 – 68^{pcf} & 20" casing ran to 146^{m} & cemented as program by 170^{bbl} cement.
- 17 – 1/2" hole size drilled to 1441^{m}, while increased Mw from 68 to 90^{pcf}, 13 – 3/8" casing ran & cemented to 1441^{m} by 780^{bb} cement.
- 12 – 1/4" hole size drilled to 3910^{m} w/ performance drlg & 3 – 15BPH loss. At depth 2354^{m} suddenly pressure dropped f/ 2850 to 1950^{psi} & string twist off from float sub & 9.18^{m} fish left in hole. Fish recovered by overshot 10 – 1/8". At depth 3117^{m}due to pressure dropped f/ 2560 to 2320^{psi} POOH string but string got stuck at 3040^{m} & free by applied 50^{klbs} overpull, POOH to surface & observed wash out on float sub above 8" PDM. Took MSFL/DLL/SONIC/CALIPER/GR/CDR & ZDL/CNC/DSL/WTS/TTRM logs, 9 – 5/8" casing ran & cemented to 3909^{m} with 1133^{bbl} cement.
- 8 – 1/2" hole size drilled to 4226^{m} with no major problem, took VDL/SBT/CCL/GR logs, string held up at 3905^{m}, tried to pass this point several times with no success, logged up from this pont to 100^{m} above 13 – 3/8" casing shoe. Due to tight hole Mw increased f/ 104 to 105^{pcf}. Took first full suite log as DLL/SONIC/MSFL/CDR from TD to 3810^{m} & second full suite log as CNT/LDT/NGS/GR from 3910^{m} to TD. 7" liner ran to 4226^{m}. Due to first dry test results, decided was made to tie back 7" liner to 3777^{m}. The well suspended with cement plug & completion string & rig released @1396.06.27.
- Re – entry the well at 1396.11.03 by retrieving completion string & ran new completion & rig released at 24:00 on 1396.12.23
- Rig less operation was performed on date 17.05.1398. tried to RIH with coiled tubing to unload the well. There was an obstruction in the well at depth 3918^{m}. Unable to pass the depth. Several runs were performed by coiled tubing and mill to remove the obstruction.
- Rig less operation for unloading and well flowing was performed on date 30.01.1400.
- Pressure monitoring was performed on tubing head spool vent port, it was 4,000^{psi}. Metal seal of tubing hanger is damaged.
- Try to run coiled tubing to TD and wash the wellbore with xylene. Run CCL/GR/MFC logging BHA in the well. It was observed 7" liner ID has been decreased to about 3inches in interval 3889 – 3911^{m}.
- Flow the well, an abnormal annulus pressure was observed.

- The other Rig less monitoring well annulus pressure on date 18.09.1400. It was annulus pressure is 5,300^{psi}. Sampling was performed from annulus fluid. It was contained with gas and hydrocarbon.

Casing Spec.	20"	K – 55, BTC, 94$^{\#}$	146^{m}
	13-3/8"	N – 80, BTC, 68$^{\#}$	1441^{m}
	9-5/8"	T - 95,TSH Blue, 53.5$^{\#}$	0 – 1164^{m}
		T - 95,TPCQ, 53.5$^{\#}$	1164 – 3909^{m}
	7"	110S, TPCQ, 29$^{\#}$	3826 – 4226
	7” Tie-back	110S, TPCQ, 29$^{\#}$	3826 – 3777

Table 2-2: AZNS-127 Casing History

Summary of Workover Operation

The AZNS – 127 workover well was De-Complete by rig OEOC – 208 on Pad 54, from "08 – May – 2024" to "18 – March – 2024" as described in the following:

De – Completion Summary:

Installed 8" poor boy degasser flare. Pressure test SP & function test TDS, MPs; Ok. Checked section B: P-Seals & ring gasket; Ok. Checked section C: BTM P-Seal; Ok. Top P-Seal & ring gasket; Not Ok. Disconnected adjustable choke of production line. L/U pump truck to XMT kill line & annulus. L/U choke manifold to production line & annulus, meanwhile preparing 200^{bbl} 106^{pcf} mud. Pressure test all line after rectify several leakages. Open Ann & monitor Press; 4700 psi. Bleed off Press to 3000 psi, Build up to3800 psi in 1 Min. Increase Press f/ 3800 to 3900 psi in 5 Min.

P/U 60 Jts 5" DP Jt by Jt, made & R/B 20 Stds to derrick. Monitored annulus pressure build up to 4800^{psi} after 6 hrs. Opened the annulus thru choke manifold to oil storage; after 6min. decreased pressure to 980^{psi}. Decreased pressure gradually to 500^{psi} in 20min. Closed the annulus; Shut in pressure 500^{psi}. Built up to 1000^{psi} in 15min. Opened the annulus, suddenly dropped to zero. After 6min closed the annulus. Opened the 2SV, top master valve & production valve. Tightened XMT cap. Connected hand pump to exit block, tried to open TR3SV. Monitored tubing pressure 3400^{psi}. Opened the well f/ TBG to oil storage tank; dropped pressure to zero in 2min. Closed the well. Built up to 3700^{psi} in 2min. L/U annulus to choke manifold. Obs. annulus pressure built up to 4300^{psi} in 3 hrs. Opened the well & pressure decreased to 600^{psi}. Closed the annulus. Pressure rapidly built up to 3300^{psi}. Opened the well again. Pressure suddenly dropped to zero. Flew the well for 10min. Closed the well, built up to 3300^{psi}. Monitored the well for1 hrs. Pressure built up to 4300^{psi}.

Due to jammed stem of inside SOV of THS in close position, installed pressure gauge on annulus to choke manifold line & pressure test same to 7000^{psi}, Ok. R/U slickline equipment. Installed slickline BOP & lubricator on XMT cap. Tried to pressure test, Obs. leakage between lubricator & BOP. Changed lubricator O-ring & tried to pressure test; No-Success. L/D previous BOP. Installed another slickline BOP & Lubricator on XMT cap. Pressure test slickline BOP & lubricator w/ 7000^{psi} to 15min; Ok. RIH w/ 3.4" gauge cutter to TBG HGR, tried to pass this depth; No-Success. Pulled out to surface & R/D slickline BOP & lubricator.

Installed 4 – 1/16" 10k lubricator on XMT. Pressure test to 7000^{psi} for 15min; Ok. RIH w/ BPV retrieving tools to TBG HGR & retrieved 4" BPV f/ TBG HGR. L/D 4 – 1/16" 10k lubricator. Due to unable to open TR3SV, decided M/U 5" DPs. R/D slickline equipment.

R/U slickline equipment. Pressure test; Ok. Fixed CL port exit block. Ran 3.4" gauge cutter to 206^{ft}, RIH w/ 3.437" protection sleeve to 190^{ft}, opened TR3SV while holding pressure to 1200^{psi}. Applied 7500^{psi} pressure in CL & opened TR3SV. Cont. running tools to 200^{ft}, took weight, tried to set protection sleeve by jarring several times; No-Success. Flew the well with 1/4" choke size & Cont. work on set protection sleeve; No-Success. Pulled out tools to 191^{ft}. Obs. stuck tools. Worked on stuck to max. 60^{lb} over-pull; No-Success. Finally flew the well by annulus & tubing. Tools got free. Cont. pulled out tools to 50^{ft} & stuck tools again. R/D lubricator. Obs. cut the slickline wire from 50^{ft} & left in hole about 3^{m} tools & 13^{m} wire. RIH w/ 2.5" LIB to 157^{ft}, POOH to surface; Obs. wire mark on same.

While wait on OEOC slickline service, RIH w/ 2.8" spear to TOF. Worked on fish & retrieved 2.45^{m} of wire after 3 attempts. RIH w/ 2.8" LIB to 157^{ft}, POOH to surface. Obs. rope socket marked on same. RIH w/ 2.8" blind box, POOH to surface. Rope socket marked on same. Welded 3.2" skirt on 2.2" JDC, RIH w/ JDC to TOF. Engaged w/ same. Tried to pull out, Obs. indication of shear JDC Shear pin. POOH to surface. Obs. sheared JDC Shear pin. Redressed w/ steel shear pin. RIH w/ JDC to TOF again. Engaged & worked on fish, Pulled to surface. Obs. retrieved 10^{ft} stem of fish. RIH w/ 2.8" LIB to TOF, POOH to surface. Obs. slickline mechanical jar on the same. RIH w/ JDC to engage w/ jar. Worked on fish, jarring up. Pulling out fish 20^{ft}. Cont. worked on stuck. RIH w/ blind box & knock on TOF, without progress. POOH to surface. RIH w/ JDC to engage w/ Jar. Worked on fish, jarring up. POOH to surface. R/D slickline equipment.

In order to work on fish w/ coiled tubing, wait on receive coiled tubing unit. While waiting on OEOC slickline service, R/U CT & M/U 2 – 1/8" dimple connector on 1 – 1/2" CT pipe & performed pull test to max 20^{klb}. Obs. failure in dimple connector, disconnected same. M/U 1 – 11/16" roll on connector on 1 – 1/2" CT pipe & performed pull test to max 20^{klb}, Ok. Installed CT riser on XMT & CT BOP. Close CT blind rams & tried to perform pressure test; No-Success. Tightened between two risers flange & tried to perform pressure test; No-Success. P/U riser & BOP; Tried to perform pressure test; No-Success. Changed 1 Jt CT riser & performed pressure test CT BOP & risers, Ok. RIH CT BHA to 48^{m} & engaged w/ the fish; Obs. stuck the fish. Worked on stuck to max 18^{klb} over-pull; No-Success. While shearing JDC Shear pin @ 2 times & POOH to surface & redressed w/ brass shear pin.

In the 3rd attempt, caught the fish & worked on same upward to 18^{klb} over-pull while bleeding off annulus pressure f/ 4800 to 2000^{psi}. Obs. fish got free, jumped same & sheared the JDC Shear Pin. POOH to surface. Cut 12^{m} 1 – 1/2" CT pipe & changed roll on connector. Performed pull test to 20^{klb}; Ok. RIH w/ 1 – 1/2" CT & 3.2" JDC to 48^{m}. Worked on fish & POOH to surface. In the last attempt caught, engaged the fish & bleed off annulus pressure f/ 4800 to 2000^{psi}. Pulled out f/ 48 to 10^{m} w/ 2 – 3^{klb} drag. Obs. took

over-pull. Worked on stuck upward to max 18^{klb}; No-Success. Then Obs. CT got free & POOH to surface. Closed swab valve & unable close TMV. Obs. sheared JDC Shear pin & TOF located in TMV. Installed 1 No 4 – 1/16" 10k gate valve on XMT. Redressed JDC shear pin to steel & M/U on CT BHA. Tightened all bolts. Performed pressure test to 6000^{psi} for 10min; Ok.

Due to refused work on fish & stopped operation from OEOC CT service, wait on CT Service to make a decision. Changed JDC shear pin to brass. RIH w/ 1 – 1/2" CT & 3.2" JDC to 11^{m} & engaged w/ the fish. Worked on stuck upward to max 18^{klb} over-pull; No-Success. Slack off 6^{klb} weight on JDC & sheared JDC Shear Pin. POOH to surface. R/D CT injector head. R/U slickline equipment & worked on fish w/ JDC jarring up several times; No-Success.

Performed injectivity test & monitored the well & checked communication between string & annulus; Ok. Pumped & direct Circ. 30^{bbl} fresh water by applying back pressure in order to check communication between string & annulus; Ok. Commenced killing the well by W&W method, "pumping pressure f/ 5500 to 3600^{psi}, annulus pressure f/5500 to 2400^{psi}". Meanwhile R/D slickline equipment & CT raiser.

In order to work on fish closed 3SV & released above pressure, M/U & RIH with 3 – 1/2" handmade JDC & tried to engage with fish several times, in the last attempt successfully engaged with fish & recovered w/ max 10^{klb} over-pull. Operate 3SV; Ok. Monitored the well pressure, observed wellhead pressure decreased f/ 500 to 300^{psi}. Bleed off to zero & remained constant. Annulus pressure decreased f/ 2800 to 2000^{psi}, bleed off to 1800^{psi} & remained constant. Performed cavity pressure test to 4000psi for 15min; Ok. Pumped & direct Circ. 125^{bbl} 137^{pcf} mud, "pumping pressure f/ 1300 – 2800^{psi} to 1700^{psi} & annulus pressure 2400^{psi}". Got 78 to 85^{pcf} gas cut mud in return. Stopped pumping.

Bleed off pressure f/ 2200^{psi} to zero step by step & monitored the pressure for any build up in each step; Negative. Wellhead pressure decreased to zero & remained constant. Pumped & direct Circ. 125^{bbl} 145^{pcf} mud. Stopped pumping. Bleed off pressure f/ 700^{psi} to zero step by step & remained constant. Pumped 20^{bbl} 135^{pcf} mud to annulus. Obs. wellhead & annulus pressure decreased to zero. Opened cap of 3/8" CIL & monitored for 1 hr. Ok. Cut the CIL. Closed 3SV & removed XMT cap. Installed TWCV in TBG HGR & test same to 5000^{psi} for 15min; Ok. N/D XMT while passing CIL thru BTM section.

N/D XMT & N/U 13 – 5/8" 10k BOP stack. Connected hydraulic, choke & kill lines. L/U cement line to BOP & performed BOP pressure test as NIOC policy; Ok. Passed CL f/ BOP stack to rig floor. Installed bell nipple. Changed annulus to C/M line, tested same; Ok. P/U & M/U 5" DP, XOS (4 – 1/2" IF * 4 – 1/2" TPCQ), 4 – 1/2" TBG HGR running tools. Retrieved TWCV. Screw TBG HGR running tools + X.O.S (4 – 1/2"

TPCQ * 4 – 1/2" IF) + 1 Jt 5" DP + 1 No. Pup Jt on TBG HGR. Performed pressure test of running tool O-ring to 1000^{psi}, Ok. F/U string, opened 3SV & monitored the well, Ok. Back off running tools & R/D same. Close 3SV. R/U Golpa flange, BOP & lubricator of slickline (OEOC Co.), performed pressure test same w/ 1000^{psi}. Obs. leakage between Golpa flange & 4 – 1/16"gate valve. R/D 4 – 1/16" gate valve, changed ring gasket, pressure tested same again to 1000^{psi}, Ok. Ran 3.4" gauge ring to 61^{m} (3SV depth), Obs. drag @59^{m}, pulled out to surface. Ran spear to 61^{m} (3SV depth) to catch possible cable in the well, pulled out to surface, Obs. No cable. Ran w/ protection sleeve to 61^{m} (3SVdepth), held 7500^{psi} pressure inside CL, set protection sleeve inside 3SV, retrieved protection sleeve & pulled out to surface. B/O slickline BOP. R/U 4 – 1/16" gate valve on Golpa flange, M/U slickline BOP & lubricator & performed pressure test same w/1000^{psi}, Ok.

Ran w/ protection sleeve to 61^{m}, held 7500^{psi} pressure inside CL, set protection sleeve inside 3SV & pulled out to surface. Ran w/ 2.230" gauge cutter to 61^{m} (3SV depth), took weight, tried to pull out, Obs. slickline string stuck, tried to pass this point for several times, No Success. Pumped 5^{bbl} 138^{pcf} mud wile working on stuck, No success. Increased CL pressure to 10,000^{psi} step by step, Obs. slickline string got free. Pulled out gauge cutter to surface & L/D same. Ran with GS pulling tools of protection sleeve & retrieved same. Pulled out protection sleeve to surface. Ran back in hole w/ protection sleeve to 61^{m} (3SV depth) & tried to set protection sleeve for several times, No Success. POOH separation sleeve to surface. Decided to extend protection sleeve, RIH w/ new protection sleeve to 61^{m}, set protection sleeve inside 3SV. RIH w/ 1 – 7/8" stem to 80^{m}, checked opening of 3SV flapper, Ok, POOH stem to surface. RIH w/ 2.230" gauge ring to 3788^{m} (first CIV depth), due to limitation of slickline cable tension decided to POOH to surface. B/O & L/D same.

R/D slickline BOP, lubricator, upper & lower sheaves. Installed X.O.S (4 – 1/16" flange * 4 – 1/2" IF box) on gate valve, pressure tested same w/ 1000^{psi}, Ok. R/D mouse hole, R/U wireline equipment, P/U & M/U CCL & sinker bar. Pumped 100^{bbl} 137^{pcf} mud in string while applying 600 – 700^{psi} pressure, back pressure in annulus. Obs. the well, Ok. B/O X.O.S (4 – 1/16" flange * 4 – 1/2" IF box) & cut box connection, welded 7" new Vam box connection on top of 4 – 1/16" flange & installed same on 4 – 1/16" gate valve. Installed wireline BOP & pressure tested same w/ 1000^{psi}, Ok. Closed Hydrill, unscrew T.H.S tie down, P/U string, recorded string weight, applied allowable over pull to 298^{klb}, got back on 211^{klb}, according to limitation, applied 4$^{klb.ft}$ torque & tried to unlatch, No Success. RIH with CCL & sinker bar to 3850^{m} (5^{m} below PKR depth), POOH CCL & Sinker bar to surface & L/D same.

Held PJSM Assembled & armed colliding tools, RIH to 3850^{m}, spaced out to 3845^{m}, performed colliding, POOH colliding tools to surface, Obs. running tools of colliding

tools & CCL left in hole. Pumped 115[bbl] 137[pcf] mud, Cont. pumping 685[bbl] 106[pcf] mud while dumping return mud, after pumping totally 800[bbl] mud, get 88[pcf] mud in return, divert return to poor boy degasser, Cont. pumping 106[pcf] mud, after pumped totally 850[bbl] mud, got 97[pcf] mud in return, all the time applied 700 – 1000[psi] back pressure in annulus. Opened T.H.S tie down screw, closed Hydrill, pressure tested tie down in open position w/ 1000[psi], Ok. B/O & R/D 4 – 1/16" gate valve, Gol. Flang, landing Jt & TBG HGR running tools. P/U & M/U TBG HGR running tools, X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) & 2 Jts DP as landing Jt, screw running tools on TBGHGR, tried to release anchor seal with maximum 120[klb] overpull & applied 5[klb.ft] torque as forward for several times, No Success. B/O & L/D 2 Jts DP, X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) & TBG HGR running tools.

P/U & M/U TBG HGR running tools, landing Jt, Gol. Flange, 4 – 1/16" G.Flange & X.O.S (4 – 1/16" flange * 7" N. Vam), performed break Circ., NL-NG. Obs. the well, Static. R/U wireline equipment (Sealand Co.) & performed pressure test of wireline BOP w/ 1000[psi], Ok. RIH w/ CCL & sinker bar to 3786[m], took weight, tried to pass this point for several times, No Success. POOH to surface, changed CCL, RIH w/ 1 – 7/8" CCL & sinker bar to 3786[m], took weight, tried to pass this point for several times, No Success. POOH to surface. L/D wireline equipment. L/D 4 – 1/16" X.O.S (4 – 1/16" flange * 7" N. Vam), 4 – 1/16" gate valve, Gol. Flange, landing Jt & TBG HGR running tools, closed shear rams & performed break circulation, NL-NG. M/U TBG HGR running tools, X.O.S (4 – 1/12" TPCQ * 4 – 1/2" IF), 2 Jts 5" DP as landing Jt & screw on TBG HGR, at the beginning of P/U string, Obs. 270[klb] on weight indicator, while pulling the string, indication of EXP.Jt movement not observed (expansion joint is in full open position). Due to condition of EXP.Jt unable to follow releasing mechanism of Anchor seal due to TBG HGR is inside of THS unable to rotate.

P/U string 30[cm] to be sure TBG HGR is not inside THS (weight: 290[klb]), set torque limit on 7[klb.ft], rotated & reciprocated string between 290 – 300[klb] weight on weight indicator to transfer TQ for several times, No Success. Released the TQ, P/U string to 400[klb] on weight indicator, to release Anchor seal, No success. Established circulation, Obs. gas cut mud in return, dumped 200[bbl] 50 – 55[pcf] cut mud, after 200[bbl], Obs. 90[pcf] mud in return, divert return to poor boy degasser. Cont. circulation while getting 90 – 80[pcf] cut mud in return & applied 400 – 1400[psi] back pressure. Stopped operation due to poor performance of transfer pump of middle, degasser, desander, desilter tank & also hopper #1 & #2 causing unable to transfer mud to suction tank. Worked on same to rectify problem while holding 1200[psi] back pressure. Resumed Circ. while applying 1200 – 1400[psi] back pressure, got 106[pcf] mud in return. Stopped pumping, Obs. the well, Static. Worked on collided point to get string free with Max. 120[klb] overpull & applying 6[klb.ft] TQ as forward, string got free, string weight on weight indicator: 195[klb]. R/D wireline,

R/U CSG running equipment, POOH completion string while L/D landing Jt, TBG HGR, fluted swage, 3SV w/ long & short flow coupling (Obs. protection sleeve was not set in profile) & totally 35 Jts 4 – 1/2" TBG, meanwhile Circ. across wellhead, NL-NG. Cont. POOH completion string while L/D 94 Jts 4 – 1/2" TBG (totally 294 Jts), while using safety clamps & Circ. across wellhead; NL-NG. Obs. last Jt of string w/ underneath coupling washout, worn & torn in coupling area. Also Obs. 5 No. of cross coupling protectors left in hole & 9 No. of cross coupling protectors parted because of eroded condition. After Jt No. 35 some of cross couplings had been lost, eroded & parted. Obs. some of asphaltene on cross coupling protectors of injection line & slips area of 4 – 1/2" TBGs so need to be washed with steam, which lead delay in POOH completion string.
P/U & M/U 7 – 7/8" Over shot (w/ 4 – 3/4" Basket Grapple Mill Control, 4 – 1/2" Basket Grapple) & Fishing BHA & RIH to 1880^{m}, NL-NG. Estimated TOF: 2706^{m}. Took weight @ 1880^{m}, tried to pass several times; No Success. Established Circ. & pass gently. Cont. wash down to 2660^{m} while trying to pass several intervals w/o wash down; No Success (Asphaltene precipitation indication); NL-NG. Meanwhile got 150^{bbl} 50^{pcf} gas cut mud & dumped same @ 2020^{m}. Established Circ. to uniform Mw in/out on 106^{pcf}, got 300^{bbl} 75 – 95^{pcf} gas cut mud w/ NL-NG. Obs. huge amount of asphaltene precipitation on shakers. Resumed wash down to 2731^{m} while took weight several depth & pass same. Confirmed top of fish @2731^{m}. Tried to catch the fish several times; No Success. Commenced POOH w/ 7 – 7/8" over shot & fishing BHA f/2731^{m} to surface. Circ. CWH; NL-NG. Obs. some scratches on outer body & edge of over shot guide.
R/U slickline equipment & ran 3.8" magnet to 2731^{m}, held up. POOH to surface. 1^{lb} junk recovered (mixed with asphaltene). R/D slickline equipment while waiting on receive 8 – 3/8" Mill. P/U & M/U 12 Jts 6 – 1/2" DC, 3 Jts 5" HWDP & RIH OEDPs to 1250^{m} Jt by Jt. Circ. CWH; NL-NG. POOH string to surface & L/D same. RIH OEDPs by 12 Jts 4 – 3/4" DC & 12 Jts 3 – 1/2" DP. POOH OEDPs to surface & R/B. While Circ. CWH; NL-NG. Difficulty B/O & L/D 1 Jt 5" Pup Jt, 5" FOSV & Circ. swage due to over torque. Tried to B/O 11" plug tester from 1 Jt 5" DP w/ 48$^{klb.ft}$ TQ several times; No Success. Set 11" wear bushing in THS.
P/U & M/U 8 – 3/8" M.M J.Mill, 1 No. 6 – 5/8" Junk Sub, 1 No. 7" Junk Sub, 6 – 1/2" D.Pin, 6 – 1/2" Bit Sub & RIH w/ 4 Stds to 2729.5^{m} & took weight on this depth. Recorded off BTM parameters. Commenced milling obstruction f/ 2729.5 to 2736^{m}; NL-NG. While got 100^{bbl} 80 – 90^{pcf} gas cut mud @ BTM's up. Cont. milling f/ 2736 to 2739^{m}; NL-NG. Circ. BTM's up, over Circ. till clean shakers w/ Hi-Rate; NL-NG. Obs. metal wool on ditch magnet & also CIL cover on shakers. Worked on junk sub several times. Commenced POOH w/ 8 – 3/8" M.M J.Mill & milling BHA f/ 2739 (TOF) to surface. Circ. CWH; NL-NG. Mill Condition: 80% used. Obs. mark of TBG as top of

fish on mill. Both junk subs full of junk including pieces of CIL, cross coupling & metal wool. Recovered 19[lbs] junk with 2 junk subs.

P/U & M/U 7 – 7/8" overshot, dressed with 4 – 1/2" Basket Grapple & 4 – 3/4" mill control & RIH to 2739[m]. Worked on Fish & tried to catch & engage with fish several times. Obs. 20 – 50[klbs] excess weight on M/D & suddenly drop excess weight. Commence POOH w/ 7 – 7/8" overshot & fishing BHA f/ 2739[m] (TOF) to surface, Circ. CWH; NL-NG. Obs. mark of TOF on 3 wickers of basket grapple & also mark of junk on outer body of guide. P/U & M/U 7 – 5/8" overshot (dressed w/ 4 – 1/2" Basket Grapple, 4 – 1/2" Mill Control & Extension sub) & RIH to 2739[m]. While W/D last 2 Stds, tried to catch & engage with fish. P/U string Obs. caught & engaged w/ fish. Worked on fish w/ 280 to 450[klbs] on M/D several times; No Success. Commence POOH overshot to surface. Obs. fishing jar mandrel packing was failed. R/U tubing running equipment, Obs. top of CILs were collapsed, cut same & bleed of pressure. Released overshot & L/D 6-1/2" DC 7-7/8" overshot. Obs. washed out TOF, 4-1/2" basket grapple, 4-1/2" mill control, inside 7-7/8" bowl & broken 1 No. basket grapple wicker.

POOH completion string totally L/D 37 jts 4-1/2" tubing. Obs. twist off string f/ tubing coupling & recovered all cross couplings & CILs. TOF depth at 3078[m]. Performed static flow check. P/U & M/U 8-3/8" M.M.J.mill used & RIH to 1170[m], performed static flow check. Closed annulus preventer & Cir. Thru C/M & uniform Mw in/out on 106[pcf], got 30[bbl] 50 – 90[pcf] gas cut mud while monitored pit level w/ NL.NG. Opened the well & performed static flow check, Ok. Resumed RIH 8-3/8" mill & milling BHA to 3078[m]. Commence milling f/ 3078.2 to 3079.3[m] while Obs. metal wool on magnet. Performed static flow check, Ok. Commenced POOH milling BHA to surface.

P/U & M/U 7-5/8" overshot & RIH to 3079.3[m]. Established Cir. & CBU w/ hi – rate, huge amount of asphaltene observed on shakers & got 20[bbl] 102[pcf] gas cut mud. Tried to catch & engaged with fish. P/U string & Obs. the fish was caught. Worked on fish w/ maximum 390[klbs] on M/D. H/L dropped to 245[klbs]. Commenced POOH fishing BHA to surface w/ NL.NG. Changed TDS elevator & POOH completion string & L/D 8 jts 4-1/2" tubing. Obs. twist of string f/ tubing coupling & also Obs. tubing couplings was worm & eroded. TOF at 3152.5[m].

P/U & M/U 7-7/8" overshot & RIH to TOF at 3152.5[m]. Caught the fish after several attempts, worked on fish w/ maximum 480[klbs] on M/D. Applied different parameters several times, Obs. H/L dropped to 245[klbs]. Commenced POOH string & fishing BHA to surface. Obs. 1 joint 4-1/2" tubing retrieved & twist of tubing coupling. Released overshot f/ 4-1/2" tubing.

P/U & M/U 8-3/8" M.M.J.mill used & RIH to 3159[m]. Took weight at this point, tried to wash down several times, no progress. Wash & ream down to 3161.2[m]. commenced milling from 3161.2 to 3162.6[m] while got 100[bbl] 90 – 100[pcf] gas cut mud w/ NL.NG.

Obs. metal wool on ditch magnet & Asphaltene on shakers. Recovered 20[lbs] metal from magnet. Worked on junk sub several times & commenced POOH milling BHA to surface. 5[lbs] metal, debris & Asphaltene f/ 2 junk subs.
P/U & M/U 7-5/8" overshot & RIH to 3162[m]. washed down last stand & CBU w/ NL.NG. tried to catch the fish. Obs. catch & engage w/ fish. Worked on fish w/ max 400[klbs] on M/D, assembled 2-1/8" CCL, 2-1/8" weight bar & 1-11/16" Wt. bar. R/U wireline equipment, installed top sheave on monkey board & installed BTM sheave to mouse hole. M/U -1/8" CCL, 2-1/8" weight bar & 1-11/16" Wt. bar on rope socket & passed same thru string. RIH CCL & Wt. Bar as dummy run to 3170[m]. took weight, tried to pass this point several times, No success. POOH to surface & R/G equipment. R/U CT & M/U 1 roll on connector & XOS & Golpa flange. M/U XOS on string & CT BOP & performed pressure test. RIH CT to 3610[m] while Obs. drag at 3158[m] & 3213[m], cleared same. Took weight at 3610[m], tried to pass this point several times w/ applying max. 2000[lb] weight, No success. POOH CT to surface.
R/U Sealand wireline equipment. RIH wireline 2-1/8" CCL & 2 No. 2-1/8" weight bar as dummy run to 3222[m], Obs. drop tension. Passed this point & RIH to 3226[m], Obs. drop tension. Tried to pass this point several times, No success. POOH to surface & L/D assembly. Ran back wireline w/ 1-11/16" CCL & 3No. 1-11/16" weight bar as dummy run to 3612[m], took weight & tried to pass several times, No success. POOH & L/D assemblies. Assembled & RIH wireline w/ 1-3/8" FPIT & CCL tools to 3600[m] meanwhile checked the string w/ applied 170[klbs] overpull. POOH wireline assey. to surface & L/D same. FPIT results as bellow:

- 1000 & 2000 & 3100[m]: 100% free
- 3200 & 330[m]: 50% free
- 3400[m]: 25% free
- 3420[m]: 10% free
- 3440 & 3450[m]: 5% free
- 3460, 3470, 3490, 3500 & 3600[m]: 0% free

Assembled & RIH w/ 1-7/8" colliding & CCL tools to 3612[m]. Obs. drop tension. POOH to 3594[m] & correlated depth w/ CCL, applied 180[klbs] overpull on string. Shot the colliding, checked & worked on string, No success. POOH wireline to surface. Again assembled & RIH w/ 1-7/8" colliding & CCL tools as run #2 to 3398[m] while correlating depth. Applied 180[klbs] overpull on string & shot the colliding. No indication Obs. worked on string, No success. POOH wireline to 3275[m], wireline got stuck, worked on stuck several times to get free. Continued POOH to surface & L/D colliding Assy. Worked on string upward w/ max. 180[klbs] at neutral point several times, No success. Assembled &

RIH wireline w/ 2-1/8" CCL & 1 No. 2-1/8" weight bar as dummy run to 3272^{m}, Obs. drop tension, tried to pass this point, No success. POOH to surface.
RIH w / 2-1/8" colliding & CCL tools as run #3 to 3290^{m}, Obs. drop tension, POOH to 3140^{m} & ran back to 3281^{m}, correlated depth & applied 200^{klbs} overpull on string, shot colliding. Obs. indication of shot, POOH wireline to surface & L/D wireline equipment. Worked on string upward w/ max. 160^{klbs} overpull f/ neutral point to max. 50^{klbs} overpull in several times, No success. RIH wireline w/ 2-1/8" colliding & CCL tools as run #4 to 3185^{m}, correlated depth & applied 200^{klbs} overpull on string, shot the colliding & Obs. indicator. POOH to surface & L/D equipment & R/D unit. Worked on string w/ max. 160^{klbs} overpull upward & downward several times, string got free. POOH string f/ 10^{m} while Obs. 80^{klbs} overpull & abnormal high pressure. CBU & balanced Mw in/out, Obs. huge amount of debris & tubular coating & Asphaltene on shakers, got 300^{bbl} cut mud min 92^{pcf}. POOH to surface, Obs. 7-5/8" overshot & retrieved 21.1^{m} of 4-1/2" tubing f/ collided point & L/D same. Estimated TOF at 3183.1^{m}.
P/U & M/U new 8-3/8" M.M.J. mill & RIH milling BHA to 3182.5^{m}. Tagged TOF. Commenced milling the fish f/ 3182.5 to 3185.8^{m}. worked on junk sub several times, pumped 60^{bbl} hi vis pill & CBU till cleaned up shakers. POOH milling BHA to surface. P/U & M/U 7-7/8" overshot & RIH to tag the fish at 3185.8^{m}, CBU (pick – up weight: 246^{klbs}, Slack weight: 242^{klbs}). Tried to engage with fish, Obs. overpull. Worked on string by applying different parameters several times, string got free. Commenced POOH the string & fishing assembly to surface. Obs. 7-7/8" overshot with 4-1/2" tubing fish neck + 8 Jts 4-1/2" tubing + 73.318 + 9.268^{m} of 4-1/2" tubing w/ CIV control line protector & L/D same. (TOF at 3270.77^{m})
P/U & M/U 8-3/8" M.M.J. mill & RIH to 3203^{m}, took weight, tried to pass this point several times, No success. W & R/D to 3269.5^{m} while CBU, Obs. 65^{pcf} cut mud w/ NL.NG. Tagged TOF & commenced milling the fish f/ 3269.5 to 3274^{m}. worked on junk sub several times, pumped hi-vis pill & CBU w/ NL.NG. POOH Milling BHA to surface & L/D same.
P/U & M/U 7-7/8" overshot & RIH to TOF at 3274^{m}. Engaged with fish & applied different parameters till string got free. POOH overshot to surface. Obs. 4-1/2" tubing fish neck, 6.7^{m} + 2 Jts 4-1/2" tubing (18.58^{m}) + cut coupling 4-1/2" tubing (0.14^{m}) w/ CIV control line protector & L/D same. P/U & M/U 8-3/8" used M.M.J mill & RIH to 3297.5^{m}. Took weight, tried to pass this point several times, NO success. Tag the fish at 3297.5^{m} & commenced milling the fish to 3298.8^{m}. POOH milling BHA to surface & L/D same. Retrieved 6^{lb} junk while Circ. cross wellhead. P/U & M/U 7-5/8" overshot & RIH to TOF. Caught the fish & worked on string by applying different parameters several times.

R/U Sealand wireline unit & RIH w/ 2-1/8" CCL & 1 No. 2-1/8" weight bar & 1 No. 1-11/16" weight bar as dummy run to 3318, 3333 & 3385^{m}. Obs. drop tension at same depths. Several tries to pass depth 3386^{m} without any succeed, POOH wireline to surface & L/D Assy. Ran wireline 2-1/8" colliding & 1-11/16" CCL to 3385^{m}, Obs. drop tension, correlated the depth & applied 180^{klbs} overpull on string. Fired the colliding shots. POOH to surface & L/d equipment. Worked on string to get free. POOH 60^{m} of string w/ 7-5/8" overshot to surface & L/D same.
P/U & M/U used 8-3/8" M.M.J mill & RIH to 3260^{m}. Washed down to 3300^{m}, hard wash down to 3306^{m} while string stall several times. Continued medium wash & ream down to 3384^{m} & hard wash & ream down to 3390^{m} while got 240^{bbl} gas cut mud & huge amount of Asphaltene & CIL cover on shakers. Commenced milling obstructions f/ 3390 to 3391^{m} w/ NL.NG. POOH the milling BHA to surface & L/D.
P/U & M/U 7-5/8" overshot & RIH to 3391^{m} to TOF, tried to engage with fish several times, No success. Commenced POOH the string & fishing BHA to surface. P/U & M/U used 8-3/8" M.M.J. mill& RIH to BTM. Took weight at 3390.7^{m}, commenced milling f/ 3390.7 to 3391.9^{m}, POOH 8-3/8" milling BHA to surface & L/D same. P/U & M/U 7-5/8" overshot & RIH to tag the fish at 3391.9^{m}. Due to tubing & casing annulus filled with junk & Asphaltene, unable to guide overshot into the fish. Commenced POOH overshot & fishing BHA to surface. P/U & M/U used 8-3/8" M.M.J. mill & RIH to 3391.9^{m}. Commenced milling the fish to 3392.7^{m}, worked on junk several times & POOH milling BHA to surface. P/U & M/U 8-3/8" M.M.J. mill & 9-5/8" scraper & RIH to 3350^{m}. POOH milling BHA & scraper to surface. R/U wireline equipment & assembled 3-5/8" CCL-GR-CAST. RIH logging equipment to 3350^{m} & logged up to 35^{m}. POOH to surface while logging f/ 3300 to 3150^{m}, R/D wireline equipment & RIH w/ OEDP to 3390^{m} while L/U CMT line & prepare for CMT job.

Side Track 8 – 1/2" Hole Size Summary:

L/U CMT line to string. Pumped 8^{bbl} 75^{pcf} brine as spacer ahead. Mixed & pumped 105^{sxs} (SV: 25.4^{bbl}, SW: 118^{pcf}) CMT "G" Delijan, pumped 1^{bbl} 75^{pcf} brine as spacer behind, displaced w/ 170^{bbl} 108^{pcf} mud. WOC to get hard, performed static flow check. POOH w/ OEDPs. P/U & M/U 8-1/2" TCI bit & RIH to 3130^{m}. Took weight, W & R/D f/ 3130 to 3170^{m} & W/D to 3310^{m}, CBU, Obs. CMT slurry on shakers. Continued W/D to 3347^{m}, took weight. D/O hard CMT to 3350^{m}. Performed positive pressure test by applying 1000^{psi} for 15min. Ok. Commenced POOH 8-1/2" TCI bit & D/O BHA to surface. (bit dull grade: 1-1-WT-A-E-X-NO-BHA).

P/U & M/U used 8-1/2" TCI bit & 9-5/8" scraper & RIH to TOC at 3350^{m}. scrap f/ 3130 to 3350^{m}, 3 times each stand. Pumped & Circ. out 50^{bbl} hi-vis pill till shakers clean.

POOH 8-1/2" TCI bit & 9-5/8" scraper to surface & L/D. (bit dull grade: 1-1-WT-A-E-X-No-BHA)

P/U 1 joint 5" DP & M/& 9-5/8" hydraulic whipstock Assy. & ran to 20^{m}, M/U UBHO & align to whipface. Continued run whipstock Assy. to 3340^{m} while F/U string w/ fresh mud & passed drift thru all tubulars. R/U wireline logging equipment & RIH w/ gyro tools 3317^{m} (UBHO depth). Oriented whipface on 90 deg. POOH gyro tools to surface & R/D wireline equipment. L/U CMT line to string & performed pressure test to 3500^{psi}, Ok. Set whipstock packer at 3339.08^{m} (bottom of whipstock Assy. at 3340^{m}) w/ 3500^{psi} stepwise. Bleed off pressure & performed 75^{klbs} overpull to shear the screws on the rugby mill, L/D CMT line. (top of whipface: 3333^{m}, Btm of Whipface: 3337^{m}). Applied 2000^{psi} pressure through the string for breaking 2 No. of nozzles plug, Ok.

Commenced milling as side track #1 with controlled parameter f/ 3333.3 to 3337.2^{m} w/ NL.NG. Continued milling to 3341.3^{m}. Confirmed KOP at 3338.6^{m} by geologist. B/R & R/D window interval f/ 3333.3 to 3341.3^{m} several times to smooth. Pumped 30^{bbl} hi-vis pill & CBU till shakers clean. Commenced POOH w/ 8-1/2" TCI bit & L/D same.

P/U new DHM Lang Fang, adjusted AKO 1.5 deg. P/U & M/U used 8-1/2" PDC bit, 6-1/8" UBHO & 6-1/2" NMDC, performed shallow test of tools, Ok. RIH to 1500^{m}, while fill up string every 10 STDs, w/ NL-NG. Performed shallow test (w/ GPM: 385, SPP: 1170^{psi}) & took signal, Ok. Cont. RIH to 3337.4^{m} (BTM of Whipface @ 3337.6^{m}), Took weight, POOH to 3332^{m} (Top of Whipface @ 3333.3^{m}), Turned string 90 Deg to right, ran back to 3339.5^{m}, took weight, POOH to 3338.5^{m}, pumped 20^{bbl} Hi-Vis Pill, RIH w/ pumping (GPM: 450, SPP: 2030^{psi}) & rotation (RPM: 20) to 3339.5^{m}, took weight, W&R to 3341.30^{m}, w/ NL-NG. Commenced directional drlg as rotation mode f/ 3341.3 to 3383.5^{m}. Due to Obs. drop inclination, decision was made to POOH, stopped drilling, POOH to 18^{m}, while Circ. cross wellhead, w/ NL-NG. Performed pressure test of 6-3/4" D.H.M & MWD tools w/ 8-1/2" PDC Bit (GPM: 400, SPP: 800^{psi}), Obs. No Signal of MWD. L/D MWD tools, POOH to surface. L/D 8-1/2"PDC Bit, dull bit grading: (1-2-BT-C-X-I-WT/HC-BHA).

P/U & M/U used 8-1/2" B.Bit #02 RR3 & RIH w/ last 6-3/4" D.H.M, P/U new 6-3/4" MWD tools & RIH to 18^{m}, align tools, P/U 6-3/4" Gyro Sub, align tools, performed shallow test (GPM: 400, SPP: 800^{psi} & GPM: 500, SPP: 1200^{psi}), Ok, Cont. RIH directional BHA w/ gyro sub to 3379^{m}. took weight, W/D to BTM at 3383.5^{m} while took survey at 3354.3^{m} (Inc.: 1.1 deg, Az.: 161.4 deg) & 3369.7^{m} (Inc.: 0.9 deg, Azi.: 145.9 deg). POOH to 3377.5^{m}, about 6^{m} off bottom. R/U logging equipment, RIH gyro tools

as wireline to 3358[m] (gyro sub depth), adjust tools face of DHM. POOH wireline to surface & R/D logging equipment. RIH to BTM, resumed drlg as slide mode f/ 3383.5 to 3390.7[m]. Continued drlg as slide mode & rotation to 3415.5[m]. CBU till shakers clean. Commenced POOH on elevator to 3330[m], with good hole condition. Cont. POOH to surface. L/D jar & gyro sub & directional BHA.

P/U new 6-3/4" DHM, adjusted AKO 1.5 deg. P/U & M/U used 8-1/2" PDC bit, M/U 6-3/4" float sub + 6-1/2" UBHO +6-1/2" NMDC, performed shallow test of BHA, Ok. Performed drlg assembly offset. RIH to 3415.5[m] while W/D last stand & filled up every 10 stds. Commenced drlg as rotation & slide mode f/ 3415.5 to 3454[m] while W & R/d every last drilled std before connections. Continued drlg as rotation & slide mode to 3522[m]. POOH with B/R to as wiper trip to 3400[m] with good hole condition. W & R/D to BTM. In order to drop inclination commenced drlg as slide mode f/ 3522 to 3524.5[m] w/ NL.NG. Continued drlg as rotation as slide mode to 3666.5[m]. While drlg string got stall & stuck several times in sliding mode drlg. POOH with B/R as wiper trip to 3530[m] & RIH back to BTM with good hole condition. Pumped & spotted 70[bbl] inhibited hi-vis pill & POOH on elevator to surface while Circ. cross wellhead w/ NL.NG & L/D BHA.

M/U used 8-1/2" PDC bit & drlg BHA w/ 8-1/2" stabilizer & RIH to 3445[m]. Took weight at this depth. RIH with R/D to 3448[m], string got stuck, worked on stuck by jarring up w/ max. 150[klbs] overpull to get free. Continued RIH with hard R/D to 3666.5[m] w/ NL.NG. commenced drlg to 3781[m] w/ NL.NG. String got stuck while B/R last drilled stand. Worked on stuck by applying torque & jerk down to get it free. Stopped drlg due to fault in TDS (frequent rotation loss while drlg). RIH back to BTM & resumed drlg f/ 3781 to 3835[m] w/ NL.NG. Commenced POOH drlg BHA to surface while CBU cross wellhead w/ NL.NG. L/D same. (bit dull grading: 2-5-BT-N-X-1/32"-CT-PR)

P/U & M/U new 8-1/2" PDC bit & RIH with drlg BHA to 3750[m] with good hole condition. Washed & R/D to BTM at 3835[m] & clear tight spots. Commenced drlg with low parameters to 3891[m]. CBU in order to take last sample while B/R last drilled stand w/ NL.NG. TD confirmed at 3891[m]. Soft B/R as wiper trip f/ TD to 3840[m], soft R/D to BTM. Circulated out 40[bbl] hi-vis pill to clean shakers. Spotted 130[bbl] inhibited hi-vis pill w/ NL.NG. Commenced POOH 8-1/2" drlg BHA on elevator to surface with good hole condition & L/D same. (bit dull grading: 1-1-BT-N-X-I-NO-TD)

7" Liner Operation Summary:

Closed SBR & VBR top rams, retrieved wear bushing, set plug tester & performed BOP test as NIOC policy, Ok. R/U 7" liner running equipment. P/U & M/U 7" liner equipment & RIH to 49^{m} while M/U swage & break Circ. to confirm float function for flow passing thru, Ok. Continued RIH to 746^{m} w/ NL.NG. Install 7" * 8-1/2" spiralizer. P/U & M/U rotational hydraulic 7" liner hanger Assy., while installing wiper plug on slick joint f/ LG, Ok. M/U circulating swage & established Circ., check POB sealing, Ok. Filled up tie back sleeve w/ oil & grease. M/U TDS on setting tools neck & continued RIH to 3890.5^{m}. Recorded P/U & S/O string weight on M/D (320 / 290^{klbs}). Dropped ball, chased the ball to seat in LDC & increased pressure to 1800^{psi}. held pressure & set 7" liner hanger at 3890.5^{m} by applying 70^{klbs} weight & applied totally 100^{klbs} weight.

L/U CMT line, M/U CMT head on string. L/U CMT line to CMT head, P/U string, increased pressure to 3100^{psi}, Obs. shear ball seat & got returned. Released setting tools, P/U string 0.8^{m} performed pull test & confirmed string is free, M/U CMT head on string. L/U CMT line to CMT head, P/U the string & established circulate thru CMT head manifold by rig pump up to 900^{psi}, while preparing CMT solution. Performed pressure test on CMT line & pump truck & batch mixer, Ok. CBU w/ NL.NG. Pumped 10^{bbl} spacer ahead by rig pump, pumped 10^{bbl} chemical wash, mixed & prepared 397^{sxs} (SV: 95^{bbl}, SW: 118^{pcf}) CMT "G" delijan, pumped 95^{bbl} CMT slurry 118^{pcf}. Displaced CMT with totally 260^{bbl} mud. Obs. shearing after pumped 177^{bbl} mud & 2600^{psi}. Obs. bumping at 3000^{psi} w/ NL.NG. Held pressure as integrity test for 10 min. Ok. Released pressure & check back flow, No return. B/O CMT line & P/U string for 1^{m}, slack off 30^{klbs} on tie back sleeve& set liner hanger top packer. L/D CMT head & check string weight, Ok.

POOH 12 stds to 2788^{m}, flushed the string. Continued POOH 5 more stds. CBU w/ NL.NG. Performed static flow check, resumed POOH w/ 7" liner hanger setting tools to surface. Filled up the well, L/D liner hanger setting tools. Set 11" wear bushing in T.H.S, P/U & M/U 8-1/2" TCI bit Kingdream & RIH to 3127^{m}, took weight at this depth, washed down w/ 320 GPM to 3129^{m}, tried to wash down w/ 10^{klbs} weight without any succeed. Commenced D/O soft to medium CMT f/ 3129 to 3139^{m}, Cont. D/O medium

to hard CMT to 3149.5[m], confirmed 7" liner lap at this depth. Pumped 50[bbl] Hi-Vis Pill & CBU till the hole got clean w/ NL-NG. POOH 8-1/2" TCI bit & D/O BHA to surface & L/D same w/ NL-NG. Bit dull grade 1-1-WT-A-E-I-No-BHA.

P/U & M/U & RIH 6-1/8" MT bit new Kingdream & C/O BHA to 3150.5[m] w/ NL-NG. while washed down, pass 7" liner lap & D/O hard CMT & 7" POB f/ 3150 to 3152.5[m], dressed & smoothed POB interval without rotation several times. Continued RIH to 3784[m], while washed & reamed down to 3840[m], D/O POB, D/O hard CMT to LDC depth at 3853.5[m]. Pumped 30[bbl] Hi-Vis Pill & CBU till the hole got clean. Commenced POOH 6-1/8" MT bit & C/O BHA to surface & L/D w/ NL-NG. Bit dull grade 7-2-BT-(N, M)-E-I-ER-BHA.

P/U & M/U 6-1/8" used Kingdream w/ scraper & RIH to BTM at 3853.5[m] while scraped inside 9-5/8" CSG & 7" liner several times. POOH to surface, B/O & L/D bit & scraping BHA.

R/U & assembled Sealand Co. CCL/GR/SBT/VDL logging tools. RIH wireline logging tools to 3145[m] & logged down to 3852[m]. Logged up f/ 3845 to 3050[m]. Ran back to 3450[m] to repeat the log. Logged up f/ 3450 to 3050[m]. Powered off tools & POOH to surface, R/D Sealand logging tools & equipment.

Dry Test Summary:

P/U & M/U OEOC 9-5/8" dry test tools for 7" liner lap & RIH to 30[m] above liner lap at 3120[m]. Used 62.4[pcf] water as cushion to 2627[m]. R/U dry test surface equipment & L/U to CMT line. Recorded P/U & S/O string weight 230/225[klbs], rotated string 6 rounds forward, S/O 25[klbs] & set 9-5/8" dry test packer at 3119[m]. Obs. hydro-spring tester valve opening indication on M/D. Monitored open period 3hrs. Obs. no bubble & annulus level was static. S/O weight & rotated string 11 rounds forward closed DCIP valve. Recorded close period for 6hrs. & Obs. no bubble. Established reverse circulate & got water cushion as same as F/U volume. Continued reverse circulation till Mw uniformed in &

out to 88^{pcf} & secured the well. After test Obs. pressure increase to 213 psi. Unset 9-5/8" dry test packer w/ normal P/U weight. B/O & L/D dry test surface equipment, meanwhile M/U 5" circulation swage & established circulate w/ NL-NG. Performed static flow check for half an hour, Static. Commenced POOH 9-5/8" dry test assy. to surface, B/O & L/D tools.

6 – 1/8" Hole Size Summary:

Changed BOP top VBR, pressure tested same to 7500^{psi} for 10min. Ok. Set 11" wear bushing in T.H.S. P/U & M/U 6" M.M.J.mill & RIH to 3853^{m}. Commenced D/O LDC & hard CMT f/ 3853 to 3891.5^{m}. Commenced drlg formation w/ 6" M.M.J.mill 3893.5^{m}. Smoothed 7" liner shoe several times. Performed flow check, Static. POOH milling BHA to surface & L/D same w/ NL-NG.

P/U & M/U 6-1/8" new PDC bit Beste & RIH to 3871^{m}, connected TDS & established circulation & displaced the well with fresh 100^{pcf} KCl-Poly-$CaCO_3$ mud until Mw uniformed on 100^{pcf} w/ NL-NG. L/U CMT unit to circulation swage, closed top rams & performed FIT by applying pressure thru string to 7500^{psi} step by step. Monitored & Recorded pressure on choke manifold. FIT @3871^{m}, EMW: 108^{pcf}

Time	03:35	03:40	03:50	03:55	04:10
Rate (bbl/min)	0.5	0.5	0.5	0.5	0.5
Volume (bbl)	1	2	2.2	2.2	2.2
Pressure (psi)	200	400	500	600	700

Resumed RIH to BTM, pumped Hi-Vis Pill & commenced drlg formation f/ 3893.5 to 4003m while B/R & R/D last drilled stand. Obs. gas cut mud & max 82% total gas. Stopped drlg & decision was made to increase Mw to 104pcf by added kill mud to system. Resumed drlg formation f/ 4003 to 4035m w/ NL-NG. POOH w/ hard B/R to 4014m. string got stuck at 4022m & made it free by applying different parameters.

Continued POOH w/ hard B/R to 3870m w/ NL-NG. RIH back while R/D & B/R every stand & checked intervals without RPM to TD. Resumed drlg formation f/ 4035 to 4140m as TD.

Pumped 30bbl Hi-Vis Pill & circulated same out. POOH as condition trip with B/R to 3891m (7" liner shoe) & performed static flow check, Static. RIH back on elevator to BTM with good hole condition. Pumped & circulated out 30bbl Hi-Vis Pill. Pumped & spotted inhibited Hi-Vis & Hi-Dens fresh mud at BTM. POOH to surface & L/D 6-1/8" bit & BHA, bit dull grade: 0-0-NO-A-X-I-NO-TD.

P/U & M/U 4-1/2" liner assy. included float shoe, liner joints, float collar, LDC & RIH to 62m. F/U string & continued RIH to 353m. P/U & M/U PPA rotational hydraulic set/Mechanical release liner hanger assy. included tie back sleeve, LG & top packer, POB & rotational hydraulic hanger. Recorded P/U & S/O hook load 12/12^{klbs}, F/U tie back sleeve w/ oil & grease & Cont RIH 4-1/2" liner on 3-1/2" & 5"DP to 4140m (btm) Set hanger. M/U CMT Head on string. L/U CMT Line to CMT Head, L/U CMT Line to CMT Head. Perform cmt job as per plan. POOH to surf. M/U 6" MMJ Mill RIH w/ 6" MMJ Mill and RIH to 3480m, W/D to 3720m. D/O cmt to 3780.5m. Circ. Btm's up. POOH to 200^{m} while L/D excess tubulars till, Cont. POOH f/ 200m to Surf. L/D excess tubulars in derrick. RIH w/ 3-1/2" J.Mill on 3-1/8" DC & 2-7/8" DP Jt by Jt to 254m, Cont. RIH w/ 3-1/2" J.Mill on 3-1/8" to 3780.5m (4-1/2" L.Lap). commence D/O med. Cmt f/ 3780.5 to 3784m, 'D/O POB, Cont. wash and ream & Clean out inside 4-1/2" LNR to 4030m, Cont. wash and ream & Clean out inside 4-1/2" LNR to 4085m. D/O Med to Hard CMT f/ 4085 to 4102m. Tag LDC @ 4102m. Pump & Circ. Out Hi-Vis Pill. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 4102 to 1150m, 'Cont. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 1150m to surface, RIH w/ open end string & 7" Scraper BHA to 917m, Due to change in plan from client side, POOH w/ OEDPs to surface, RIH w/ Clean out BHA to 450m, Cont. RIH w/ Clean out BHA to 4100m, Cont. Wash & Ream Dn to 4102m, D/O Pump Dn Plug, Wiper Plug & 4-1/2" LNR LDC. 'Cont. D/O Hard CMT to 4134m, Circ. Out Hi-Vis Pill, Spot 20 bbl Non-Solid Brine, POOH to 3100m, Cont. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 3100m to

surface, Change top rams to 4-1/2" x 7" VBR, Press test same, Run CCL/GR/RBT Wireline logging tools to 4120m, POOH to 1900m, Cont. POOH w/ CCL/GR/RBT Wireline logging tools to surface. Run CCL/ 2-7/8" Gun as dummy run to 4120m. RIH w/ 3-1/2" Mill & Clean Out BHA to 2550m, String plug, work on same & Unplug, RIH w/ clean out BHA to 4134m, Circ. Out tandem pill, Spot Non-Solid Brine, POOH to 3350m, Cont. POOH w/ 3-1/2" M.M. J.Mill & Clean out BHA f/ 3350m to surface. Run CCL & 2-7/8" Blank Gun as dummy run to 4134m. RIH w/ 2-7/8" WCP HSD Gun Assy as Run #1 to 4124.7m. Fire the gun. POOH to 100m. Wait on daylight. Perform 2nd run of perforation f/ 4120.7 - 4117.7m to & f/ 4092 to 4088.2 m, Wait on daylight, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 4 to 3587m, Try to pass this depth, RIH w/ Clean out BHA to 1600m, Meanwhile Slip & Cut Drlg line @ 450m, Cont. RIH w/ Clean out BHA f/ 1600 to 4128m, CBU, Spot Non-Solid Brine, POOH to 1750m, Cont. POOH w/ 3-1/2" M.M J.Mill f/ 1750m to surface, Run dummy run to 4080.2m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #4 to 4080.2m. Fire the gun. POOH to 100m. Wait on daylight, Cont. wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4075m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #5 & #6 as per plan. Fire the gun. POOH to Run #6 to 100m. Wait on daylight, 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #7 & #8 as per plan. Fire the gun. POOH to Run #8 to 100m. Wait on daylight. 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 9 & # 10 as per plan. Fire the gun. POOH to Run # 10 to 100m. Wait on daylight. 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 11 & # 12 as per plan. Fire the gun. POOH to Run # 12 to 100m. Wait on daylight. 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 3989.65m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 13. Fire the gun. POOH to surface. R/D WL unit. Press. Test BOP. P/U and M/U 3-1/2" J.Mill and RIH to 4050m

Cont. RIH w/ 3-1/2" M.M.J.Mill and RIH to 4134m (btm) Circ. Btm's up. Spud 25 bbl Comp. fluid @ btm. POOH to 2300m,Cont. POOH f/2300m to surface while L/D exceess tubulars. Perform CCL logging. RIH & L/D 6 Jts 3-1/2" DP & 15 Jts 4-3/4" DC, 'RIH w/ 7" Scrapper to 3286m. Scrap PKR interval 3286 to 3325m. Cont RIH to 3777m. Pump Hi-Vis. Circ. Btm's up, Pump Hi-Vis. Circ. Btm's up. Pump 60 bbl Hi-Vis pill (104pcf, Ferobar free) + 20 bbl Hi-Vis pill (82 pcf brine) followed by Synthetic Completion Fluid and displace the well, Obs. The well, static. POOH and L/D DPs jt by jt to 2700m, 'Cont. POOH w/ 7" scrapper while L/D Jt by Jt f/ 2700 to 770m, Cont. POOH w/ 7" f/ 770m to surface, R/U slick line & TBG running equip. Start RIH 3-1/2" X 4-1/2" comp string to 73m, Cont. RIH 3-1/2" X 4-1/2" comp string to 406m. Run 2.8" gauge ring. Work on rectify lakages of pump truck lines, Cont. RIH 3-1/2" X 4-1/2" comp string to 1470m. While press. Test string @ 406 & 1224m; OK, Cont. RIH w/ Totally 284 Jts 4-1/2" TBG (18.9#, L-80, HSM-3) & 22 Jts 3-1/2" TBG (10.2#, L-80, HSM-3) to 2932m. While Press test completion string @ 2010 & 2932m; "OK". 'Run completion string to 3257m, M/U TR3SV Assy & Perform P/T TR3SV, RIH to 3274m, Perform P/T Completion string & inflow test of flapper, Cont. Run to 3325m, Perform P/T completion string, 'Change L/J, P/T Completion string; OK. Set Protection Sleeve. Ret. check valve f/ SSD. Set Plug in No-Go. Due to Client Request, Decided to repeat string Press test. Ret. Plug f/ No-Go. Set Check valve in SSD. P/T Completion string; OK. Ret. check valve f/ SSD. RIH w/ Plug to top of SSD. In order to record upward Wt.; Obs. Suddenly drop tension. POOH to surface; Obs. About 830m Slick line wire, rope socket, 12 ft stem, 6 ft Mech. Jar, Running tools & 2.75" Plug left in hole. Due to lack of stem; wait on Rec. stem. 'Run LIB, Run 2.8" Wire grab w/o no catch wire, Ret. P.Sleeve, Run 3.2" Wire grab; Recover 2 ft wire, Run 3.2" 3 leg wire grab; Recover fish w/o plug, Set P.Sleeve, Run 2.8" Wire grab. 'Ret. Plug, Run Gauge ring to 3330m, Set Plug & Prong in No-Go, Run Shifting tool to SSD, Set PKR, Perform Pull test, Land TBG HGR, P/T String to 5000 psi, 'P/T tie downs, Ann, Ret. P.Sleeve, Back off & L/D L.J, Install TWCV & P/T same against SBR, N/D BOP Stack, N/U BMV & Related P/T,

'N/U XMT, P/T same, Ret. TWCV, Set P.Sleeve, Ret. Prong & Plug, Ret. P.Sleeve, Install BPV, Close All XMT Valve except BMV & Secure the well. Released the rig.

AZNS – 127 Well Sketch

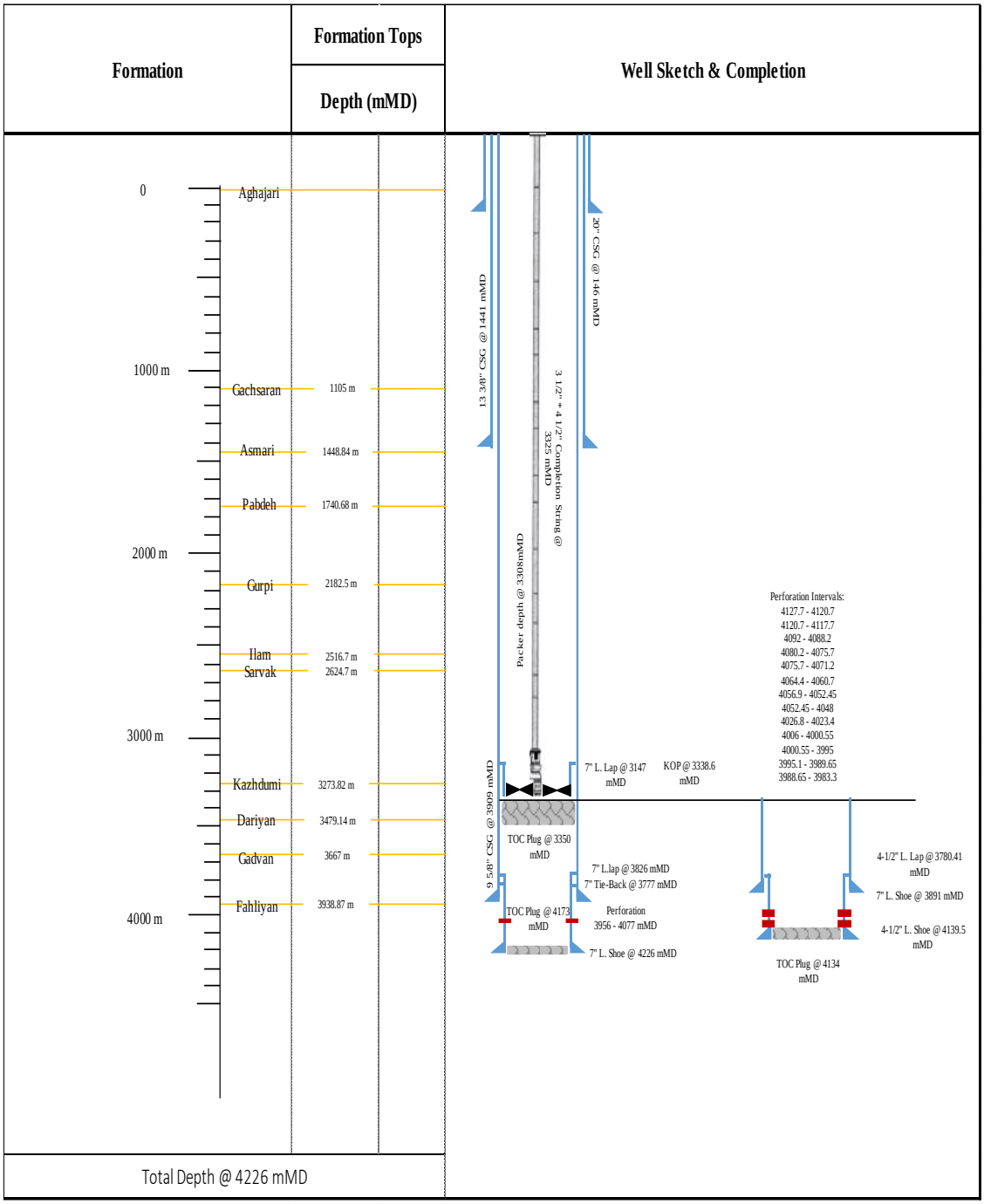


Figure 2-1: AZNS-127 Well Sketch

2.3 AZNS – 127 ROP Chart

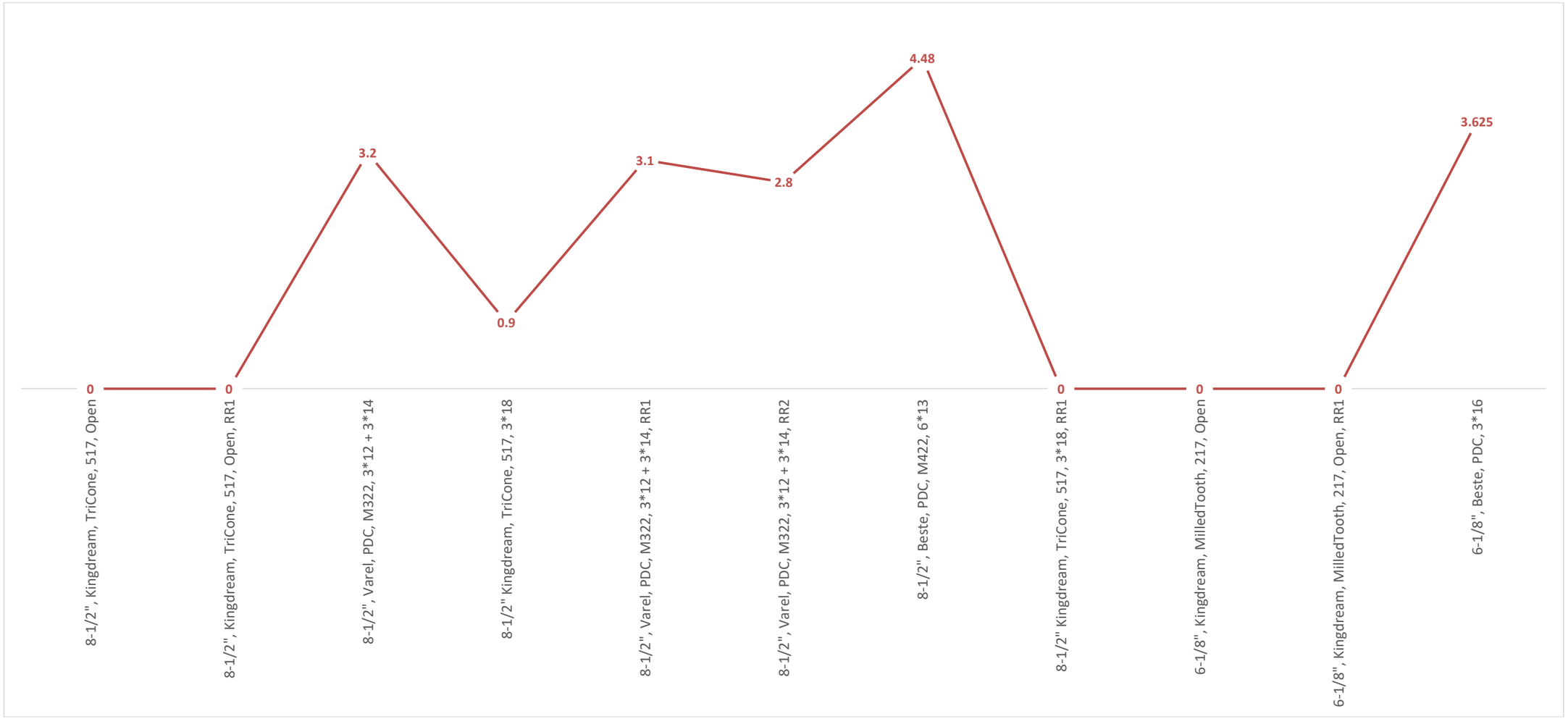


Figure 2-2: AZNS-127 ROP Chart

2.4 Bit Record Table

| | | |

2.5 Bit Parameters Plot

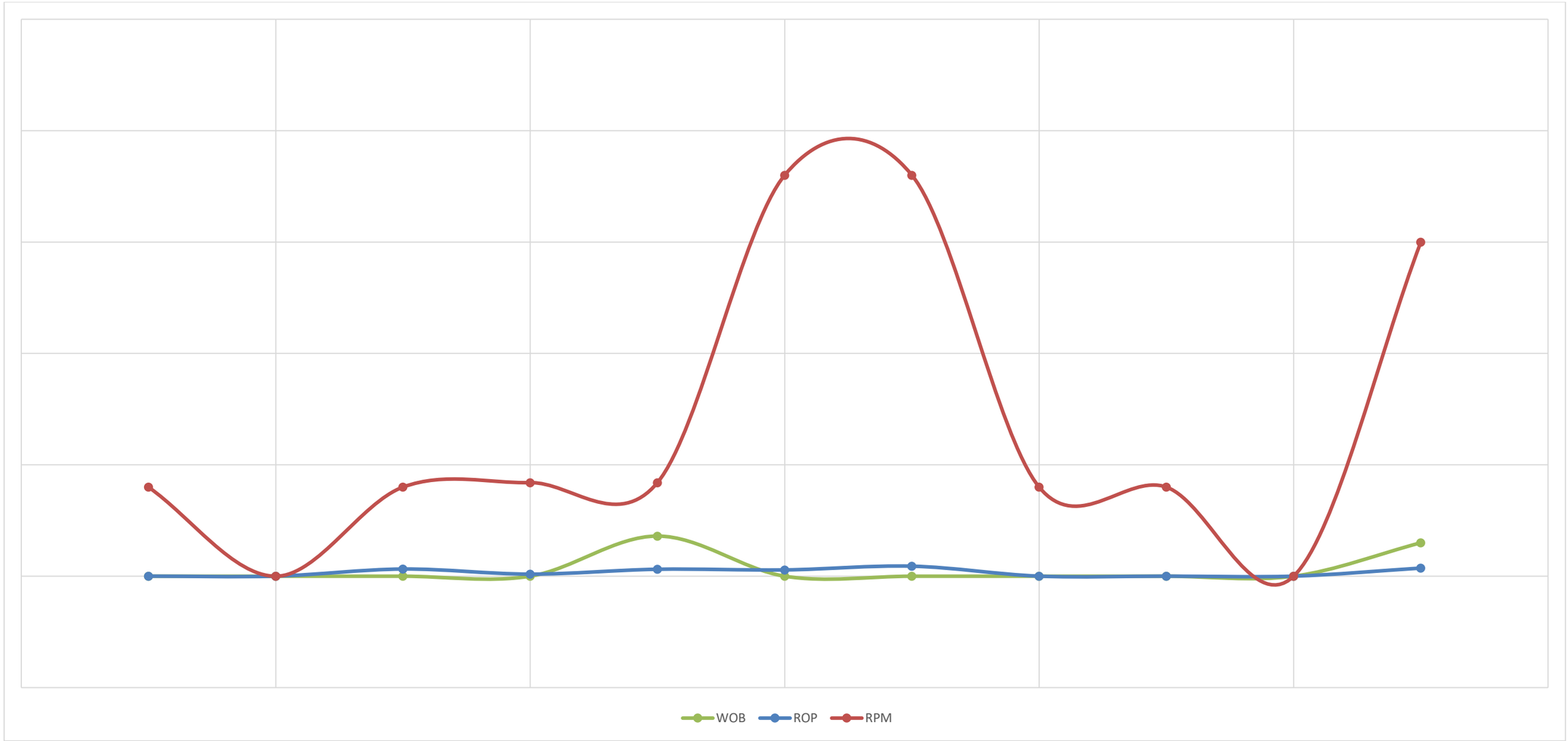


Figure 2-3: AZNS-127 Bit Parameters Plot

3. Time Analysis Report

3.1 Time vs. Depth Chart

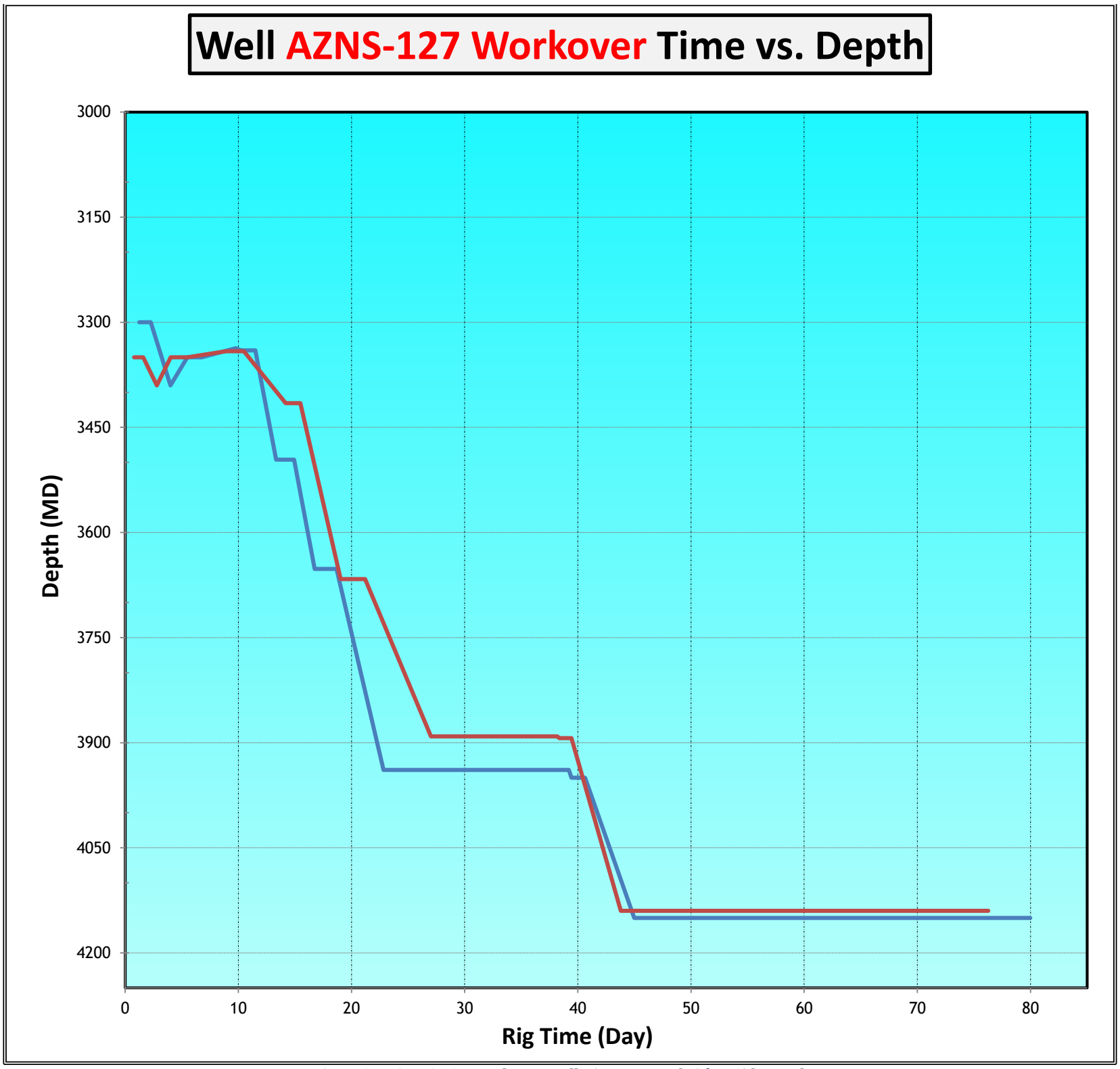


Figure 3-1: AZNS-127 Workover Well Time vs. Depth After Side Track

3.2 Total Time Break-Down of Operation

WorkOver Section

De-Complete, Re-Complete Operation TBD of Well: AZNS-127 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)	Actual Depth (meters)	Time Duration w/ contingency (Actual	Time Duration w/ contingency (Actual	Total Duration w/ contingency (Actual
Work-over													
Held PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC. (Actual: 20/02/1403 f/ 03:00 to 20/02/1403 10:00 & 21/02/1403 00:30 to 10:00)	Trip	4226	48	2.00	2.0	2.2	52.8	2.20	2.20		16.5	0.69	0.69
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.(Actual: 19/02/1403 f/ 12:00 to 20/02/1403 15:00)	Wellhead	4226	48	2.00	4.0	2.2	52.8	2.20	4.4		20.0	0.83	1.5
Secure the well by close 3SV and X-Mass Tree Valves, Remove 7-1/16" XMT cap, R/U Slick line and perform pressure test,RIH w/Gauge Cutter to top of 3SV, RIH & set protection sleeve, RIH w/Gauge Cutter to guide shoe.Note: If unable to pass to the bottom, follow contingency plan #1.(Actual: 20/02/1403 15:00 to 26/02/1403 21:00) Note:Chemical Wash May be Needed while RIH W/CT.	Trip	4226	40	1.67	5.7	1.8	44.0	1.83	6.2		110.5	4.60	6.1
try to displace well with heavy mud weight through SSD, and/or R/U CT and RIH to btm.displace the well with heavy weitgh mud .R/U Slick Line, Retrieve Protection Sleeve, Close 3SV, R/D S/L. (Actual: 24/02/1403 18:00 to 27/02/1403 22:00 & 29/02/1403 00:00 to 02/03/1403 13:00)	Trip	4226	50	2.08	7.8	2.3	55.0	2.29	8.5		164.0	6.83	13.0
Set TWCV inside tubing hanger & pressure test, N/D XMT & N/U BOPs, open tie down screws,perform BOP pressure Test (Actual: 27/01/1403 22:00 to 26/01/1403 21:00) Note: if observe any leakage or damage on THS, refer to next sequence	Wellhead	4226	60	2.50	10.3	2.8	66.0	2.75	11.3		26.0	1.08	14.0
Retrieve TWCV, R/U casing running equipment, RIH with TBH retrieving tool & engage with TBH, apply over pull/Rotation to un-latch the seal assembly. circulate to balance mud weight and flow check before pulling out. (Actual: 02/03/1403 13:00 to 04/03/1403 12:00) Note: if observe any damage on THS, follow contingency plan#2 Note: If unable to un-latch seal assembly, follow contingency plan #3.	Completion	4226	50	2.08	12.3	2.3	55.0	2.29	13.6		47.0	1.96	16.0
Pull out the completion string from anchor seal and lay down completion string precautionary due to probable trapped pressure below plug & Prong. While POOH monitor the well and annulus to make sure the well is safe and over balance. (Actual: 04/03/1403 12:00 to 17/03/1403 18:00 & 18/03/1403 00:00 to 11/04/1403 24:00)	Trip	4226	90	3.75	16.1	4.1	99.0	4.13	17.7		910.5	37.94	53.9
Perform BOP Test. (Actual: 17/03/1403 18:00 to 17/03/1403 24:00)	BOP	4226	6	0.25	16.3	0.3	6.6	0.28	18.0		10.0	0.42	54.4

Table 3-1: AZNS-127 WorkOver Time Breakdown

Side Track Section

De-Complete, Re-Complete Operation TBD of Well: AZNS-127 Work-over (Side Tracking)	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Actual Depth (m)	Time Duration (Actual Hours)	Time Duration (Actual Days)	Total Time Duration (Actual Days)
Work-over												
RIH with 8 1/2" bit and 9 5/8" scraper, scrap 9-5/8" casing prior to Corrosion Log, POOH to surface. (Actual: 12/04/1403 00:00 to 12/04/1403 21:00)	Trip		SV	3300		30	1.25	1.3	3350.0	19.0	0.79	55.2
Change Ram, Install shooting nipple, R/U WL Eq. RIH WL with Corrosion Log to cheack 9-5/8" Csg. , POOH and R/D Logging Eq, change Ram. (Actual: 12/04/1403 21:00 to 13/04/1403 16:00)	Log		SV	3300		24	1.00	2.3	3350.0	19.0	0.79	56.0
R/U & run 5" & 3 1/2" open end drill pipes to 3390m, MD, set 25 Bbls cement as balance plug. POOH (Actual: 13/04/1403 16:00 to 14/04/1403 21:00)	CMT		SV	3390		42	1.75	4.0	3390.0	29.0	1.21	57.2
RIH 8-1/2" bit & dress cement to 3350m, MD, Perform Positive Pressure Test CMT Plug,POOH. (Actual: 14/04/1403 21:00 to 16/04/1403 02:30)	D/O		SV	3350		36	1.50	5.5	3350.0	29.5	1.23	58.4
RIH with 8 1/2" bit and 9 5/8" scraper, scrap whipstock setting depth,& Displace mud in hole by 85 pcf mud, observe the well, POOH. (Actual: 16/04/1403 02:30 to 17/04/1403 14:00)	Side Track		SV	3350		30	1.25	6.8	3350.0	35.5	1.48	59.9
M/U & R.I.H. with 9 5/8" permanent hydraulic set Whipstock (suitable for 9-5/8" casing, 53.5#),Set same & open window @ 3337m. (Actual: 17/04/1403 14:00 to 20/04/1403 24:00)	Side Track		SV	3337		72	3.00	9.8	3341.3	82.0	3.42	63.3
Drill 3m in formation to 3340m,POOH to Surface (Actual: 21/04/1403 00:00 to 21/04/1403 16:00)	Side Track	3	SV	3340		12	0.50	10.3	3341.3	16.0	0.67	64.0
BOP Test.	BOP Test		GD	3340		6	0.25	10.5	3341.3		0.00	64.0
P/U & M/U New 8-1/2" Rock Bit and Directional BHA, RIH to Btm. (Actual: 21/04/1403 16:00 to 22/04/1403 13:30)	Trip		SV	3340		24	1.00	11.5	3341.3	21.5	0.90	64.9
Cont. Drilling 8-1/2" hole f/ 3340m to 3496m. (Actual: 22/04/1403 13:30 to 26/04/1403 08:30)	Drilling	156	SV KZ DR	3496	4	44	1.83	13.3	3415.5	89.0	3.71	68.6
Perform Hole condition POOH to surface. (Actual: 26/04/1403 08:30 to 26/04/1403 24:00)	Hole Cond/Circ		DR	3496		20	0.83	14.2	3415.5	15.5	0.65	69.2
P/U & M/U New 8-1/2" Rock Bit and Directional BHA, RIH to Btm. (Actual: 27/04/1403 00:00 to 15:30)	Trip		SV	3496		18	0.75	14.9	3415.5	15.5	0.65	69.9
Cont. Drilling 8-1/2" hole f/ 3496m to 3652m(DOP). (Actual: 27/04/1403 15:30 to 31/04/1403 08:00)	Drilling	156	DR GD	3652	4	44.0	1.83	16.8	3666.5	85.5	3.56	73.4
Perform Hole condition POOH to Surface to Change BHA (Actual: 31/04/1403 08:00 to 01/05/1403 05:30)	Drilling		DR GD	3652		20.0	0.83	17.6	3666.5	24.5	1.02	74.5
BOP Test.	BOP Test		GD	3652		6.0	0.25	17.8	3666.5		0.00	74.5
P/U & M/U New 8-1/2" PDC Bit and BHA, RIH to Btm. (Actual: 01/05/1403 05:30 to 02/05/1403 09:00)	Trip		GD	3652		20.0	0.83	18.7	3666.5	27.5	1.15	75.6
Cont.D.Drilling 8-1/2" hole f/ 3652m to section TD @ 3939m.(Actual: 02/05/1403 09:00 to 08/05/1403 04:300)	Drilling	287	GD	3939	3	100.0	4.17	22.8	3891.0	139.5	5.81	81.4
CBU, Perform condition trip to last CSG Shoe prior to Run Liner, RIH to Btm, CBU, Spot pill POOH to surface, L/D PDC Bit. (Actual: 08/05/1403 04:30 to 08/05/1403 24:00)	Hole Cond/Circ		GD	3939		36.0	1.50	24.3	3891.0	19.5	0.81	82.2
BOP Test. (Actual: 09/05/1403 00:00 to 09/05/1403 04:00)	BOP Test		GD	3939		10.0	0.42	24.8	3891.0	4.0	0.17	82.4
R/U CSG running gear, Run 7" Liner to Btm, R/D CSG running gear. (Actual: 09/05/1403 04:00 to 10/05/1403 06:00)	CSG		GD	3939		42	1.75	26.5	3891.0	26.0	1.08	83.5
Perform 7" Liner cementing operation. (Actual: 10/05/1403 06:00 to 10/05/1403 14:30)	CMT		GD	3939		12	0.50	27.0	3891.0	8.5	0.35	83.8
POOH w/ L.Hanger setting tool to 2800m, Flush string & CBU. (Actual: 10/05/1403 14:30 to 10/05/1403 16:00)	CSG		GD	3939		6	0.25	27.3	3891.0	1.5	0.06	83.9
POOH to surface & L/D L.Hanger setting tool. (Actual: 10/05/1403 16:00 to 10/05/1403 24:00)	CSG		GD	3939		16	0.67	27.9	3891.0	8.0	0.33	84.2
P/U & M/U 8-1/2" MT Bit BHA, RIH to L.Lap @ +/- 3150m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH. (Actual: 11/05/1403 00:00 to 11/05/1403 24:00)	D/Out		GD	3939		36	1.50	29.4	3891.0	24.0	1.00	85.2
L/D excess 5" DPs/HWDP.	Trip		GD	3939		12	0.50	29.9	3891.0		0.00	85.2
P/U & M/U 4-3/4" DCs & 3-1/2" DPs/HWDPS, R/Back same in derrick.	Trip		GD	3939		12	0.50	30.4	3891.0		0.00	85.2
P/U & M/U 5-7/8" MT Bit BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 3915m while D/Out any cement, work on junk sub and CBU. Perform 7" Liner CIT. (80% of Burst press) with 2000 psi,for 15 min, (If not obs bumping).CBU & POOH to Surface. (Actual: 12/05/1403 00:00 to 14/05/1403 03:00)	D/Out		GD	3939		48	2.00	32.4	3891.0	51.0	2.13	87.4
RIH with 5-7/8" MT Bit and tandem scraper (7" & 9-5/8") to LDC depth, Scrap inside 7" Liner and 9-5/8" casing before Dry test, CBU, POOH to Surface. (Actual: 14/05/1403 03:00 to 15/05/1403 06:00)	Testing		GD	3939		45	1.88	34.3	3891.0	27.0	1.13	88.5
BOP Test. (Actual: 18/05/1403 01:00 to 18/05/1403 03:30)	BOP Test		GD	3939		6.0	0.25	34.5	3891.0	2.5	0.10	88.6
RIH with 9-5/8" Dry test BHA, RIH to L.Lap @ +/- 3150m, Perform 7" Liner lap Dry Test (based on Gadvan Reservoir Datum Pressure), POOH. (Actual: 15/05/1403 18:00 to 18/05/1403 01:00)	Testing		GD	3939		60	2.50	37.0	3891.0	55.0	2.29	90.9
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform CMT quality Logging of 7" Liner from 7" liner LDC @ 3915 m to 7" Liner Lap @ 3150 m (SBT or /CBL/VDL), R/D Logging Eq, change Ram. (Actual: 15/05/1403 06:00 to 15/05/1403 18:00)	Log		GD	3939		24	1.00	38.0	3891.0	12.0	0.50	91.4
P/U & M/U 6-1/8" MT Bit and C/Out BHA, RIH to LDC depth. Displace 7" Liner volume with 100 pcf drilling fluid.	Trip		GD	3939		20	0.83	38.9	3891.0		0.00	91.4
Cont' D/Out CMT, Shoe and Rat Hole. (Actual: 18/05/1403 03:30 to 19/05/1403 08:30)	Hole Cond/Circ		GD	3939		8	0.33	39.2	3891.0	29.0	1.21	92.6
Total Duration of 8-1/2" Hole Section:							***39.21***	***39.2***				***92.6***
Drilling 6-1/8" hole to 3950m. (Actual: 19/05/1403 08:30 to 19/05/1403 12:30)	Drilling	11	FH	3950	2	5.5	0.23	39.44	3893.5	4.0	0.17	92.8
CBU and balance Mw.,perform FIT with Required Pressure 720 psi (EMW: 108pcf, should be equal to max. planned Mw for this hole section plus 2 pcf),POOH to Surface (Actual: 19/05/1403 12:30 to 20/05/1403 09:00 & 20/05/1403 23:00 to 21/05/1403 04:30)	Trip		FH	3950		23	0.96	40.40	3893.5	26.0	1.08	93.8
BOP Test.	BOP Test		FH	3950		6	0.25	40.65	3893.5		0.00	93.8
RIH w/ 6-1/8 PDC bit & Drilling 6-1/8" hole f/ 3950m to well TD @ 4150m. (Actual: 20/05/1403 09:00 to 20/05/1403 23:00 & from 21/05/1403 04:30 to 25/05/1403 09:00)	Drilling	200	FH	4150	3	104	4.33	44.98	4140.0	104.5	4.35	98.2
CBU, Perform condition trip to last CSG Shoe prior to Liner Running, RIH to Btm, CBU, Spot pill. (Actual: 25/05/1403 09:30 to 25/05/1403 16:00 & from 18:00 to 24:00)	Hole Cond/Circ		FH	4150		12	0.50	45.48	4140.0	12.5	0.52	98.7
POOH to surface. (Actual: 26/05/1403 00:00 to 26/05/1403 15:30)	Trip		FH	4150		18	0.75	46.23	4140.0	15.5	0.65	99.4
BOP Test.	BOP Test		FH	4150		6	0.25	46.48	4140.0		0.00	99.4
R/U 4-1/2" Liner running Eq., Run 4-1/2" liner (+/- 311m LNR) to Btm @ 4150m, R/U CMT lines, Set L.Hanger +/- 3793m (100m overlap), release L.Hanger setting tool. (Actual: 26/05/1403 15:30 to 27/05/1403 20:30)	CSG		FH	4150		36	1.50	47.98	4140.0	29.0	1.21	100.6
Cement 4-1/2" Liner as per program. (Actual: 27/05/1403 20:30 to 28/05/1403 03:00)	CMT		FH	4150		8	0.33	48.31	4140.0	6.5	0.27	100.8
POOH w/ L.Hanger setting tool to 3500m, Flush string & CBU. (Actual: 28/05/1403 03:00 to 28/05/1403 05:30)	CSG		FH	4150		6	0.25	48.56	4140.0	2.5	0.10	100.9
POOH to surface & L/D L.Hanger setting tool. (Actual: 28/05/1403 05:30 to 28/05/1403 17:00)	CSG		FH	4150		12	0.50	49.06	4140.0	11.5	0.48	101.4
P/U & M/U 6-1/8" MT Bit BHA, RIH to L.Lap @ +/- 3793m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH. (Actual: 28/05/1403 17:00 to 30/05/1403 02:00)	D/O		FH	4150		40	1.67	50.73	4140.0	33.0	1.38	102.8
L/D excess 3-1/2" DPs/HWDP. (Actual: 30/05/1403 02:00 to 30/05/1403 09:00)	Trip		FH	4150		8	0.33	51.06	4140.0	7.0	0.29	103.1
P/U & M/U 3-1/2" DCs & 2-7/8" DPs/HWDPS, R/Back same in derrick.	Trip		FH	4150		12	0.50	51.56	4140.0		0.00	103.1
P/U & M/U 3-1/8" Mill BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 4126m while D/Out any cement, work on junk sub and CBU. Perform4-1/2" Liner CIT. (80% of Burst press) with 2000 psi,for 15 min, (If not obs bumping).CBU, POOH to Surface. (Actual: 30/05/1403 13:00 to 03/06/1403 07:00 & from 03/06/1403 13:30 to 07/06/1403 11:00 & from 08/06/1403 07:00 to 10/06/1403 24:00 & from 12/06/1403 15:00 to 17:30 & from 19:30 to 14/06/1403 06:00)	D/O		FH	4150		60	2.50	54.06	4140.0	270.5	11.27	114.4
BOP Test. (Actual: 30/05/1403 09:00 to 30/05/1403 13:00)	BOP Test		FH	4150		6	0.25	54.31	4140.0	4.0	0.17	114.5
RIH with 3-1/8" Mill and 4-1/2" scraper to LDC depth, Scrap inside 4-1/2" Liner , CBU, POOH to Surface. (Actual: 03/06/1403 07:00 to 03/06/1403 13:30)	Trip		FH	4150		40	1.67	55.98	4140.0	6.5	0.27	114.8
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform CMT quality Logging of 4-1/2" Liner from LDC depth @ 4126 m to 4-1/2" Liner Lap @ 3839 m (SBT or CBL/VDL), R/D Logging Eq, change Ram.	Log		FH	4150		24	1.00	56.98	4140.0		0.00	114.8
P/U & M/U 3-1/8" Mil and C/Out BHA, RIH to LDC depth. Displace 4-1/2" Liner volume with Non Solid fluid, POOH	Trip		FH	4150		36	1.50	58.48	4140.0		0.00	114.8
P/U & M/U W/L Perforation Assy, RIH to perforation depth, correlate depth with WL and perforate Guns as program.POOH to Surface. (Actual: 07/06/1403 11:00 to 08/06/1403 07:00 & from 10/06/1403 10:00 to 12/06/1403 15:00 & from 14/06/1403 06:00 to 19/06/1403 19:30)	Log		FH	4150		48	2.00	60.48	4140.0	206.5	8.60	123.4
BOP Test. (Actual: 19/06/1403 19:30 to 20/06/1403 02:00)	BOP Test		FH	4150		6	0.25	60.73	4140.0	8.5	0.35	123.8
P/U & M/U 3-1/8" Mill BHA, RIH to top of LDC depth while smoothing Perforated intervals. (Actual: 20/06/1403 02:00 to 22/06/1403 12:00)	Trip		FH	4150		60	2.50	63.23	4140.0	58.0	2.42	126.2
Change Ram, Install shooting nipple, R/U WL Eq. RIH WL with CCL/GR Log to confirm Perforated intervals, POOH and R/D Logging Eq, change Ram. (Actual: 22/06/1403 12:00 to 22/06/1403 20:30)	Log		FH	4150		18	0.75	63.98	4140.0	8.5	0.35	126.5
RIH with 3-1/8" mill & clean out BHA (7" scraper included) to bottom , displace the well with completion fluid (non-solid) $CaBr_2$, POOH & L/D DP's. (Actual: 22/06/1403 20:30 to 22/06/1403 24:00 & from 23/06/1403 00:00 to 26/06/1403 11:00)	Trip		FH	4150		72	3.00	66.98	4140.0	86.5	3.60	130.1
BOP Test.	BOP Test		FH	4150		6	0.25	67.23	4140.0		0.00	130.1

Table 3-2: AZNS-127 Side Track Time Breakdown

Completion Section

Run 10K completion string as per program. NOTE: Annulus pressure test during completion operation is related to corrosion log result. (Actual: 26/06/1403 11:00 to 02/07/1403 24:00)	Completion		FH	4150		168	7.00	74.23	4140.0	181.0	7.54	137.7
Set BPV inside TBH, N/D BOP's & install new 10K XMT. (Actual: 03/07/1403 00:00 to 03/07/1403 24:00)	Completion		FH	4150		48	2.00	76.23	4140.0	24.0	1.00	138.7
Retrive BPV, install TWCV and Pressure test the XMT. (Actual: 04/07/1403 00:00 to 04/07/1403 04:30)	Completion		FH	4150		12	0.50	76.73	4140.0	4.5	0.19	138.9
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig. (Actual: 04/07/1403 04:30 to 04/07/1403 14:00)	Completion		FH	4150		8	0.33	77.06	4140.0	9.5	0.40	139.3
Retrieve TWCV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
RIH with S/L, retrieve protection sleeve.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion		FH	4150		Rig Less	0.00	77.06	4140.0		0.00	139.3
Total Duration of 6-1/8" Hole Section:							*37.9*	*77.1*				*139.3*
NOTE:												
Total Duration for Cut & Slip DL in all hole section. (Actual: 25/05/1403 16:00 to 18:00 & from 12/06/1403 17:30 to 19:30)	Cut & Slip			4150		12	0.50	77.6	4140.0	4.0	0.17	139.4
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service			4150		58	2.42	80.0			0.00	139.4
Total Duration of AZNS-127 Work-over Well:								80.0				139.4

Table 3-3: AZNS-127 Completion Time Breakdown

3.3 NPT Break-Down

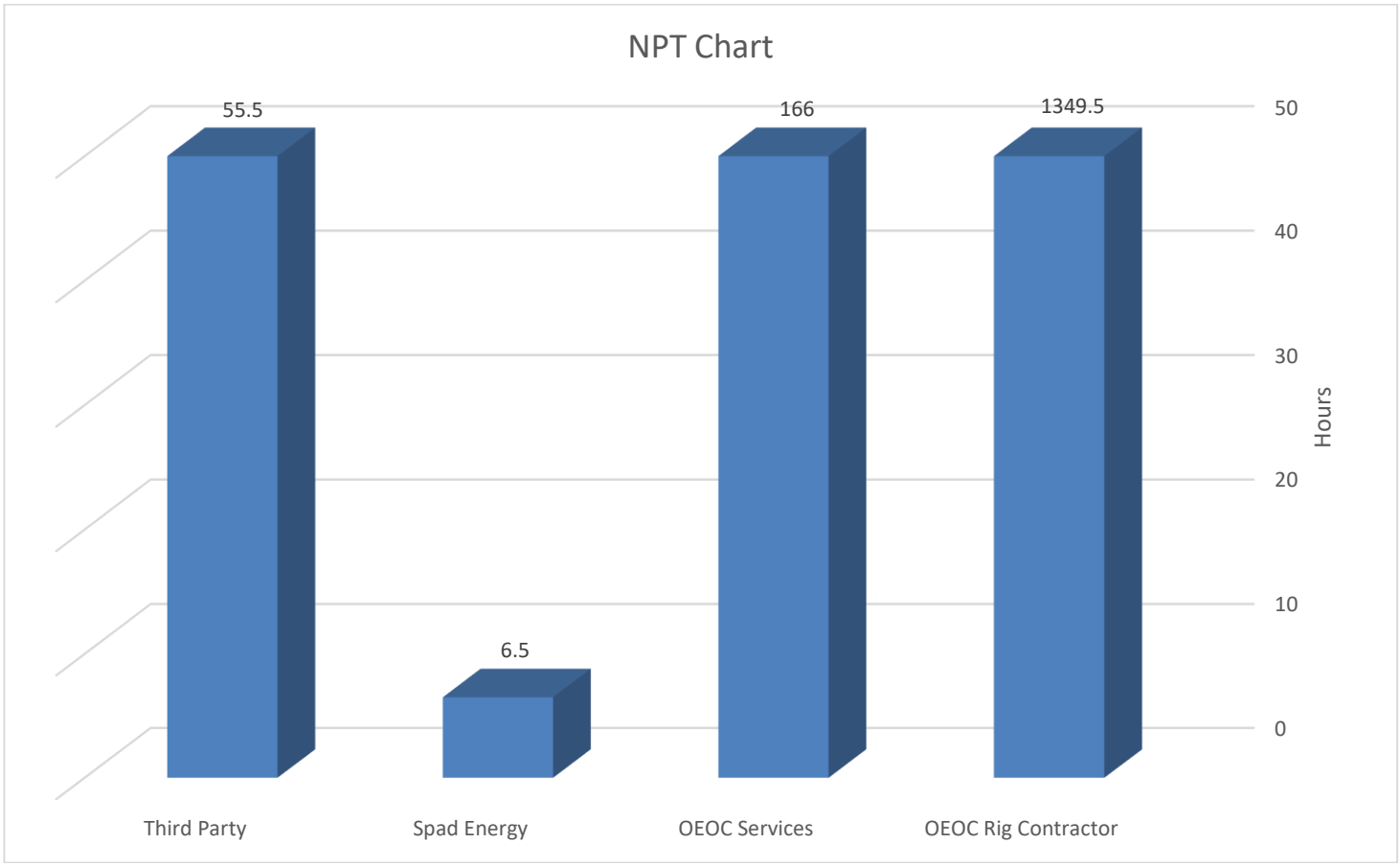


Figure 3-2: AZNS-127 NPT Break-Down Chart

Company	Date	Problem event	Lost Time (hrs)
OEOC Services	1403. 02. 21	Wellhead, Wait on Rec. XMT 2SV cap.	2
		Slickline, Left tools & wire in hole.	10
	1403. 02. 22		24
	1403. 02. 23		24
	1403. 02. 24		24
	1403. 02. 25		24
	1403. 02. 26		24
	1403. 02. 27		24
	1403. 02. 31	Wireline, Wait on daylight.	5
	1403. 03. 01		5
OEOC Rig	1403. 03. 02	Not balance mud weight in & out&try to release anchor seal no succ.	24
	1403. 03. 03	Try release anchor seal no succ.	24
	1403. 03. 04		9
	1403. 03. 07	Delay in operation.	3
	1403. 03. 08		2
		Wait on Rec. 8 – 3/8” J.mill	9
	1403. 03. 09		20
	1403. 03. 10	Recover string operation longer.	24
	1403. 03. 11		24

	1403. 03. 12		24
	1403. 03. 13		24
	1403. 03. 14		24
	1403. 03. 15		24
	1403. 03. 16		24
	1403. 03. 17		24
	1403. 03. 18		24
	1403. 03. 19		24
	1403. 03. 20		24
	1403. 03. 21		24
	1403. 03. 22		24
	1403. 03. 23		24
	1403. 03. 24		24
	1403. 03. 25		24
	1403. 03. 26		24
	1403. 03. 27		24
	1403. 03. 28		24
	1403. 03. 29		24
	1403. 03. 30		24
	1403. 03. 31		24
	1403. 04. 01		24

	1403. 04. 02		**24**
	1403. 04. 03		**24**
	1403. 04. 04		**24**
	1403. 04. 05		**24**
	1403. 04. 06		**24**
	1403. 04. 07		**24**
	1403. 04. 08		**24**
	1403. 04. 09		**24**
	1403. 04. 10		**24**
	1403. 04. 11		**24**
	1403. 04. 12		**24**
	1403. 04. 13		**24**
OEOC Rig	**1403. 04. 19**	Work on TDS.	**21**
	1403. 04. 20		**24**
	1403. 04. 22	Trip for change BHA.	**1**
	1403. 04. 23		**24**
	1403. 04. 24		**12**
	1403. 04. 29	ROP less than program.	**8**
	1403. 04. 30		**7**
	1403. 05. 02	RIH to BM time more than program.	**9**
	1403. 05. 03	ROP less than program.	**7**

	1403. 05. 04		**12**
		Work on TDS.	**3**
	1403. 05. 05	ROP less than program.	**12**
	1403. 05. 06		**7**
Spad Energy Co.	**1403. 05. 10**	Delay in operation, Leakage from pump truck, Plug CMT silo, Unable to turn on batch mixer.	**3**
OEOC Rig	**1403. 05. 12**	Bad weather & wind speed.	**7**
	1403. 05. 23	Wash pipe failure.	**2**
		Work on mud condition.	**10.5**
	1403. 05. 24		**1**
Spad Energy Co.	**1403. 05. 27**	Delay in operation, Leakage from pump truck.	**3.5**
OEOC Rig	**1403. 05. 30**	Bad weather & wind speed.	**1**
		Delay in operation.	**10**
	1403. 05. 31	String plugged.	**2**
	1403. 06. 03	Out off program.	**12**
	1403. 06. 08	Not pass string according to settling.	**23**
	1403. 06. 09		**24**
	1403. 06. 10	Not pass logging tools according to settling, wait on daylight.	**14**
	1403. 06. 11		**11**
	1403. 06. 12		**18**
	1403. 06. 13		**24**
	1403. 06. 14		**22**

	1403. 06. 15		**14**
	1403. 06. 16		**13**
	1403. 06. 17		**13**
	1403. 06. 18		**13.5**
	1403. 06. 19		**11.5**
OEOC Rig	**1403. 06. 23**	Bad weather & wind speed.	**4**
Third Party	**1403. 06. 27**	Pump truck failure.	**8.5**
	1403. 06. 28		**3**
OEOC Rig	**1403. 06. 30**	Delay for pressure test of completion string.	**5**
	1403. 06. 31		**6**
		Change program due to unsuccessful pressure test of completion string.	**10**
Third Party		Test truck failure.	**3**
		Sl tools left in hole.	**5**
	1403. 07. 01		**24**
	1403. 07. 02		**12**
Total NPT (hrs.)			**1577.5**
Total NPT (days)			**65.73**

Table 3-4: AZNS-127 NPT Time BreakDown

4. Geological & Logging Reports

4.1 Logging and Testing Reports

Run	Date	Depth Interval (m)	Log Suite
1		3050 – 3849	SBT – VDL – GR – CCL (7” liner)
2		3750 – 4118.8	RBT – VDL – GR – CCL (4 – 1/2” liner)
3		3355.15	Gyro + Motor (9 – 5/8” CSG, 5 shots)
4		3317	Gyro – Whipstock (9 – 5/8” CSG, 6 shots)
5	28 – 02 – 2024	Surface to 3030	Dummy run [CCL + 2* Gun]
6	01 – 03 – 2024	4076 to 3930	SBT – VDL – GR – CCL (repetition)
7	01 – 03 – 2024	4075 to 2860	CBIL – GR – CCL [liner 7"]
8	03 – 03 – 2024	4061 to surface	CCL – GR (perforation check)

Table 4-1: AZNS-127 Logging Information

5. Bottom Hole Assembly Record

Tool Description	Length (m)	Cum. Length (m)
BHA No. #1		
Overshot	1.18	1.18
6 – 1/2” DC	9.60	10.78
Fishing Jar	2.96	13.74
5” HWDP	56.02	69.76
Total Length (m)		**69.76**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #2		
8 – 3/8” Junk Mill	1.15	1.15
6 – 5/8” Junk Sub	1.08	2.23
7” Junk Sub	0.91	3.14
6 – 1/2” D. Pin	0.67	3.81

6 – 1/2” DC	109.38	114.42
5” HWDP	83.55	197.97
Total Length (m)		**197.97**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #3		
7 – 7/8” Overshot	1.18	1.18
6 – 1/2” DC	9.13	10.31
Fishing Jar	2.96	13.27
5" HWDP	83.55	96.82
Total Length (m)		**96.82**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-

Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #4		
7 – 5/8” Overshot	2.16	2.16
6 – 1/2” DC	9.13	11.29
Fishing Jar	2.96	14.25
5" HWDP	83.55	97.80
Total Length (m)		**97.80**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #5		
8 – 3/8” J. Mill	1.15	1.15
6 – 1/2” Bit Sub	1.23	2.38
6 – 1/2” DC	82.11	84.49
5" HWDP	30.63	115.12

Total Length (m)		**115.12**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #6		
7 – 5/8” Overshot	1.18	1.18
5” HWDP	83.55	84.73
Total Length (m)		**84.73**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #7		

7 – 5/8” Overshot	1.18	1.18
5” HWDP	83.55	84.73
Total Length (m)		**84.73**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #8		
8 – 3/8” Junk Mill	1.15	1.15
6 – 5/8” Junk Sub	1.08	2.23
7” Junk Sub	0.91	3.14
D. Pin	0.67	3.81
6 – 1/2” Bit Sub	1.23	5.04
6 – 1/2” DC	82.11	87.15
5” HWDP	27.97	115.12
Total Length (m)		**115.12**
Drilling & Mud Properties		
WOB (lb)		-

RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #9		
7 – 5/8” Overshot	1.18	1.18
5” HWDP	83.55	84.73
Total Length (m)		**84.73**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #10		
8 – 3/8” Junk Mill	1.14	1.14
6 – 5/8” Junk Sub	1.08	2.22
D. Pin	0.67	2.89

7” Junk Sub	0.91	3.80
6 – 1/2” Bit Sub	1.23	5.03
6 – 1/2” DC	109.38	114.41
5” HWDP	83.55	197.96
Total Length (m)		**197.96**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #11		
7 – 5/8” Overshot	1.18	1.18
5” HWDP	83.55	84.73
Total Length (m)		**84.73**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-

Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #12		
8 – 3/8” Junk Mill	1.14	1.14
6 – 5/8” Junk Sub	0.72	1.86
7” Junk Sub	0.91	2.77
D. Pin	0.67	3.44
6 – ½” Bit Sub	1.23	4.67
6 – ½” DC	109.38	114.05
5” HWDP	83.55	197.60
Total Length (m)		**197.60**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #13		
7 – 5/8” Overshot	1.18	1.18

5” HWDP	92.83	94.01
Total Length (m)		**94.01**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #14		
8 – 3/8” Junk Mill	1.14	1.14
6 – 5/8” Junk Sub	0.72	1.86
7” Junk Sub	0.91	2.77
D. Pin	0.67	3.44
6 – ½” Bit Sub	1.23	4.67
6 – ½” DC	109.38	114.05
5” HWDP	83.55	197.60
Total Length (m)		**197.60**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-

Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #15		
7 – 5/8” Overshot	1.18	1.18
5” HWDP	92.83	94.01
Total Length (m)		**94.01**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #16		
8 – 3/8” Junk Mill	1.14	1.14
6 – 5/8” Junk Sub	0.72	1.86
7” Junk Sub	0.91	2.77
D. Pin	0.67	3.44

6 – ½” Bit Sub	1.23	4.67
6 – ½” DC	109.38	114.05
5” HWDP	83.55	197.60
Total Length (m)		**197.60**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #17		
7 – 5/8” Overshot	1.18	1.18
5” HWDP	92.83	94.01
Total Length (m)		**94.01**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-

Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #18		
8 – 3/8” Junk Mill	1.125	1.125
6 – 5/8” Junk Sub	0.72	1.84
7” Junk Sub	0.91	2.75
D. Pin	0.67	3.42
6 – ½” Bit Sub	1.23	4.65
6 – ½” DC	109.38	114.03
5” HWDP	83.55	197.58
Total Length (m)		**197.58**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #19		
7 – 5/8” Overshot	1.18	1.18
5” HWDP	92.83	94.01

Total Length (m)		94.01
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		106.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #20		
8 – 3/8” Junk Mill	1.125	1.125
6 – 5/8” Junk Sub	0.72	1.84
7” Junk Sub	0.91	2.75
D. Pin	0.67	3.42
6 – ½” Bit Sub	1.23	4.65
6 – ½” DC	109.38	114.03
5” HWDP	83.55	197.58
Total Length (m)		**197.58**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-

SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		108.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #21		
8 – 3/8” Junk Mill	1.13	1.13
9 – 5/8” Scraper	1.37	2.50
6 – ½” Bit Sub	1.23	3.73
5” HWDP	83.55	87.28
Total Length (m)		**87.28**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		108.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #22		
8 – ½” B. Bit	0.24	0.24
6 – ½” Bit Sub	1.23	1.47
6 – ½” DC	82.11	83.58

5” HWDP	83.55	167.13
Total Length (m)		**167.13**
Drilling & Mud Properties		
WOB (lb)		0 – 15
RPM		30 – 40
Torque (klb)		1 – 2
SPP (psi)		300 – 900
Flow (gpm)		150 – 300
Mw (pcf)		108.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #23		
8 – ½” B. Bit	0.24	0.24
9 – 5/8” Scraper	1.37	1.61
6 – ½” Bit Sub	1.23	2.84
5” HWDP	83.55	86.39
Total Length (m)		**86.39**
Drilling & Mud Properties		
WOB (lb)		
RPM		
Torque (klb)		
SPP (psi)		400 – 2500
Flow (gpm)		150 – 600

Mw (pcf)		85.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #24		
8 – 5/16” Whipstock	7.13	7.13
8 – ½” Three Mill	4.24	11.37
Running Tool	1.33	12.70
5” DP	9.55	22.25
Gyro Sub	0.75	23.00
6 – ½” DC	82.45	105.45
5” HWDP	83.55	189.00
Total Length (m)		**189.00**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		85.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #25		
8 – ½” B. Bit	0.24	0.24
6 – ¾” PDM	8.24	8.48

6 – ½" Float Sub	0.42	8.90
6 – ½" UBHO	0.84	9.74
6 – ½" NMDC	8.62	18.36
Gyro Sub	0.855	19.22
5" HWDP	223.11	242.33
D. Pin	0.37	242.69
6 – ½" Jar	6.40	249.09
5" HWDP	27.31	276.40
Total Length (m)		**276.40**
Drilling & Mud Properties		
WOB (lb)		10 – 20
RPM		Motor
Torque (klb)		0 – 4
SPP (psi)		1750 – 2200
Flow (gpm)		400 – 470
Mw (pcf)		85.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #26		
8 – ½" PDC Bit	0.30	0.30
6 – ¾" PDM	7.96	8.26
6 – ¾" Float Sub	0.42	8.68
6 – ½" UBHO	0.84	9.52

6 – ½" NMDC	8.62	18.14
5" HWDP	223.11	241.25
D. Pin	0.37	241.62
6 – ½" Jar	6.40	248.02
5" HWDP	27.31	275.33
Total Length (m)		**275.33**
Drilling & Mud Properties		
WOB (lb)		1 – 20
RPM		Motor
Torque (klb)		0 – 7
SPP (psi)		1950 – 2500
Flow (gpm)		390 – 470
Mw (pcf)		87.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #27		
8 – ½" B. Bit	0.30	0.30
Bit Sub	1.23	1.53
6 – ½" DC	18.04	19.57
Stabilizer	1.53	21.10
6 – ½" DC	27.54	48.64
5" HWDP	111.69	160.33
D. Pin	0.67	161.00

6 – ½” Jar	6.52	167.52
5” HWDP	83.38	250.90
Total Length (m)		**250.90**
Drilling & Mud Properties		
WOB (lb)		1 – 20
RPM		90 – 170
Torque (klb)		2 – 9
SPP (psi)		1900 – 2500
Flow (gpm)		440 – 500
Mw (pcf)		87.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #28		
8 – ½” PDC Bit	0.275	0.275
Bit Sub	1.23	1.51
6 – ½” DC	18.04	19.55
Stabilizer	1.53	21.08
6 – ½” DC	27.54	48.62
5” HWDP	111.69	160.31
D. Pin	0.67	160.975
6 – ½” Jar	6.52	167.50
5” HWDP	83.38	250.88
Total Length (m)		**250.88**

Drilling & Mud Properties		
WOB (lb)		5 – 18
RPM		150 – 180
Torque (klb)		4 – 6
SPP (psi)		2000 – 2400
Flow (gpm)		450 – 480
Mw (pcf)		87.00
BHA No. #29		
8 – ½" B. Bit	0.24	0.24
Bit Sub	1.23	1.47
6 – ½" DC	82.11	83.58
5" HWDP	111.69	195.27
Total Length (m)		**195.27**
Drilling & Mud Properties		
WOB (lb)		0 – 10
RPM		30 – 40
Torque (klb)		1 – 2
SPP (psi)		1200
Flow (gpm)		320
Mw (pcf)		88.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #30		

6 – 1/8" BTM Bit	0.18	0.18
4 – ¾" Bit Sub	0.99	1.17
4 – ¾" DC	192.78	193.95
Total Length (m)		**193.95**
Drilling & Mud Properties		
WOB (lb)		2 – 18
RPM		30 – 60
Torque (klb)		1 – 3
SPP (psi)		500 – 2000
Flow (gpm)		120 – 300
Mw (pcf)		88.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #31		
6 – 1/8" MT Bit	0.18	0.18
7" Scraper	1.01	1.19
Bit Sub	0.99	2.18
4 – ¾" DC	110.39	112.57
3 – ½" DP	605.82	718.39
X.O.S	1.12	719.51
D. Pin	0.69	720.20
9 – 5/8" Scraper	1.37	721.57
Bit Sub	1.23	722.80

Total Length (m)		**722.80**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		88.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #32		
Bottom BT (1st memory gauge)	1.26	1.26
Anchor Pipe	1.53	2.79
X.O.S (4-1/2 IF box * 3-1/2 FH pin)	0.80	3.59
Middle Packer Element	1.33	4.92
9 – 5/8" Above Packer	0.76	5.68
X.O.S (3-1/2 IF box * 4-1/2 IF pin)	0.80	6.48
X.O.S (3-1/2 IF box * 3-1/2 IF pin)	0.40	6.88
RH Release Safety Joint	0.73	7.61
Hydraulic jar	1.00	8.61
Middle BT (2nd memory gauge)	1.25	9.86

Haydrospring	1.56	11.42
D.C.I.P	1.64	13.06
X.O.S (4-1/2 IF box * 3-1/2 IF pin)	0.25	13.31
5" DP	9.56	22.87
Reverse Sub	0.4	23.27
6 – ½" DC	82.11	105.38
Total Length (m)		**105.38**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		88.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #33		
6 – ½" Junk Mill	0.49	0.49
Junk Sub	0.92	1.41
4 – ¾" DC	192.78	194.19
3 – ½" HWDP	83.63	277.82
3 – ½" DP	577.78	855.60
X.O.S	1.12	856.72

D. Pin	0.69	857.41
Bit Sub	1.23	858.64
Total Length (m)		**858.64**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		89.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #34		
6 – 1/8" PDC Bit	0.23	0.23
Bit Sub	0.86	1.09
4 – ¾" DC	18.05	19.14
Stabilizer	1.37	20.51
4 – ¾" DC	174.73	195.24
4 – ¾" Jar	6.84	202.08
4 – ¾" DC	27.56	229.64
3 – ½" HWDP	83.63	313.27
3 – ½" DP	808.85	1122.12
X.O.S	1.12	1123.24

Total Length (m)		1123.24
Drilling & Mud Properties		
WOB (lb)		4 – 16
RPM		120 – 140
Torque (klb)		1 – 7
SPP (psi)		2200 – 2750
Flow (gpm)		215 – 250
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #35		
6” Junk Mill	0.50	0.50
Bit Sub	0.86	1.36
4 – ¾” DC	220.34	221.70
3 – ½” DP	576.92	798.62
X.O.S	1.12	799.74
Total Length (m)		**799.74**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-

Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #36		
3 – ½" Junk Mill	0.44	0.44
Bit Sub	0.80	1.24
3 – 1/8" DC	158.05	159.29
2 – 7/8" DP	237.77	397.06
X.O.S	0.69	397.75
3 – ½" DP	28.52	426.27
D. Pin	0.28	426.55
Bit Sub	0.86	427.41
3 – ½" DP	548.40	975.81
X.O.S	1.12	976.93
Total Length (m)		**976.93**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**

BHA No. #37		
3 – ½” Junk Mill	0.44	0.44
Bit Sub	0.80	1.24
3 – 1/8” DC	158.05	159.29
2 – 7/8” DP	237.77	397.06
X.O.S	0.69	397.75
3 – ½” DP	28.52	426.27
D. Pin	0.28	426.55
Bit Sub	0.86	427.41
3 – ½” DP	577.39	1004.80
X.O.S	1.12	1005.92
Total Length (m)		**1005.92**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #38		
Bit Sub	0.80	0.80

D. Pin	0.28	1.08
7” Scraper	1.01	2.09
Bit Sub	0.86	2.95
3 – ½” HWDP	83.63	86.58
3 – ½” DP	576.92	663.50
X.O.S	1.12	664.62
Total Length (m)		**664.62**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00

6. Casing Information

Casing Spec.	20"	K – 55, 94^{ppf}, BTC	0 – 155^{m}
	13-3/8"	N – 80, 68^{ppf}, BTC	0 – 1441^{m}
	9-5/8"	T – 95, 53.5^{PPF}, TPCQ	0 – 3909^{m}
	7" (main hole)	C110, 29^{PPF}, TPCQ	3826 – 4226^{m}
	7" Tie-back	C110, 29^{PPF}, TPCQ	3777 – 3826^{m}
	7" (side track)	110SS, 29^{PPF}, TPG2	3147 – 3891^{m}
	4.5 (side track)	110SS, 13.5^{PPF}, TPG2	3780.41 – 4139.5^{m}

Table 6-1: AZNS-127 Final Casing Data

7. Cement Job Report

7.1 7” Liner Cementing Operation Information

Operation Type: 7” Liner cementing job

Hole Size: 8 – 1/2”

Liner Depth: 3890^{m}

Total Volume:

- Pump 10^{bbl} chemical wash 62.4^{pcf}
- Pump 10^{bbl} spacer 90^{pcf}
- Pump 95^{bbl} cement slurry 118^{pcf}

Total Displacement Volume: Displace with 260^{bbl} mud

7.2 Engineering Data

Casing Data				
MD (m)	OD (in)	Weight (lb/ft.)	ID (in)	Grade
3890	7	29	6.184	S – 110
Slurry Data				
Name	Volume (bbl)	Density (pcf)	Remark	
Cement Slurry	95	118	-	

Table 7-1: AZNS-127 Engineering Data

7.3 Execution of Operation

Item	Description	Qty
1	Cementing pump truck & related connections (PD – 03)	1 ea.
2	Cementing pump truck & related connections (PD – 05)	1 ea.
3	Batch mixer	1 ea.
4	Air compressor (PD – C – 02)	1 ea.
5	Silo (PD – S – 07)	1 ea.

Table 7-2: AZNS-127 Equipment Used

Item	Description	Qty
1	Cementing Supervisor	1 per.
2	Cementing Operator	2 per.
3	Batch Mixer Operator	1 per.
4	Cementing Helper	5 per.
5	Batch Mixer Helper	1 per.

Table 7-3: AZNS-127 Operation Crew

7.4 Operation Ticket

FIELD SERVICES TICKET

SPAD ENERGY

ماهان صنعت آریا EPC CONTRACTOR

Client:	MSA	Contract No:	Invoice No:
Service Company:	SPAD ENERGY		
Field : AZADGAN	Well No: AZNS-127	Rig No: OEOC 208	Date: 31-jul-2024
Division:		Period:	26-jul-2024 till 31-jul-2024
Service Requested:	PERSONEL-EQUIPMENT liner 7" Cementing Operation		

ITEM	Description	Amount		STB		OPT		Total
		Qty	Unit	Qty	Unit	Qty	Unit	
1	Cementing Supervisor	1	per	0	Day	1	Day	1
3	Cementing Operator	2	per	0	Day	1	Day	2
4	Helper	5	per	0	Day	1	Day	5
	Batch mixer Operator	1	per	0	Day	1	Day	1
	Batch mixer Helper	1	per	0	Day	1	Day	1
	Batch mixer	1	UNIT	0	Day	1	Day	1
5	PumpTruck PD - 03	1	UNIT	0	Day	1	Day	1
6	PumpTruck PD - 05	1	UNIT	0	Day	1	Day	1
7	Silo PD-S-06	1	UNIT	5	Day	1	Day	6
	Air Compressor PD-C-02	1	UNIT	0	Day	1	Day	1
	MOB/ DMOB Batch mixer	1	UNIT	1				
12	MOB/ DMOB PumpTruck PD - 05	1	UNIT	1				
13	MOB/ DMOB Silo PD-S-06	1	UNIT	1				
14	MOB/ DMOB Air Compressor	1	UNIT	1				

SPAD Energy supervisor:

Client Representative:

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
MSA
EPDS CONTRACTOR Date: 31 July
Rig No: 208/AZNS-127

	DATE		
On location	26	jul	2024
Operation Start	31	jul	2024
End of operation	31	jul	2024
Left Location	31	jul	2024
On Base			

Remark:

به دلیل عدم توجیه برنامه از سوی کارفرما
جریمه کانگ گردیده است

* فاقد دستگاه حسابات حرکتی گردیده است -
* تاسیسات کمی متوقف به بررسی تاسیسات در دفتر تهران شرکت ماهان صنعت آریا می باشد.
- در تاریخ ۱۰، ۱۵، ۲۰۲۴ به میزان 900 لیتر گازوئیل جهت مصرف دستگاه تحویل شرکت اسپادانرژی گردیده است.
- در تاریخ 31-July کارکرد دستگاه کلی صورت Standby و دیگر صورت operation مورد تایید می باشد.

-3HRS NPT consider due to delay in operation, Leakye From pump truck during Displace, Plugged cmt silo before mixing & unable turn on Batch mixer before mixing.

7.5 Job Log & Summary

JOB SUMMARY

Client:	MSA	Rig No:	208	JOB PERIOD DATE	
Service Company:	SPAD ENERGY	Well No:	AZNS-127	From:	31-Jul-2024
Field:	S. AZADEGAN	Hole Section:	8 1/2"	To:	31-Jul-2024
Service Requested:	CEMENTING Liner 7"				

JOB LOG

Date	Time	Volume (bbl)	Density (pcf)	Rate (bpm)	Pressure (psi)	Description
31-Jul-24	8:00					Start Mix Spacer
31-Jul-24	9:00					Start mix solution
31-Jul-24	10:15					End of mix the solution
31-Jul-24	10:20					Safty and operation meeting
31-Jul-24	10;35				3000	Test the line and both unit W500/3000 psi
31-Jul-24	11:35	10		5		Pump 10 bbl spacer by rig pump
31-Jul-24	11:45	10	62.4	4	500	Pump 10 bbl Chemical wash by CMT unit
31-Jul-24	11:50	95	118	4	700	Start mix CMT slurry 118 pcf by CMT unit
31-Jul-24	12:30	95	118	4		End of pumping after 95 bbl CMT slurry 118pcf
31-Jul-24	12:30	260	87	6	500-2900	Start Displace by CMT unit
31-Jul-24	14:20	260	87	6	500-2900	End of Displace the CMT slurry W/ 260 bbl mud
31-Jul-24	15:00					End of the CMT job & wash CMT unit

Material Chemical wash

Consume			Mixing Information		
Materials	Unit	QTY for Job		Spacer Volume	10 bbl
Chemical Wash	gal	50		Spacer Density	62.4 pcf
Anti foam	gal	2			
			Total Dead Volume		
			0 bbl		

Material Spacer

Consume			Mixing Information		
Materials	Unit	QTY for Job		Spacer Volume	40 bbl
Chemical Spacer SE-P03	lb	440		Spacer Density	90 pcf
Antifoam(SE-A20)	gal	4			
Surfactant	gal	80	Total Dead Volume		
			30 bbl		

Material CMT Slurry

Consume			Mixing Information		
Materials	Unit	QTY for Job	Total Mixing Water in CMT unit	Cement Slurry Volume	95 bbl
Class "G" Cement Delijan	sx	298	59.23 bbl	Cement Slurry Density	118 pcf
Silica Flour	lb	10,901.00	Total Mixing Fluid in CMT unit	Slurry Yield	1.7921 Cu ft/sks
Retarder SE-R-02	lb	390.00	63.7 bbl	Mixing Water Required	7.0219 gal / sks
Dispersant SE-D031	gal	22.00		Mixing Fluid Required	7.5517 gal / sks
FLC-F05	lb	54.00			
Anti gas. G20	gal	71.00			
Micro silica	lb	886.00	Total Dead Volume		
Antifoam(SE-A20)	gal	11	0		

Client Representative: ماهان صنعت آریا Drilling Supervisor South Azadegan Project Rig No: 208/AZNS 127 Date: 31 July

SPAD ENERGY supervisor:

7.6 Job Graph

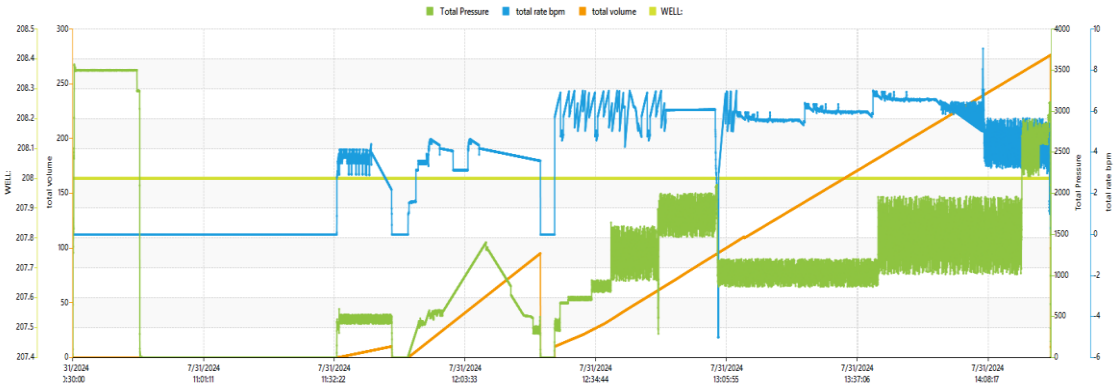


Figure 7-1: AZNS-127 7" Liner Job Graph

7.7 Operation Instructions

Pre – Job Preparation

- Bulk plant R/U & preparations (prepare the checklist).
- Cementing pump truck R/U & preparations (prepare the checklist).
- Preparting mixing fluid (solution) with cooperation of the rig crew).
- Cementing high pressure line R/U to the rig floor (secure them with safety cable).
- Setting up HSE measurments for the operation.

Pre – Job Preparation

- M/U liner cementing head on string (dart plug already loaded), L/U CMT line to CMT head manifold.
- Pressure test CMT lines to 3000psi, Ok.
- Pump 10bbl chemical wash 62.4pcf by CMT pump truck.
- Pumped 10bbl spacer 62.4pcf by rog pump.
- Mixed & pumped 298sxs CMT "G" Delijan & 10901lb Silica Flour (S.V: 95bbl, S.W: 118pcf) as slurry w/ 4bpm / 700psi.
- Released pump down plug.
- Displaced w/ totally 260bbl mud, bleed of pressure.
- P/U & checked the string weight, Ok. B/O & L/D CMT manifold.
- POOH 10 Stds, flush string & CBU.

7.8 Pumping Schedule

Name	Flow Rate (bb/min)	Volume (bbl)	Stage Time	Comments
Chemical Wash	5	10	2	-
Spacer	4	40	10	-
Slurry	4	95	24	-
Displacement	6	260	44	-

Table 7-4: AZNS-127 Pumping Schedule

7.9 Amount of Material & Used in Operation

Slurry Type				
Wt: 118^{pcf}	S.V: 95^{bbl}	Mixed W. V: 59.23^{bbl}	Mixed F. V: 63.7^{bbl}	S.Y: 1.7921 ft^2/sk
Row	**Material Name**	**Consumbles**	**Unit**	**Description**
1	Cement G class	298	sks	-
2	Silica Flour	10901	lb	-
3	Retarder SE-R-02	390	lb	-
4	Dispersant SE-D-31	22	gal	-
5	FLC SE-F-05	54	lb	-
6	Anti gas SE-G-20	71	gal	-
7	Micro Silica	886	lb	-
8	Anti foam SE-A20	11	gal	-

Table 7-5: AZNS-127 Material Used Table

7.10 4 – 1/2” Liner Cementing Operation Information

Operation Type: 4 – 1/2” Liner cementing job

Hole Size: 5 – 7/8”

Liner Depth: 4150^{m}

Total Volume:

- Pump 15^{bbl} chemical wash 62.4^{pcf} as spacer
- Pump 32^{bbl} cement slurry 118^{pcf}

Total Displacement Volume: Displace with 202^{bbl} mud

7.11 Engineering Data

Casing Data				
MD (m)	OD (in)	Weight (lb/ft.)	ID (in)	Grade
4140	4 – 1/2	13.5	3.92	SS – 110
Slurry Data				
Name	Volume (bbl)	Density (pcf)	Remark	
Cement Slurry	32.5	118	-	

Table 7-6: AZNS-127 Engineering Data

7.12 Execution of Operation

Item	Description	Qty
1	Cementing pump truck & related connections (PD – 03)	1 ea.
2	Cementing pump truck & related connections (PD – 05)	1 ea.
3	Batch mixer	1 ea.
4	Air compressor (PD – C – 04)	1 ea.
5	Silo (PD – S – 05)	1 ea.

Table 7-7: AZNS-127 Equipment Used

Item	Description	Qty
1	Cementing Supervisor	1 per.
2	Cementing Operator	2 per.
3	Batch Mixer Operator	1 per.
4	Cementing Helper	4 per.
5	Batch Mixer Helper	1 per.

Table 7-8: AZNS-127 Operation Crew

7.13 Operation Ticket

FIELD SERVICES TICKET

Client:	MSA	Contract No:	Invoice No:
Service Company:	SPAD ENERGY		
Field : AZADGAN	Well No: AZNS-127	Rig No: OEOC 208	Date: 18-Aug-2024
Division:		Period:	15-Aug-2024 till 18-Aug-2024
Service Requested:	PERSONEL-EQUIPMENT liner 4.5" Cementing Operation		

ITEM	Description	Amount		STB		OPT		Total
		Qty	Unit	Qty	Unit	Qty	Unit	
1	Cementing Supervisor	1	per	0	Day	1	Day	1
2	Cementing Operator	2	per	0	Day	1	Day	2
3	Helper	4	per	0	Day	1	Day	4
4	Batch mixer Operator	1	per	0	Day	1	Day	1
5	Batch mixer Helper	1	per	0	Day	1	Day	1
6	Batch mixer	1	UNIT	0	Day	1	Day	1
7	PumpTruck PD - 03	1	UNIT	0	Day	1	Day	1
8	PumpTruck PD - 05	1	UNIT	0	Day	1	Day	1
9	Silo PD-S-06	1	UNIT	3	Day	1	Day	4
10	Air Compressor PD-C-04	1	UNIT	0	Day	1	Day	1
11	MOB/ DMOB Batch mixer	1	UNIT	1				
12	MOB/ DMOB PumpTruck PD - 05	1	UNIT	1				
13	MOB/ DMOB Silo SE-S-07	1	UNIT	1				
14	MOB/ DMOB Air Compressor	1	UNIT	1				

SPAD Energy supervisor:

Client Representative: ماهان صنعت آریا Drilling Supervisor South Azadegan Project Date: Rig No: 106/AZNS127

EPDS CONTRACTOR Date: 18, Aug

	DATE		
On location	15	Aug	2024
Operation Start	17	Aug	2024
End of operation	18	Aug	2024
Left Location	18	Aug	2024
On Base			

Remark:

- 2HRS NPT consider for spad energy due to daily inoperation leakage from line pump truck.

7.14 Job Log & Summary

JOB SUMMARY

Client:	MSA	Rig No:	208	JOB PERIOD DATE	
Service Company:	SPAD ENERGY	Well No:	AZNS-127	From:	17-Aug-2024
Field:	S. AZADEGAN	Hole Section:	6 1/8"	To:	18-Aug-2024
Service Requested:	CEMENTING Liner 4 1/2"				

JOB LOG

Date	Time	Volume (bbl)	Density (pcf)	Rate (bpm)	Pressure (psi)	Description
17-Aug-24	23:45					Start mix solution
18-Aug-24	0:15					End of mix the solution
18-Aug-24	0:30					Safty and operation meeting
18-Aug-24	0:40				3000	Test the line and both unit W500/3000 psi
18-Aug-24	1:30	10				Pump 10 bbl Chemical wash by CMT unit
18-Aug-24	2:15	32				Start mix CMT slurry 118 pcf by CMT unit
18-Aug-24	2:30	32	118	4	400	End of pumping after 32 bbl CMT slurry 118pcf
18-Aug-24	2:35	202		6		Start Displace by CMT unit
18-Aug-24	3:20	202		6		End of Displace the CMT slurry W/ 202 bbl mud
18-Aug-24	3:45					End of the CMT job & wash CMT unit

Material Chemical wash

Consume			Mixing Information		
Materials	Unit	QTY for Job		Spacer Volume	15 bbl
Chemical Wash	gal	50		Spacer Density	62.4 pcf
Anti foam	gal	2			
			Total Dead Volume		
			0 bbl		

Material CMT Slurry

Consume			Mixing Information		
Materials	Unit	QTY for Job	Total Mixing Water in CMT unit	Cement Slurry Volume	32 bbl
Class "G" Cement Delijan	sx	103.32	17 bbl	Cement Slurry Density	118 pcf
Silica Flour	lb	3,784.70	Total Mixing Fluid in CMT unit	Slurry Yield	1.7670 Cu ft/sks
Retarder SE-R-02	lb	108.49	18.1 bbl	Mixing Water Required	6.6995 gal / sks
Dispersant SE-D031	gal	3.62		Mixing Fluid Required	7.3642 gal / sks
FLC-F09	lb	15.50			
Anti gas. G20	gal	31.00			
Antifoam(SE-A20)	gal	3.1	Total Dead Volume		
			0		

Dumped Material: 284 lb of silica four + 18sx of cement which is total of 1mt of blended cmt & silica flour

Client Representative: ماهان صنعت آریا Drilling Supervisor South Azadegan Project MSA Rig No: 208/Azns127 EPDS CONTRACTOR Date: 18 Aug

SPAD ENERGY supervisor:

-2HRS NPT consider For spad Energy due to dealy in operation & leakge From line &pump truck.

7.15 Job Graph

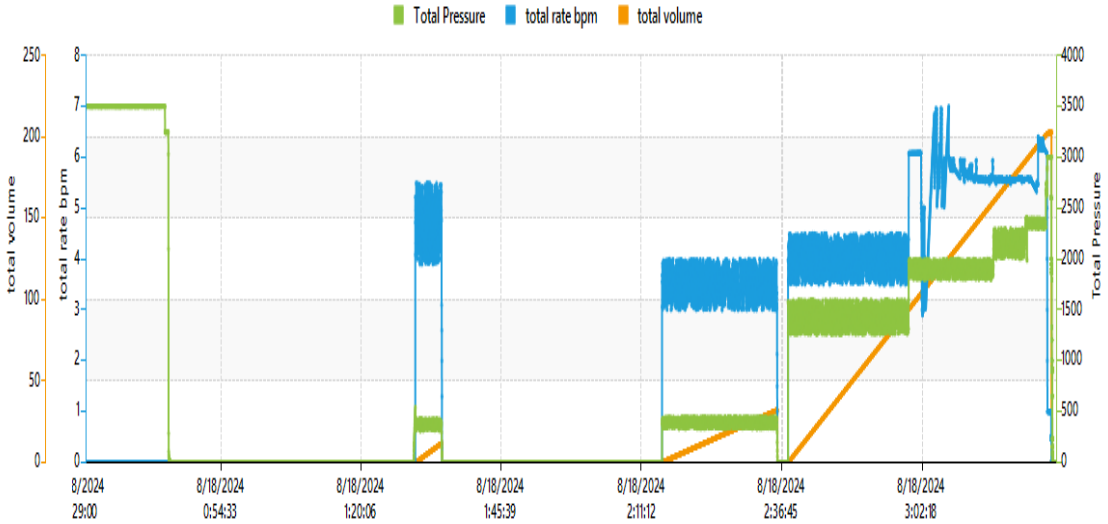


Figure 7-2: AZNS-127 4 – 1/2" Liner Job Graph

7.16 Operation Instructions

Pre – Job Preparation

- Bulk plant R/U & preparations (prepare the checklist).
- Cementing pump truck R/U & preparations (prepare the checklist).
- Preparting mixing fluid (solution) with cooperation of the rig crew).
- Cementing high pressure line R/U to the rig floor (secure them with safety cable).
- Setting up HSE measurments for the operation.

Pre – Job Preparation

- Loaded pump down plug, M/U CMT head manifold & pup joint on string, CBU while L/U CMT line to CMT head manifold.
- Pressure test CMT lines to 3000^{psi}, Ok.
- Pump 10^{bbl} chemical wash 62.4^{pcf} by CMT pump truck.
- Mixed & pumped 103.32^{sxs} CMT "G" Delijan (S.V: 32^{bbl}, S.W: 118^{pcf}) as slurry w/ 4 bpm / 400^{psi}.
- Released pump down plug.
- Displaced w/ totally 202^{bbl} mud, bleed of pressure.
- P/U & checked the string weight, Ok. B/O & L/D CMT manifold.
- POOH 10 Stds, flush string & CBU.

7.17 Pumping Schedule

Name	Flow Rate (bb/min)	Volume (bbl)	Stage Time	Comments
Chemical Wash	4	15	4	-
Slurry	5	32	4	-
Displacement	6	202	39	-

Table 7-9: AZNS-127 Pumping Schedule

7.18 Amount of Material & Used in Operation

Slurry Type				
Wt: 118^{pcf}	S.V: 32^{bbl}	Mixed W. V: 17^{bbl}	Mixed F. V: 18.1^{bbl}	S.Y: 1.7670 ft^2/sk
Row	**Material Name**	**Consumbles**	**Unit**	**Description**
1	Cement G class	103.32	sks	-
2	Silica Flour	3784.7	lb	-
3	Retarder SE-R-02	108.49	lb	-
4	Dispersant SE-D-31	3.62	gal	-
5	FLC SE-F-09	15.5	lb	-
6	Anti gas SE-G-20	31	gal	-
7	Anti foam SE-A20	3.1	gal	-

Table 7-10: AZNS-127 Material Used Table

8. Completion Report

8.1 Completion Tubing Information

Tubing Size (in)	Weight (ppf)	Grade	Ultimate Strengths	Connection	Collapse Pressure (psi)	Burst Pressure (psi)	Interval (m)
4 – 1/2	18.9	L – 80	233^{klb}	HSM3	13830	13380	0 – 3078.74
3 – 1/2	10.2	L – 80	440^{klb}	HSM3	12120	11560	3078.74– 3325

Table 8-1: AZNS-85 Tubing Data

8.2 Completion Tools Specification Checklist

Item	Equipment Description	
1	**CONTROL LINE "WEIQIXIN":**	
	THINKNESS:	0.065"
	WORK PRESSURE:	15000 PSI
	O.D.:	0.25"
	TYPE:	SEAMLESS
	MATERIAL:	INC 825
	LENGTH:	120m/Roll
2	**UPPER FLOW COUPLING " WEIQIXIN" :**	
	OD:	**5.5"**
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	6 FT
	MATERIAL:	13CR-110KSI
	LOWER FLOW COUPLING " WEIQIXIN" :	
	OD:	5.550"
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	4 FT
	MATERIAL:	13CR-110KSI
3	**TUBING RETRIEVABL SAFETY VALVE " WEIQIXIN" :**	
	SETTING DEPTH:	64.19^{m}
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	325 DEG F

	MAX O.D.:	7.5"
	MIN. I.D.:	3.437"
	PROFILE:	AF
	SELF-EQUALIZING FEATURE	HYDRAULIC CONTROL
	MATERIAL:	13CR-110KSI
	THREADS:	4-1/2" 18.9 PPF HSM3
	PROTECTION SLEEVE	
	PROFILE	AF
	MIN ID	2.9"
4	**TRAVELING JOINT " WEIQIXIN" :**	
	STROKE LENGTH	10 FT
	SHEAR PIN SET:	15000LB
	ELASTOMER:	AFLAS
	WORKING PRESSURE:	10000 psi
	TEMPERATURE:	300 DEG F
	TOOL OD:	5.315"
	TOOL ID:	2.922"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	TRANSFER TORQUE FEATURE	YES
5	**SLIDING SIDE DOOR " WEIQIXIN" :**	
	PROFILE:	X
	MAX.OD:	4.551"
	MIN ID:	2.750"
	PRESSURE RATING:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	OPEN METHOD	DOWNWARD OPEN
6	**COMPLETION PACKER " WEIQIXIN" :**	
	TYPE:	PERMANENT
	CASING :	7" 32 PPF
	OD:	5.689"
	ID:	2.922"
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	ELEMENT MATERIAL:	AFLAS
	4.0" ANCHOR SEAL	
	MAX OD (PACKING):	4"
	ID:	2.922"
	WORKING PRESSURE:	10000PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B
	TYPE:	RATCH LATCH

7	**LANDING NIPPLE " WEIQIXIN" :**	
	PROFILE TYPE:	XN
	WORKING PRESSURE:	10000 PSI
	MAX O.D.:	4.5"
	SEAL BORE:	2.75”
	MIN ID:	2.635"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	TEMPERATURE:	300 DEG F
	MATERIAL:	13CR-110KSI
8	**RE-ENTRY GUIDE SHOE " WEIQIXIN" :**	
	OD:	4.5"
	MIN ID:	2.922"
	TOP CONNECTION	3-1/2" 10.20 PPF HSM3 BOX
	MATERIAL:	13CR-110KSI
9	**3-1/2" CHEMICAL INJECTION" WEIQIXIN" :**	
	TYPE	TUBING RETRIVABLE
	MAX OD	5.00"
	MIN ID	2.922"
	MATERIAL:	13CR-110KSI
	TEMPERATURE	300 DEG F
	CONNECTION	3-1/2" 10.20 PPF HSM3 B*P
	CIV CONNECTION	NPT 3/8" BOX FOR CIV
	1.00"CIV VALVE	
	TYPE	DUAL CHECK VALVE WITH RUPTURE DISC
	WORKING PRESSURE	15000PSI
	MARERIAL	13CR-110KSI
	TEMPERATURE:	300 DEG F
	TOP CONNECTION	3/8" AUTOCLAVE
	BOTTOM CONNECTION	3/8" NPT
	RUPTURE DISC PRESSURE	6000PSI

Table 8-2: AZNS-127 Equipment Checklist

Item	Name	Description
1	**Tubing Hanger**	11" BOWL x 4-1/2" 18.9 PPF BOX TPCQ-EX CONNECTION BOTTOM & 4-1/2" 4-2G ACME BOX TOP WITH 4-1/2" EXTENDED NECK, WITH ONE CONTINUOUS 1/4" NPT C.LINE EXIT FOR SSSV, ONE 1/4" LINE EXIT FOR DHPG, INSTALLATION COMPLETE WITH RELATED FITTINGS AND PLUGS, WITH TWO CHANNEL FOR 3/8" CHEMICAL, C/W ALIGNMENT PREPATION FOR ESP. BPV: 4" TYPE H THREAD PREPARATION, THREE FS SEAL TO PACK OFF THE UPPER NECK AND TWO S SEAL TO PACK OFF THE BODY OF HANGER, MATERIAL: 410SS. M.C: FF-0.5 T.C: U PSL:3G PR:2 COMPANY: MSP Serial Number: 2909423F XC230002
2	**Fluted Adaptor**	4 1/2" 18.9 PPF, L-80 HSM3 PIN * TPCQ EX PIN, OD: 5.5'' ID: 3.64'' SLOTS: 6 * 1 INCH WIDTH

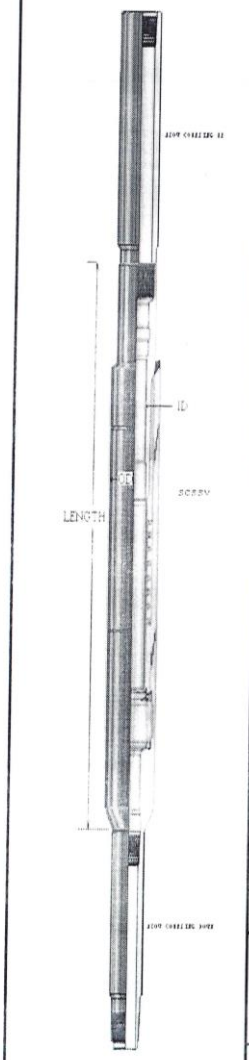
Table 8-3: AZNS-127 Equipment Checklist

8.3 Assembling Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

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SCSSV Test Report



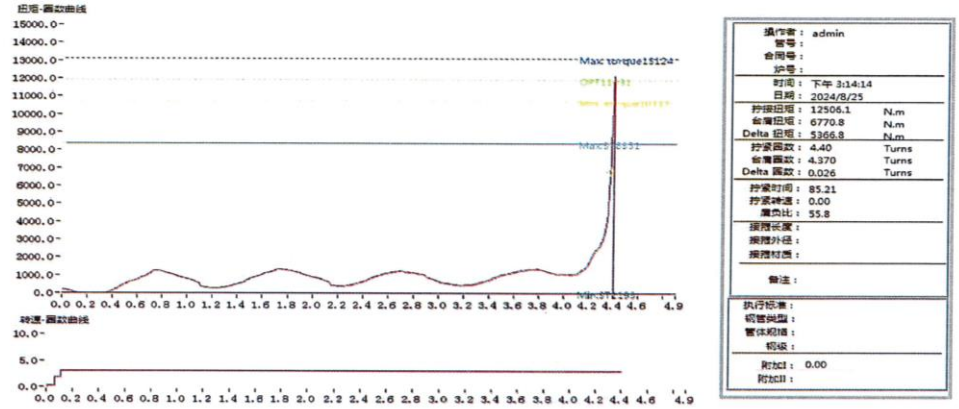
Type:	210W 343-AF04	
Serial Number:	2308 23038-05	
Connection Size:	4-1/2" 18.90 PPF HSM3 BOX × PIN	
Working Pressure	10,000Psi	
OD:	7.500"	
ID:	3.437"	
Length:	1.782 m	
Material:	S13CR-110 KSI	
Hydraulic Control Pressure	Full Open	Full Closed
	1800 Psi	800 Psi
8,000 Psi Hydraulic Control Pressure No Drop in 15 Minutes	ok	
8,000 Psi Pressure No Drop in 15 Minutes	ok	
8,000 Psi flapper Pressure No Drop in 15 Minutes	ok	
Top Connection	Length:	1.623 m
Top Flow Coupling:	Serial Number:	2307 Y230414-05
	Length:	1.726 m
Bottom Flow Coupling:	Serial Number:	2307 Y230415-05
	Length:	1.118 m
Bottom Connection:	Length:	1.624m
Well Name and Location:	AZNS-127	

MSA Project Manager:
Name/Title:
Date:

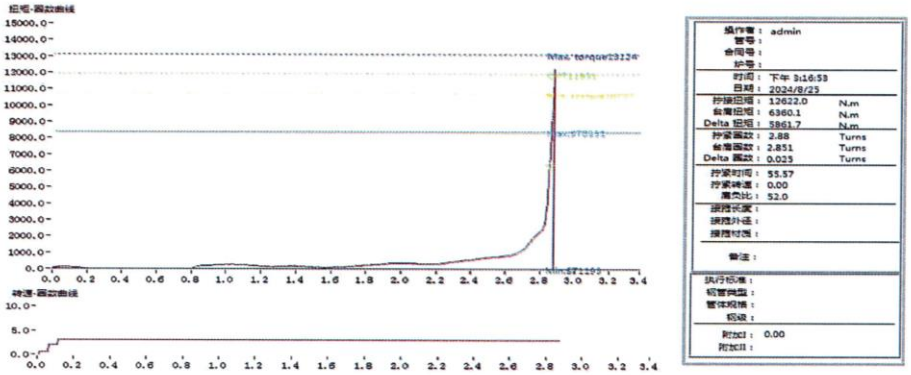
WQX:
Name/Title:
Date: 2024.9.14

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EPDS CONTRACTOR Date:

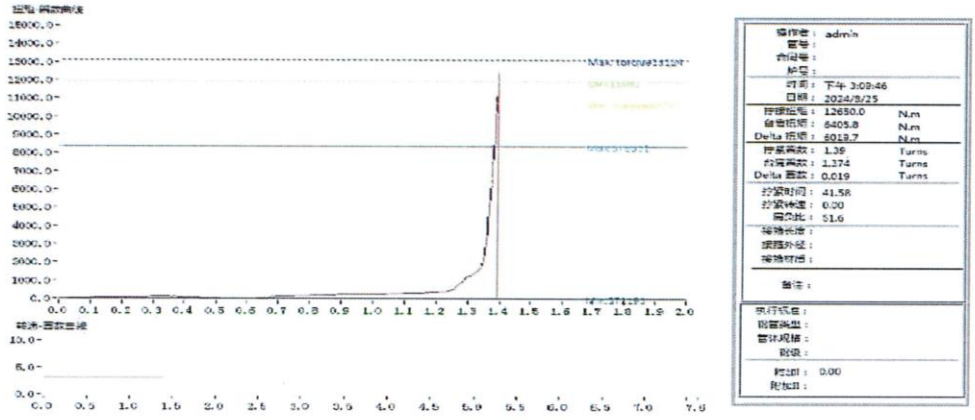
Figure 8-1: AZNS-127 TRSSSV Test Report



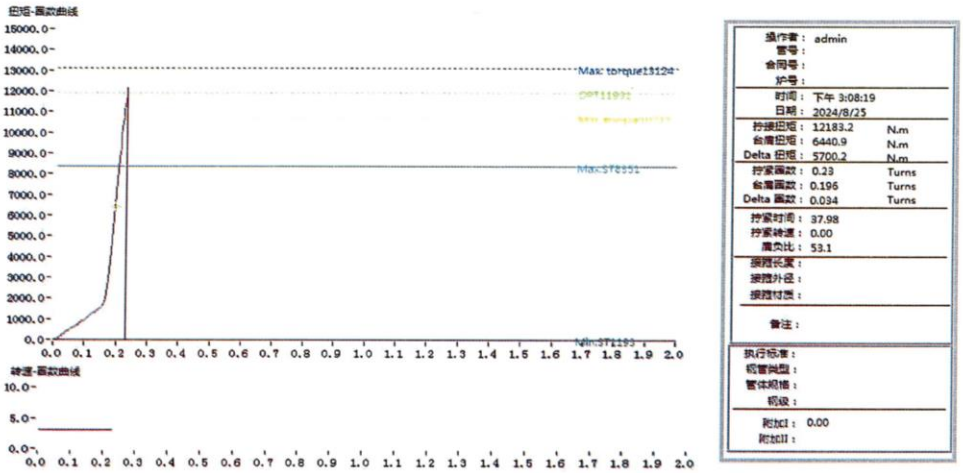
Top pup joint-top flow coupling



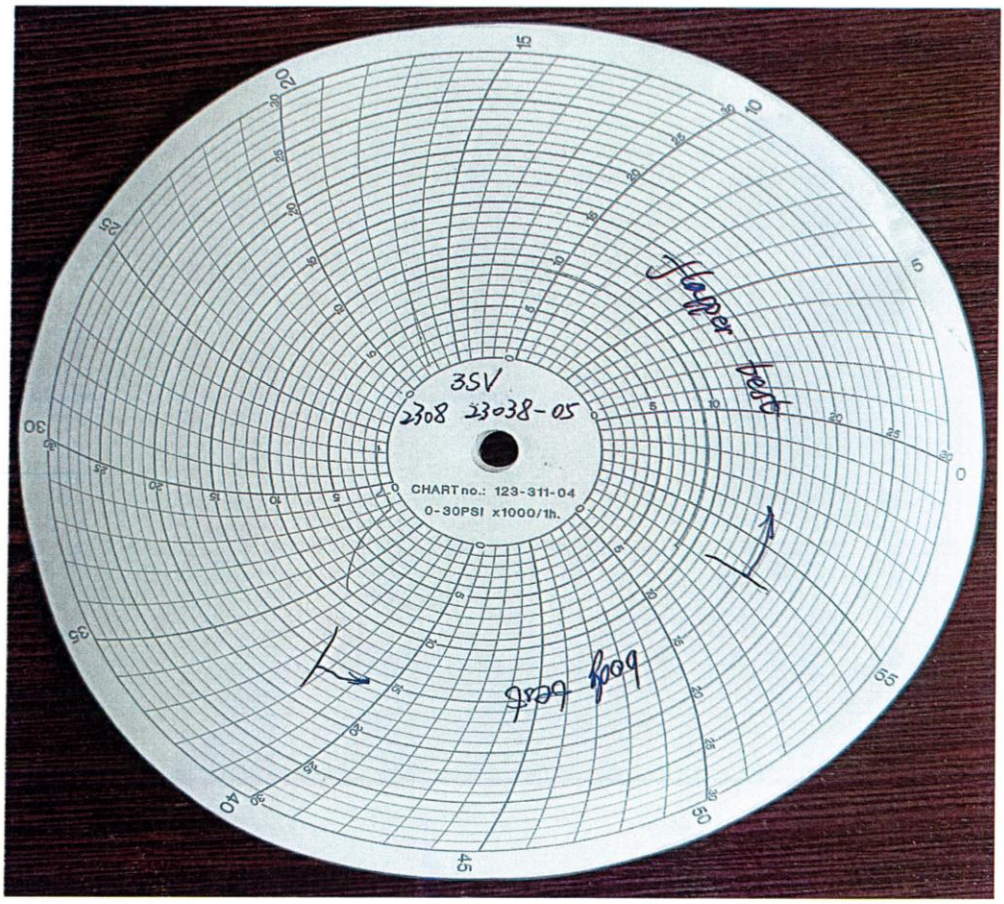
Top flow coupling – 3SV



3SV - bottom flow coupling



Bottom flow coupling - bottom pup joint



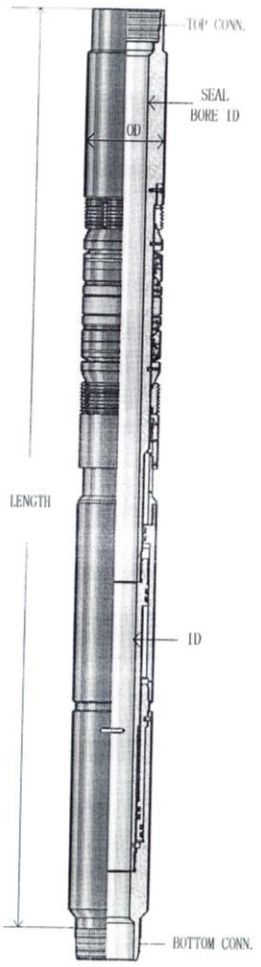
Pressure test curve

Figure 8-2: AZNS-127 TRSSSV Connections & Body Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

WEIQIXIN

WSDP Hydraulic Set Permanent Packer Test Report



Type:	336W150-000	
Serial Number:	2308 23062-03	
Connection Size	LEFT HAND X 3-1/2" 10.20 PPF HSM3 PIN	
Working Pressure	10,000 Psi	
OD:	5.910"	
ID:	2.922"	
Length:	4.440m	
Material:	S13CR-110KSI	
Temperature:	300 DEG F	
Element :	Aflas	
Setting Pressure:	3,500 Psi	
Seal Bore Size:	Upper Seal Bore ID:4.00" & Lower Seal Bore ID:2.922"	
Top Connection:	Length:	1.557m
Bottom Connection:	Length:	1.553 m
Well Name and Location:	AZNS-127	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024. 9. 19

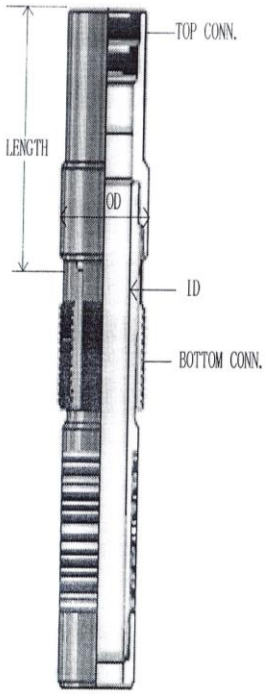
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South Azadegan Project
Rig No:
EPDS CONTRACTOR Date:

Figure 8-3: AZNS-127 Permanent Packer Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

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Model WK Anchor Tubing Seal Report



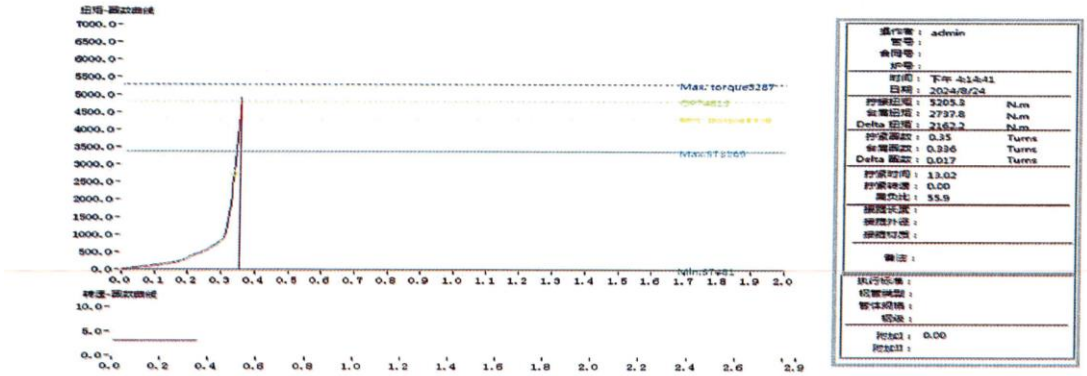
Type:	360W150-000
Serial Number:	2308 23063-07
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX X MULE SHOE GUIDE
OD:	5.689"
ID:	2.922"
Seal OD:	4.000"
Working pressure:	10,000Psi
Length:	230mm
Material:	S13CR-110 KSI
Well Name and Location	AZNS-127

MSA Project Manager:
Name/Title:
Date:

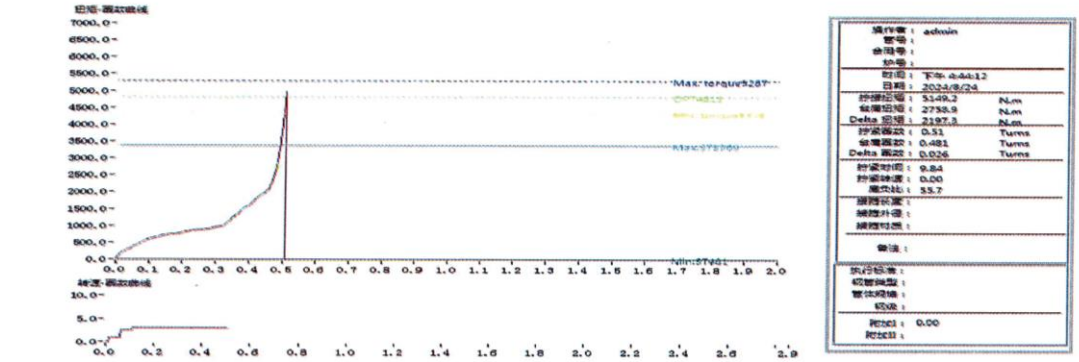
WQX:
Name/Title:
Date: 2024.9.17

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APDC CONTRACTOR Date:

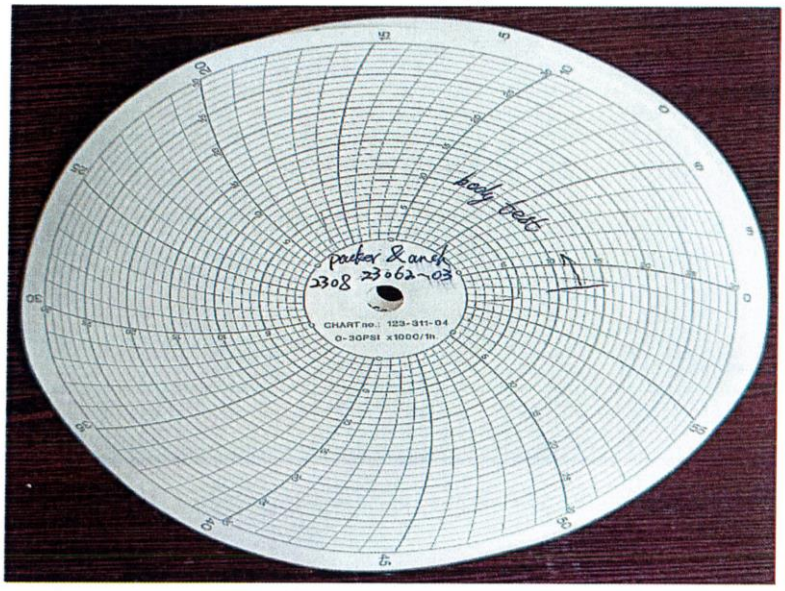
Figure 8-4: AZNS-127 Anchor Tubing Seal Test Report



Top pup joint – anchor seal



Packer – bottom pup joint



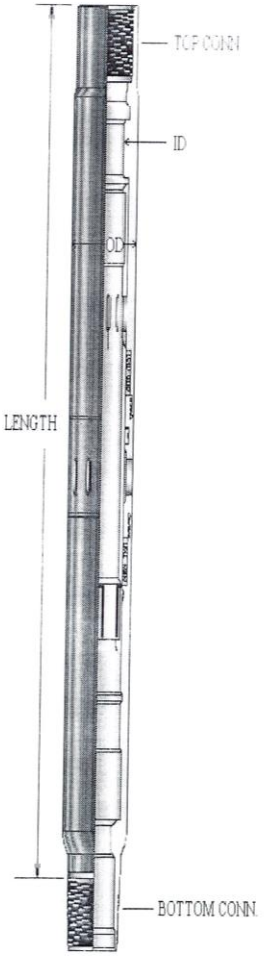
Pressure test curve

Figure 8-5: AZNS-127 Permanent Packer Connections & Body Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

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Sliding Side Door Test Report



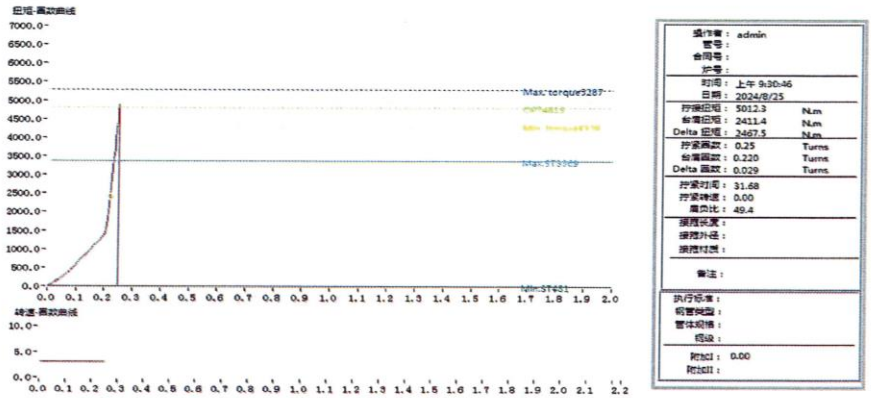
Type:	750W275-X39	
Serial Number:	2308 23039-05	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX× PIN	
Working Pressure	10,000 Psi	
OD	4.551"	
ID:	2.750"	
Seal Bore ID :	2.750"	
Length:	1.196m	
Temperature	300 DEG F	
Material:	S13CR-110KSI	
Profile:	X	
Shifting Direction:	DOWNWARD OPEN	
Top Connection:	Length:	1.541 m
Bottom Connection:	Length:	9.512 m
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
8,000 Psi blanking plug Pressure No Drop in 15 Minutes	Ok	
Well Name and Location:	AZNS-127	

MSA Project Manager:
Name/Title:
Date:

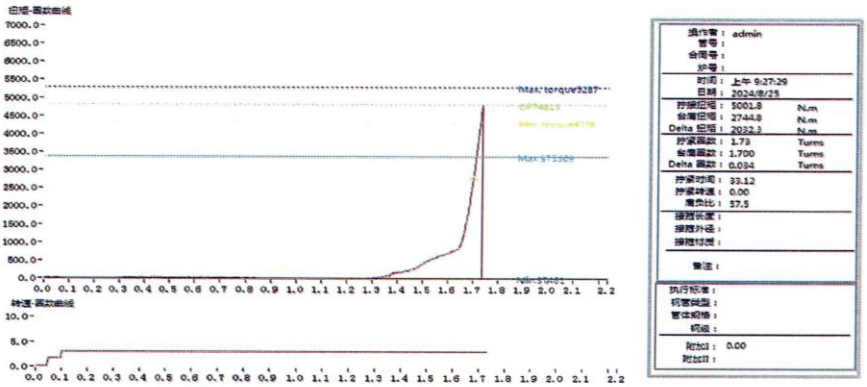
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MSA
Rig No:
EPDS CONTRACTOR Date:

WQX:
Name/Title:
Date: 2024.9.19

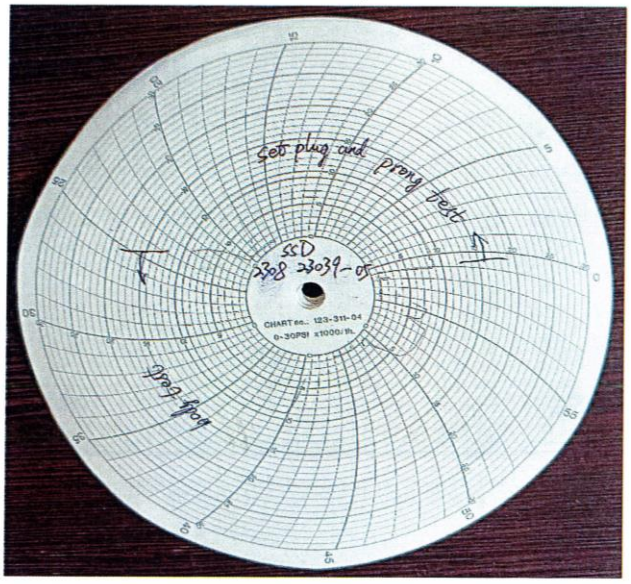
Figure 8-6: AZNS-127 SSD Test Report



Top pup joint – SSD



SSD – bottom joint



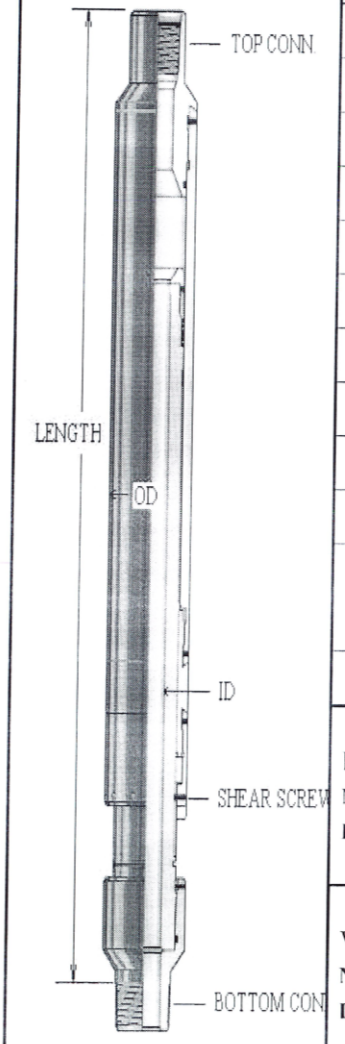
Pressure test curve

Figure 8-7: AZNS-127 SSD Connections & Body Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

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Expansion Joint Test Report



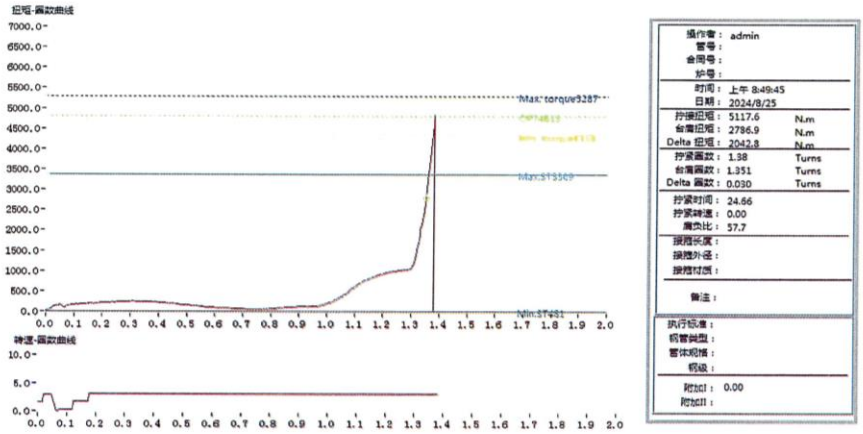
Type:	370W350-030	
Serial Number:	2308 23040-05	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX× PIN	
Working Pressure	10,000 Psi	
OD:	5.315"	
ID:	2.922"	
Length:	5.782 m	
Material	S13CR-110KSI	
Share Screw:	3	
Top Connection:	Length:	1.557 m
Bottom Connection:	Length:	1.529 m
8,000 Psi Pressure No Drop in 15 Minutes	ok	
Well Name and Location:	AZNS-127	

MSA Project Manager:
Name/Title:
Date:

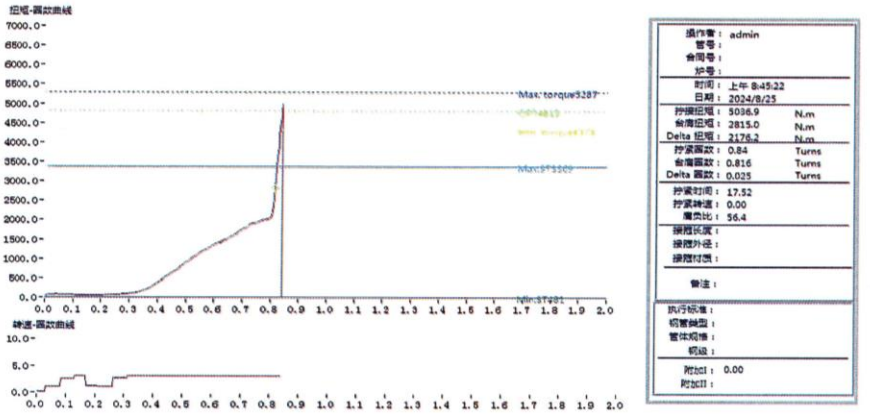
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Name/Title:
Date: 2024.9

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EPDS CONTRACTOR Date:

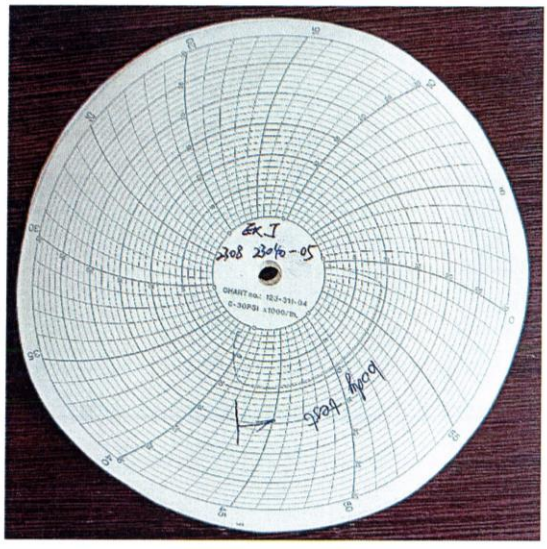
Figure 8-8: AZNS-127 EX Jt Test Report



Top pup joint – EX.J



EX.J – bottom pup joint



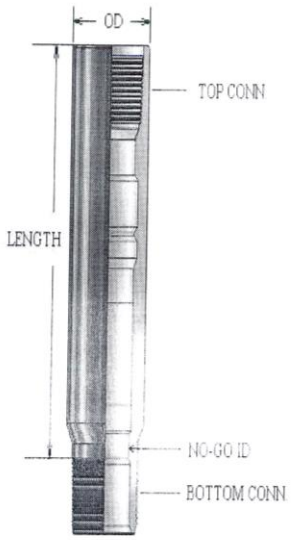
Pressure test curve

Figure 8-9: AZNS-127 EX Jt Connections & Body Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Landing Nipple Test Report



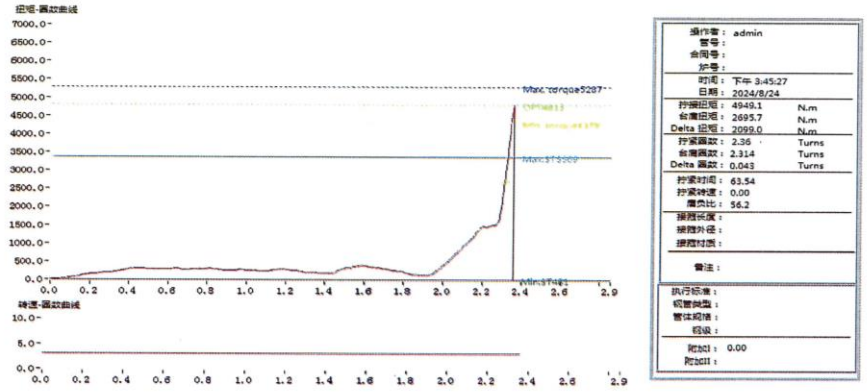
Type:	706W275-N24	
Serial Number:	2307 Y230416-05	
Connection Size:	3-1/2"10.22 PPF HSM3 BOX × PIN	
OD:	4.500"	
ID:	2.750"	
NO-GO ID:	2.635"	
Length:	0.378m	
Material:	S13CR-110 KSI	
Top Connection:	Length:	1.535m
8,000 Psi check valve Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-127	

MSA Project Manager:
Name/Title:
Date:

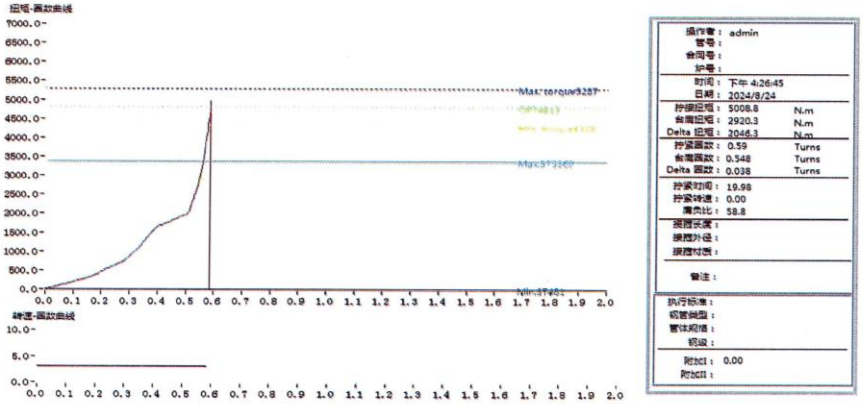
WQX:
Name/Title:
Date: 2024.9.19

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Drilling Supervisor
South Azadegan Project
Rig No
EPDS CONTRACTOR Date:

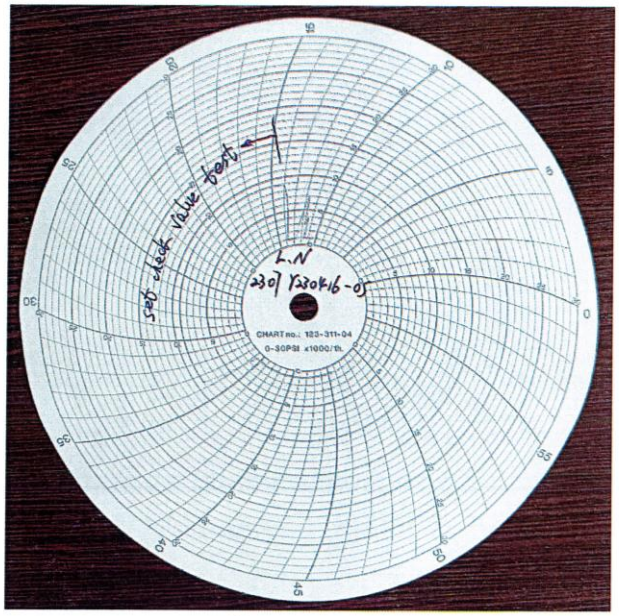
Figure 8-10: AZNS-127 Landing Nipple Test Report



Top point – L.N



L.N – Muleshoe



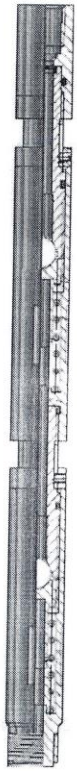
Pressure test curve

Figure 8-11: AZNS-127 Landing Nipple Connections & Body Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2309 23104-06
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	310mm
Material:	S13CR-110 KSI
8,000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-127

MSA Project Manager:
Name/Title:
Date:

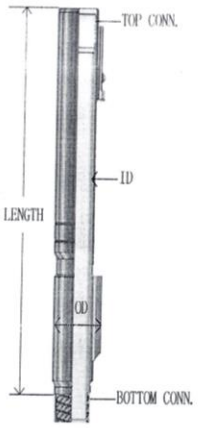
WQX:
Name/Title:
Date: 2024.9.19

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
Date:

Figure 8-12: AZNS-127 First CIV Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

Chemical Injection Mandrel Test Report



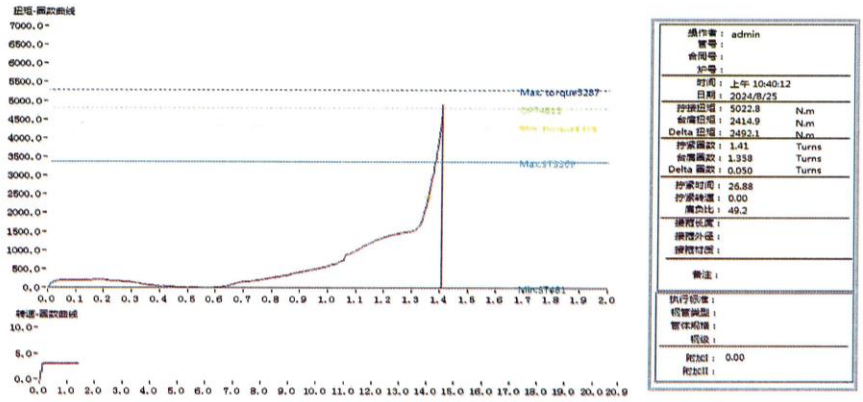
Type:	800W350-028	
Serial Number:	2309 Y230826-01	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	15,000Psi	
Temperature:	300 DEG F	
Length:	0.728m	
Top Connection:	Length:	1.558 m
Bottom Connection:	Length:	9.519 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-127	

MSA Project Manager:
Name/Title:
Date:

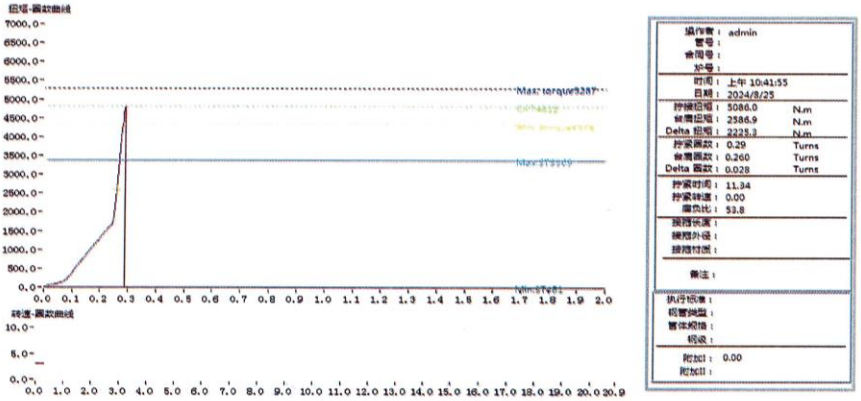
WQX:
Name/Title:
Date: 2024.9.19

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
EPDS CONTRACTOR Date:

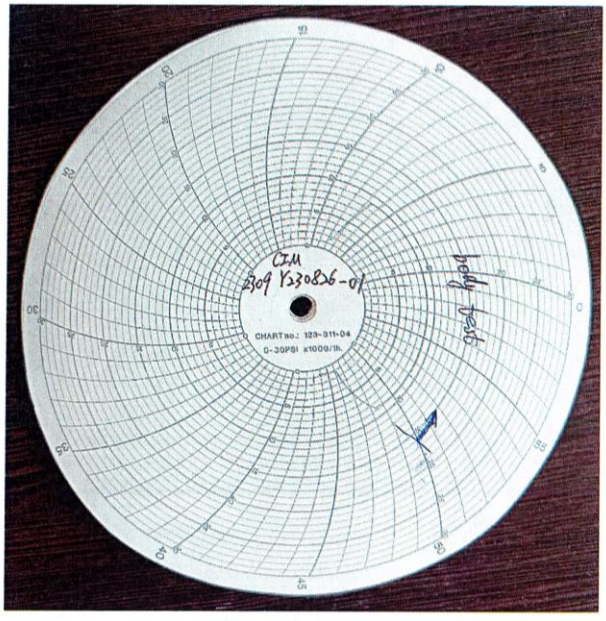
Figure 8-13: AZNS-127 First CIM Test Report



Top pup joint - CIM



CIM – bottom joint



Pressure test curve

Figure 8-14: AZNS-127 First CIM Connections & Body Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2309 23104-10
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	310mm
Material:	S13CR-110 KSI
8,000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-127

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.9.19

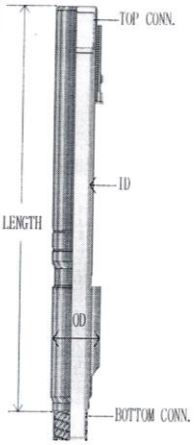
ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
EPDS CONTRACTOR Date:

Figure 8-15: AZNS-127 Second CIV Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Mandrel Test Report



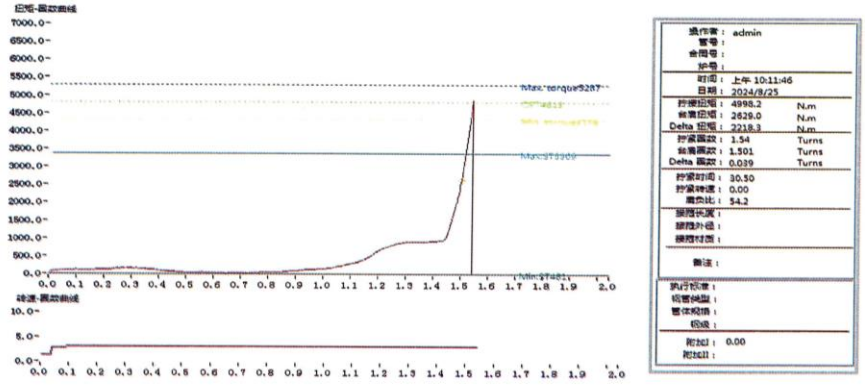
Type:	800W350-028	
Serial Number:	2309 Y230826-02	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	15.000Psi	
Temperature:	300 DEG F	
Length:	0.728m	
Top Connection:	Length:	1.518 m
Bottom Connection:	Length:	9.524 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-127	

MSA Project Manager:
Name/Title:
Date:

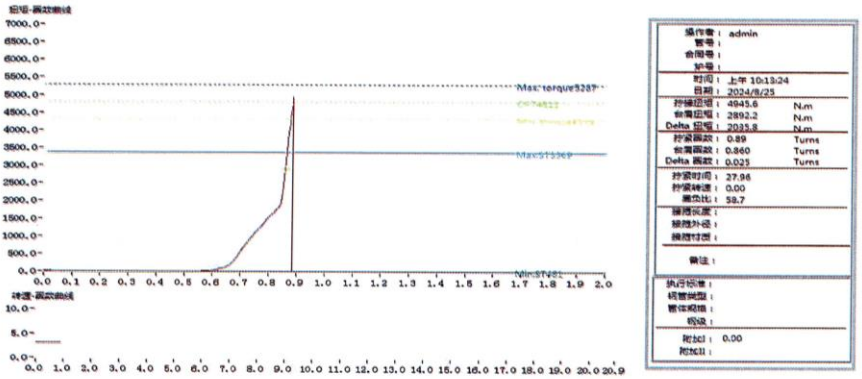
ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
EPDS CONTRACTOR Date:

WQX:
Name/Title:
Date: 2024.9.19

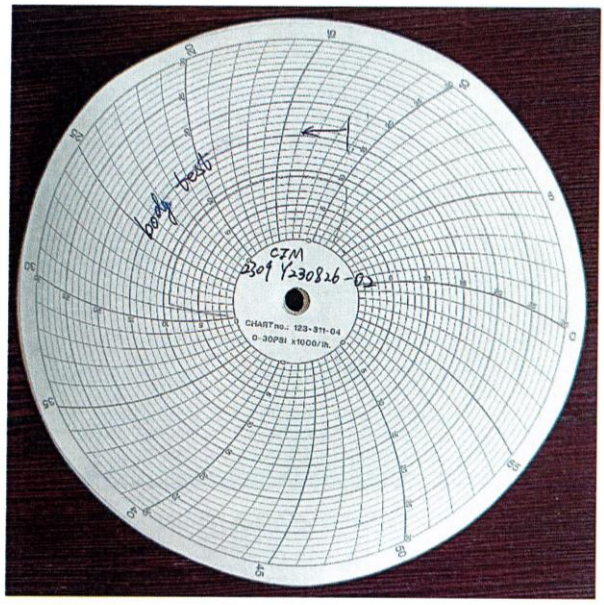
Figure 8-16: AZNS-127 Second CIM Test Report



Top pup joint - CIM



CIM – bottom joint



Pressure test curve

Page 18 of 18

Figure 8-17: AZNS-127 Second CIM Connections & Body Pressure Test Graph

8.4 Completion Operation Checklist

Item	Operation Name	Remark	Check Mark✓
1	Scraper Trip	Packer depth should be scraped	√
2	Pre-job preparation	Pre-connection All Down Hole Equipment With Upper & Lower Pup Joints On Confirmed Thread-makeup at Workshop, And Pressure Test Prior To Being Transported To Well Site.	√
		Retrieve Wear Bushing, M/U Jet Sub And Flush Bop Stack & Well Head With Fresh Water.	√
		All Crossovers, Pup Joints For Space Out Measured Put In Right Place.	√
		Lift Control Line And Protectors On Rig Floor.	√
		Hold A Pre-Job Safety Meeting For Running Upper Completion String.	√
3	Assembling and RIH	M/U 3-1/2" Re-entry Guide Shoe And 3-1/2" BTM No Go Landing Nipple	√
		Check The Conditions Landing Nipple To Make Sure There Is No Damage, Observe There Is No Any Trash Trapped Inside.	√
		Assemble Landing Nipple.	√
		RIH 1 Joint Tubing As Per Tally	√
		P/U And M/U Hydraulic-Set PERMANENT Production Packer.	√
		Much More Care Should Be Taken When Packer Is Passing Well Head And Liner Hanger While RIH,	√
		Assemble Ratch Latch If Packer	-
		Slack Off Packer Assembly, Double Check The Quantity Of Shear Pins,	√

		Remove The Tape On The Element, Check If Any Damages On It.	
		RIH 1 Joint Tubing As Per Tally	√
		M/U Pup Joint And SSD (close Position) To Tubing String With check valve Inside.	√
		Pressure Test Tubing String 7000psi/15min Every 50 JTs.	√
		RIH Completion String and M/U traveling joint as per running tally.	√
		M/U 2 NO. CIV as per running tally& Press Test to 2000 Psi.	-
		Cont. Running Tubing as per Tally while Install Cross protector for the 2 NO. 3/8"control line.	-
		M/U TRSSSV	√
		Install control line to TRSSSV and pressure test control line 8000psi x 15mins. And then keep 5000psi pressure while running to ensure TRSSSV open. Install TRSSSV protector for the 1/4" control line. Continue to run 4-1/2" tubing as per running tally.	√
4	M/U and Space out Tubing Hanger	Pick up & M/U tubing hanger with double pin crossover. M/U landing joint. Check string weight upward and downward.	√
		Roughly cut control line about 2 meters above tubing hanger, wrap 2-3 turns and then feed through Tubing Hanger.	√
		Check string weight upward and downward and Space out Tubing Hanger 1.5 m (Half stroke) above tubing head spool	√
Item	**Operation Name**	**Remark**	**Check Mark✓**
5	Set	R/Up Slick Line. Remove Check valve inside SSD	√

	production packer	RIH and Set check vlve in Landing Nipple. POOH to surface.	√
		Pressurize inside tubing to 500psi to ensure traveling joint is full open. Increase Pressure gradually to SET PACKER. Increase pressure to 7000 Psi to Test String.	√
7	Land Tubing hanger	Move down completion string and apply required weight to shear traveling joint load pins and Land Tubing hanger in T.H.S. Slack off completely & Record the weight.	√
		Drain the BOP stack through the gate valves of tubing head before landing hanger in THS. Leave the valves open.	√
		Tie Down Screws on THS should be fully open before Operation	√
8	N/D BOP & N/U X-mas tree	Tighten tie down screws.	√
		Test inside string w/ 5000 psi, 15 min.	√
		R/Up Slick Line. RIH with slick line and Ret. Check valve from Landing Nipple. POOH to surface.	√
		Test Annulus w/ 2000 Psi 15min.	√
		Close BOP ram and test above tubing hanger (annulus between BOP and landing joint) to 1000psi.	√
		B/O & L/D landing joint.	√
		Set 4” 2Way in TBG Hanger & Test it, Back off and L/D Running Tool.	√
		N/D BOP.	√
		Feed control line, through head bonnet.	√
		Check hanger for any damage and pressure test control line to 5000psi	√

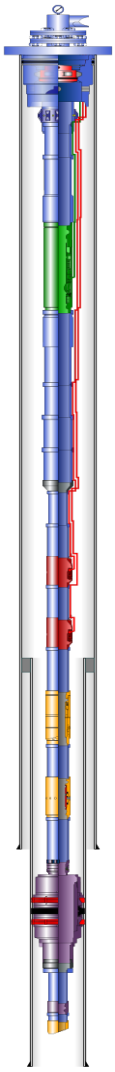
		Pressure test chemical injection line & rapture the disc after installing Bonnet.	-
		N/U x-mass tree as program.(Bottom Master)	√
		Perform required tests for X-mass tree. (Bottom Master)	√
		Connect from X-mass tree kill valve to the pump unit and test X-Mas tree body to 1000/*7500* psi for 15 min against TWCV.	√
		Remove 4" two way, install 4" one way	√
		Close TRSSSV	√
		Release the rig. Clean the well site.	√

Table 8-4: AZNS-127 Operation Checklist

8.5 Completion Sketch & Tally

AZNS-127 Completion Sketch

Contractor	MSA / OEOC	Liner & Tubing Data					Client	PEDEC / PPL
Field	AZADEGAN	Description	7"Liner	4 1/2" Liner	4 1/2" Tubing	3 1/2" Tubing	7'' Liner Lap:	3147
Well Number:	AZNS-127	OD[in]	7.00	4.50	4.50	3.50	7'' Liner Shoe:	3891
Rig :	208-OEOC	Nominal weight [lb/ft]	29.00	13.50	18.90	10.20	4.5'' Liner Lap:	3780.5
RTE-GLE	14.45	Grade	110SS	110SS	L-80	L-80	4.5" Liner Shoe:	4139.5
RTE-THF	11.33	Thread	TP-G2	TP-G2	HSM3	HSM3	Total Depth:	4140
Rig Spudded Date :	8/5/2024	Inside diameter [in]	6.184	3.920	3.640	2.915	X-mass Tree:	MSP-T Type
End of Operation		API Drift Diameter	6.059	3.795	3.515	2.797	Size & Rating:	4-1/16" 10K

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m) F/ RTE	TOP(m) F/ RTE
	1	Elevation (Floor - TBG Head Flange) + Running			11.788	11.788	
	2	TBG HANGER MSP, 10K, 11" * 4 1/2'' ACME Box up * 4 1/2'' TPCQ-EX Box Down , M/C: FF-0.5 T/C: UPSL: 3G PR: 2, 4" Type ''H'' Two Way	5.118	3.64	0.520	12.308	11.788
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* Super 13CR-110KSI, Weiqixin Co	6.1	3.64	0.964	13.272	12.308
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.567	60.839	13.272
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.624	62.463	60.839
	6	4-1/2" FLOW COUPLING,18.9#,Super 13CR-110KSI,HSM3 , 6 FT, Weiqixin Co	5.5	3.64	1.726	64.189	62.463
	7	TRSSV-10 TBG Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , Super 13CR-110KSI, 10K WP With AF Profile, Flapper Type, Weiqixin Co	7.5	3.437	1.783	65.972	64.189
	8	4-1/2" FLOW COUPLING, 18.9#,Super 13CR-110KSI,HSM3 , 4' FT, Weiqixin Co	5.5	3.64	1.117	67.089	65.972
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.624	68.713	67.089
	10	318 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	3019.032	3087.745	68.713
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.119	3088.864	3087.745
	12	17 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	160.982	3249.846	3088.864
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.560	3251.406	3249.846
	14	3 1/2" CIV TBG Retrievable Chemical Injection Mandrel, HSM3 , 10.2#,Super 13CR-110KSI w/ check valve, dual type & rupture disc, Weiqixin Co	5	2.922	0.730	3252.136	3251.406
	15	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.524	3261.660	3252.136
	16	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.517	3263.177	3261.660
	17	3 1/2" CIV TBG Retrievable Chemical Injection Mandrel, HSM3 , 10.2#,Super 13CR-110KSI w/ check valve, dual type & rupture disc, Weiqixin Co	5	2.922	0.730	3263.907	3263.177
	18	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.529	3273.436	3263.907
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.557	3274.993	3273.436
	20	3-1/2" EXP. Joint,HSM3 , Super 13CR-110KSI, 10'STK (Half Open) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 lbs, Weiqixin Co	5.135	2.922	7.260	3282.253	3274.993
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.530	3283.783	3282.253
	22	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.528	3293.311	3283.783
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.556	3294.867	3293.311
	24	3-1/2" Non-Elastomeric Sliding Side Door, Super 13CR-110KSI, HSM3 With Type X Selective Profile (Profile: 2.750" X, Downward open), Weiqixin Co	4.551	2.75	1.196	3296.063	3294.867
	25	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.519	3305.582	3296.063
	26	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.508	3307.090	3305.582
	27	3-1/2" Anchor seal assembly,HSM3, 10.2#,Super 13CR-110KSI, 4" Packing OD, Weiqixin Co	4.000	2.922	0.230	3307.320	3307.090
	28	7' Hydraulic set permanent production PKR, model WSDP to be run in 7.00" 29-32 PPF CSG, Super 13CR-110KSI, UPPER SEAL BORE: 4", Weiqixin Co	5.689	2.922	4.443	3311.763	3307.320
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.250	2.922	1.537	3313.300	3311.763
	30	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.524	3322.824	3313.300
	31	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.561	3324.385	3322.824
	32	3-1/2" Bottom No-Go Landing,HSM3 , 10.2#, Super 13CR-110KSI W/ Seal Bore 2.75" Type XN, Weiqixin Co	4.5	2.562	0.374	3324.759	3324.385
	33	3-1/2" Wirline Re-entry guide, 10.2 #,HSM3 Con,Super 13CR-110KSI, Weiqixin Co	4.25	2.922	0.241	3325.000	3324.759
		END OF COMPLETION STRING					3325.000

TR3SV	65.972	64.189
Traveling Joint	3282.253	3274.993
CIV 1	3252.136	3251.406
CIV 2	3263.907	3263.177
SSD	3296.063	3294.867
7'' HYDRAULIC SET PERMANENT PRODUCTION PACKER	3311.763	3307.320
NO-GO Landing Nipple	3324.759	3324.385
Entry Guide	3325.000	3324.759
End of String	3325.000	

Figure 8-18: AZNS-127 Completion Sketch

MSA

Field :	Azadegan		Well NO:	AZNS127	Rig NO:	OEOC208	Drig CO:	OEOC	
PEDEC, PPL			Completion String						
Grade	ID "	Drift "	Weight lb/ft	Burst psi	Collaspse psi	Yield klbs	Coupling "	TQ ft x lb	Nominal Dia. OD
L-80	2.922	2.797	10.2	11,560.0	12,120.0	233	3.962	3200-3550-3900	3-1/2"
L-80	3.640	3.515	18.9	13,380.0	13,820.0	440	5.137	7920-8800-9680	4-1/2"
BTM NO GO Depth:	3324.39		Formation Name :		Fahliyan	Hook Weight :			58 k lb
BTM of String	3325.00		Formation Top :		3989	VOLUME (TOTAL)			136.85
Packer Depth :	3307.32					Mud Weight (pcf) :			104

Number on jt	Grade	Run In No.	Jt Length (m)	Cum Length (m)	Dist to RT (m)	M/D weight (Klbs)	VOL(BBL)	Remarks	
								joint no	IN / OUT
					3325.000	58.00			
3-1/2" Wirline Re-entry guide, 10.2 #,HSM3 Con,Super 13CR-110KSI, Weiqixin Co	3-1/2" Mule Shoe		0.241	0.24	3324.759	58.01	0.01		
3-1/2" Bottom No-Go Landing,HSM3 , 10.2#, Super 13CR-110KSI W/ Seal Bore 2.75" Type XN, Weiqixin Co	3-1/2" BTM NO GO		0.374	0.62	3324.385	58.02	0.02		
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.561	2.18	3322.824	58.06	0.06		
#A4	TBG 3-1/2", 10.2#, L-80,HSM3	1	9.524	11.70	3313.300	58.31	0.32	1	
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.537	13.24	3311.763	58.35	0.36		
7" Hydraulic set permanent production PKR, model WSDP to be run in 7.00" 29-32 PPF CSG, Super 13CR-110KSI, UPPER SEAL BORE: 4", Weiqixin Co	PKR		4.443	17.68	3307.320	58.47	0.48		
3-1/2" Anchor seal assembly,HSM3, 10.2#,Super 13CR-110KSI, 4" Packing OD, Weiqixin Co	Ratch Lach		0.230	17.91	3307.090	58.47	0.49		
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.508	19.42	3305.582	58.51	0.53		
#A1	TBG 3-1/2", 10.2#, L-80,HSM3	2	9.519	28.94	3296.063	58.76	0.79	2	
3-1/2" Non-Elastomeric Sliding Side Door, Super 13CR-110KSI, HSM3 With Type X Selective Profile (Profile: 2.750" X, Downward open), Weiqixin Co	SSD		1.196	30.13	3294.867	58.79	0.82		
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.556	31.69	3293.311	58.83	0.86		
#A5	TBG 3-1/2", 10.2#, L-80,HSM3	3	9.528	41.22	3283.783	59.09	1.12	3	
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.530	42.75	3282.253	59.13	1.16		
3-1/2" EXP. Joint,HSM3 , Super 13CR-110KSI, 10'STK (Fully Open) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 lbs, Weiqixin Co	Traveling Jt		7.260	50.01	3274.993	59.32	1.36		
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.557	51.56	3273.436	59.36	1.40		
#A2	TBG 3-1/2", 10.2#, L-80,HSM3	4	9.529	61.09	3263.907	59.61	1.66	4	
3 1/2" CIV TBG Retrievable Chemical Injection Mandrel, HSM3 , 10.2#,Super 13CR-110KSI w/ check valve, dual type & rupture disc, Weiqixin Co	CIV #1		0.730	61.82	3263.177	59.63	1.68		
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.517	63.34	3261.660	59.67	1.72		
#A3	TBG 3-1/2", 10.2#, L-80,HSM3	5	9.524	72.86	3252.136	59.92	1.98	5	
3 1/2" CIV TBG Retrievable Chemical Injection Mandrel, HSM3 , 10.2#,Super 13CR-110KSI w/ check valve, dual type & rupture disc, Weiqixin Co	CIV #2		0.730	73.59	3251.406	59.94	2.00		
3-1/2" Pup Jt,HSM3 , 10.2#, L-80, Type-1	Pup Jt 3-1/2", 10.2#,HSM3		1.560	75.15	3249.846	59.98	2.05		Protector
#A6	TBG 3-1/2", 10.2#, L-80,HSM3	6	9.504	84.66	3240.342	60.23	2.30	6	Protector

#A7	TBG 3-1/2", 10.2#, L-80,HSM3	7	9.517	94.18	3230.825	60.48	2.56	7	Protector
#A8	TBG 3-1/2", 10.2#, L-80,HSM3	8	9.062	103.24	3221.763	60.72	2.81	8	Protector
#A9	TBG 3-1/2", 10.2#, L-80,HSM3	9	9.528	112.77	3212.235	60.97	3.07	9	Protector
#A10	TBG 3-1/2", 10.2#, L-80,HSM3	10	9.063	121.83	3203.172	61.21	3.32	10	Protector
#A11	TBG 3-1/2", 10.2#, L-80,HSM3	11	9.520	131.35	3193.652	61.46	3.57	11	Protector
#A12	TBG 3-1/2", 10.2#, L-80,HSM3	12	9.524	140.87	3184.128	61.71	3.83	12	Protector
#A13	TBG 3-1/2", 10.2#, L-80,HSM3	13	9.521	150.39	3174.607	61.96	4.09	13	Protector
#A14	TBG 3-1/2", 10.2#, L-80,HSM3	14	9.525	159.92	3165.082	62.21	4.35	14	Protector
#A15	TBG 3-1/2", 10.2#, L-80,HSM3	15	9.525	169.44	3155.557	62.46	4.61	15	Protector
#A16	TBG 3-1/2", 10.2#, L-80,HSM3	16	9.533	178.98	3146.024	62.72	4.87	16	Protector
#A17	TBG 3-1/2", 10.2#, L-80,HSM3	17	9.516	188.49	3136.508	62.97	5.13	17	Protector
#A18	TBG 3-1/2", 10.2#, L-80,HSM3	18	9.527	198.02	3126.981	63.22	5.39	18	Protector
#A19	TBG 3-1/2", 10.2#, L-80,HSM3	19	9.524	207.54	3117.457	63.47	5.65	19	Protector
#A20	TBG 3-1/2", 10.2#, L-80,HSM3	20	9.535	217.08	3107.922	63.72	5.91	20	Protector
#A21	TBG 3-1/2", 10.2#, L-80,HSM3	21	9.524	226.60	3098.398	63.97	6.17	21	Protector
#A22	TBG 3-1/2", 10.2#, L-80,HSM3	22	9.534	236.14	3088.864	64.22	6.43	22	Protector
X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	XOS 4-1/2", HSM3 B X 3-1/2" HSM3, P	22	1.119	237.26	3087.745	64.28	6.46		
#1	TBG 4-1/2", 18.9#, L-80,HSM3	23	9.572	246.83	3078.173	64.74	6.86	23	Protector
#2	TBG 4-1/2", 18.9#, L-80,HSM3	24	9.525	256.35	3068.648	65.21	7.26	24	Protector
#3	TBG 4-1/2", 18.9#, L-80,HSM3	25	9.526	265.88	3059.122	65.67	7.67	25	Protector
#4	TBG 4-1/2", 18.9#, L-80,HSM3	26	9.550	275.43	3049.572	66.14	8.07	26	Protector
#5	TBG 4-1/2", 18.9#, L-80,HSM3	27	9.527	284.96	3040.045	66.61	8.47	27	Protector
#6	TBG 4-1/2", 18.9#, L-80,HSM3	28	9.544	294.50	3030.501	67.07	8.87	28	Protector
#7	TBG 4-1/2", 18.9#, L-80,HSM3	29	9.541	304.04	3020.960	67.54	9.28	29	Protector
#8	TBG 4-1/2", 18.9#, L-80,HSM3	30	9.551	313.59	3011.409	68.00	9.68	30	Protector
#9	TBG 4-1/2", 18.9#, L-80,HSM3	31	9.441	323.03	3001.968	68.46	10.08	31	Protector
#10	TBG 4-1/2", 18.9#, L-80,HSM3	32	9.531	332.56	2992.437	68.93	10.48	32	Protector
#11	TBG 4-1/2", 18.9#, L-80,HSM3	33	9.539	342.10	2982.898	69.40	10.88	33	Protector
#12	TBG 4-1/2", 18.9#, L-80,HSM3	34	9.528	351.63	2973.370	69.86	11.29	34	Protector
#13	TBG 4-1/2", 18.9#, L-80,HSM3	35	9.524	361.15	2963.846	70.33	11.69	35	Protector
#14	TBG 4-1/2", 18.9#, L-80,HSM3	36	9.530	370.68	2954.316	70.79	12.09	36	Protector
#15	TBG 4-1/2", 18.9#, L-80,HSM3	37	9.526	380.21	2944.790	71.26	12.49	37	Protector
#16	TBG 4-1/2", 18.9#, L-80,HSM3	38	9.553	389.76	2935.237	71.72	12.90	38	Protector
#17	TBG 4-1/2", 18.9#, L-80,HSM3	39	9.532	399.30	2925.705	72.19	13.30	39	Protector
#18	TBG 4-1/2", 18.9#, L-80,HSM3	40	9.558	408.85	2916.147	72.65	13.70	40	Protector / Press Test
#19	TBG 4-1/2", 18.9#, L-80,HSM3	41	9.529	418.38	2906.618	73.12	14.11	41	Protector
#20	TBG 4-1/2", 18.9#, L-80,HSM3	42	9.532	427.91	2897.086	73.58	14.51	42	Protector
#21	TBG 4-1/2", 18.9#, L-80,HSM3	43	9.532	437.45	2887.554	74.05	14.91	43	Protector
#22	TBG 4-1/2", 18.9#, L-80,HSM3	44	9.571	447.02	2877.983	74.52	15.31	44	Protector
#23	TBG 4-1/2", 18.9#, L-80,HSM3	45	9.190	456.21	2868.793	74.97	15.70	45	Protector
#24	TBG 4-1/2", 18.9#, L-80,HSM3	46	9.434	465.64	2859.359	75.43	16.10	46	Protector
#25	TBG 4-1/2", 18.9#, L-80,HSM3	47	9.526	475.17	2849.833	75.89	16.50	47	Protector

#26	TBG 4-1/2", 18.9#, L-80,HSM3	48	9.534	484.70	2840.299	76.36	16.91	48	Protector
#27	TBG 4-1/2", 18.9#, L-80,HSM3	49	9.530	494.23	2830.769	76.82	17.31	49	Protector
#28	TBG 4-1/2", 18.9#, L-80,HSM3	50	9.531	503.76	2821.238	77.29	17.71	50	Protector
#29	TBG 4-1/2", 18.9#, L-80,HSM3	51	9.532	513.29	2811.706	77.75	18.11	51	Protector
#30	TBG 4-1/2", 18.9#, L-80,HSM3	52	9.524	522.82	2802.182	78.22	18.52	52	Protector
#31	TBG 4-1/2", 18.9#, L-80,HSM3	53	9.528	532.35	2792.654	78.68	18.92	53	Protector
#32	TBG 4-1/2", 18.9#, L-80,HSM3	54	9.525	541.87	2783.129	79.15	19.32	54	Protector
#33	TBG 4-1/2", 18.9#, L-80,HSM3	55	9.525	551.40	2773.604	79.61	19.72	55	Protector
#34	TBG 4-1/2", 18.9#, L-80,HSM3	56	9.531	560.93	2764.073	80.08	20.13	56	Protector
#35	TBG 4-1/2", 18.9#, L-80,HSM3	57	9.531	570.46	2754.542	80.54	20.53	57	Protector
#36	TBG 4-1/2", 18.9#, L-80,HSM3	58	9.534	579.99	2745.008	81.01	20.93	58	Protector
#37	TBG 4-1/2", 18.9#, L-80,HSM3	59	9.531	589.52	2735.477	81.48	21.33	59	Protector
#38	TBG 4-1/2", 18.9#, L-80,HSM3	60	9.578	599.10	2725.899	81.94	21.74	60	Protector
#39	TBG 4-1/2", 18.9#, L-80,HSM3	61	9.534	608.64	2716.365	82.41	22.14	61	Protector
#40	TBG 4-1/2", 18.9#, L-80,HSM3	62	9.531	618.17	2706.834	82.87	22.54	62	Protector
#41	TBG 4-1/2", 18.9#, L-80,HSM3	63	9.544	627.71	2697.290	83.34	22.95	63	Protector
#42	TBG 4-1/2", 18.9#, L-80,HSM3	64	9.528	637.24	2687.762	83.80	23.35	64	Protector
#43	TBG 4-1/2", 18.9#, L-80,HSM3	65	9.382	646.62	2678.380	84.26	23.74	65	Protector
#44	TBG 4-1/2", 18.9#, L-80,HSM3	66	9.161	655.78	2669.219	84.71	24.13	66	Protector
#45	TBG 4-1/2", 18.9#, L-80,HSM3	67	9.574	665.36	2659.645	85.18	24.54	67	Protector
#46	TBG 4-1/2", 18.9#, L-80,HSM3	68	9.528	674.88	2650.117	85.64	24.94	68	Protector
#47	TBG 4-1/2", 18.9#, L-80,HSM3	69	9.578	684.46	2640.539	86.11	25.34	69	Protector
#48	TBG 4-1/2", 18.9#, L-80,HSM3	70	9.538	694.00	2631.001	86.58	25.74	70	Protector
#49	TBG 4-1/2", 18.9#, L-80,HSM3	71	9.534	703.53	2621.467	87.04	26.15	71	Protector
#50	TBG 4-1/2", 18.9#, L-80,HSM3	72	9.527	713.06	2611.940	87.51	26.55	72	Protector
#51	TBG 4-1/2", 18.9#, L-80,HSM3	73	9.566	722.63	2602.374	87.97	26.95	73	Protector
#52	TBG 4-1/2", 18.9#, L-80,HSM3	74	9.244	731.87	2593.130	88.42	27.34	74	Protector
#53	TBG 4-1/2", 18.9#, L-80,HSM3	75	9.521	741.39	2583.609	88.89	27.75	75	Protector
#54	TBG 4-1/2", 18.9#, L-80,HSM3	76	9.543	750.93	2574.066	89.36	28.15	76	Protector
#55	TBG 4-1/2", 18.9#, L-80,HSM3	77	9.530	760.46	2564.536	89.82	28.55	77	Protector
#56	TBG 4-1/2", 18.9#, L-80,HSM3	78	9.539	770.00	2554.997	90.29	28.95	78	Protector
#57	TBG 4-1/2", 18.9#, L-80,HSM3	79	9.138	779.14	2545.859	90.73	29.34	79	Protector
#58	TBG 4-1/2", 18.9#, L-80,HSM3	80	9.529	788.67	2536.330	91.20	29.74	80	Protector
#59	TBG 4-1/2", 18.9#, L-80,HSM3	81	9.568	798.24	2526.762	91.67	30.15	81	Protector
#60	TBG 4-1/2", 18.9#, L-80,HSM3	82	9.523	807.76	2517.239	92.13	30.55	82	Protector
#61	TBG 4-1/2", 18.9#, L-80,HSM3	83	9.534	817.30	2507.705	92.60	30.95	83	Protector
#62	TBG 4-1/2", 18.9#, L-80,HSM3	84	9.545	826.84	2498.160	93.06	31.35	84	Protector
#63	TBG 4-1/2", 18.9#, L-80,HSM3	85	9.527	836.37	2488.633	93.53	31.76	85	Protector
#64	TBG 4-1/2", 18.9#, L-80,HSM3	86	9.526	845.89	2479.107	93.99	32.16	86	Protector
#65	TBG 4-1/2", 18.9#, L-80,HSM3	87	9.553	855.45	2469.554	94.46	32.56	87	Protector
#66	TBG 4-1/2", 18.9#, L-80,HSM3	88	9.541	864.99	2460.013	94.92	32.97	88	Protector
#67	TBG 4-1/2", 18.9#, L-80,HSM3	89	9.523	874.51	2450.490	95.39	33.37	89	Protector
#68	TBG 4-1/2", 18.9#, L-80,HSM3	90	9.552	884.06	2440.938	95.86	33.77	90	Protector
#69	TBG 4-1/2", 18.9#, L-80,HSM3	91	9.403	893.47	2431.535	96.31	34.17	91	Protector
#70	TBG 4-1/2", 18.9#, L-80,HSM3	92	9.574	903.04	2421.961	96.78	34.57	92	Protector
#71	TBG 4-1/2", 18.9#, L-80,HSM3	93	9.526	912.57	2412.435	97.25	34.97	93	Protector
#72	TBG 4-1/2", 18.9#, L-80,HSM3	94	9.538	922.10	2402.897	97.71	35.38	94	Protector
#74	TBG 4-1/2", 18.9#, L-80,HSM3	95	9.542	931.65	2393.355	98.18	35.78	95	Protector
#75	TBG 4-1/2", 18.9#, L-80,HSM3	96	9.531	941.18	2383.824	98.64	36.18	96	Protector
#76	TBG 4-1/2", 18.9#, L-80,HSM3	97	9.575	950.75	2374.249	99.11	36.59	97	Protector
#77	TBG 4-1/2", 18.9#, L-80,HSM3	98	9.525	960.28	2364.724	99.58	36.99	98	Protector
#78	TBG 4-1/2", 18.9#, L-80,HSM3	99	9.572	969.85	2355.152	100.04	37.39	99	Protector
#79	TBG 4-1/2", 18.9#, L-80,HSM3	100	9.455	979.30	2345.697	100.51	37.79	100	Protector
#80	TBG 4-1/2", 18.9#, L-80,HSM3	101	9.565	988.87	2336.132	100.97	38.20	101	Protector

#81	TBG 4-1/2", 18.9#, L-80,HSM3	102	9.578	998.45	2326.554	101.44	38.60	102	Protector
#82	TBG 4-1/2", 18.9#, L-80,HSM3	103	8.895	1007.34	2317.659	101.87	38.98	103	Protector
#83	TBG 4-1/2", 18.9#, L-80,HSM3	104	9.424	1016.77	2308.235	102.33	39.38	104	Protector
#84	TBG 4-1/2", 18.9#, L-80,HSM3	105	9.403	1026.17	2298.832	102.79	39.77	105	Protector
#85	TBG 4-1/2", 18.9#, L-80,HSM3	106	9.532	1035.70	2289.300	103.26	40.17	106	Protector
#86	TBG 4-1/2", 18.9#, L-80,HSM3	107	9.576	1045.28	2279.724	103.73	40.58	107	Protector
#87	TBG 4-1/2", 18.9#, L-80,HSM3	108	9.552	1054.83	2270.172	104.19	40.98	108	Protector
#88	TBG 4-1/2", 18.9#, L-80,HSM3	109	9.532	1064.36	2260.640	104.66	41.39	109	Protector
#89	TBG 4-1/2", 18.9#, L-80,HSM3	110	9.542	1073.90	2251.098	105.12	41.79	110	Protector
#90	TBG 4-1/2", 18.9#, L-80,HSM3	111	9.527	1083.43	2241.571	105.59	42.19	111	Protector
#91	TBG 4-1/2", 18.9#, L-80,HSM3	112	9.467	1092.90	2232.104	106.05	42.59	112	Protector
#92	TBG 4-1/2", 18.9#, L-80,HSM3	113	9.530	1102.43	2222.574	106.52	42.99	113	Protector
#93	TBG 4-1/2", 18.9#, L-80,HSM3	114	9.526	1111.95	2213.048	106.98	43.39	114	Protector
#94	TBG 4-1/2", 18.9#, L-80,HSM3	115	9.529	1121.48	2203.519	107.45	43.80	115	Protector
#95	TBG 4-1/2", 18.9#, L-80,HSM3	116	9.570	1131.05	2193.949	107.91	44.20	116	Protector
#96	TBG 4-1/2", 18.9#, L-80,HSM3	117	9.523	1140.57	2184.426	108.38	44.60	117	Protector
#97	TBG 4-1/2", 18.9#, L-80,HSM3	118	9.568	1150.14	2174.858	108.85	45.01	118	Protector
#98	TBG 4-1/2", 18.9#, L-80,HSM3	119	9.550	1159.69	2165.308	109.31	45.41	119	Protector
#99	TBG 4-1/2", 18.9#, L-80,HSM3	120	9.544	1169.24	2155.764	109.78	45.81	120	Protector
#100	TBG 4-1/2", 18.9#, L-80,HSM3	121	9.526	1178.76	2146.238	110.24	46.22	121	Protector
#101	TBG 4-1/2", 18.9#, L-80,HSM3	122	9.543	1188.31	2136.695	110.71	46.62	122	Protector
#102	TBG 4-1/2", 18.9#, L-80,HSM3	123	9.536	1197.84	2127.159	111.17	47.02	123	Protector
#103	TBG 4-1/2", 18.9#, L-80,HSM3	124	9.534	1207.38	2117.625	111.64	47.42	124	Protector
#104	TBG 4-1/2", 18.9#, L-80,HSM3	125	9.568	1216.94	2108.057	112.11	47.83	125	Protector
#105	TBG 4-1/2", 18.9#, L-80,HSM3	126	9.580	1226.52	2098.477	112.58	48.23	126	Protector
#106	TBG 4-1/2", 18.9#, L-80,HSM3	127	9.531	1236.05	2088.946	113.04	48.64	127	Protector
#107	TBG 4-1/2", 18.9#, L-80,HSM3	128	9.532	1245.59	2079.414	113.51	49.04	128	Protector
#108	TBG 4-1/2", 18.9#, L-80,HSM3	129	9.528	1255.11	2069.886	113.97	49.44	129	Protector
#109	TBG 4-1/2", 18.9#, L-80,HSM3	130	9.146	1264.26	2060.740	114.42	49.83	130	Protector
#110	TBG 4-1/2", 18.9#, L-80,HSM3	131	9.448	1273.71	2051.292	114.88	50.23	131	Protector
#111	TBG 4-1/2", 18.9#, L-80,HSM3	132	9.529	1283.24	2041.763	115.34	50.63	132	Protector
#112	TBG 4-1/2", 18.9#, L-80,HSM3	133	9.547	1292.78	2032.216	115.81	51.03	133	Protector
#113	TBG 4-1/2", 18.9#, L-80,HSM3	134	9.526	1302.31	2022.690	116.28	51.43	134	Protector
#114	TBG 4-1/2", 18.9#, L-80,HSM3	135	9.532	1311.84	2013.158	116.74	51.84	135	Protector
#115	TBG 4-1/2", 18.9#, L-80,HSM3	136	9.554	1321.40	2003.604	117.21	52.24	136	Protector
#116	TBG 4-1/2", 18.9#, L-80,HSM3	137	9.548	1330.94	1994.056	117.67	52.64	137	Protector
#117	TBG 4-1/2", 18.9#, L-80,HSM3	138	9.555	1340.50	1984.501	118.14	53.05	138	Protector
#118	TBG 4-1/2", 18.9#, L-80,HSM3	139	9.540	1350.04	1974.961	118.61	53.45	139	Protector
#119	TBG 4-1/2", 18.9#, L-80,HSM3	140	9.529	1359.57	1965.432	119.07	53.85	140	Protector
#120	TBG 4-1/2", 18.9#, L-80,HSM3	141	9.528	1369.10	1955.904	119.54	54.25	141	Protector
#121	TBG 4-1/2", 18.9#, L-80,HSM3	142	9.424	1378.52	1946.480	120.00	54.65	142	Protector
#122	TBG 4-1/2", 18.9#, L-80,HSM3	143	9.531	1388.05	1936.949	120.46	55.05	143	Protector
#123	TBG 4-1/2", 18.9#, L-80,HSM3	144	9.546	1397.60	1927.403	120.93	55.46	144	Protector
#124	TBG 4-1/2", 18.9#, L-80,HSM3	145	9.534	1407.13	1917.869	121.39	55.86	145	Protector
#125	TBG 4-1/2", 18.9#, L-80,HSM3	146	9.549	1416.68	1908.320	121.86	56.26	146	Protector
#126	TBG 4-1/2", 18.9#, L-80,HSM3	147	9.529	1426.21	1898.791	122.32	56.67	147	Protector
#127	TBG 4-1/2", 18.9#, L-80,HSM3	148	9.542	1435.75	1889.249	122.79	57.07	148	Protector
#128	TBG 4-1/2", 18.9#, L-80,HSM3	149	9.185	1444.94	1880.064	123.24	57.46	149	Protector
#129	TBG 4-1/2", 18.9#, L-80,HSM3	150	9.115	1454.05	1870.949	123.68	57.84	150	Protector
#130	TBG 4-1/2", 18.9#, L-80,HSM3	151	9.565	1463.62	1861.384	124.15	58.25	151	Protector
#131	TBG 4-1/2", 18.9#, L-80,HSM3	152	9.531	1473.15	1851.853	124.62	58.65	152	Protector
#132	TBG 4-1/2", 18.9#, L-80,HSM3	153	9.138	1482.29	1842.715	125.06	59.03	153	Protector
#133	TBG 4-1/2", 18.9#, L-80,HSM3	154	9.438	1491.72	1833.277	125.52	59.43	154	Protector
#134	TBG 4-1/2", 18.9#, L-80,HSM3	155	9.544	1501.27	1823.733	125.99	59.84	155	Protector
#135	TBG 4-1/2", 18.9#, L-80,HSM3	156	9.524	1510.79	1814.209	126.45	60.24	156	Protector

#136	TBG 4-1/2", 18.9#, L-80,HSM3	157	9.579	1520.37	1804.630	126.92	60.64	157	Protector
#137	TBG 4-1/2", 18.9#, L-80,HSM3	158	9.547	1529.92	1795.083	127.39	61.05	158	Protector
#138	TBG 4-1/2", 18.9#, L-80,HSM3	159	9.532	1539.45	1785.551	127.85	61.45	159	Protector
#139	TBG 4-1/2", 18.9#, L-80,HSM3	160	9.564	1549.01	1775.987	128.32	61.85	160	Protector
#140	TBG 4-1/2", 18.9#, L-80,HSM3	161	9.579	1558.59	1766.408	128.79	62.26	161	Protector
#141	TBG 4-1/2", 18.9#, L-80,HSM3	162	9.213	1567.81	1757.195	129.24	62.65	162	Protector
#142	TBG 4-1/2", 18.9#, L-80,HSM3	163	9.534	1577.34	1747.661	129.70	63.05	163	Protector
#143	TBG 4-1/2", 18.9#, L-80,HSM3	164	9.528	1586.87	1738.133	130.17	63.45	164	Protector
#144	TBG 4-1/2", 18.9#, L-80,HSM3	165	9.531	1596.40	1728.602	130.63	63.85	165	Protector
#145	TBG 4-1/2", 18.9#, L-80,HSM3	166	9.544	1605.94	1719.058	131.10	64.26	166	Protector
#146	TBG 4-1/2", 18.9#, L-80,HSM3	167	9.532	1615.47	1709.526	131.56	64.66	167	Protector
#147	TBG 4-1/2", 18.9#, L-80,HSM3	168	9.543	1625.02	1699.983	132.03	65.06	168	Protector
#148	TBG 4-1/2", 18.9#, L-80,HSM3	169	9.568	1634.59	1690.415	132.50	65.47	169	Protector
#149	TBG 4-1/2", 18.9#, L-80,HSM3	170	9.526	1644.11	1680.889	132.96	65.87	170	Protector
#150	TBG 4-1/2", 18.9#, L-80,HSM3	171	9.527	1653.64	1671.362	133.43	66.27	171	Protector
#151	TBG 4-1/2", 18.9#, L-80,HSM3	172	9.407	1663.05	1661.955	133.89	66.67	172	Protector
#152	TBG 4-1/2", 18.9#, L-80,HSM3	173	9.529	1672.57	1652.426	134.35	67.07	173	Protector
#153	TBG 4-1/2", 18.9#, L-80,HSM3	174	9.458	1682.03	1642.968	134.81	67.47	174	Protector
#154	TBG 4-1/2", 18.9#, L-80,HSM3	175	9.568	1691.60	1633.400	135.28	67.87	175	Protector
#155	TBG 4-1/2", 18.9#, L-80,HSM3	176	9.526	1701.13	1623.874	135.75	68.28	176	Protector
#156	TBG 4-1/2", 18.9#, L-80,HSM3	177	9.532	1710.66	1614.342	136.21	68.68	177	Protector
#157	TBG 4-1/2", 18.9#, L-80,HSM3	178	9.532	1720.19	1604.810	136.68	69.08	178	Protector
#158	TBG 4-1/2", 18.9#, L-80,HSM3	179	9.536	1729.73	1595.274	137.14	69.48	179	Protector
#159	TBG 4-1/2", 18.9#, L-80,HSM3	180	9.531	1739.26	1585.743	137.61	69.89	180	Protector
#160	TBG 4-1/2", 18.9#, L-80,HSM3	181	9.531	1748.79	1576.212	138.07	70.29	181	Protector
#161	TBG 4-1/2", 18.9#, L-80,HSM3	182	9.533	1758.32	1566.679	138.54	70.69	182	Protector
#162	TBG 4-1/2", 18.9#, L-80,HSM3	183	9.433	1767.75	1557.246	139.00	71.09	183	Protector
#163	TBG 4-1/2", 18.9#, L-80,HSM3	184	9.409	1777.16	1547.837	139.46	71.49	184	Protector
#164	TBG 4-1/2", 18.9#, L-80,HSM3	185	9.186	1786.35	1538.651	139.91	71.87	185	Protector
#165	TBG 4-1/2", 18.9#, L-80,HSM3	186	9.537	1795.89	1529.114	140.37	72.28	186	Protector
#166	TBG 4-1/2", 18.9#, L-80,HSM3	187	9.413	1805.30	1519.701	140.83	72.67	187	Protector
#167	TBG 4-1/2", 18.9#, L-80,HSM3	188	9.567	1814.87	1510.134	141.30	73.08	188	Protector
#168	TBG 4-1/2", 18.9#, L-80,HSM3	189	9.532	1824.40	1500.602	141.76	73.48	189	Protector
#169	TBG 4-1/2", 18.9#, L-80,HSM3	190	9.528	1833.93	1491.074	142.23	73.88	190	Protector
#170	TBG 4-1/2", 18.9#, L-80,HSM3	191	9.529	1843.46	1481.545	142.70	74.29	191	Protector
#171	TBG 4-1/2", 18.9#, L-80,HSM3	192	9.532	1852.99	1472.013	143.16	74.69	192	Protector
#172	TBG 4-1/2", 18.9#, L-80,HSM3	193	9.533	1862.52	1462.480	143.63	75.09	193	Protector
#173	TBG 4-1/2", 18.9#, L-80,HSM3	194	9.525	1872.05	1452.955	144.09	75.49	194	Protector
#174	TBG 4-1/2", 18.9#, L-80,HSM3	195	9.528	1881.57	1443.427	144.56	75.90	195	Protector
#175	TBG 4-1/2", 18.9#, L-80,HSM3	196	9.527	1891.10	1433.900	145.02	76.30	196	Protector
#176	TBG 4-1/2", 18.9#, L-80,HSM3	197	9.533	1900.63	1424.367	145.49	76.70	197	Protector
#177	TBG 4-1/2", 18.9#, L-80,HSM3	198	9.533	1910.17	1414.834	145.95	77.10	198	Protector
#178	TBG 4-1/2", 18.9#, L-80,HSM3	199	9.531	1919.70	1405.303	146.42	77.51	199	Protector
#179	TBG 4-1/2", 18.9#, L-80,HSM3	200	9.530	1929.23	1395.773	146.88	77.91	200	Protector
#180	TBG 4-1/2", 18.9#, L-80,HSM3	201	9.522	1938.75	1386.251	147.35	78.31	201	Protector
#181	TBG 4-1/2", 18.9#, L-80,HSM3	202	9.536	1948.29	1376.715	147.81	78.71	202	Protector
#182	TBG 4-1/2", 18.9#, L-80,HSM3	203	9.529	1957.81	1367.186	148.28	79.12	203	Protector
#183	TBG 4-1/2", 18.9#, L-80,HSM3	204	9.548	1967.36	1357.638	148.74	79.52	204	Protector
#184	TBG 4-1/2", 18.9#, L-80,HSM3	205	9.533	1976.90	1348.105	149.21	79.92	205	Protector
#185	TBG 4-1/2", 18.9#, L-80,HSM3	206	9.541	1986.44	1338.564	149.68	80.32	206	Protector
#186	TBG 4-1/2", 18.9#, L-80,HSM3	207	9.531	1995.97	1329.033	150.14	80.73	207	Protector
#187	TBG 4-1/2", 18.9#, L-80,HSM3	208	9.533	2005.50	1319.500	150.61	81.13	208	Protector
#188	TBG 4-1/2", 18.9#, L-80,HSM3	209	9.526	2015.03	1309.974	151.07	81.53	209	Protector / Press Test
#189	TBG 4-1/2", 18.9#, L-80,HSM3	210	9.534	2024.56	1300.440	151.54	81.93	210	Protector
#190	TBG 4-1/2", 18.9#, L-80,HSM3	211	9.531	2034.09	1290.909	152.00	82.34	211	Protector

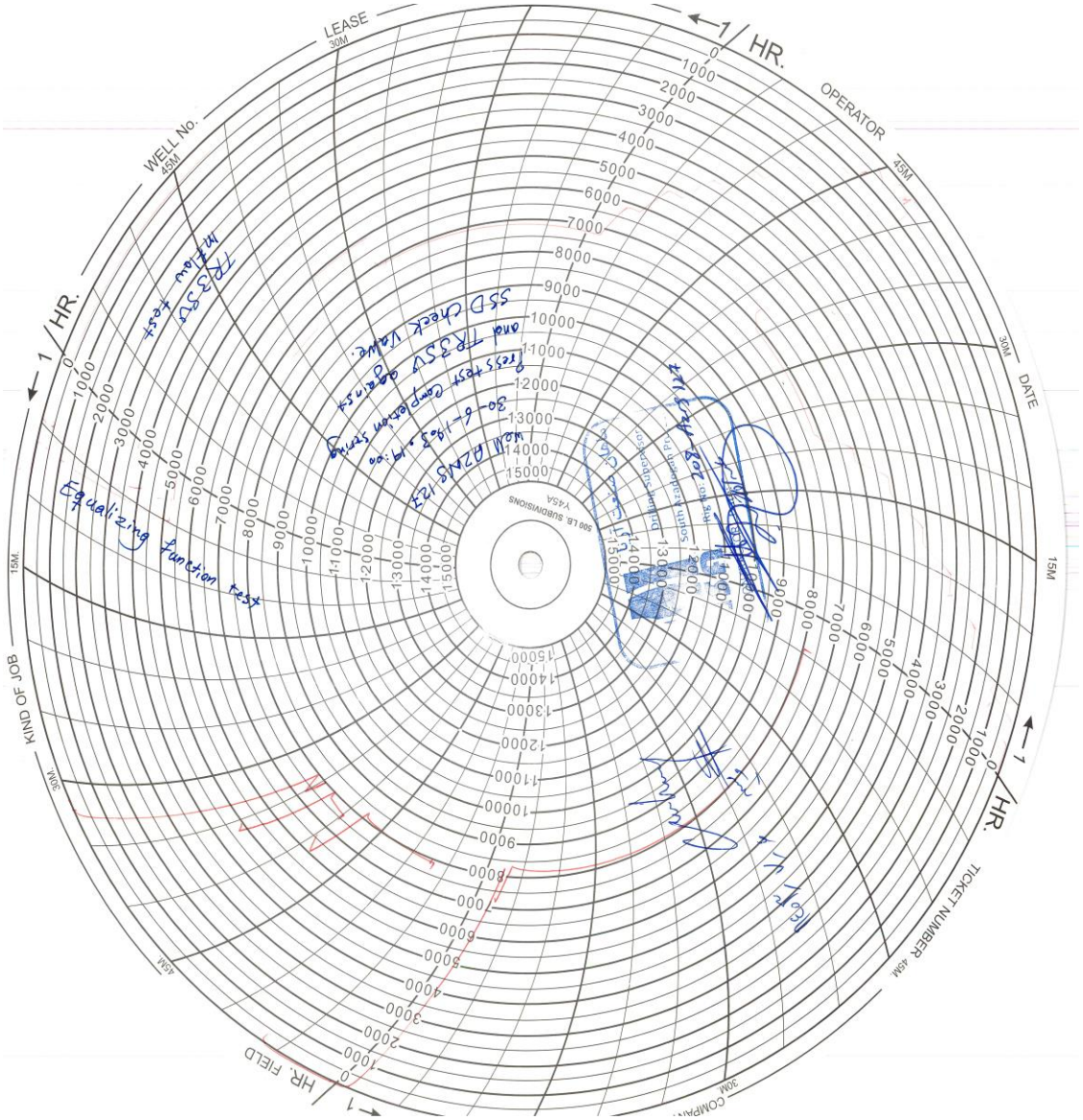
#191	TBG 4-1/2", 18.9#, L-80,HSM3	212	9.531	2043.62	1281.378	152.47	82.74	212	Protector
#192	TBG 4-1/2", 18.9#, L-80,HSM3	213	9.551	2053.17	1271.827	152.93	83.14	213	Protector
#193	TBG 4-1/2", 18.9#, L-80,HSM3	214	9.545	2062.72	1262.282	153.40	83.55	214	Protector
#194	TBG 4-1/2", 18.9#, L-80,HSM3	215	9.547	2072.27	1252.735	153.87	83.95	215	Protector
#195	TBG 4-1/2", 18.9#, L-80,HSM3	216	9.199	2081.46	1243.536	154.32	84.34	216	Protector
#196	TBG 4-1/2", 18.9#, L-80,HSM3	217	9.473	2090.94	1234.063	154.78	84.74	217	Protector
#197	TBG 4-1/2", 18.9#, L-80,HSM3	218	9.526	2100.46	1224.537	155.24	85.14	218	Protector
#198	TBG 4-1/2", 18.9#, L-80,HSM3	219	9.530	2109.99	1215.007	155.71	85.54	219	Protector
#199	TBG 4-1/2", 18.9#, L-80,HSM3	220	9.431	2119.42	1205.576	156.17	85.94	220	Protector
#200	TBG 4-1/2", 18.9#, L-80,HSM3	221	9.528	2128.95	1196.048	156.63	86.34	221	Protector
#201	TBG 4-1/2", 18.9#, L-80,HSM3	222	9.528	2138.48	1186.520	157.10	86.75	222	Protector
#202	TBG 4-1/2", 18.9#, L-80,HSM3	223	9.530	2148.01	1176.990	157.56	87.15	223	Protector
#203	TBG 4-1/2", 18.9#, L-80,HSM3	224	9.317	2157.33	1167.673	158.02	87.54	224	Protector
#204	TBG 4-1/2", 18.9#, L-80,HSM3	225	9.324	2166.65	1158.349	158.47	87.93	225	Protector
#205	TBG 4-1/2", 18.9#, L-80,HSM3	226	9.418	2176.07	1148.931	158.93	88.33	226	Protector
#206	TBG 4-1/2", 18.9#, L-80,HSM3	227	9.526	2185.60	1139.405	159.40	88.73	227	Protector
#207	TBG 4-1/2", 18.9#, L-80,HSM3	228	9.532	2195.13	1129.873	159.86	89.14	228	Protector
#208	TBG 4-1/2", 18.9#, L-80,HSM3	229	9.531	2204.66	1120.342	160.33	89.54	229	Protector
#209	TBG 4-1/2", 18.9#, L-80,HSM3	230	9.527	2214.19	1110.815	160.80	89.94	230	Protector
#210	TBG 4-1/2", 18.9#, L-80,HSM3	231	9.418	2223.60	1101.397	161.25	90.34	231	Protector
#211	TBG 4-1/2", 18.9#, L-80,HSM3	232	9.413	2233.02	1091.984	161.71	90.74	232	Protector
#212	TBG 4-1/2", 18.9#, L-80,HSM3	233	9.526	2242.54	1082.458	162.18	91.14	233	Protector
#213	TBG 4-1/2", 18.9#, L-80,HSM3	234	9.423	2251.97	1073.035	162.64	91.54	234	Protector
#214	TBG 4-1/2", 18.9#, L-80,HSM3	235	9.531	2261.50	1063.504	163.10	91.94	235	Protector
#215	TBG 4-1/2", 18.9#, L-80,HSM3	236	9.530	2271.03	1053.974	163.57	92.34	236	Protector
#216	TBG 4-1/2", 18.9#, L-80,HSM3	237	9.416	2280.44	1044.558	164.03	92.74	237	Protector
#217	TBG 4-1/2", 18.9#, L-80,HSM3	238	9.410	2289.85	1035.148	164.49	93.14	238	Protector
#218	TBG 4-1/2", 18.9#, L-80,HSM3	239	9.429	2299.28	1025.719	164.95	93.54	239	Protector
#219	TBG 4-1/2", 18.9#, L-80,HSM3	240	9.434	2308.72	1016.285	165.41	93.93	240	Protector
#220	TBG 4-1/2", 18.9#, L-80,HSM3	241	9.529	2318.24	1006.756	165.88	94.34	241	Protector
#221	TBG 4-1/2", 18.9#, L-80,HSM3	242	9.530	2327.77	997.226	166.34	94.74	242	Protector
#222	TBG 4-1/2", 18.9#, L-80,HSM3	243	9.568	2337.34	987.658	166.81	95.14	243	Protector
#223	TBG 4-1/2", 18.9#, L-80,HSM3	244	9.568	2346.91	978.090	167.28	95.55	244	Protector
#224	TBG 4-1/2", 18.9#, L-80,HSM3	245	9.304	2356.21	968.786	167.73	95.94	245	Protector
#225	TBG 4-1/2", 18.9#, L-80,HSM3	246	9.154	2365.37	959.632	168.18	96.33	246	Protector
#226	TBG 4-1/2", 18.9#, L-80,HSM3	247	9.430	2374.80	950.202	168.64	96.72	247	Protector
#227	TBG 4-1/2", 18.9#, L-80,HSM3	248	9.539	2384.34	940.663	169.10	97.13	248	Protector
#228	TBG 4-1/2", 18.9#, L-80,HSM3	249	9.546	2393.88	931.117	169.57	97.53	249	Protector
#229	TBG 4-1/2", 18.9#, L-80,HSM3	250	9.532	2403.42	921.585	170.03	97.93	250	Protector
#230	TBG 4-1/2", 18.9#, L-80,HSM3	251	9.534	2412.95	912.051	170.50	98.34	251	Protector
#231	TBG 4-1/2", 18.9#, L-80,HSM3	252	9.555	2422.50	902.496	170.97	98.74	252	Protector
#232	TBG 4-1/2", 18.9#, L-80,HSM3	253	9.578	2432.08	892.918	171.43	99.14	253	Protector
#233	TBG 4-1/2", 18.9#, L-80,HSM3	254	9.529	2441.61	883.389	171.90	99.55	254	Protector
#234	TBG 4-1/2", 18.9#, L-80,HSM3	255	9.532	2451.14	873.857	172.36	99.95	255	Protector
#235	TBG 4-1/2", 18.9#, L-80,HSM3	256	9.552	2460.70	864.305	172.83	100.35	256	Protector
#236	TBG 4-1/2", 18.9#, L-80,HSM3	257	9.548	2470.24	854.757	173.30	100.76	257	Protector
#237	TBG 4-1/2", 18.9#, L-80,HSM3	258	9.425	2479.67	845.332	173.76	101.15	258	Protector
#238	TBG 4-1/2", 18.9#, L-80,HSM3	259	9.328	2489.00	836.004	174.21	101.55	259	Protector
#239	TBG 4-1/2", 18.9#, L-80,HSM3	260	9.535	2498.53	826.469	174.68	101.95	260	Protector
#240	TBG 4-1/2", 18.9#, L-80,HSM3	261	9.193	2507.72	817.276	175.13	102.34	261	Protector
#241	TBG 4-1/2", 18.9#, L-80,HSM3	262	9.533	2517.26	807.743	175.59	102.74	262	Protector
#242	TBG 4-1/2", 18.9#, L-80,HSM3	263	9.549	2526.81	798.194	176.06	103.14	263	Protector
#243	TBG 4-1/2", 18.9#, L-80,HSM3	264	9.528	2536.33	788.666	176.52	103.55	264	Protector
#244	TBG 4-1/2", 18.9#, L-80,HSM3	265	9.134	2545.47	779.532	176.97	103.93	265	Protector
#245	TBG 4-1/2", 18.9#, L-80,HSM3	266	9.410	2554.88	770.122	177.43	104.33	266	Protector

#246	TBG 4-1/2", 18.9#, L-80,HSM3	267	9.146	2564.02	760.976	177.88	104.72	267	Protector
#247	TBG 4-1/2", 18.9#, L-80,HSM3	268	9.576	2573.60	751.400	178.34	105.12	268	Protector
#248	TBG 4-1/2", 18.9#, L-80,HSM3	269	9.342	2582.94	742.058	178.80	105.51	269	Protector
#249	TBG 4-1/2", 18.9#, L-80,HSM3	270	9.142	2592.08	732.916	179.25	105.90	270	Protector
#250	TBG 4-1/2", 18.9#, L-80,HSM3	271	9.531	2601.62	723.385	179.71	106.30	271	Protector
#251	TBG 4-1/2", 18.9#, L-80,HSM3	272	9.527	2611.14	713.858	180.18	106.71	272	Protector
#252	TBG 4-1/2", 18.9#, L-80,HSM3	273	9.529	2620.67	704.329	180.64	107.11	273	Protector
#253	TBG 4-1/2", 18.9#, L-80,HSM3	274	9.419	2630.09	694.910	181.10	107.51	274	Protector
#254	TBG 4-1/2", 18.9#, L-80,HSM3	275	9.533	2639.62	685.377	181.57	107.91	275	Protector
#255	TBG 4-1/2", 18.9#, L-80,HSM3	276	9.528	2649.15	675.849	182.03	108.31	276	Protector
#256	TBG 4-1/2", 18.9#, L-80,HSM3	277	9.251	2658.40	666.598	182.48	108.70	277	Protector
#257	TBG 4-1/2", 18.9#, L-80,HSM3	278	9.415	2667.82	657.183	182.94	109.10	278	Protector
#258	TBG 4-1/2", 18.9#, L-80,HSM3	279	9.532	2677.35	647.651	183.41	109.50	279	Protector
#259	TBG 4-1/2", 18.9#, L-80,HSM3	280	9.530	2686.88	638.121	183.87	109.90	280	Protector
#260	TBG 4-1/2", 18.9#, L-80,HSM3	281	9.438	2696.32	628.683	184.33	110.30	281	Protector
#261	TBG 4-1/2", 18.9#, L-80,HSM3	282	9.536	2705.85	619.147	184.80	110.71	282	Protector
#262	TBG 4-1/2", 18.9#, L-80,HSM3	283	9.326	2715.18	609.821	185.25	111.10	283	Protector
#263	TBG 4-1/2", 18.9#, L-80,HSM3	284	9.423	2724.60	600.398	185.71	111.50	284	Protector
#264	TBG 4-1/2", 18.9#, L-80,HSM3	285	9.529	2734.13	590.869	186.18	111.90	285	Protector
#265	TBG 4-1/2", 18.9#, L-80,HSM3	286	9.528	2743.66	581.341	186.65	112.30	286	Protector
#266	TBG 4-1/2", 18.9#, L-80,HSM3	287	9.536	2753.20	571.805	187.11	112.70	287	Protector
#267	TBG 4-1/2", 18.9#, L-80,HSM3	288	9.545	2762.74	562.260	187.58	113.11	288	Protector
#268	TBG 4-1/2", 18.9#, L-80,HSM3	289	9.530	2772.27	552.730	188.04	113.51	289	Protector
#269	TBG 4-1/2", 18.9#, L-80,HSM3	290	9.525	2781.80	543.205	188.51	113.91	290	Protector
#270	TBG 4-1/2", 18.9#, L-80,HSM3	291	9.533	2791.33	533.672	188.97	114.31	291	Protector
#271	TBG 4-1/2", 18.9#, L-80,HSM3	292	9.543	2800.87	524.129	189.44	114.72	292	Protector
#272	TBG 4-1/2", 18.9#, L-80,HSM3	293	9.528	2810.40	514.601	189.90	115.12	293	Protector
#273	TBG 4-1/2", 18.9#, L-80,HSM3	294	9.523	2819.92	505.078	190.37	115.52	294	Protector
#274	TBG 4-1/2", 18.9#, L-80,HSM3	295	9.531	2829.45	495.547	190.83	115.92	295	Protector
#275	TBG 4-1/2", 18.9#, L-80,HSM3	296	9.523	2838.98	486.024	191.30	116.33	296	Protector
#276	TBG 4-1/2", 18.9#, L-80,HSM3	297	9.549	2848.53	476.475	191.77	116.73	297	Protector
#277	TBG 4-1/2", 18.9#, L-80,HSM3	298	9.388	2857.91	467.087	192.22	117.13	298	Protector
#278	TBG 4-1/2", 18.9#, L-80,HSM3	299	9.531	2867.44	457.556	192.69	117.53	299	Protector
#279	TBG 4-1/2", 18.9#, L-80,HSM3	300	9.529	2876.97	448.027	193.15	117.93	300	Protector
#280	TBG 4-1/2", 18.9#, L-80,HSM3	301	9.399	2886.37	438.628	193.61	118.33	301	Protector
#281	TBG 4-1/2", 18.9#, L-80,HSM3	302	9.335	2895.71	429.293	194.07	118.72	302	Protector
#282	TBG 4-1/2", 18.9#, L-80,HSM3	303	9.587	2905.29	419.706	194.54	119.13	303	Protector
#283	TBG 4-1/2", 18.9#, L-80,HSM3	304	9.548	2914.84	410.158	195.00	119.53	304	Protector
#284	TBG 4-1/2", 18.9#, L-80,HSM3	305	9.530	2924.37	400.628	195.47	119.93	305	Protector
#285	TBG 4-1/2", 18.9#, L-80,HSM3	306	9.551	2933.92	391.077	195.93	120.34	306	Protector / Press Test
#287	TBG 4-1/2", 18.9#, L-80,HSM3	307	9.529	2943.45	381.548	196.40	120.74	307	Protector
#288	TBG 4-1/2", 18.9#, L-80,HSM3	308	9.536	2952.99	372.012	196.87	121.14	308	Protector
#289	TBG 4-1/2", 18.9#, L-80,HSM3	309	9.529	2962.52	362.483	197.33	121.54	309	Protector
#290	TBG 4-1/2", 18.9#, L-80,HSM3	310	9.535	2972.05	352.948	197.80	121.95	310	Protector
#291	TBG 4-1/2", 18.9#, L-80,HSM3	311	9.533	2981.59	343.415	198.26	122.35	311	Protector
#292	TBG 4-1/2", 18.9#, L-80,HSM3	312	9.552	2991.14	333.863	198.73	122.75	312	Protector
#293	TBG 4-1/2", 18.9#, L-80,HSM3	313	9.532	3000.67	324.331	199.19	123.16	313	Protector
#294	TBG 4-1/2", 18.9#, L-80,HSM3	314	9.343	3010.01	314.988	199.65	123.55	314	Protector
#295	TBG 4-1/2", 18.9#, L-80,HSM3	315	9.443	3019.46	305.545	200.11	123.95	315	Protector
#296	TBG 4-1/2", 18.9#, L-80,HSM3	316	9.531	3028.99	296.014	200.58	124.35	316	Protector
#297	TBG 4-1/2", 18.9#, L-80,HSM3	317	9.428	3038.41	286.586	201.04	124.75	317	Protector
#298	TBG 4-1/2", 18.9#, L-80,HSM3	318	9.415	3047.83	277.171	201.50	125.15	318	Protector
#299	TBG 4-1/2", 18.9#, L-80,HSM3	319	9.567	3057.40	267.604	201.96	125.55	319	Protector
#300	TBG 4-1/2", 18.9#, L-80,HSM3	320	9.531	3066.93	258.073	202.43	125.95	320	Protector

#301	TBG 4-1/2", 18.9#, L-80,HSM3	321	9.555	3076.48	248.518	202.89	126.36	321	Protector
#302	TBG 4-1/2", 18.9#, L-80,HSM3	322	9.408	3085.89	239.110	203.35	126.75	322	Protector
#303	TBG 4-1/2", 18.9#, L-80,HSM3	323	9.159	3095.05	229.951	203.80	127.14	323	Protector
#304	TBG 4-1/2", 18.9#, L-80,HSM3	324	9.528	3104.58	220.423	204.27	127.54	324	Protector
#305	TBG 4-1/2", 18.9#, L-80,HSM3	325	9.568	3114.15	210.855	204.73	127.95	325	Protector
#306	TBG 4-1/2", 18.9#, L-80,HSM3	326	9.529	3123.67	201.326	205.20	128.35	326	Protector
#307	TBG 4-1/2", 18.9#, L-80,HSM3	327	9.527	3133.20	191.799	205.66	128.75	327	Protector
#308	TBG 4-1/2", 18.9#, L-80,HSM3	328	9.531	3142.73	182.268	206.13	129.15	328	Protector
#309	TBG 4-1/2", 18.9#, L-80,HSM3	329	9.531	3152.26	172.737	206.59	129.56	329	Protector
#310	TBG 4-1/2", 18.9#, L-80,HSM3	330	9.532	3161.80	163.205	207.06	129.96	330	Protector
#311	TBG 4-1/2", 18.9#, L-80,HSM3	331	9.530	3171.33	153.675	207.53	130.36	331	Protector
#312	TBG 4-1/2", 18.9#, L-80,HSM3	332	9.532	3180.86	144.143	207.99	130.76	332	Protector
#313	TBG 4-1/2", 18.9#, L-80,HSM3	333	9.532	3190.39	134.611	208.46	131.17	333	Protector
#314	TBG 4-1/2", 18.9#, L-80,HSM3	334	9.153	3199.54	125.458	208.90	131.55	334	Protector
#315	TBG 4-1/2", 18.9#, L-80,HSM3	335	9.146	3208.69	116.312	209.35	131.94	335	Protector
#316	TBG 4-1/2", 18.9#, L-80,HSM3	336	9.533	3218.22	106.779	209.81	132.34	336	Protector
#317	TBG 4-1/2", 18.9#, L-80,HSM3	337	9.552	3227.77	97.227	210.28	132.75	337	Protector
#318	TBG 4-1/2", 18.9#, L-80,HSM3	338	9.538	3237.31	87.689	210.75	133.15	338	Protector
#319	TBG 4-1/2", 18.9#, L-80,HSM3	339	9.437	3246.75	78.252	211.21	133.55	339	Protector
#320	TBG 4-1/2", 18.9#, L-80,HSM3	340	9.539	3256.29	68.713	211.67	133.95	340	Protector
Pup Jt 4 1/2",HSM3 ,18.9#, L-80, Type-1	Pup Jt 4 1/2",HSM3		1.624	3257.91	67.089	211.75	134.02		
4-1/2" FLOW COUPLING, 18.9#,Super 13CR-110KSI,HSM3 , 4' FT, Weiqixin Co	Short Flow Coupling		1.117	3259.03	65.972	211.81	134.07		
TRSSV-10 TBG Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , Super 13CR-110KSI, 10K WP With AF Profile, Flapper Type, Weiqixin Co	TRSSV		1.783	3260.81	64.189	211.89	134.14		
4-1/2" FLOW COUPLING,18.9#,Super 13CR-110KSI,HSM3 , 6 FT, Weiqixin Co	Long Flow Coupling		1.726	3262.54	62.463	211.98	134.21		
Pup Jt 4 1/2",HSM3 ,18.9#, L-80, Type-1	Pup Jt 4 1/2",HSM3		1.624	3264.16	60.839	212.06	134.28		
#321	TBG 4-1/2", 18.9#, L-80,HSM3	341	9.531	3273.69	51.308	212.52	134.69	341	Protector / Press Test
#322	TBG 4-1/2", 18.9#, L-80,HSM3	342	9.417	3283.11	41.891	212.98	135.08	342	Protector
#323	TBG 4-1/2", 18.9#, L-80,HSM3	343	9.531	3292.64	32.360	213.45	135.49	343	Protector
#324	TBG 4-1/2", 18.9#, L-80,HSM3	344	9.535	3302.18	22.825	213.91	135.89	344	Protector
#325	TBG 4-1/2", 18.9#, L-80,HSM3	345	9.553	3311.73	13.272	214.38	136.29	345	Protector
4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* Super 13CR-110KSI, Weiqixin Co	Fluted Swage		0.964	3312.69	12.308	214.43	136.33		
TBG HANGER MSP, 10K, 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: UPSL: 3G PR: 2, 4" Type "H" Two Way	TBG HGR		0.520	3313.21	11.788	214.45	136.35		
TBG Hanger Running Tool	TBG Hanger Running Tool		0.230	3313.44	11.558	214.46	136.36		
L/Joint	L/Joint		11.598	3325.04	-0.040	215.03	136.85		Final Press Test

Figure 8-19: AZNS-127 Completion Tally

8.6 Completion String Pressure Test



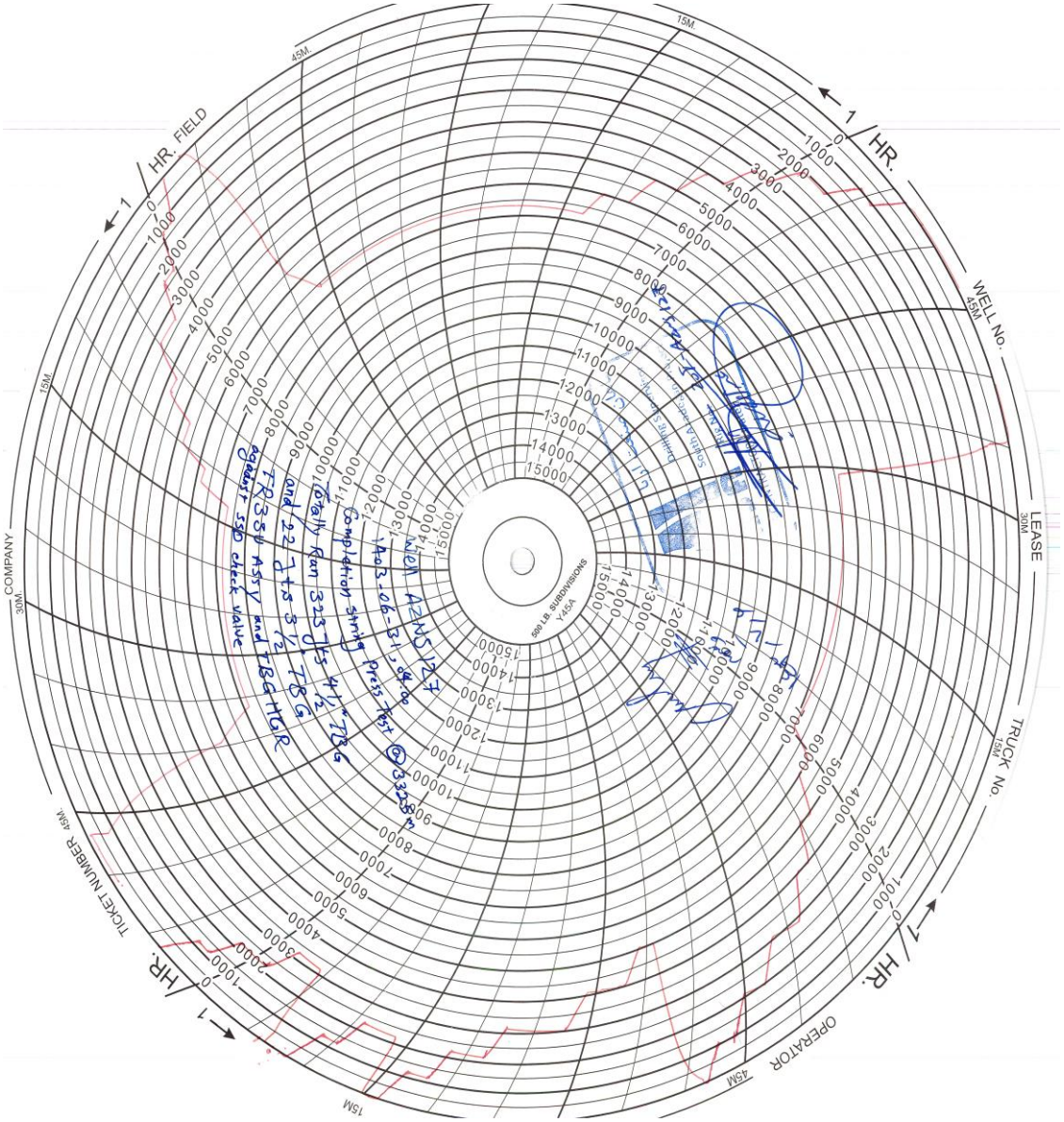


Figure 8-20: AZNS-127 Completion String & TRSSSV Pressure Test Graph

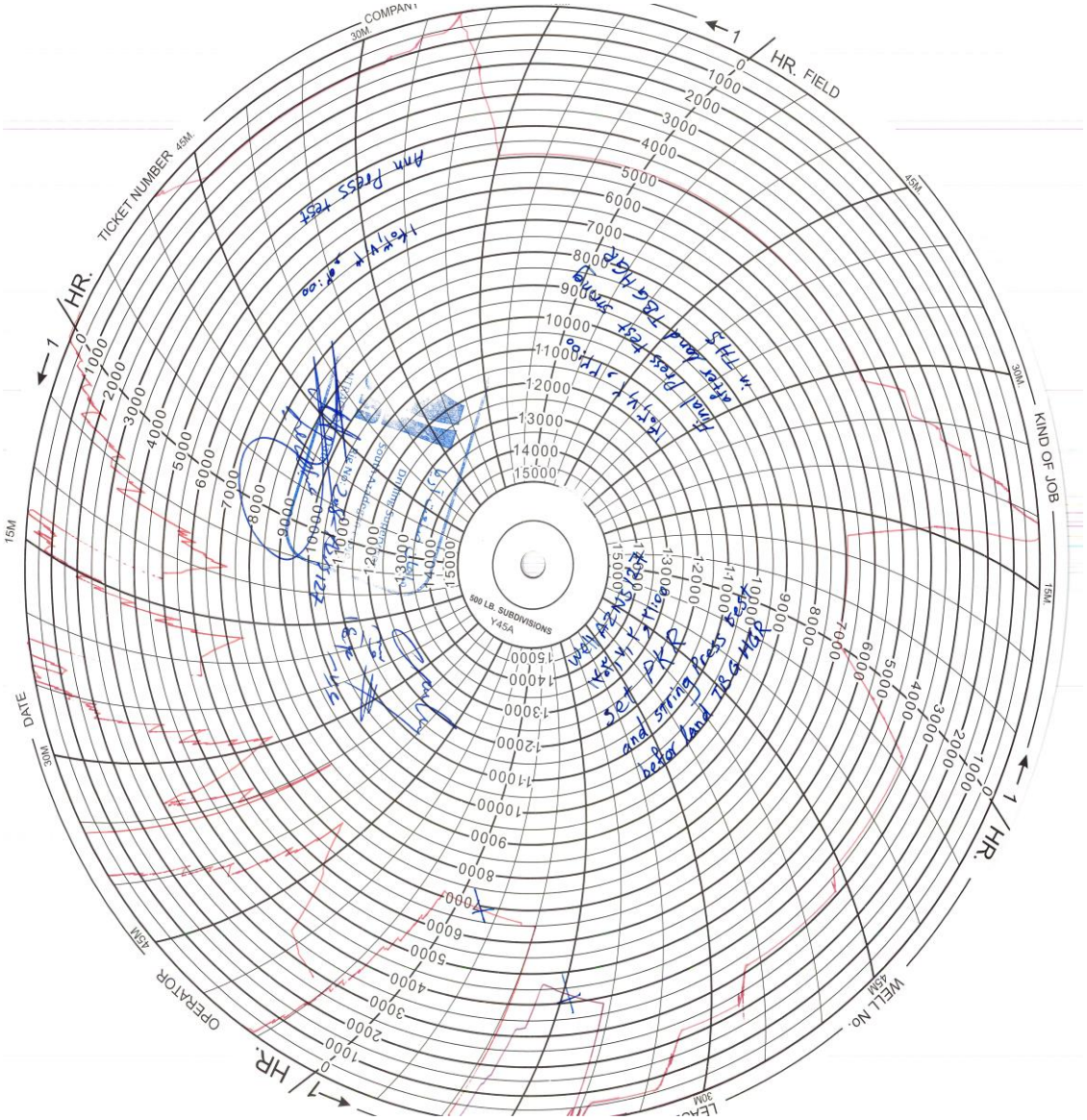


Figure 8-21: AZNS-127 Setting Packer, Annulus & Final Completion String Pressure Test

9. Wellhead Report

9.1 Final Well Status

Well Head

Item	Specification (size, pressure rating, M/C)	Test Pressure	Test Duration
C.H.H	20-3/4'' 3K, DD-0.5 PSL1, PR1	Already Installed	
C.H.S	20-3/4'' 3K * 13-5/8'' 5K , DD-0.5, PSL1, PR1	Already Installed	
MSP Drilex T.H.S.	13-5/8'' 5K * 11''-10K, FF-0.5, PSL3, PR2	Already Installed	
MSP Drilex X-MASS TREE(BTM)	4-1/16''-10K, FF-0.5, PSL3G, PR2	8000 psi	15 MIN

Wellbore Completion Fluids

Well bore	Fluid Type	Weight (pcf)
Cased Hole	Synthetic Fluid	104 – 105

Well Head Pressure Recor

Annulus	Pressure (psi)	Date / Time
Wellhead	0	1403.06.26
Tubing- 9 5/8"	0	1403.06.26
9 5/8" – 13 3/8"	0	1403.06.26
13 3/8" – 20"	0	1403.06.26

Barriers

Barrier Type	Depth	Pressure Test	Date Tested
Close TRSSSV	61.19^{m}	500 psi (inflow test after finish flow the well)	1403.06.30

Temporary Installed Equipment:

After Completing the well, Plug & Prong have been installed in No-Go landing nipple

Table 9-1: AZNS-127 Final Well Status

9.2 Wellhead & X-mass tree Specification

Well Type	Specification	Check and Remark
MSP Drilex Well Head 10,000 Psi for Fahliyan Wells	**TUBING HEAD ADAPTOR** BOTTOM STUDDED FLANGE 11", 10000 PSI, BX158 TOP STUDDED FLANGE 4-1/16" 10000 PSI BX155 FULL BORE PREPARED WITH TWO S-SEAL AND ONE METAL TO METAL SEAL TO PACK TOP NECK OF TUBING HANGER C/W RELATED TEST AND INJECTION PORTS PREPARTION. C/W TWO CONTINUOUS FEED THROUGH TYPE 1/4" NPT CONTROL LINE EXITS PREPARTION (ONE FOR 1/4" TRSSSV CL AND ONE FOR 1/4" DOWNHOLE PRESSURE GAUGE) AND TWO 3/8" NPT CONTINUOUS FEED THROUGH TYPE LINE PORTS FOR CHEMCIAL INJECTION LINE EXIT INCLUDING RELATED SWAGE LOCK FITTINGS C/WITH ONE 4-1/16", 10K WP FULL BORE MANUAL GATE VALVE FORGED BODY AND BONNET, ALLOY STAINLESS STEEL 410SS GATE & SEAT AISI 410 COATED WITH TUNGSTEN CARBIDE USING HVOF METHOD STEM: INCONEL 718 WITH BACK SEAT CAPABILITY, NON RISING STEM METAL TO METAL SEAL BETWEEN SEAT AND GATE C/W HAND WHEEL FORGED BODY ALLOY STEEL AISI 410 MAXIMUM HEIGHT: 800MM M/C:FF-0.5 T/C:U PSL:3G PR:2 CROSS T-TYPE SOLID BLOCK CROSS T-TYPE SOLID BLOCK, BOTTOM FLANGED 4-1/16" API 10000 PSI WP TOP FLANGED STUDDED 4-1/16" API 10000 PSI WP FOR TREE CAP WITH TWO STUDDED SIDE OUTLETS 4-1/16" 10000PSI WP ELEVATED AT 900 DEG AND TWO 4-1/16", 10000 PSI MANUAL GATE VALVE AT THE TOP AND BOTTOM AND ONE 4-1/16", 10000 PSI WP SAFETY VALVE EQUIPPED WITH HYDRAULIC WIRE CUT	

	ACTUATOR AS UPPER MASTER VALVE MATERIAL OF BODY,BONNET: AISI 410 SEAT, GATE: AISI 410SS HARD FACED WITH TUNGSTEN CARBIDE UASING HVOP METHOD MATERIAL OF STEM: INCONEL 718 M/C:FF-0.5 T/C:U PSL:3G PR:2G. MANUAL GATE VALVE API 4-1/16" 10000 PSI WP. FLANGED ENDS AND FULL BORE METAL TO METAL SEAL BETWEEN BODY/SEAT/GATE WITH BACK SEAT CAPABILITY, NON RISING STEM C/W HANDWHEELS MATERIAL OF BODY,BONNET,SEAT,GATE: AISI 410SS MATERIAL OF STEM: INCONEL 718 GATE&SEAT HARD FACED WITH TUNGSTEN CARBIDE BY HOVF METHOD M/C:FF-0.5 T/C:U PSL:3G PR:2 X-MAS TREE CAP 4-1/16" API 10000 PSI WP, FLANGED BOTTOM (4-1/16", BORE) 8-3/8"-4ACME-2G THREAD TOP, BLANKING PLUG TAPPED WITH 1/2" NPT THREAD, FORGED BODY, MATERIAL: STAINLESS STEEL 410 M/C:FF-0.5 T/C:U PSL:3G PR:2	

Table 9-2: AZNS-127 Equipment Specification

9.3 Wellhead Schematic



Figure 9-1: AZNS-127 Wellhead Sketch

9.4 X-mass Tree Specification (Section D)

Item	Equipment	M/C	T/C	PSL	Qty
1	BTM Master Valve 11” * 4 1/16” 10K	FF-0.5	U	3G	1
2	TOP Master Valve 4 1/16” 10K	FF-0.5	U	3G	1
3	Surface Safety Valve 4 1/16” 10K	FF-0.5	U	3G	1
4	Deeping Valve 4 1/16” 10K	FF-0.5	U	3G	1
5	Wing gate valve-1 (Kill Valve) 4 1/16" 10K	FF-0.5	U	3G	1
6	Wing gate valve-2 (production valve) 10K	FF-0.5	U	3G	1
7	41/16" Choke valve 4 1/16" 10K	FF-0.5	U	3G	1
8	Christ Mas Tree	FF-0.5	U	3G	1

Table 9-3: AZNS-127 Section "D" Specification

9.5 X-mass Tree Pressure Test Report (Section D – befor install)

Item	Test	Check Mark and Result
1	Pressure test for the kill Valve W/ 1000 & 8000 Psi / 15 min	OK
2	Pressure test for the SSV W/ 1000 & 8000 Psi / 15 min	OK
3	Pressure test for the top master valve W/ 1000 & 8000 Psi / 15 min	OK
4	Pressure test for Swab valve W/ 1000 & 8000 Psi / 15 min	OK
5	Pressure test for the Production valve W/ 1000 & 8000 Psi / 15 min	OK

Table 9-4: AZNS-127 Section "D" Test Report Befor Installation

9.6 X-mass Tree Pressure Test Report (Section D – after install)

Item	Test	Check mark and Result
1	Pressure test X-MAS TREE solid block and bottom section W/1000 &8000 psi for 15min	OK
2	Pressure test for the SSV Exit block W/ 8000 Psi / 15 min	OK
3	Pressure test neck of tubing hanger metal seal W/1000 &8000 psi for 15min	OK

4	Pressure test for Swab valve W/ 4500 Psi / 15 min	OK
5	Pressure test for Production valve W/4500 Psi / 15 min	OK
6	Pressure test cavity of FS.seal and Ring Gasket W 8000 Psi / 15 min	OK
7	Pressure test CIV line to 2000psi	OK

Table 9-5: AZNS-127 Section "D" Test Report After Installation

9.7 Wellhead Related Test Reports

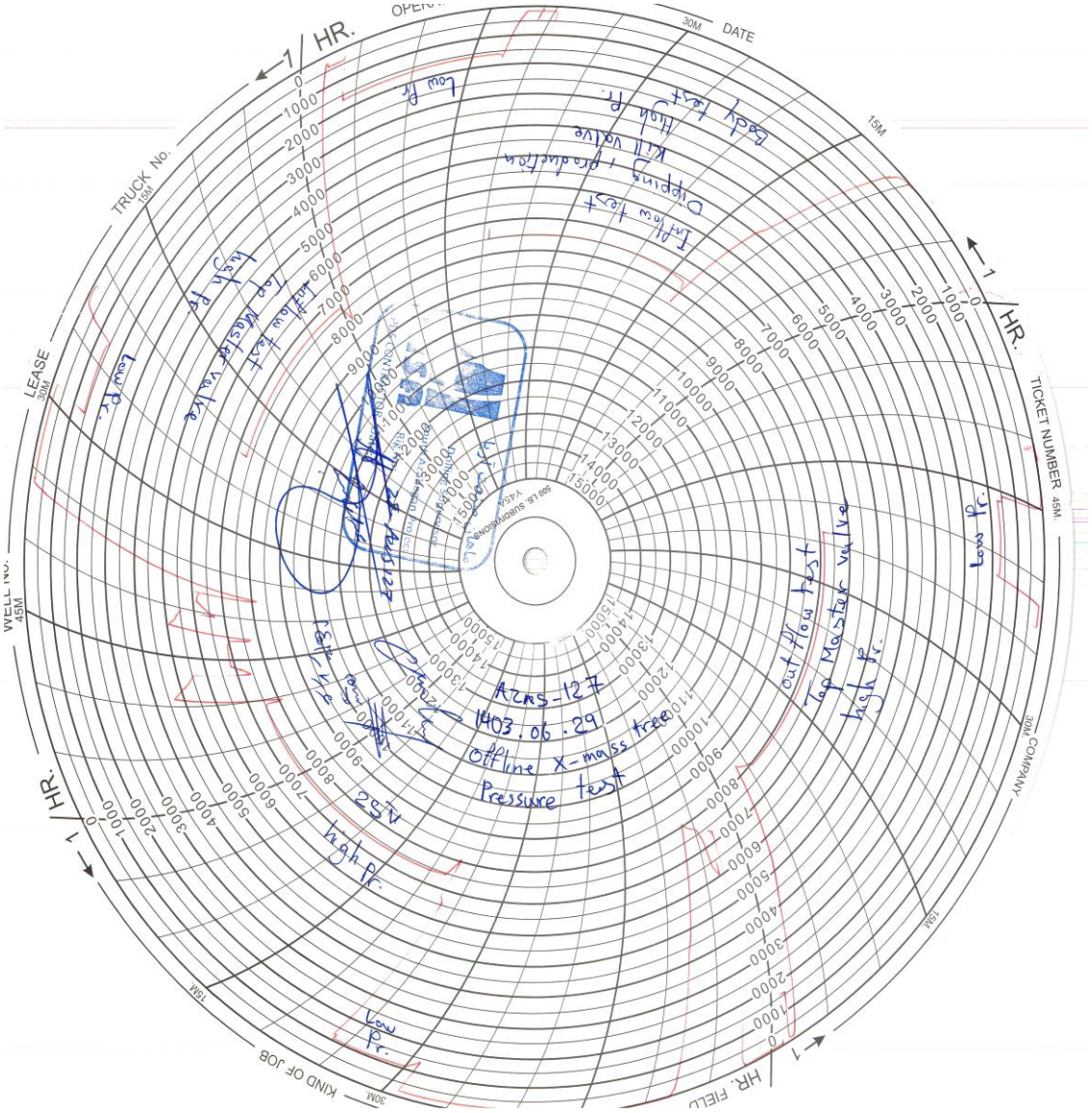


Figure 9-2: AZNS-127 X-Mass Tree Body Pressure Test Graph Before Installation

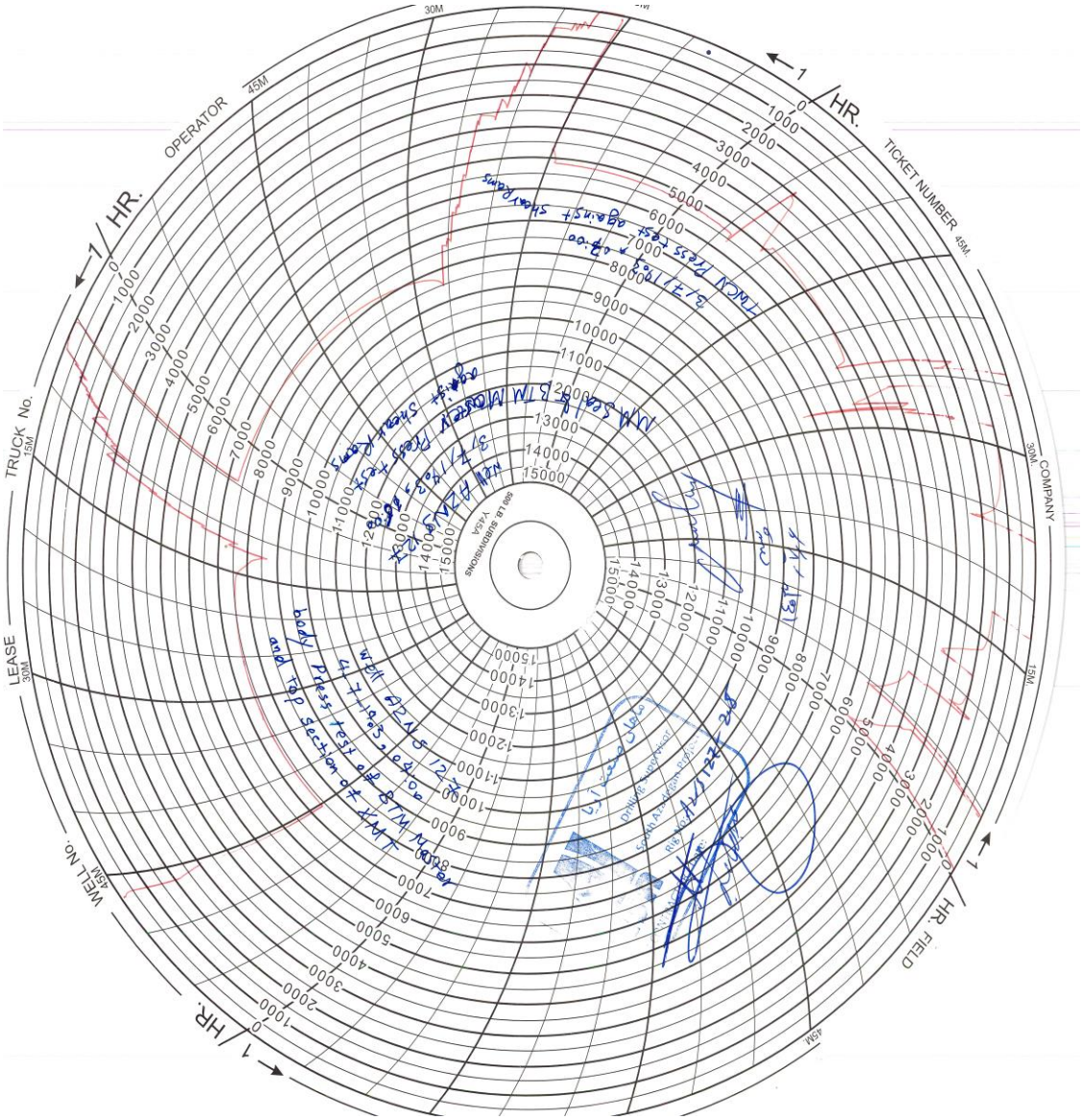


Figure 9-3: AZNS-127 X-Mass Tree Body Pressure Test Graph After Installation

تاریخ: 1403/07/03

پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست سطحی تاج چاه 127 آزادگان جنوبی، قبل از نصب در دکل 208 عملیات اکتشاف نفت

بدینوسیله در مورخه 1403/06/30 تست‌های سطحی تاج چاه X-Mass Tree Solid Block "T" Type 4 – 1/16" 10k, MSP DRILEX مرتبط به چاه شماره 127 آزادگان جنوبی با حضور نماینده محترم کارفرما (پتروپارس) در محل دستگاه 208 عملیات اکتشاف نفت، قبل از نصب بر روی تاج چاه به شرح ذیل انجام پذیرفت:

- Body Pressure Test of X – Mass Tree Solid Block w/ 1000psi, 5min & 8000psi, 15min "OK" S/N: 0269167CXC230001
- Inflow Pressure Test of 4 – 1/16" Production (WING) Valve w/ 1000psi, 5min & 8000psi, 15min "OK". S/N: 002929BXC230006
- Inflow Pressure Test of 4 – 1/16" Dipping (SWAB) Valve w/ 1000psi, 5min & 8000psi, 15min "OK". S/N: 008929KXC230001
- Inflow Pressure Test of 4 – 1/16" Kill Valve w/ 1000psi, 5min & 8000psi, 15min "OK" S/N: 002929BXC230004
- Inflow Pressure Test of 4 – 1/16" Surface Safety Valve w/ 1000psi, 5min & 8000psi, 15min "OK". S/N: SF230004
- Inflow & Out Flow Pressure Test of 4 – 1/16" Top Master Valve w/ 1000psi, 5min & 8000psi, 15min "OK". S/N: 008929LXC23001
- Pressure Test of 2SV Actuator w/ 2500psi, 15min "OK".

– گراف مربوط به تست فشار پیوست صورتجلسه گردید.

ناظر عملیات حفاری

شرکت ماهان صنعت آریا

ماهان صنعت آریا

Drilling Supervisor

South Azadegan Project

Rig No: 208 Azns127

EPDS CONTRACTOR Date:

نماینده محترم تکمیل چاه کارفرما

شرکت نفت کیش

۱۴۰۳/۷/۴

Figure 9-4: AZNS-127 Wellhead MOM Befor Installation

تاریخ: 1403/07/04

پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست‌های پس از نصب تاج و ولهد چاه 127 آزادگان جنوبی بعد از نصب در دکل 208 عملیات اکتشاف نفت

بدینوسیله در مورخه 1403/07/03 تست‌های تاج چاه X-Mass Tree Solid Block "T" Type 4 – 1/16" 10k, MSP و ولهد چاه شماره 127 آزادگان جنوبی با حضور نماینده محترم تکمیل چاه کارفرما و نماینده عملیات حفاری ماهان صنعت آریا در محل دستگاه حفاری 208 عملیات اکتشاف نفت بعد از نصب بر روی چاه به شرح ذیل انجام پذیرفت:

DRILEX

- Body Pressure Test of X – Mass Tree Solid Block w/ 10k 11" * 4 – 1/16" 10k Tubing Head Adaptor (BTM MASTER VALVE) 10k MSP/DRILEX w/ 8000^{psi} for 15min, "OK". S/N: 0269167CXC230001
- Pressure Test TBG HGR Metal Seal to 800^{psi} for 5min, 8000^{psi} for 15min, "OK".
- Pressure Test F–Seal & Ring Gasket between T.H.S & Tubing Head Adaptor (BTM MASTER VALVE) to 8000^{psi} for 15min, "OK".
- Pressure Test Neck of TBG HGR F-Seal to 8000^{psi} for 15min, "OK".
- Pressure Test 1/4" Control Line (CL) to 8000^{psi} for 15min, "OK".
- Pressure Test 3/8" Control Injection Lines (CIL) to 2000^{psi} "OK".

– فشار تمامی دالیزها بررسی گردید که فشار آنها صفر بود.
– بعد از اتمام نصب تاج چاه درون آویزه لوله مغزی Two Way Check Valve جهت انگیزش چاه خارج گردید و BPV نصب گردید.
– خروجی کنترل لاین شیر ایمنی درون چاهی در جهت شیر اصلی پایینی می‌باشد و همچنین CIV شماره یک در جهت ویست پیت و CIV شماره دو در خلاف جهت ویست پیت می‌باشد و همچنین بر روی دو CIV و کنترل لاین پلاگ کپ نصب گردید.
– گراف مربوط به تست فشار پیوست صورتجلسه گردید.

نماینده محترم تکمیل چاه کارفرما
شرکت نفت کیش

ناظر عملیات حفاری
شرکت ماهان صنعت آریا

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No: 208-AZNS 127

Figure 9-5: AZNS-127 Wellhead MOM After Installation

10. Drilling Fluid Information

10.1 Summary

Hole size (in)	Casing size (in)	CSG Depth (m MD)	Drilling Fluid system	Mud Properties
26”	20’’	146 M	Already has been drilled.	
17 ½”	13 3/8”	1441 M	Already has been drilled.	
12 ¼”	9 5/8”	3909 M	Already has been drilled.	
8 ½’’	7” Liner	3777 M To 4226 M	Already has been drilled.	
8 1/2” Side-Track	7” Liner	3147 M To 3890 M	KCL Poly Mud	MW: 85 – 87 PCF
6 1/8” Side-Track	4 ½” Liner	3780.5 M To 4140 M	KCL Poly Mud	MW: 100 – 104 PCF

Table 10-1: AZNS-127 Summary of Fluid

10.2 Interval Remark

WorkOver Section

The OEOC208 was spud at AZN127, RIH with protection sleeve, stuck tools and work on same and retrieved wire, work on fish with CT & Performed injectivity test, try to Kill the well & set the protection sleeve and ran 2.230-gauge cutter to 61 m, took weight, POOH to surface.

Run w/ new protection sleeve & set same inside 3SV, RIH w/ 2.230" gauge ring to 3788m (first CIV depth), POOH to surface. R/U wire line equipment, pump 120 bbl 137 PCF mud in string, try to unlatch Anchor Seal, "No Success". RIH with CCL & sinker bar to 3850 m, POOH to surface.

Pumped 106 PCF mud and try to release anchor seal several times, "No Success" several times, establish circulation, Obs. Gas cut mud in return, Cont. circulation, get 106 pcf mud in return. Work on collided point, string got free, R/D wire line, R/U CSG running equipment, POOH completion string totally 294 jts.

RIH w/ 7- 7/8" O. Shot & Fishing BHA to 2731 m, try to Catch fish several times with No-Succ. POOH w/ 7- 7/8" Over shot & Fishing BHA f/ 2731m to surface. Run 8 3.8" magnet to 2731m, POOH to surface; 1 lb junk recovered.

RIH w/ 8- 3/8" milling BHA to 2729.5m, Milling obstruction f/ 2729.5 to 2739 m, POOH to surface. RIH w/ 7-7/8" Overshot to 2739m, try to catch the fish, POOH to surface.

RIH w/ 7-5/8" Overshot & Fishing BHA to 2739, Catch & Engage w/ fish, Work on fish Try to Free the String from Collided Point; No Succ. Establish Circulation to Uniform MW and POOH to surface, L/D Totally 37 Jts (339m) 4-1/2" TBG.

RIH w/ 8-3/8" M.M J.Mill, Milling BHA to 3078 m, CBU, Milling f/ 3078.2 to 3079.3 m, Pump & Circ. Out 85 bbl Hi-Vis Pill, POOH to surface.

RIH w/ 7-5/8" Overshot to 3079.3m, CBU, catch fish & POOH to surface, L/D 8Jts 4-1/2" TBG.

RIH w/ 7-5/8" Overshot to 3152m. CBU, Catch Fish, POOH to surface, 1 Jt of 4-1/2" TBG Retrieved.

RIH w/ 8-3/8" M.M J. Mill, Milling BHA to 3161.2 m, milling to 3162.6 and POOH to surface.

RIH w/ 7-5/8" Overshot to 3162m, catch fish, R/U W/L Equip, RIH W/ CCL & Wt. bar as dummy run to 3170m, Took Wt, POOH to surface.

RIH 1-1/2"C.TBG to 3610m, Took Weight, try to Pass this Point in Several Times, No Succ., Pump 100 bbl Hi-Vis Pill, POOH 1-1/2"C.TBG to surface.

R/U wireline and RIH w/ 2-1/8" Dummy run to 3222m. Try to pass, no success. RIH w/ 1-11/16" Dummy run to 3612m, try to pass, no success. Assemble and RIH w/1-3/8" FPIT to 3600m, POOH to surface.

RIH Wireline w/ 1-7/8"Colliding Tools to 3612m, Obs. Drop Tension, POOH to 3594m, apply 180Klb Over pull on String, Shoot the Colliding, Work on String, No Succ., POOH, Run Back Wireline w/ 1-7/8"Colliding Tools as Run#02 to 3398m, Shoot the Colliding, POOH to 3275m, Stuck Tools, Work on it, Got Free, POOH to surface, Obs. Shoot the Colliding, Work on String Upward & at Neutral Point in Several Times, No Succ., RIH Wireline w/ 2-1/8"CCL as Dummy Run to 3272m, Try to Pass this Point,

No Succ., POOH, RIH Wireline w/ 2-1/8"Colliding & CCL Tools as Run#03 to 3272m, Fire Colliding Shot, POOH, Again RIH Wireline w/ New 2-1/8"Colliding & CCL Tools as Run#04 to 3185m, Fire Colliding Shot, POOH, R/D Wireline Equip., Work on String in Several Times, No Succ.

Cont. Work on String w/ Apply Different Parameters in Several Times, String Got Free, Circ. Btm's up, POOH to Surf., Obs. 7-5/8"Over Shot & Retrieve 21.1m of 4-1/2"TBG f/ Collided Shot Point, L/D Same (Top of Fish is 3183.1m).

RIH New 8-3/8"M.M J. Mill to 3182.5m, Tag Top of Fish, Start Milling to 3185.8m, POOH to surface.

RIH 5-7/8"Overshot to 3185.8m, Tag Top of Fish, Engage w/ Fish, Work on it in Several Times, Got Free, POOH String for 13m While Obs. 30 klbs Over pull, Cont. POOH to surface, Obs. 7-5/8" Over Shot W/4-1/2" TBG Fish Neck,2.383 + 8 Jts 4-1/2" TBG +,73.318m + 9.268m.

RIH W/8-3/8"M.MJ. Mill to 3269.5m, Tag Top of Fish, Start Milling to 3274m, POOH to surface.

RIH 5-7/8"Overshot to 3274m, Tag Top of Fish, Engage w/ Fish, Work on it in Several Times, Cont. POOH to surface.

RIH W/8-3/8"M.M J. Mill f/ 2200 to 3297.5m, Tag Top of Fish, Start Milling to 3298.8 m, POOH to surface.

Cont. RIH 5-7/8"Overshot to 3298m, Tag Top of Fish, Engage w/ Fish, Work on it in Several Times, R/U & Run Wire Line W/ 2-1/8" Colliding & 1-11/16" CCL to 3385M, Cont. POOH w/ 7-5/8" Fishing assembly to Surface & L/D Fish, Ret. 10 Jts 4-1/2" TBG.

RIH w/ 8-3/8" M.M J. Mill to 3390m, Milling the fish to 3392.7 m, POOH to surface.

RIH w/ 8-3/8" M.M J. Mill, 9-5/8" Scraper to 3350m and POOH to surface.

Performed CAST Log.

Performed CMT plug as per program.

RIH w/ 8-1/2" TCI Bit &D/O BHA to 3130m, Wash & Ream Dn f/ 3130m

to 3170m & Wash Dn to 3347m, D/O Hard CMT to 3350m while CBU @ 3310, 3349 & 3350m. Perform Positive Press test. POOH to surface.

Based on client decision kill the well with 106 PCF Salt/Polymer mud was used for commencing this interval.

The MW increased up to by adding Limestone Powder & Hematite up to 108 PCF at end of this interval.

Treated frequently with essential chemical such as lime as Alkalinity controller to adjust pH, Starch LV for control FL, Soda Ash for control hardness & Xanthan gum as Viscosifier.

8 – 1/2” Hole Section

RIH with Scraper to 3350 m, displaced the well with 85 PCF KCL poly mud, performed flow check the well and POOH to surface.

Run hydraulic set Whipstock to 3339.08 m and set whip stock, Commenced milling as slide track to 3341.3 m, and then POOH to surface.

RIH with 8 ½" PDC bit to 3341.3 m, commenced drilling formation to 3383.5 m, POOH to surface.

RIH with 8 ½" B.BIT to 3383.5 m, POOH to 3375.5 m, performed logging job, RIH and resumed drilling slide & rotary mode to 3415.5 m, POOH to surface.

M/UP new BHA & RIH to bottom, resumed drilling 8 1/2" hole section slide & rotary mode to 3522 m, POOH as wiper trip to 3400 m and run back to bottom, resumed drilling to 3666.5 m, POOH on elevator to surface.

RIH with new BHA to bottom while stock string 3588 m, work on stock & increased MW to 87 PCF, commenced drilling 8 ½" hole section to 3835 m, POOH to surface.

RIH with 8 ½" PDC bit to bottom, resumed drilling 8 ½" hole section 3891 m, performed wiper trip and POOH to surface.

Run 7" liner and performed CMT Job as per program successfully.

The fresh KCl/Polymer mud was used for starting this section.

it was tried to control mud properties by using all available solid control equipment such as shale shakers, centrifuge was applied as per requirement in order to control MW. Hole cleaning was improved by pumping tandem pill high vis-Low Vis pills.

It was tried to reach optimum shale inhibition by using 1.5--5 wt.% KCl as main shale inhibitor by decreasing the tendency of to absorb, swell and slough-off.

The MW increased up to by adding Limestone Powder up to 87 pcf at end of this section.

SCE such as Shakers, mud cleaner and centrifuges were running to discard LGS while drilling this section.

Treated frequently with essential chemical such as Caustic Soda as Alkalinity controller to adjust pH, Soda Ash for control hardness & XC Polymer as Viscosifier.

6 – 1/8" Hole Section

RIH with 6 1/8" MT BIT & clean out BHA to 3845 m, D/O hard CMT to 3853 m (LDC), POOH to surface while Performed CIT test.

Performed scraping operation.

Performed logging Job.

Performed dry test.

RIH with 6" J. MILL to 3853 m, commenced D/O CMT to 3891 m, commenced drilling formation to 3893.5 m, POOH to surface.

RIH with 6 1/8" PDC bit to 3893.5 m while displaced the well with 100 PCF Kcl/Poly Mud @ 3870 m & performed FIT test, commenced drilling formation to 4003 m, increased MW to 104 PCF, resumed drilling formation to 4140 m, pumped Hi-Vis pill & performed condition trip then POOH to surface with good hole condition.

R/UP 4 ½" liner running EQP and run 4 ½" liner as per program successfully.

RIH with 6" J. MILL to 3720, D/O CMT to 3780.5 and POOH to surface.

RIH with 3 ½" J. MILL to 3780.5 m (L.LAP), commenced D/O CMT & clean out to 4134 m and POOH to surface.

Performed logging Job.

Performed clean out job with 3 ½" J. MILL.

Performed perforation Job.

10.3 Drilling Fluid type

FLUID PROPERTIES RECAP					
KCL Pol Mud		Drilling start	Range		Drilling End
MW (min)	pcf	106	106	108	108
Funnel Viscosity	sec/qt	44	41	52	41
PV	cps	21	22	39	22
YP	lb/100sq.ft	15	6	17	6
pH		9.5	9.3	11.3	11.3
Solid	%	38	38	39	39
Cl (NaCl)	mg/L	180,000	180000	185000	180,000
Total Hardness	mg/L	1,120	1120	4000	4,000
Ca Hardness	mg/L	800	800	2200	2,200

Table 10-2: AZNS-127 Fluid Properties Recap Table

10.4 Material Consumption Analysis

Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
Barite	33	3,300	108,900	lbs	990.00	ppb
Bentonite	1	2,200	2,200	lbs	2.53	ppb
Caustic Soda	33	55	1,815	lbs	0.39	ppb
Hematite	130	3,300	429,000	lbs	493.10	gpb
Lime	128	55	7,040	lbs	1.52	ppb
Limestone	579	2,200	1,273,800	lbs	274.43	ppb
Salt (NaCl)	187	2,200	411,400	lbs	88.63	ppb
Starch LV	477	55	26,235	lbs	5.65	ppb
Soda ASH	40	55	2,200	lbs	0.47	ppb
Sodium Bicarbonate	10	55	550	lbs	0.12	ppb
Xanthan Gum	40	55	2,200	lbs	0.47	ppb

Initial Volume	0.0 bbl
Total Volume Built	4,641.7 bbl
Volume received	0.0 bbl
Volume Returned	0.0 bbl
Total Volume Consumed	3,271 bbl
Volume Transfer to Next Section	1,370.7 bbl
Meter Drilled	0 m

Table 10-3: AZNS-127 Material Consumption Analysis Table

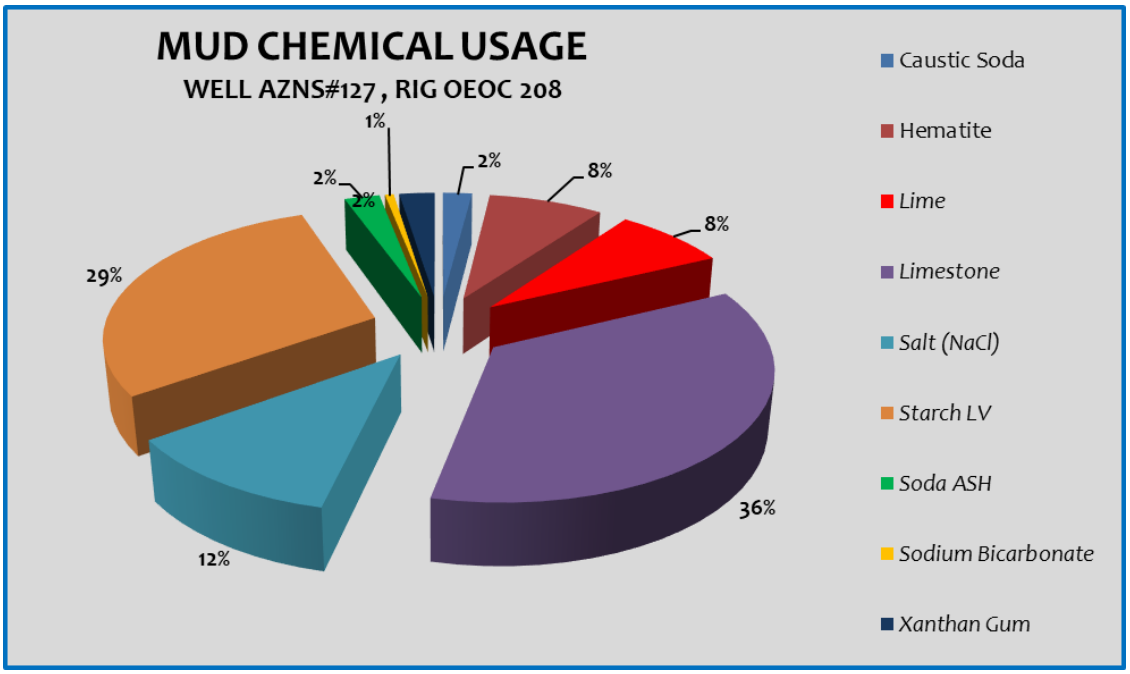


Figure 10-1: AZNS-127 Mud Chemical Usage

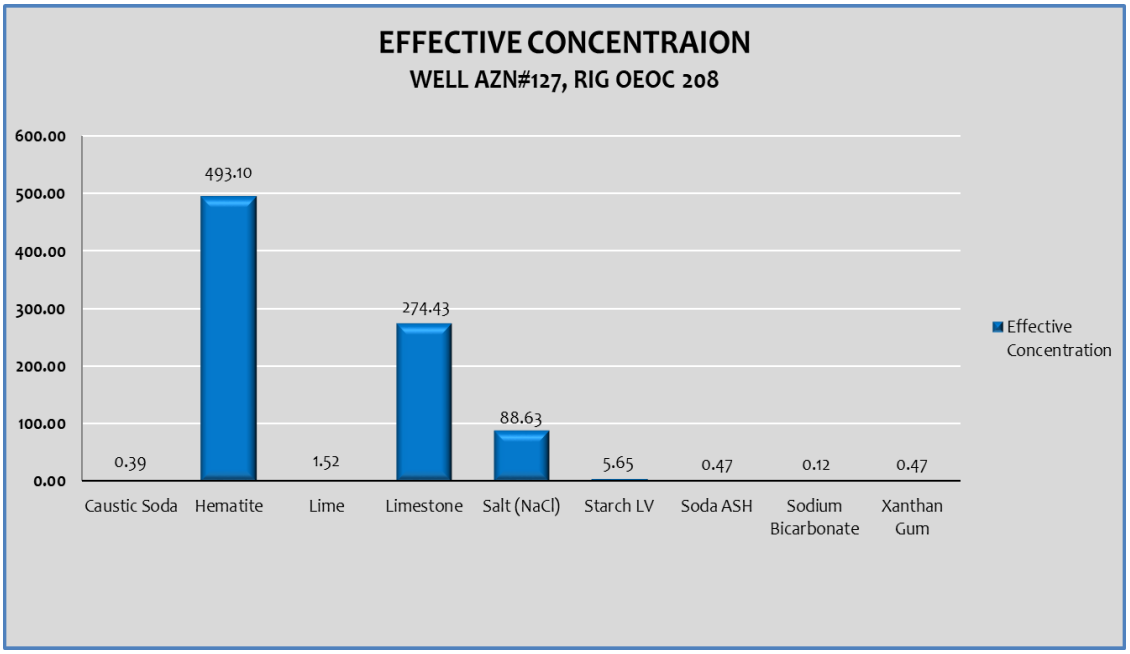


Figure 10-2: AZNS-127 Effective Concentration

10.5 Volume Analysis

WELL AZN#127, RIG OEOC 208 VOLUME ANALYSIS		
Initial Volume	bbls	0.0
Received Volume	bbls	0.0
Volume Return	bbls	0.0
Volume Built	bbls	4,641.7
Surface Losses/Dumped	bbls	3,271.0
Formation Losses	bbls	0.0
Total Losses	bbls	3,271.0
Volume Transfer Out	bbls	1,371

Table 10-4: AZNS-127 Volume Analysis Table

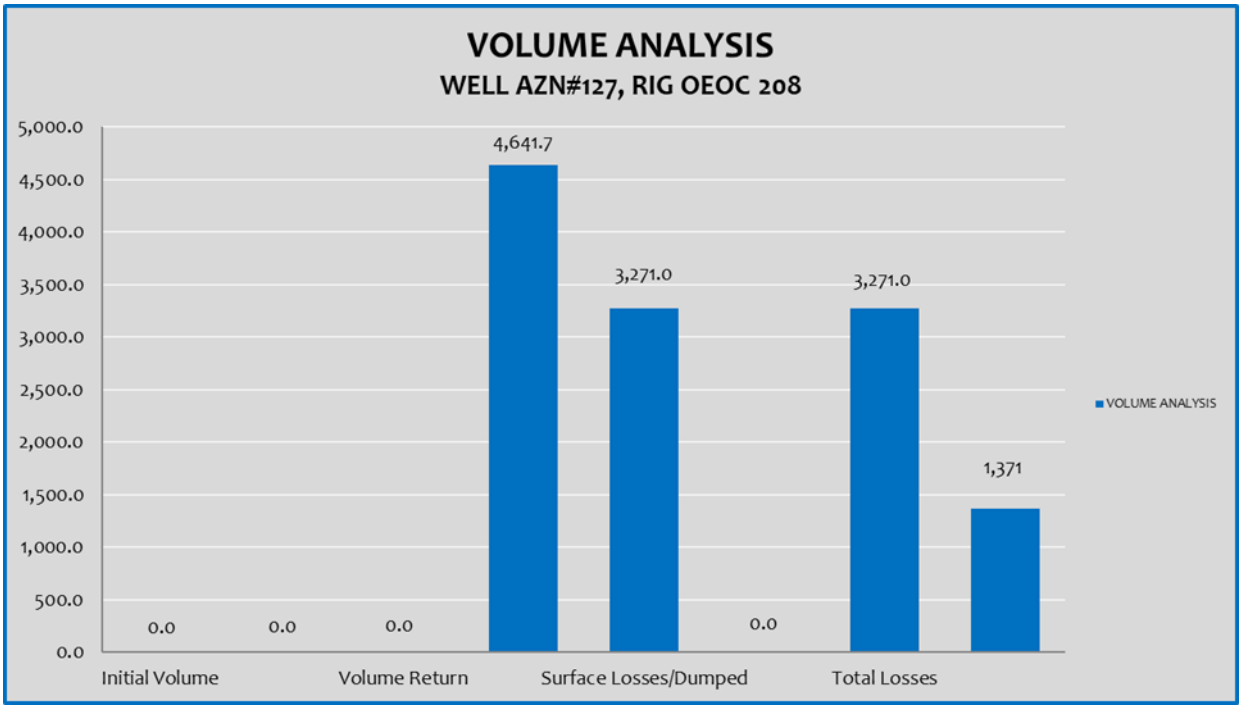


Figure 10-3: AZNS-127 Volume Analysis Chart

10.6 Daily volume Breakdown

Date/DMR.N		Volume of Built	Surface Losses	Formation Losses
8-May-24	01	361.7	0	0
9-May-24	02	520.0	0	0
10-May-24	03	547.8	0	0
11-May-24	04	4.1	0	0
12-May-24	05	0.0	0	0
13-May-24	06	141.5	0	0
14-May-24	07	428.2	400	0
15-May-24	08	455.8	230	0
16-May-24	09	31.5	140	0
17-May-24	10	288.4	10	0
18-May-24	11	29.3	0	0
19-May-24	12	17.7	23	0
20-May-24	13	336.6	120	0
21-May-24	14	134.2	400	0
22-May-24	15	10.2	0	0
23-May-24	16	307.0	200	0
24-May-24	17	75.0	20	0
25-May-24	18	0.0	0	0
26-May-24	19	0.0	0	0
27-May-24	20	278.8	168	0
28-May-24	21	3.9	35	0
29-May-24	22	0.1	24	0
30-May-24	23	0.3	24	0
31-May-24	24	0.1	32	0
1-Jun-24	25	0.2	0	0
2-Jun-24	26	150.7	105	0
3-Jun-24	27	1.1	0	0
4-Jun-24	28	295.2	65	0
5-Jun-24	29	0.3	75	0
6-Jun-24	30	0.2	131	0
7-Jun-24	31	0.2	43	0
8-Jun-24	32	0.3	75	0
9-Jun-24	33	0.3	42	0
10-Jun-24	34	0.1	36	0
11-Jun-24	35	0.3	36	0
12-Jun-24	36	0.1	46	0
13-Jun-24	37	0.0	32	0
14-Jun-24	38	0.1	12	0
15-Jun-24	39	80.3	22	0
16-Jun-24	40	0.2	44	0
17-Jun-24	41	0.1	25	0

18-Jun-24	42	6.0	37	0
19-Jun-24	43	0.5	51	0
20-Jun-24	44	0.0	24	0
21-Jun-24	45	0.1	22	0
22-Jun-24	46	13.4	35	0
23-Jun-24	47	0.0	11	0
24-Jun-24	48	0.0	37	0
25-Jun-24	49	0.2	17	0
26-Jun-24	50	0.0	30	0
27-Jun-24	51	0.2	25	0
28-Jun-24	52	46.7	29	0
29-Jun-24	53	0.1	27	0
30-Jun-24	54	0.1	31	0
1-Jul-24	55	0.2	18	0
2-Jul-24	56	0.0	12	0
3-Jul-24	57	0.0	37	0
4-Jul-24	58	72.3	186	0
5-Jul-24	59	0.0	27	0
Total (bbl)		Volume of Built	Surface Losses	Formation Losses
		4,641.7	3,271	0

Table 10-5: AZNS-127 Daily Volume Break Down Table

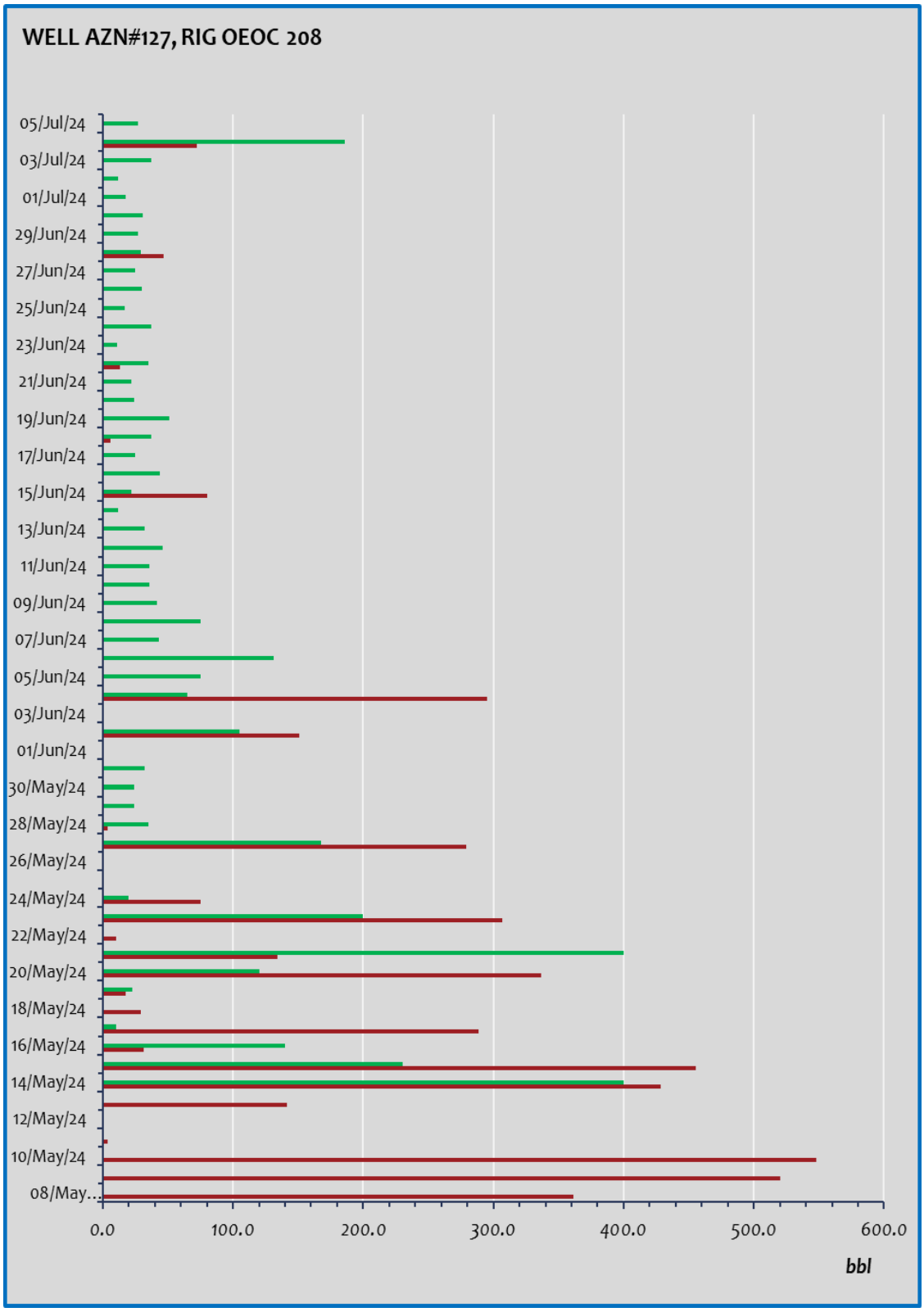


Figure 10-4: AZNS-127 Daily Volume Break Down Chart

10.7 Loss Distribution

Date/DMR.N		Volume of Built	Dump/Contamination	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
8-May-24	01	361.7	0	0	0	0	0	0	0	0	0
9-May-24	02	520.0	0	0	0	0	0	0	0	0	0
10-May-24	03	547.8	0	0	0	0	0	0	0	0	0
11-May-24	04	4.1	0	0	0	0	0	0	0	0	0
12-May-24	05	0.0	0	0	0	0	0	0	0	0	0
13-May-24	06	141.5	0	0	0	0	0	0	0	0	0
14-May-24	07	428.2	400	0	0	0	0	0	0	0	0
15-May-24	08	455.8	230	0	0	0	0	0	0	0	0
16-May-24	09	31.5	140	0	0	0	0	0	0	0	0
17-May-24	10	288.4	10	0	0	0	0	0	0	0	0
18-May-24	11	29.3	0	0	0	0	0	0	0	0	0
19-May-24	12	17.7	23	0	0	0	0	0	0	0	0
20-May-24	13	336.6	120	0	0	0	0	0	0	0	0
21-May-24	14	134.2	400	0	0	0	0	0	0	0	0
22-May-24	15	10.2	0	0	0	0	0	0	0	0	0
23-May-24	16	307.0	200	0	0	0	0	0	0	0	0
24-May-24	17	75.0	0	10	0	0	0	0	0	0	10
25-May-24	18	0.0	0	0	0	0	0	0	0	0	0
26-May-24	19	0.0	0	0	0	0	0	0	0	0	0
27-May-24	20	278.8	150	18	0	0	0	0	0	0	0
28-May-24	21	3.9	0	0	5	0	30	0	0	0	0
29-May-24	22	0.1	0	0	6	0	10	0	0	0	8
30-May-24	23	0.3	0	18	6	0	0	0	0	0	0
31-May-24	24	0.1	0	9	6	0	17	0	0	0	0
1-Jun-24	25	0.2	0	0	0	0	0	0	0	0	0
2-Jun-24	26	150.7	100	5	0	0	0	0	0	0	0
3-Jun-24	27	1.1	0	0	0	0	0	0	0	0	0
4-Jun-24	28	295.2	0	5	0	0	10	0	30	0	20
5-Jun-24	29	0.3	0	5	5	0	0	0	20	0	45
6-Jun-24	30	0.2	0	30	6	0	10	0	25	0	60
7-Jun-24	31	0.2	0	5	8	0	0	0	0	0	30
8-Jun-24	32	0.3	0	8	7	0	10	0	25	0	25
9-Jun-24	33	0.3	0	5	12	0	10	0	0	0	15
10-Jun-24	34	0.1	0	4	12	0	0	0	0	0	20
11-Jun-24	35	0.3	0	4	12	0	0	0	20	0	0
12-Jun-24	36	0.1	0	4	12	0	0	0	0	0	30
13-Jun-24	37	0.0	0	0	12	0	0	0	20	0	0
14-Jun-24	38	0.1	0	0	12	0	0	0	0	0	0
15-Jun-24	39	80.3	0	0	12	0	0	0	0	0	10
16-Jun-24	40	0.2	0	4	10	0	10	0	0	0	20
17-Jun-24	41	0.1	0	3	12	0	10	0	0	0	0

18-Jun-24	42	6.0	0	0	12	0	25	0	0	0	0
19-Jun-24	43	0.5	0	5	12	0	14	0	0	0	20
20-Jun-24	44	0.0	0	0	12	0	12	0	0	0	0
21-Jun-24	45	0.1	0	3	11	0	8	0	0	0	0
22-Jun-24	46	13.4	0	5	12	0	18	0	0	0	0
23-Jun-24	47	0.0	0	0	11	0	0	0	0	0	0
24-Jun-24	48	0.0	0	0	12	0	25	0	0	0	0
25-Jun-24	49	0.2	0	5	12	0	0	0	0	0	0
26-Jun-24	50	0.0	0	0	10	0	15	0	0	0	5
27-Jun-24	51	0.2	0	8	12	0	5	0	0	0	0
28-Jun-24	52	46.7	0	5	11	0	5	0	0	0	8
29-Jun-24	53	0.1	0	5	12	0	0	0	0	0	10
30-Jun-24	54	0.1	0	0	12	0	4	0	15	0	0
1-Jul-24	55	0.2	0	0	11	0	0	0	0	0	7
2-Jul-24	56	0.0	0	0	12	0	0	0	0	0	0
3-Jul-24	57	0.0	0	0	12	0	0	0	20	0	5
4-Jul-24	58	72.3	80	4	12	0	0	0	80	0	10
5-Jul-24	59	0.0	0	0	12	0	0	0	15	0	0
Total (bbl)		Volume of Built	Dump/Contaminate	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
		4,641.7	1,853	177	365	0	248	0	270	0	358.0

Note: Prepared 890.8 bbl KCL poly mud (DMR#57 & 59) that is consider in next section.

Table 10-6: AZNS-127 Loss Distribution

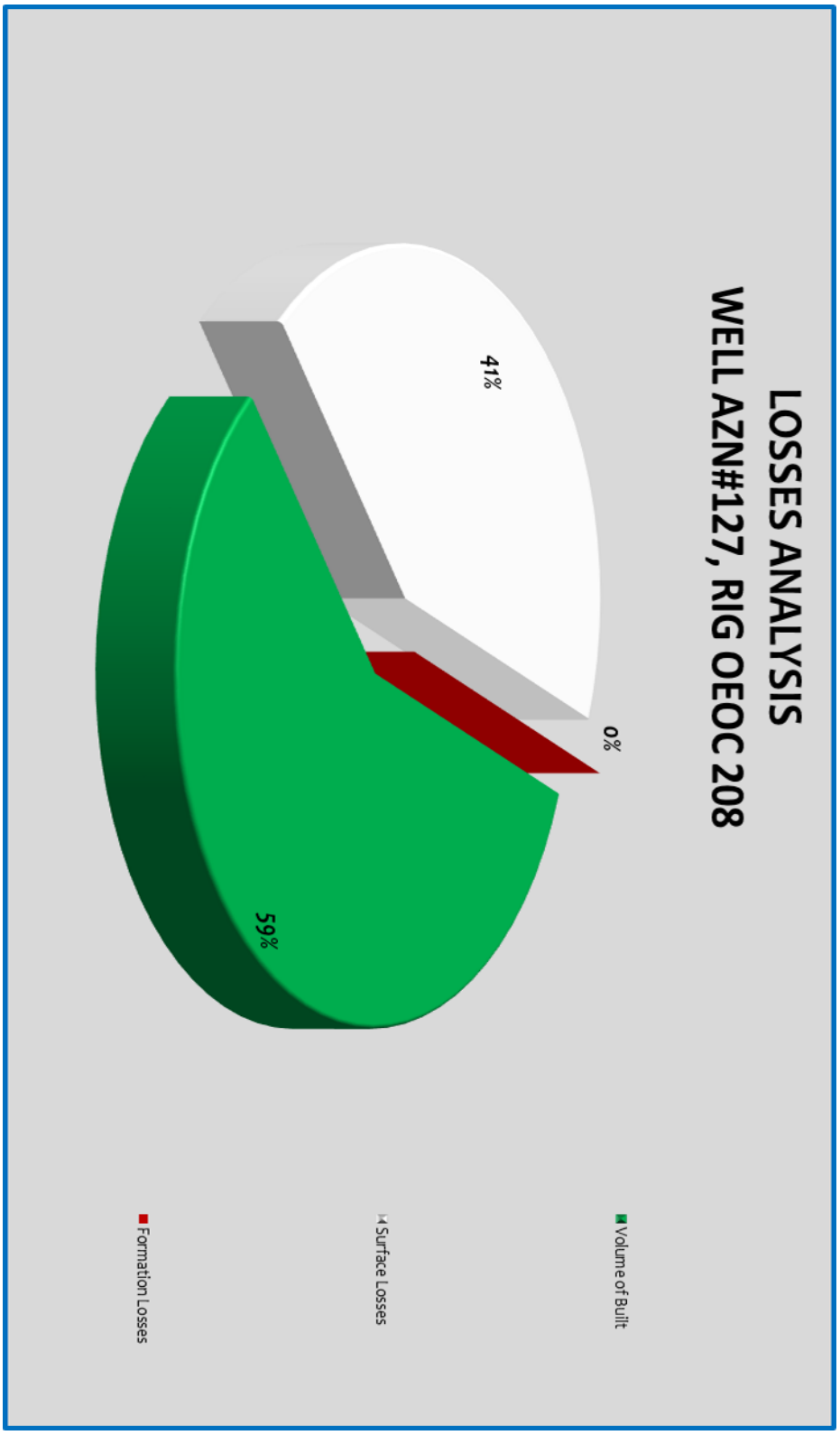


Figure 10-5: AZNS-127 Loss Analysis

10.8 Interval Fluid Properties Recap

Date	DEPTH	Density	VIS	PV	YP	Water Loss	pH	Solid %	Chloride	Calcium	Total Hardness	Mud Type
8-May-24	4226											
9-May-24	4226											
10-May-24	4226	106	47	36	15	14	9.5	38	180,000	800	1120	Salt/Poly Mud
11-May-24	4226	106	48	36	17	14	9.5	38	180,000	800	1120	Salt/Poly Mud
12-May-24	4226	106	48	36	17	14	9.5	38	180,000	800	1120	Salt/Poly Mud
13-May-24	4226	106	48	36	17	14	9.5	38	180,000	800	1120	Salt/Poly Mud
14-May-24	4226	106	48	36	17	14	9.4	38	182,000	800	1120	Salt/Poly Mud
15-May-24	4226	106	48	36	17	14	9.4	38	180,000	800	1120	Salt/Poly Mud
16-May-24	4226	106	47	36	17	14	9.5	38	180,000	800	1120	Salt/Poly Mud
17-May-24	4226	106	47	36	17	14	9.5	38	180,000	800	1120	Salt/Poly Mud
18-May-24	4226	106	47	36	17	14	9.5	38	180,000	800	1120	Salt/Poly Mud
19-May-24	4226	106	47	36	17	14	9.5	38	180,000	800	1120	Salt/Poly Mud
20-May-24	4226	106	47	36	17	14	9.4	38	180,000	800	1120	Salt/Poly Mud
21-May-24	4226	106	52	35	14	20	9.3	38	180,000	920	1200	Salt/Poly Mud
22-May-24	4226	106	47	32	14	22	9.4	38	180,000	920	1200	Salt/Poly Mud
23-May-24	4226	106	47	37	15	22	9.4	38	180,000	1000	1280	Salt/Poly Mud
24-May-24	4226	106	47	37	14	22	9.4	38	180,000	1000	1280	Salt/Poly Mud
25-May-24	4226	106	47	37	14	N.C	9.4	38	180,000	1120	1360	Salt/Poly Mud
26-May-24	4226	106	47	37	14	N.C	9.4	38	180,000	1120	1360	Salt/Poly Mud
27-May-24	4226	106	48	37	15	N.C	9.5	38	180,000	1280	1400	Salt/Poly Mud
28-May-24	4226	106	48	37	15	N.C	9.5	38	180,000	1280	1400	Salt/Poly Mud
29-May-24	4226	106	48	37	15	N.C	9.5	38	180,000	1280	1400	Salt/Poly Mud
30-May-24	4226	106	46	36	14	N.C	9.5	38	180,000	1280	1400	Salt/Poly Mud
31-May-24	4226	106	46	35	14	N.C	9.5	38	180,000	1280	1400	Salt/Poly Mud
1-Jun-24	4226	106	47	36	15	N.C	9.5	38	180,000	1280	1400	Salt/Poly Mud
2-Jun-24	4226	106	47	36	15	N.C	9.5	38	180,000	1280	1400	Salt/Poly Mud
3-Jun-24	4226	106	45	34	15	20	9.8	38	180,000	1600	1800	Salt/Poly Mud
4-Jun-24	4226	106	45	31	15	18	9.7	38	180,000	1600	1800	Salt/Poly Mud
5-Jun-24	4226	106	46	34	16	18	9.5	38	180,000	1600	1800	Salt/Poly Mud
6-Jun-24	4226	106	45	33	14	20	9.5	38	180,000	1600	1800	Salt/Poly Mud
7-Jun-24	4226	106	45	33	14	20	9.7	38	180,000	1600	1800	Salt/Poly Mud
8-Jun-24	4226	106	45	33	14	20	9.6	38	180,000	1600	1800	Salt/Poly Mud
9-Jun-24	4226	106	45	33	14	20	9.6	38	180,000	1600	1800	Salt/Poly Mud
10-Jun-24	4226	106	45	34	14	22	9.5	38	180,000	1680	1840	Salt/Poly Mud
11-Jun-24	4226	106	47	38	16	20	9.5	38	180,000	1680	1840	Salt/Poly Mud
12-Jun-24	4226	106	47	38	16	18	9.6	38	180,000	1680	1840	Salt/Poly Mud
13-Jun-24	4226	106	47	38	16	18	9.5	38	180,000	1680	1840	Salt/Poly Mud
14-Jun-24	4226	106	47	38	16	18	9.5	38	180,000	1680	1840	Salt/Poly Mud
15-Jun-24	4226	108	49	39	17	17	9.6	39	180,000	1680	1840	Salt/Poly Mud
16-Jun-24	4226	108	48	38	17	17	9.7	39	180,000	1680	1840	Salt/Poly Mud
17-Jun-24	4226	108	46	38	16	17	9.7	39	180,000	1680	1840	Salt/Poly Mud
18-Jun-24	4226	108	46	36	16	17	9.5	39	180,000	1680	1840	Salt/Poly Mud
19-Jun-24	4226	108	47	36	16	14	9.7	39	185,000	1680	1840	Salt/Poly Mud
20-Jun-24	4226	108	47	36	16	14	9.5	39	180,000	1680	1840	Salt/Poly Mud
21-Jun-24	4226	108	47	36	16	14	9.5	39	180,000	1680	1840	Salt/Poly Mud
22-Jun-24	4226	108	46	36	14	14	9.5	39	180,000	1680	1840	Salt/Poly Mud
23-Jun-24	4226	108	47	36	14	14	9.5	39	180,000	1680	1840	Salt/Poly Mud
24-Jun-24	4226	108	47	36	14	14	9.5	39	180,000	1680	1840	Salt/Poly Mud
25-Jun-24	4226	108	46	35	14	14	9.6	39	180,000	1680	1840	Salt/Poly Mud
26-Jun-24	4226	108	46	35	14	14	9.6	39	180,000	1680	1840	Salt/Poly Mud
27-Jun-24	4226	108	47	35	15	16	9.4	39	180,000	1680	1840	Salt/Poly Mud
28-Jun-24	4226	108	47	35	15	16	9.5	39	180,000	1680	1840	Salt/Poly Mud
29-Jun-24	4226	108	48	36	15	16	9.5	39	180,000	1680	1840	Salt/Poly Mud
30-Jun-24	4226	108	48	36	15	16	9.4	39	180,000	1680	1840	Salt/Poly Mud
1-Jul-24	4226	108	48	36	15	16	9.4	39	180,000	1680	1840	Salt/Poly Mud
2-Jul-24	4226	108	48	36	15	16	9.5	39	180,000	1680	1840	Salt/Poly Mud
3-Jul-24	4226	108	48	36	15	16	9.4	39	180,000	1680	1840	Salt/Poly Mud
4-Jul-24	4226	108	48	36	15	16	9.3	39	180,000	1680	1840	Salt/Poly Mud
5-Jul-24	4226	108	41	22	6	N.C	11.3	39	180,000	2200	4000	Salt/Poly Mud

Table 10-7: AZNS-127 Fluid Properties Recap

10.9 Solid Control Disscusion

Rig OEOC 208 has 3 shale shakers were dressed with:

Shaker#1: 120 Mesh Screen

Shaker#2: 120 Mesh Screen

Shaker#3: 120 Mesh Screen

WELL AZN#85, RIG OEOC 201	
Solid Control Equipment Working Hr.s	
Shaker # 01	240
Shaker # 02	320
Shaker # 03	646

Table 10-8: AZNS-127 Solid Control Desscusion

10.10 8 – 1/2” Hole Section Drilling Fluid Types

FLUID PROPERTIES RECAP					
KCL Pol Mud		Drilling start	Range		Drilling End
MW (min)	pcf	85	85	87	87
Funnel Viscosity	sec/qt	41	40	43	43
PV	cps	18	15	19	19
YP	lb/100sq.ft	17	15	17	16
API Fluid loss	cc	3	2	3	2.2
pH		9.4	9.3	9.6	9.5
KCL	%Wt	4.2	4	5.0	5
Cl (NaCl)	mg/L	170,000	165000	173000	170,000
Total Hardness	mg/L	1,680	1680	1800	1,800
Ca Hardness	mg/L	1,200	1080	1400	1,400

Table 10-9: AZNS-127 8-1/2” Fluid Properties Recap Table

10.11 8 1/2” Interval Material Consumption Analysis

Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
Citric Acid	1	55	55	lbs	0.02	ppb
Caustic Soda	26	55	1,430	lbs	0.40	ppb
Lime	128	55	7,040	lbs	1.95	ppb
Limestone	224	2,200	492,800	lbs	136.37	ppb
Salt (NaCl)	81	2,200	178,200	lbs	49.31	ppb
Starch LV	484	55	26,620	lbs	7.37	ppb
Soda ASH	45	55	2,475	lbs	0.68	ppb
Sodium Bicarbonate	3	55	165	lbs	0.05	ppb
KCL	26	2,200	57,200	lbs	15.83	ppb
Starch HT	99	55	5,445	lbs	1.51	ppb
Xanthan Gum	38	55	2,090	lbs	0.58	ppb

Initial Volume	1,371.0 bbl
Total Volume Built	3,613.7 bbl
Volume received	105.0 bbl
Volume Returned	0.0 bbl
Total Volume Consumed	2,719 bbl
Volume Transfer to Next Section	2,370.7 bbl
Meter Drilled	552 m

Table 10-10: AZNS-127 8-1/2” Interval Material Consumption Analysis

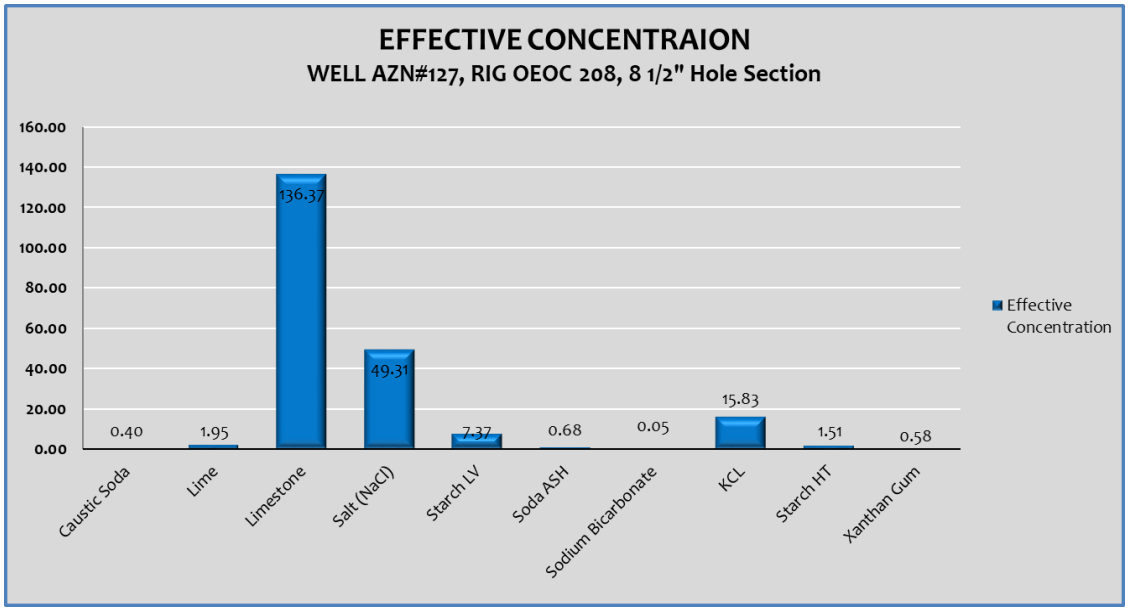


Figure 10-6: AZNS-127 8 -1/2” Effective Concentration

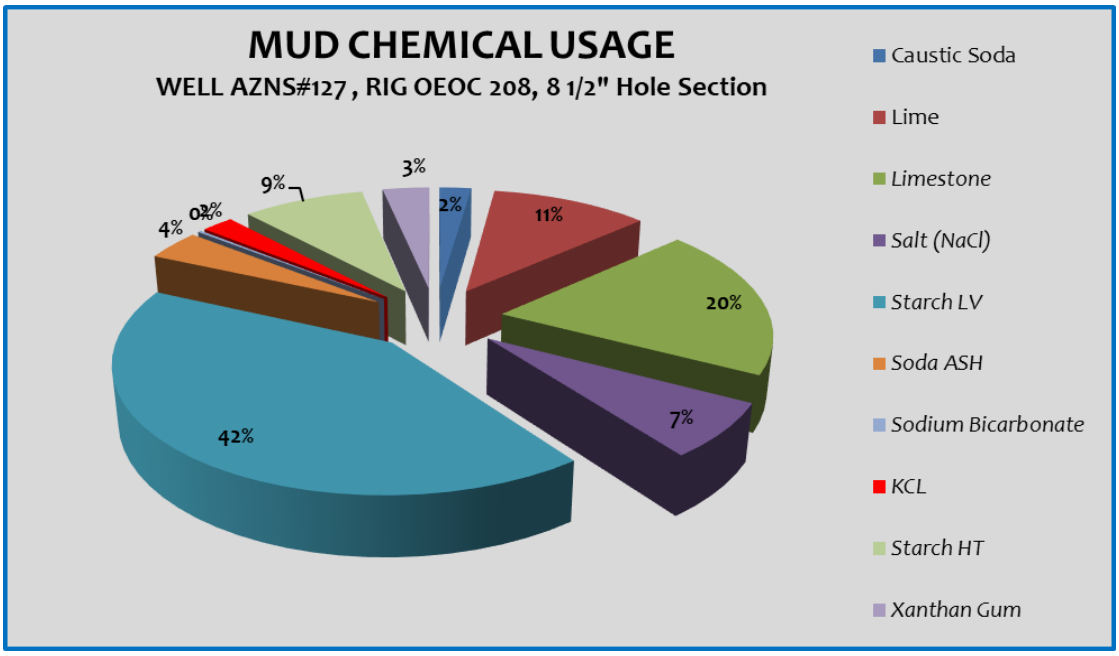


Figure 10-7: AZNS-127 8-1/2" Mud Chemical Usage

10.12 8 1/2" Interval Volume Analysis

WELL AZN#127, RIG OEOC 208 VOLUME ANALYSIS		
Initial Volume	bbls	1,371.0
Received Volume	bbls	105.0
Volume Return	bbls	0.0
Volume Built	bbls	3,613.7
Surface Losses/Dumped	bbls	2,719.0
Formation Losses	bbls	0.0
Total Losses	bbls	2,719.0
Volume Transfer Out	bbls	2,370.7

Table 10-11: AZNS-127 8-1/2" Interval Volume Analysis

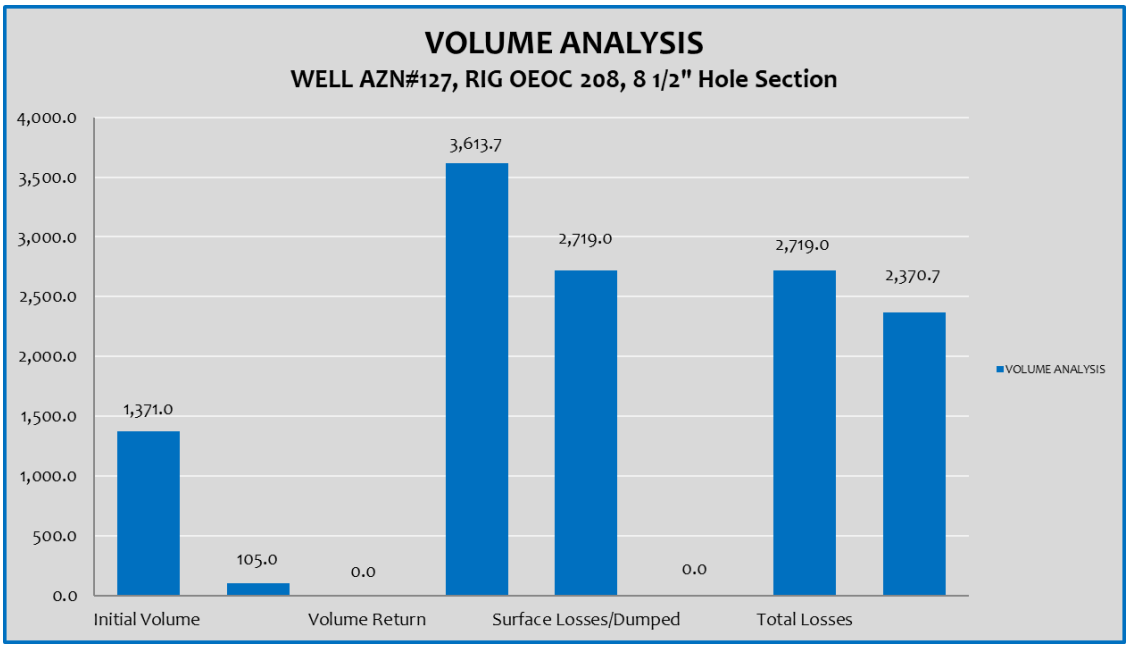


Figure 10-8: AZNS-127 8-1/2" Interval Volume Analysis

10.13 8 1/2” Interval Daily Volume Breakdown

Date/DMR.N		Volume of Built	Surface Losses	Formation Losses
6-Jul-24	60	230.8	262	0
7-Jul-24	61	172.4	47	0
8-Jul-24	62	186.5	12	0
9-Jul-24	63	108.8	494	0
10-Jul-24	64	18.7	177	0
11-Jul-24	65	0.5	15	0
12-Jul-24	66	111.9	17	0
13-Jul-24	67	96.4	67	0
14-Jul-24	68	3.1	101	0
15-Jul-24	69	110.6	51	0
16-Jul-24	70	0.1	115	0
17-Jul-24	71	132.6	34	0
18-Jul-24	72	76.0	82	0
19-Jul-24	73	72.2	87	0
20-Jul-24	74	174.9	96	0
21-Jul-24	75	0.0	154	0
22-Jul-24	76	62.8	53	0
23-Jul-24	77	149.9	52	0
24-Jul-24	78	111.7	89	0
25-Jul-24	79	92.7	92	0
26-Jul-24	80	205.4	133	0
27-Jul-24	81	92.2	115	0
28-Jul-24	82	146.6	65	0
29-Jul-24	83	0.0	74	0
30-Jul-24	84	0.0	55	0
31-Jul-24	85	15.9	55	0
1-Aug-24	86	350.2	125	0
Total (bbl)		Volume of Built	Surface Losses	Formation Losses
		3,613.7	2,719	0

Table 10-12: AZNS-127 8-1/2” Interval Daily Volume Breakdown

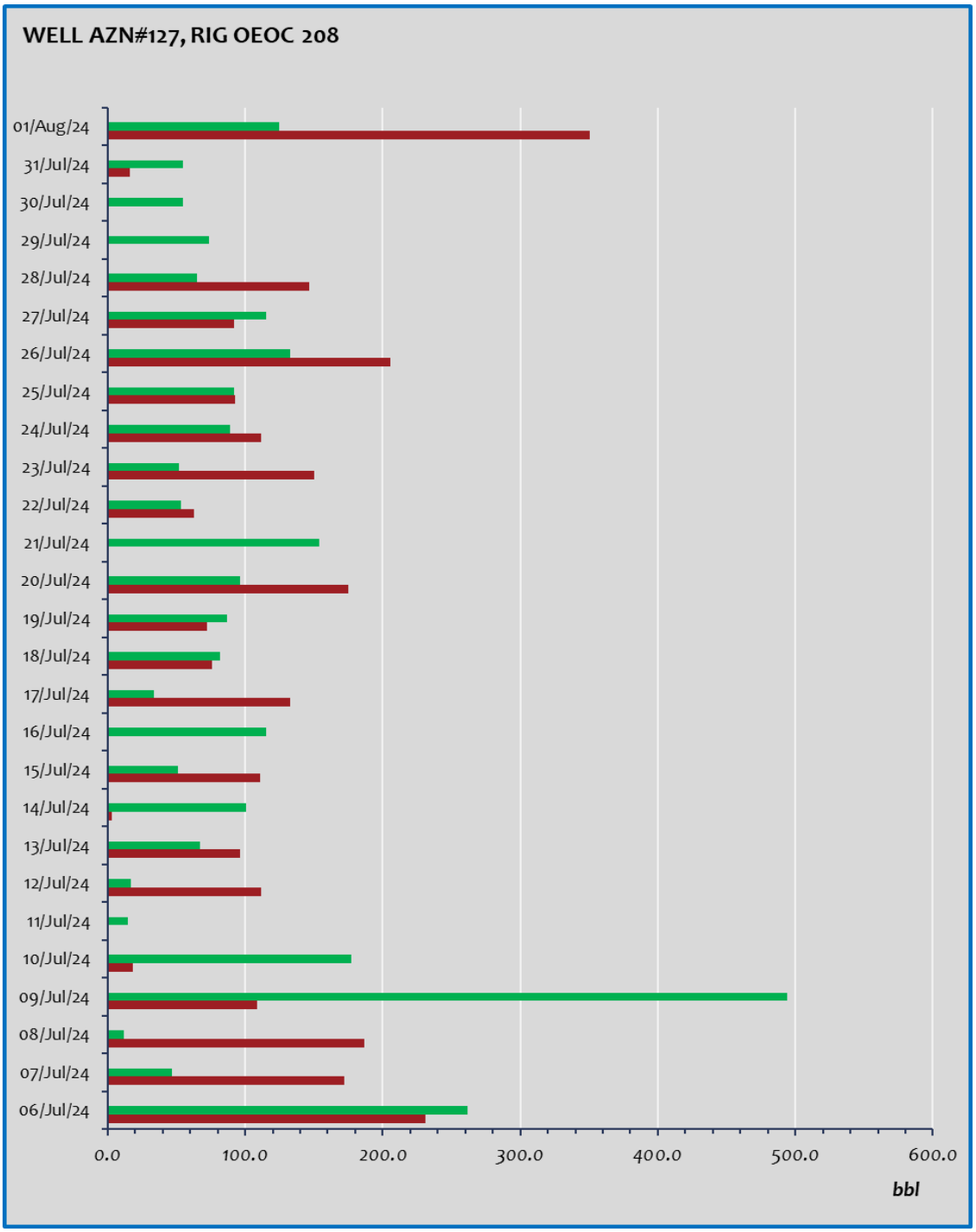


Figure 10-9: AZNS-127 8-1/2" Interval Daily Volume Breakdown

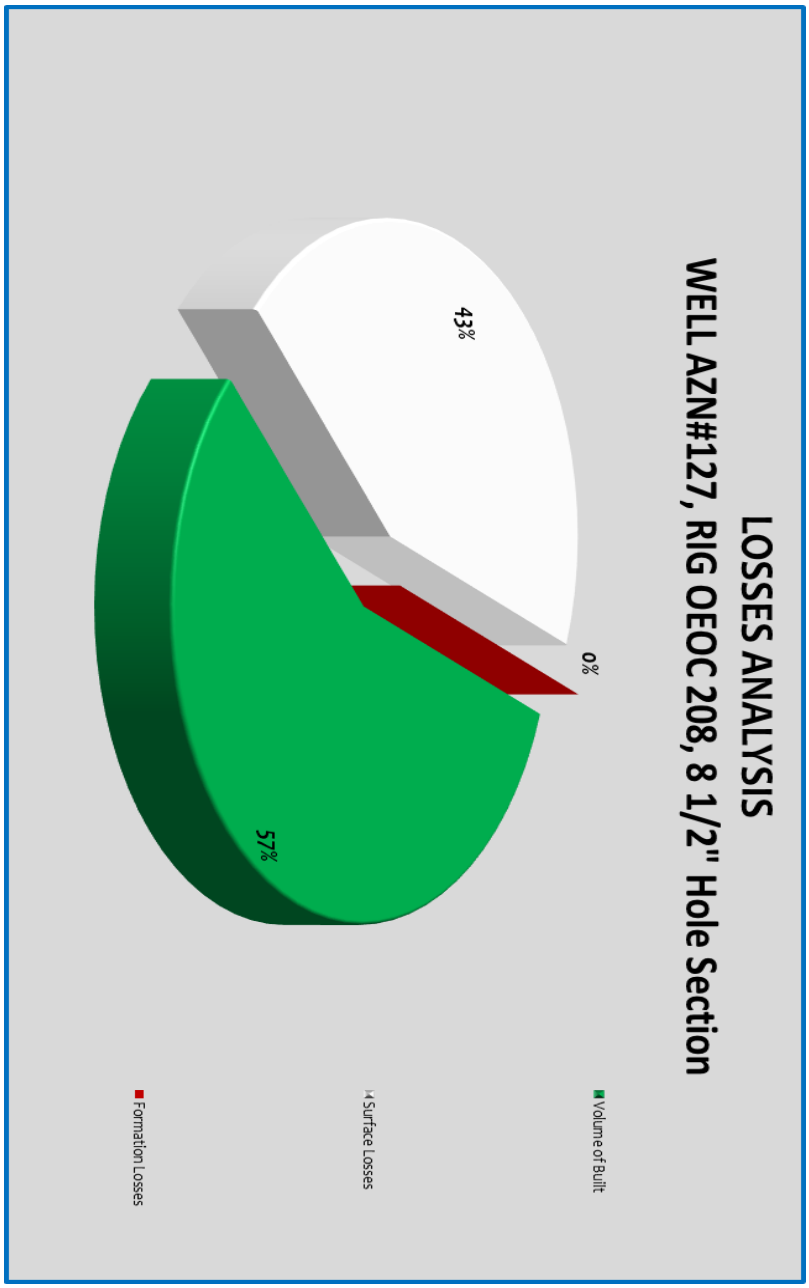


Figure 10-10: AZNS-127 8-1/2" Losses Analysis

10.14 8 – 1/2” Interval Loss Distribution

Date/DMR.N		Volume of Built	Dump/Contamination	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
6-Jul-24	60	230.8	200	0	12	0	0	0	20	0	30
7-Jul-24	61	172.4	0	0	11	0	0	0	24	0	12
8-Jul-24	62	186.5	0	0	12	0	0	0	0	0	0
9-Jul-24	63	108.8	474	8	12	0	0	0	0	0	0
10-Jul-24	64	18.7	160	5	12	0	0	0	0	0	0
11-Jul-24	65	0.5	0	5	10	0	0	0	0	0	0
12-Jul-24	66	111.9	0	6	11	0	0	0	0	0	0
13-Jul-24	67	96.4	0	0	12	0	40	0	15	0	0
14-Jul-24	68	3.1	75	0	11	10	0	0	0	0	5
15-Jul-24	69	110.6	0	15	12	24	0	0	0	0	0
16-Jul-24	70	0.1	0	5	12	0	80	0	10	0	8
17-Jul-24	71	132.6	0	10	12	7	0	0	0	0	5
18-Jul-24	72	76.0	0	24	13	30	0	0	0	0	15
19-Jul-24	73	72.2	0	24	13	35	0	0	0	0	15
20-Jul-24	74	174.9	0	24	15	35	0	0	22	0	0
21-Jul-24	75	0.0	0	18	12	24	75	0	0	0	25
22-Jul-24	76	62.8	0	10	13	0	10	0	0	0	20
23-Jul-24	77	149.9	0	12	0	30	0	0	0	0	10
24-Jul-24	78	111.7	0	24	15	35	0	0	0	0	15
25-Jul-24	79	92.7	0	36	16	40	0	0	0	0	0
26-Jul-24	80	205.4	0	58	15	40	0	0	0	0	20
27-Jul-24	81	92.2	0	25	15	25	15	0	0	0	35
28-Jul-24	82	146.6	0	30	15	0	0	0	20	0	0
29-Jul-24	83	0.0	0	24	15	0	20	0	0	0	15
30-Jul-24	84	0.0	0	0	15	0	0	0	40	0	0
31-Jul-24	85	15.9	0	0	15	0	10	0	20	0	10
1-Aug-24	86	350.2	70	20	15	0	10	0	10	0	0
Total (bbl)		Volume of Built	Dump/Contaminate	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
		3,613.7	979	383	341	335	260	0	181	0	240.0

Note: Prepared 890.8 bbl KCL poly mud (DMR#57 & 59) that is consider in 8 1/2" hole section.

Table 10-13: AZNS-127 8-1/2” Interval Loss Distribution

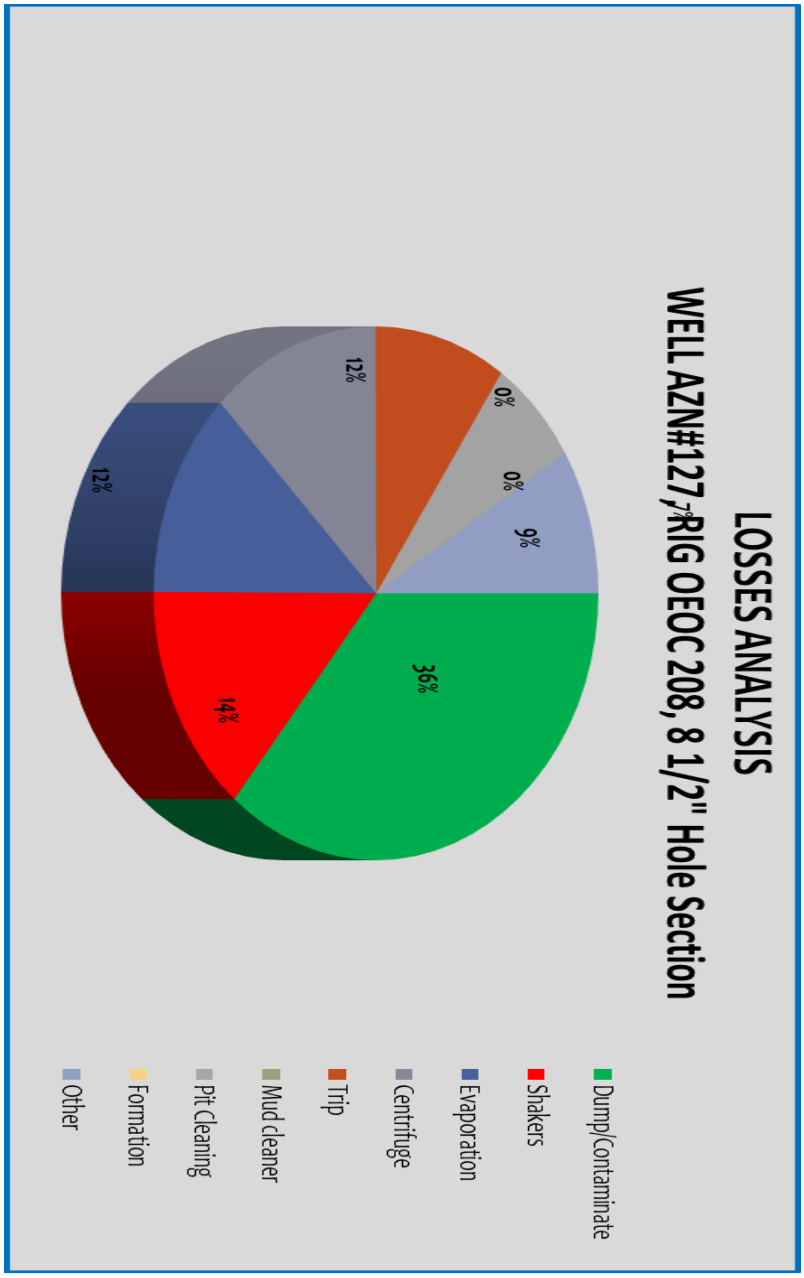


Figure 10-11: AZNS-127 8-1/2" Losses Analysis

10.15 8 – 1/2” Interval Fluid Properties Recap

Date	DEPTH	Density	VIS	PV	YP	Water Loss	pH	Solid %	Chloride	KCL (%Wt)	Calcium	Total Hardness	Mud Type
6-Jul-24	4226	85	41	14	11	3	9.5	19	165000		1200	1600	Salt/ Poly Mud
7-Jul-24	4226	85	41	14	11	3	9.5	19	165000		1200	1600	Salt/ Poly Mud
8-Jul-24	4226	85	41	14	11	3	9.4	19	165000		1200	1600	Salt/ Poly Mud
9-Jul-24	4226	85	40	14	11	3	9.4	19	165000	4.2	1000	1600	KCL Poly Mud
10-Jul-24	4226	85	41	15	13	3	9.3	19	180,000	4.2	1080	1600	KCL Poly Mud
11-Jul-24	4226	85	42	17	14	3	9.6	19	180,000	4.2	1120	1600	KCL Poly Mud
12-Jul-24	3383.5	85	41	18	17	3	9.4	19	170,000	4.2	1200	1680	KCL Poly Mud
13-Jul-24	3383.5	85	41	18	17	2.6	9.5	20	173,000	4.2	1080	1680	KCL Poly Mud
14-Jul-24	3390.7	85	41	16	15	2.8	9.3	20	165,000	4.2	1200	1680	KCL Poly Mud
15-Jul-24	3407.5	85	40	15	16	2.8	9.4	20	165,000	4.2	1280	1720	KCL Poly Mud
16-Jul-24	3454	85	40	15	16	2.8	9.4	20	165,000	4.2	1280	1720	KCL Poly Mud
17-Jul-24	3407.5	85	41	17	15	2.8	9.4	20	168,000	5	1280	1720	KCL Poly Mud
18-Jul-24	3518	85	41	16	15	2.8	9.4	20	168,000	5	1280	1800	KCL Poly Mud
19-Jul-24	3572	85	42	16	15	2.8	9.5	20	170,000	5	1320	1800	KCL Poly Mud
20-Jul-24	3638	85	43	17	16	2.8	9.5	20	168,000	5	1400	1800	KCL Poly Mud
21-Jul-24	3666	85	43	17	16	2.8	9.5	20	168,000	5	1400	1800	KCL Poly Mud
22-Jul-24	3666	87	45	18	17	2.8	9.5	21	168,000	5	1400	1800	KCL Poly Mud
23-Jul-24	3715	87	43	18	17	2.8	9.6	21	170,000	5	1320	1740	KCL Poly Mud
24-Jul-24	3761	87	42	18	16	2.4	9.4	21	170,000	5	1320	1740	KCL Poly Mud
25-Jul-24	3791	87	42	18	16	2.4	9.6	21	170,000	5	1400	1800	KCL Poly Mud
26-Jul-24	3828	87	42	18	15	2.2	9.5	21	170,000	5	1400	1800	KCL Poly Mud
27-Jul-24	3835	87	43	19	16	2.2	9.4	21	170,000	5	1400	1800	KCL Poly Mud
28-Jul-24	3875	87	42	19	15	2.2	9.5	21	170,000	5	1400	1800	KCL Poly Mud
29-Jul-24	3891	87	43	19	16	2.2	9.5	21	170,000	5	1400	1800	KCL Poly Mud
30-Jul-24	3891	87	44	20	17	2.2	9.5	21	170,000	5	1400	1800	KCL Poly Mud
31-Jul-24	3891	87	45	20	18	2.2	9.5	21	170,000	5	1400	1800	KCL Poly Mud
1-Aug-24	3891	87	41	20	12	3	10.5	21	170,000	5	2000	2600	KCL Poly Mud
Drilling Start													
Drilling End													

Table 10-14: AZNS-127 8-1/2” Interval Fluid Properties

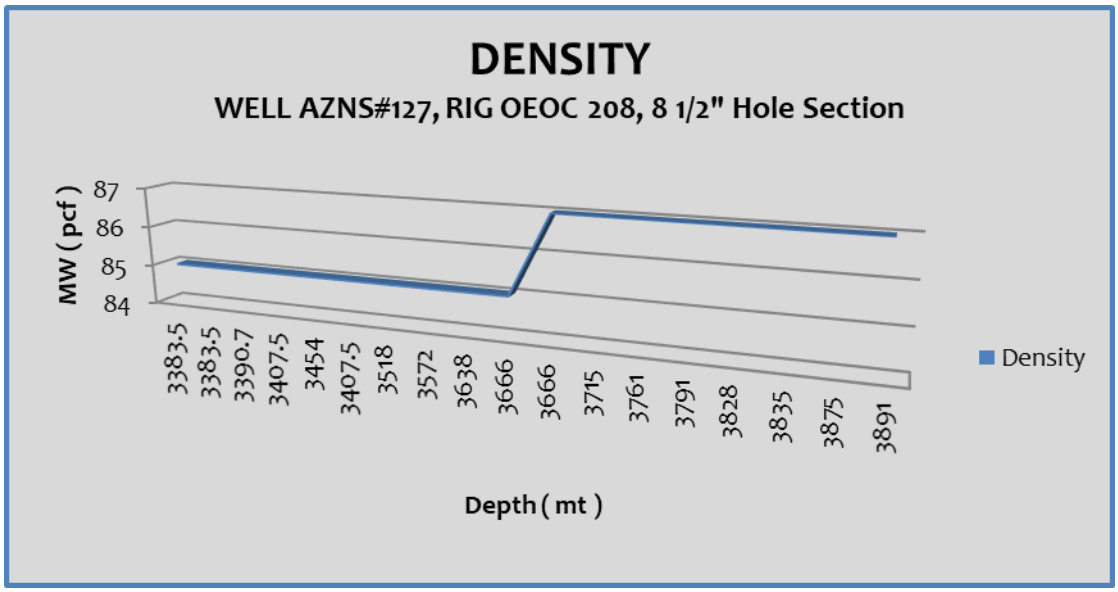


Figure 10-12: AZNS-127 8-1/2" Density Graph

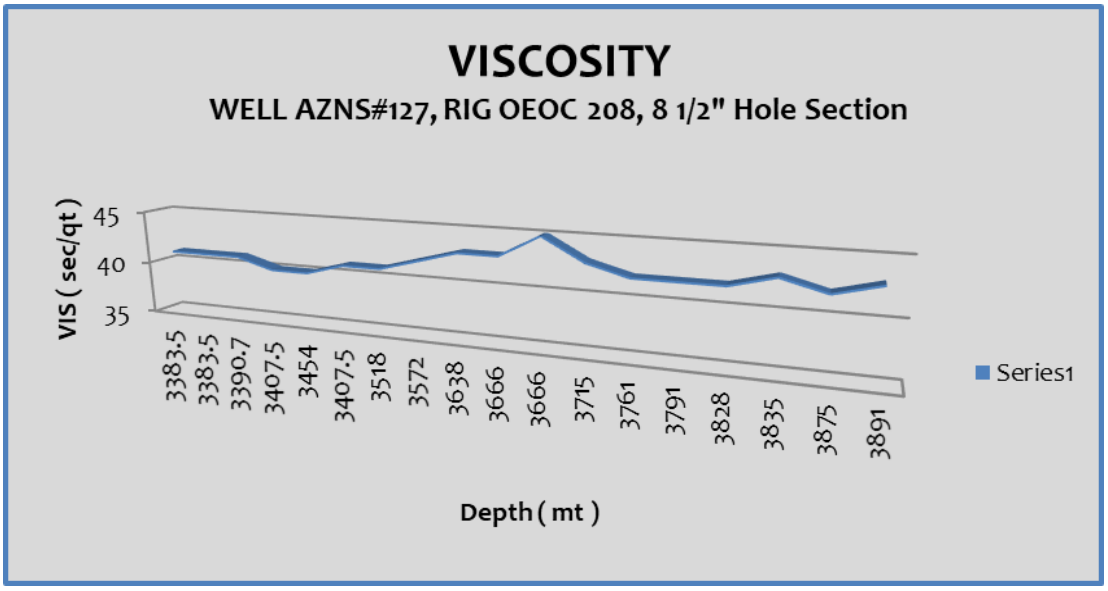


Figure 10-13: AZNS-127 8-1/2" Viscosity Graph

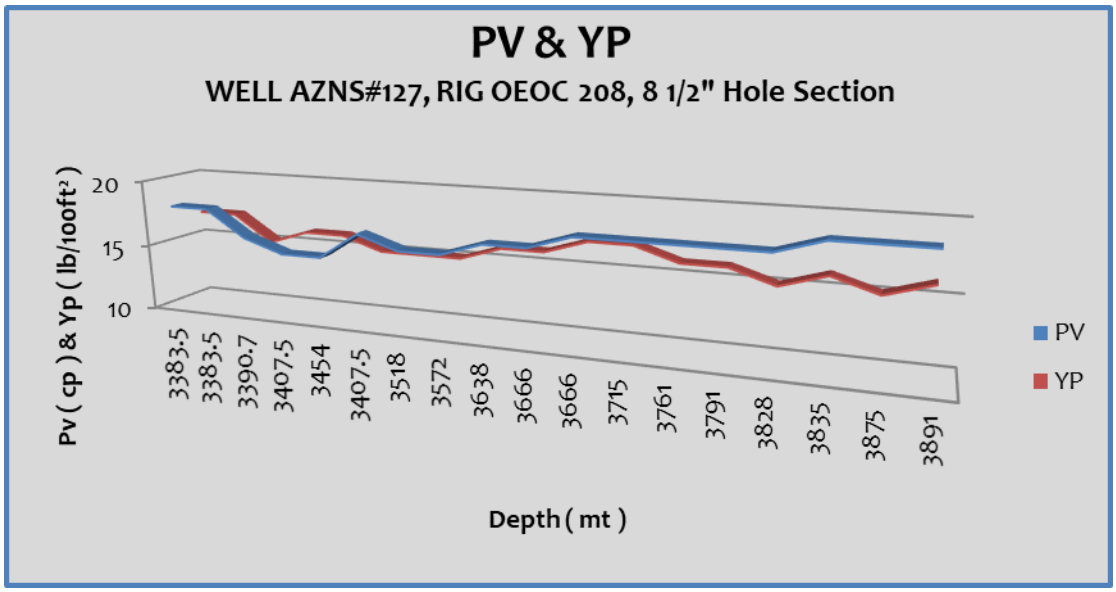


Figure 10-14: AZNS-127 8-1/2" PV & YP Graph

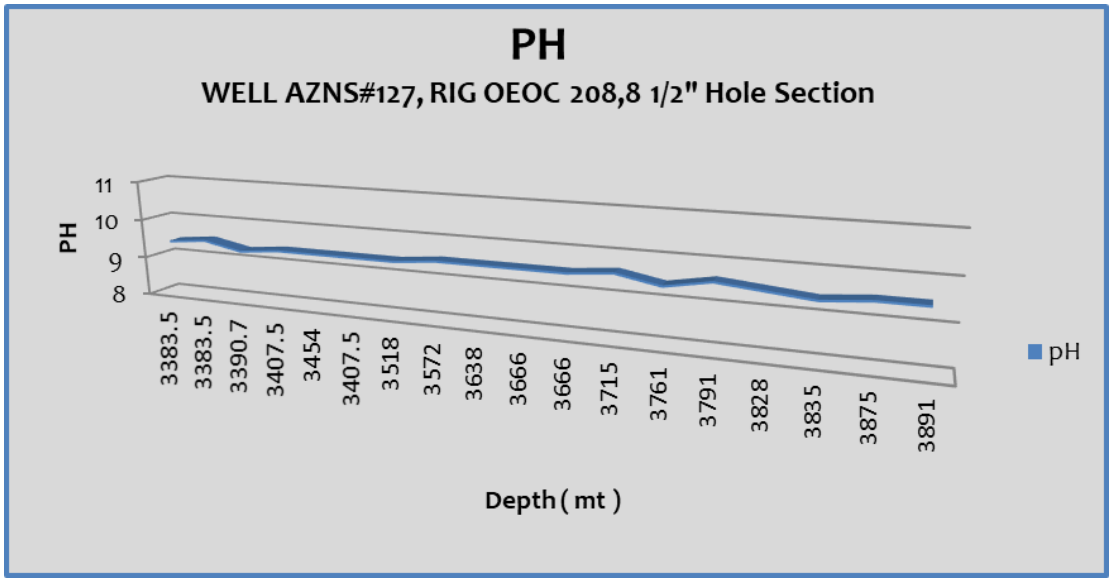


Figure 10-15: AZNS-127 8-1/2" PH Graph

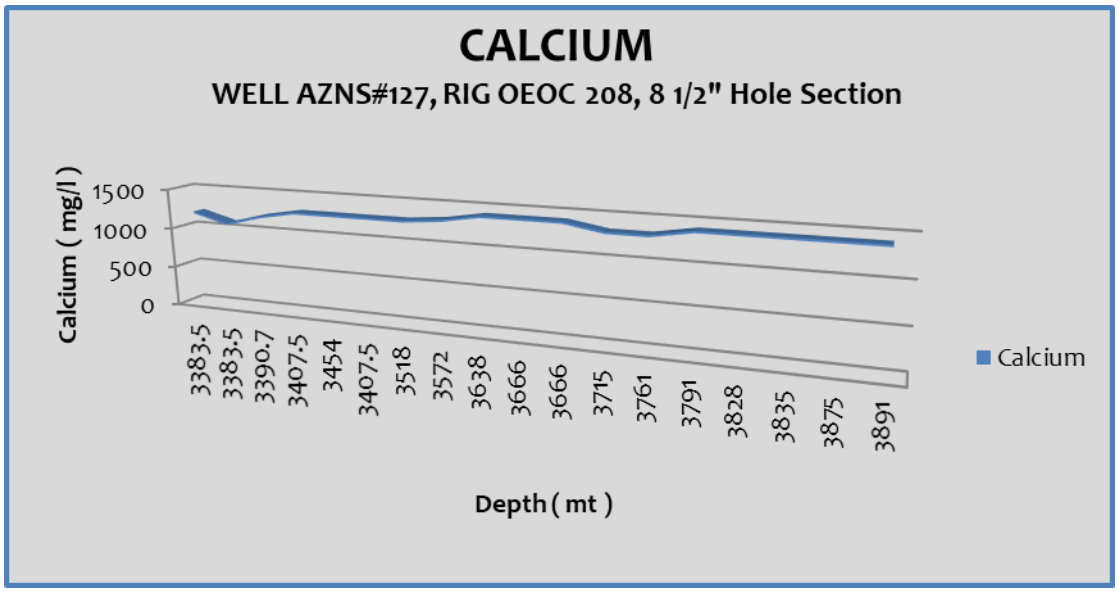


Figure 10-16: AZNS-127 8-1/2" Calcium Graph

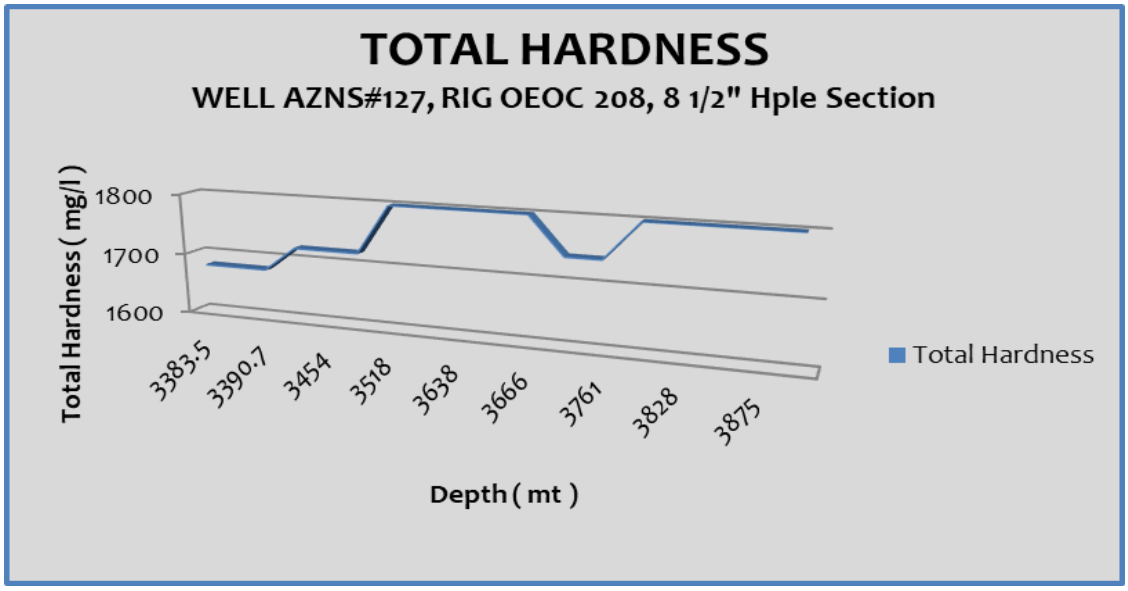


Figure 10-17: AZNS-127 8-1/2" Total Hardness Graph

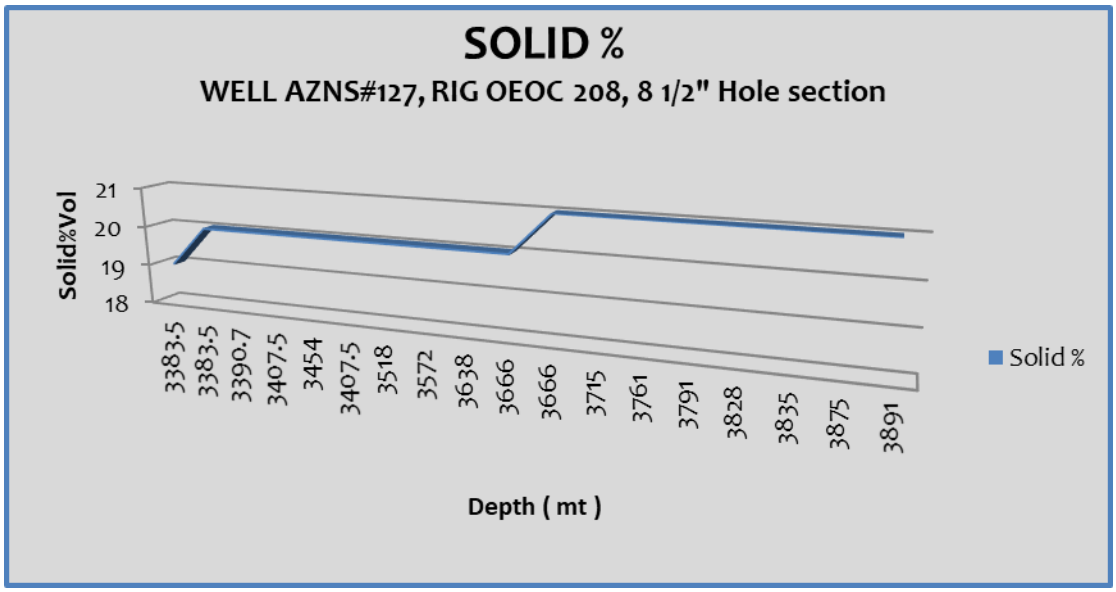


Figure 10-18: AZNS-127 8-1/2" Solid% Graph

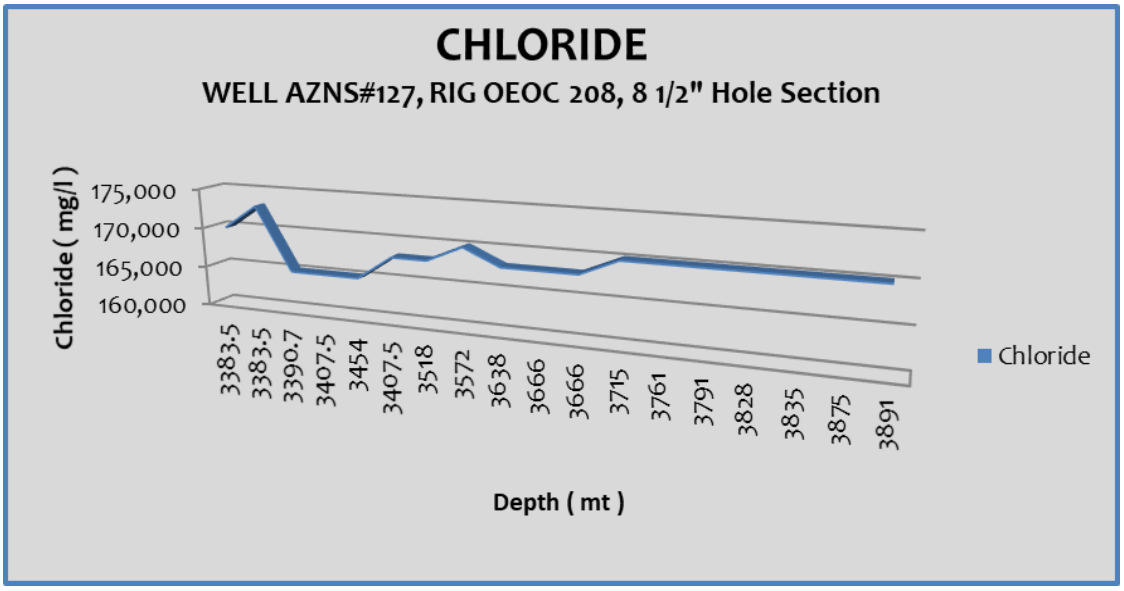


Figure 10-19: AZNS-127 8-1/2" Chloride Graph

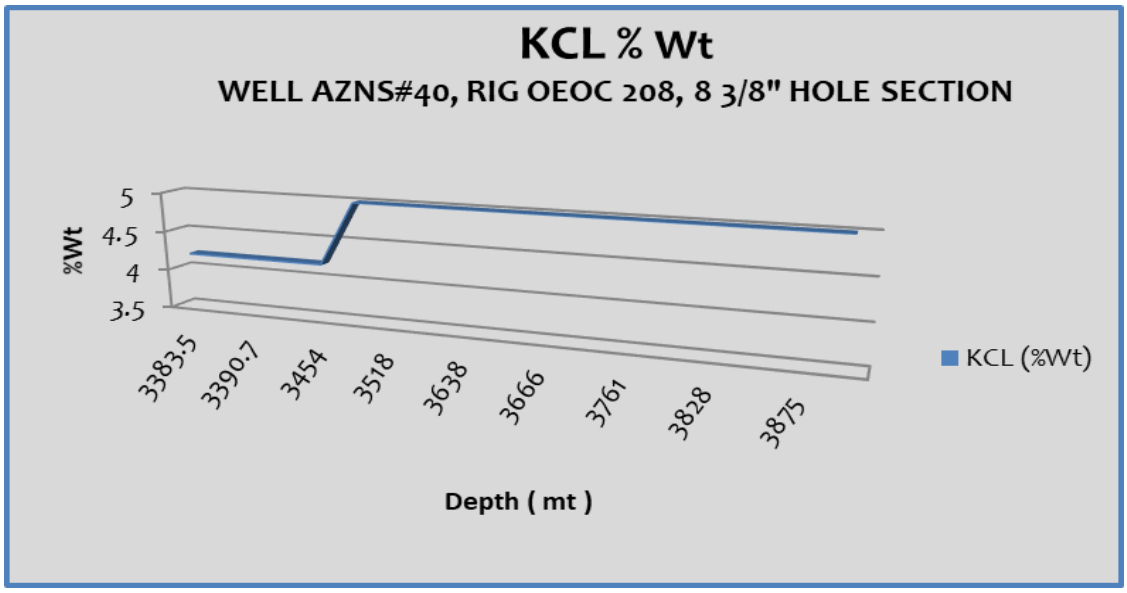


Figure 10-20: AZNS-127 8-1/2" KCl % Wt Graph

10.16 Solid Control Disscusion

Rig OEOC 208 has 3 shale shakers were dressed with:

Shaker#1: 170 Mesh Screen

Shaker#2: 200 Mesh Screen

Shaker#3: 170 Mesh Screen

WELL AZN#85, RIG OEOC 201	
Solid Control Equipment Working Hr.s	
Shaker # 01	176
Shaker # 02	394
Shaker # 03	470

Table 10-15: AZNS-127 8-1/2" Solid Control Desscusion

10.17 6 – 1/8” Hole Section Drilling Fluid Types

FLUID PROPERTIES RECAP					
KCL Pol Mud		Drilling start	Range		Drilling End
MW (min)	pcf	100	100	104	104
Funnel Viscosity	sec/qt	47	47	56	56
PV	cps	31	31	42	42
YP	lb/100sq.ft	18	15	18	16
API Fluid loss	cc	3	2	4	2.4
pH		9.4	9.4	9.7	9.7
KCL	%Wt	5	5	5.0	5
Cl (NaCl)	mg/L	180,000	180,000	186,000	180,000
Total Hardness	mg/L	1,400	1,680	1,480	1,360
Ca Hardness	mg/L	2,080	920	2,080	1,920

Table 10-16: AZNS-127 6-1/8” Fluid Properties Recap Table

10.18 6-1/8” Interval Material Consumption Analysis

Products	Units Used	Unit wt lbs	Total wt	Effective Concentration
Citric Acid	30	55	1,650 lbs	0.92 ppb
Caustic Soda	13	55	715 lbs	0.92 ppb
Hematite	41	3,300	133,980 lbs	42.57 gpb
Lime	28	55	1,540 lbs	0.49 ppb
Limestone	152	2,200	334,400 lbs	106.26 ppb
Salt (NaCl)	18	2,200	39,600 lbs	50.98 ppb
Starch LV	121	55	6,655 lbs	8.57 ppb
Soda ASH	23	55	1,265 lbs	0.40 ppb
Sodium Bicarbonate	23	55	1,265 lbs	0.70 ppb
KCL	1	2,200	2,200 lbs	2.83 ppb
Starch HT	139	55	7,645 lbs	2.43 ppb
Xanthan Gum	28	55	1,540 lbs	0.86 ppb

Initial Volume	2,370.7 bbl
Total Volume Built	776.8 bbl
Volume received	0.0 bbl
Volume Returned	0.0 bbl
Total Volume Consumed	1,617 bbl
Volume Transfer to Next Section	1,530.5 bbl
Meter Drilled	247 m

Table 10-17: AZNS-127 6-1/8” Interval Material Consumption Analysis

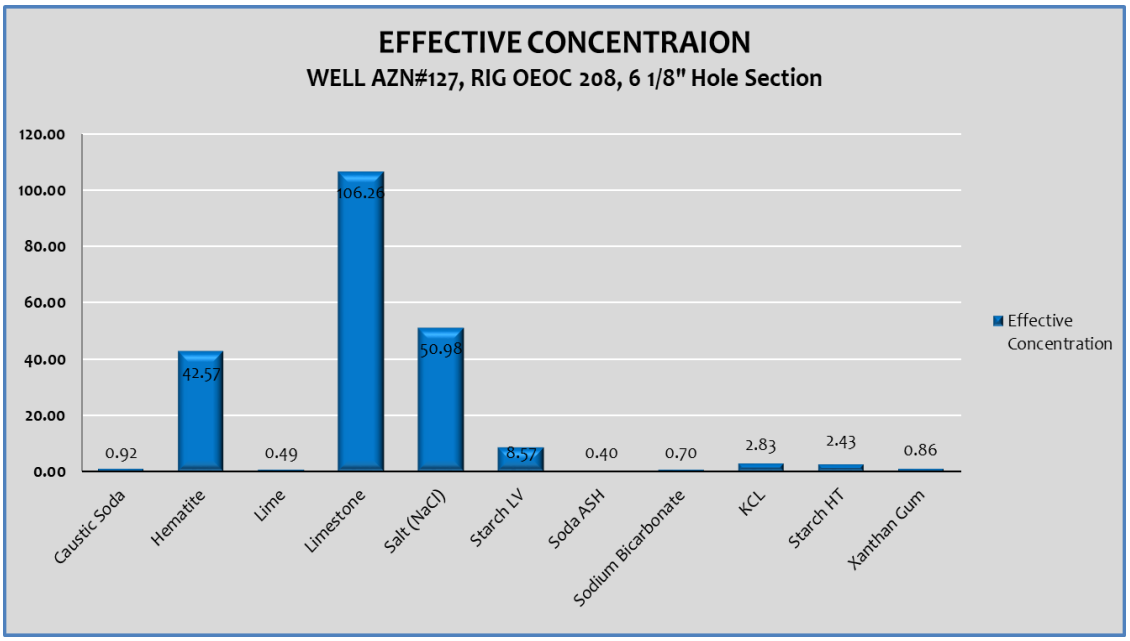


Figure 10-21: AZNS-127 6-1/8" Effective Concentration

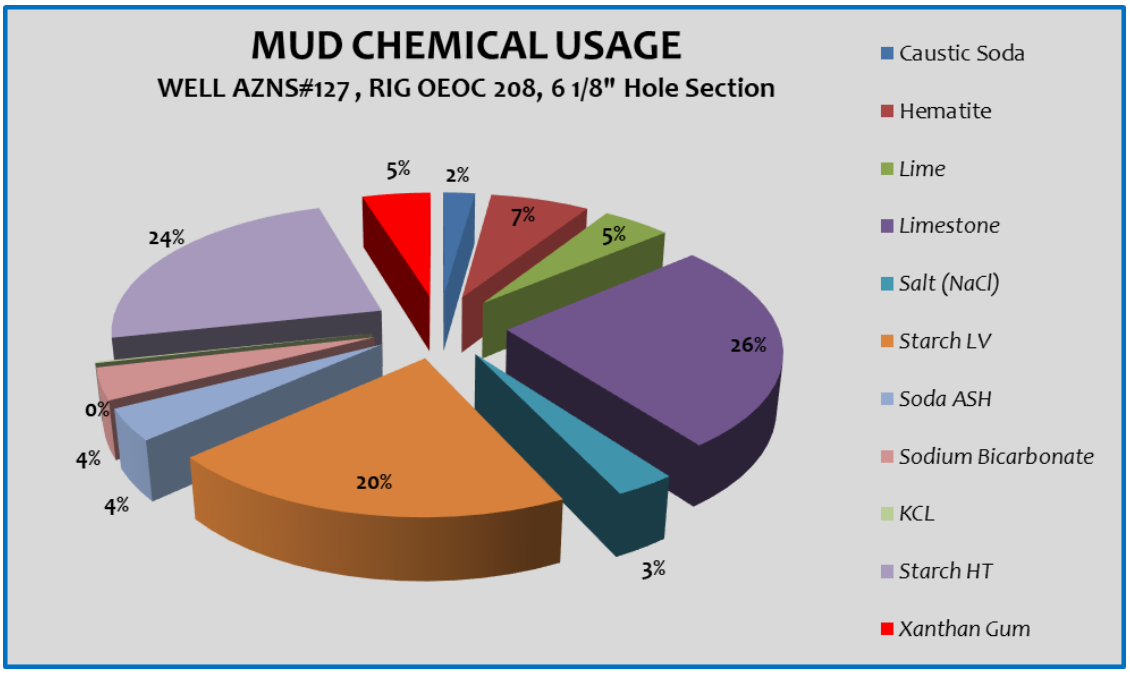


Figure 10-22: AZNS-127 6-1/8" Mud Chemical Usage

10.19 6-1/8” Interval Volume Analysis

WELL AZN#127, RIG OEOC 208 VOLUME ANALYSIS		
Initial Volume	bbls	2,370.7
Received Volume	bbls	0.0
Volume Return	bbls	0.0
Volume Built	bbls	776.8
Surface Losses/Dumped	bbls	1,617.0
Formation Losses	bbls	0.0
Total Losses	bbls	1,617.0
Volume Transfer Out	bbls	1,530.5

Table 10-18: AZNS-127 6-1/8” Interval Volume Analysis

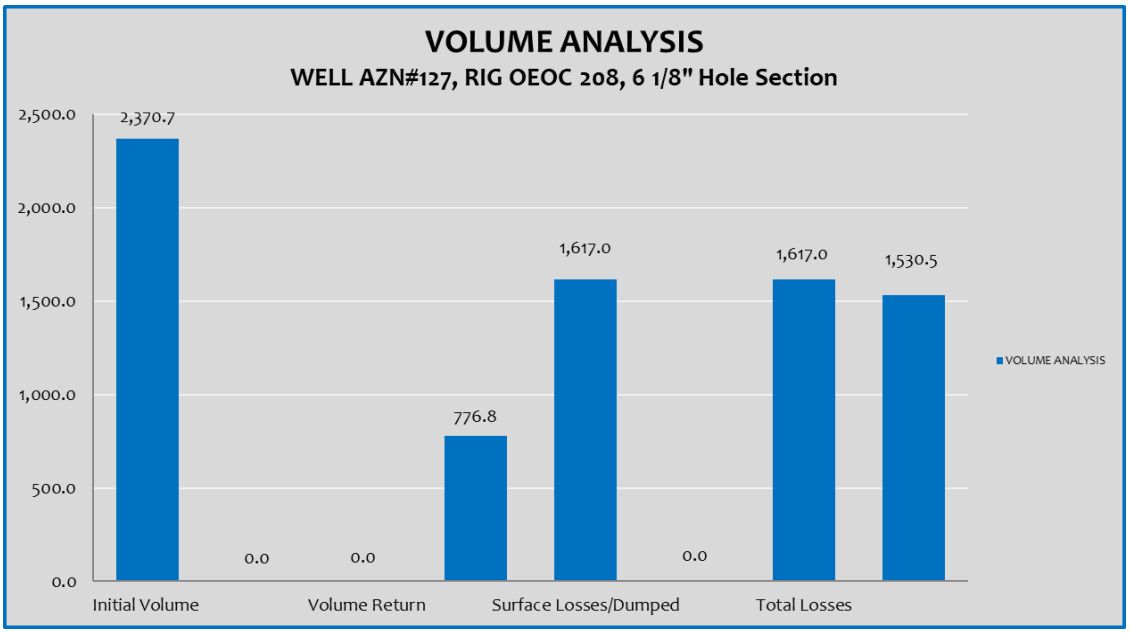


Figure 10-23: AZNS-127 6-1/8” Interval Volume Analysis

10.20 6-1/8” Interval Daily Volume Breakdown

Date/DMR.N		Volume of Built	Surface Losses	Formation Losses
2-Aug-24	87	0.9	15	0
3-Aug-24	88	1.5	58	0
4-Aug-24	89	0.3	34	0
5-Aug-24	90	0.0	22	0
6-Aug-24	91	0.0	118	0
7-Aug-24	92	9.4	48	0
8-Aug-24	93	39.0	32	0
9-Aug-24	94	0.3	35	0
10-Aug-24	95	60.4	37	0
11-Aug-24	96	46.7	29	0
12-Aug-24	97	34.1	26	0
13-Aug-24	98	179.5	36	0
14-Aug-24	99	36.2	30	0
15-Aug-24	100	16.9	30	0
16-Aug-24	101	0.1	96	0
17-Aug-24	102	0.0	224	0
18-Aug-24	103	0.0	42	0
19-Aug-24	104	0.8	32	0
20-Aug-24	105	0.0	5	0
21-Aug-24	106	0.0	62	0
22-Aug-24	107	0.8	8	0
23-Aug-24	108	0.2	12	0
24-Aug-24	109	0.0	7	0
25-Aug-24	110	137.4	38	0
26-Aug-24	111	0.7	12	0
27-Aug-24	112	9.3	37	0
28-Aug-24	113	0.0	67	0
29-Aug-24	114	188.5	13	0
30-Aug-24	115	12.8	32	0
31-Aug-24	116	0.0	21	0
1-Sep-24	117	0.0	86	0
2-Sep-24	118	0.7	6	0
3-Sep-24	119	0.1	52	0
4-Sep-24	120	0.0	89	0
5-Sep-24	121	0.0	6	0
6-Sep-24	122	0.0	8	0
7-Sep-24	123	0.0	6	0
8-Sep-24	124	0.0	8	0
9-Sep-24	125	0.2	5	0
10-Sep-24	126	0.0	5	0
11-Sep-24	127	0.0	44	0
12-Sep-24	128	0.0	21	0
13-Sep-24	129	0.0	23	0
Total (bbl)		Volume of Built	Surface Losses	Formation Losses
		776.8	1,617	0

Table 10-19: AZNS-127 6-1/8” Interval Daily Volume Breakdown

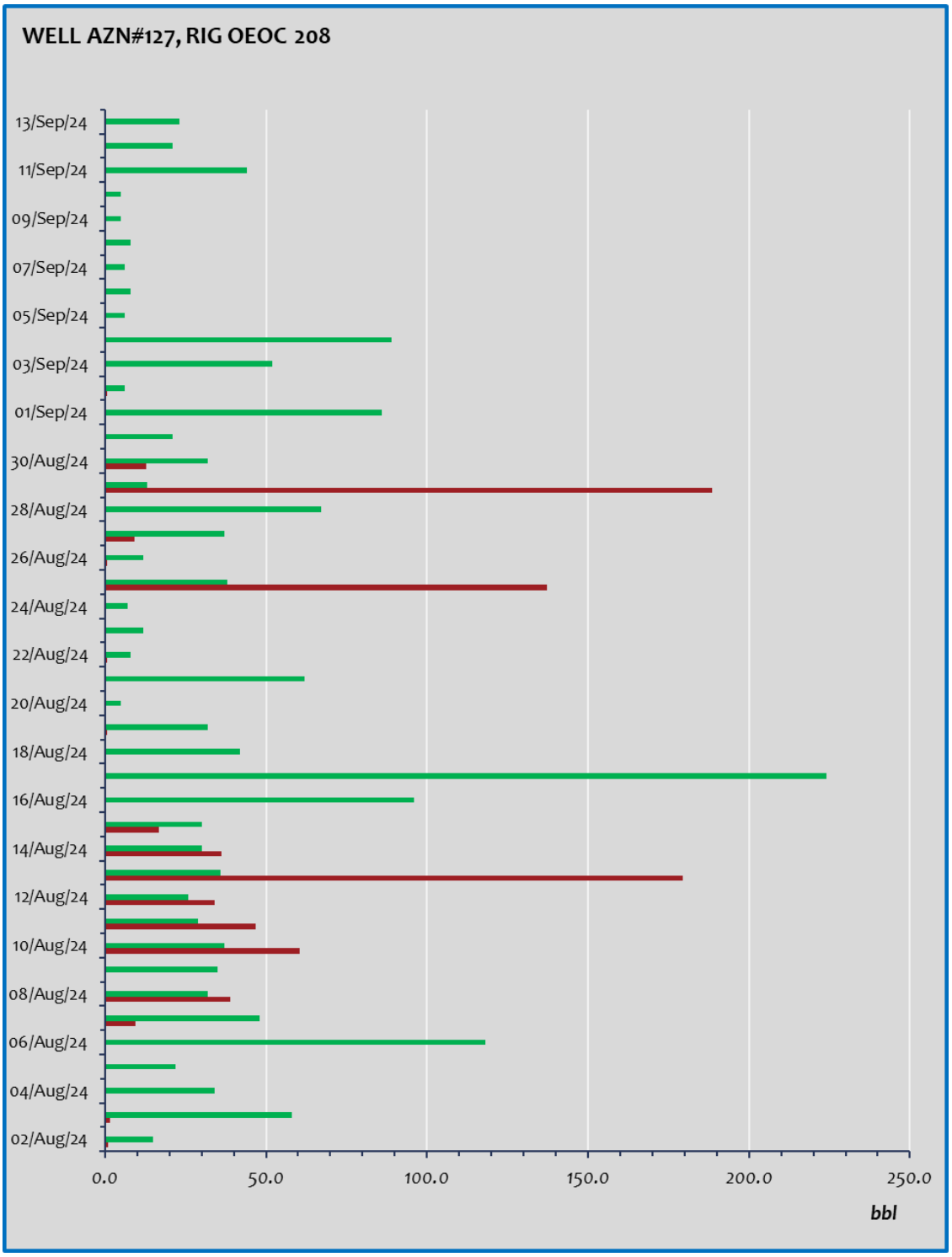


Figure 10-24: AZNS-127 6-1/8" Interval Daily Volume Breakdown

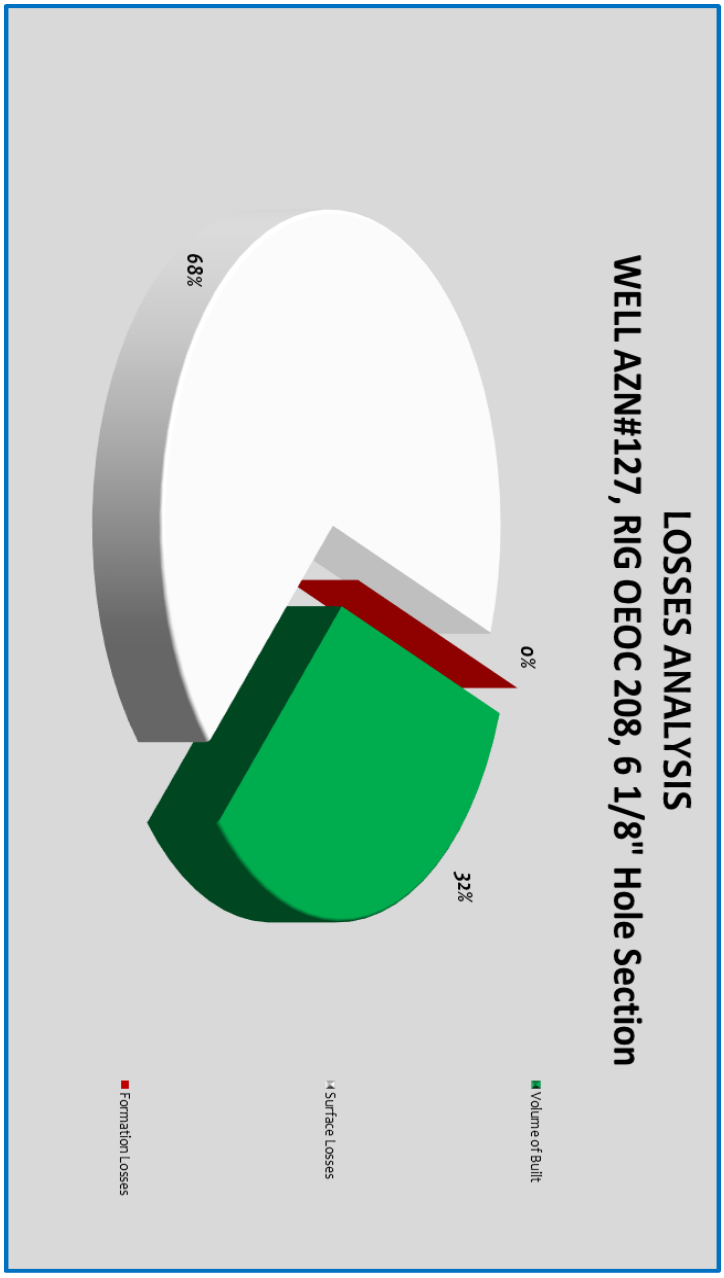


Figure 10-25: AZNS-127 6-1/8" Losses Analysis

10.21 6-1/8" Interval Loss Distribution

Date/DMR.N		Volume of Built	Dump/Contamination	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
2-Aug-24	87	0.9	0	0	15	0	0	0	0	0	0
3-Aug-24	88	1.5	0	10	15	0	6	0	12	0	15
4-Aug-24	89	0.3	0	0	18	0	4	0	0	0	12
5-Aug-24	90	0.0	0	0	18	0	4	0	0	0	0
6-Aug-24	91	0.0	100	0	18	0	0	0	0	0	0
7-Aug-24	92	9.4	20	0	20	0	4	0	0	0	4
8-Aug-24	93	39.0	0	5	15	0	0	0	12	0	0
9-Aug-24	94	0.3	0	15	12	0	0	0	8	0	0
10-Aug-24	95	60.4	0	0	12	0	0	0	10	0	15
11-Aug-24	96	46.7	0	16	13	0	0	0	0	0	0
12-Aug-24	97	34.1	0	16	10	0	0	0	0	0	0
13-Aug-24	98	179.5	0	15	10	3	0	0	0	0	8
14-Aug-24	99	36.2	0	18	12	0	0	0	0	0	0
15-Aug-24	100	16.9	0	12	10	0	0	0	0	0	8
16-Aug-24	101	0.1	0	0	18	0	38	0	25	0	15
17-Aug-24	102	0.0	212	0	12	0	0	0	0	0	0
18-Aug-24	103	0.0	30	0	7	0	5	0	0	0	0
19-Aug-24	104	0.8	0	0	7	0	25	0	0	0	0
20-Aug-24	105	0.0	0	0	5	0	0	0	0	0	0
21-Aug-24	106	0.0	40	0	7	0	0	0	15	0	0
22-Aug-24	107	0.8	0	0	8	0	0	0	0	0	0
23-Aug-24	108	0.2	0	0	7	0	5	0	0	0	0
24-Aug-24	109	0.0	0	0	7	0	0	0	0	0	0
25-Aug-24	110	137.4	0	0	8	0	0	0	30	0	0
26-Aug-24	111	0.7	0	0	7	0	0	0	0	0	5
27-Aug-24	112	9.3	0	0	7	0	25	0	0	0	5
28-Aug-24	113	0.0	0	0	7	0	55	0	0	0	5
29-Aug-24	114	188.5	0	0	8	0	0	0	0	0	5
30-Aug-24	115	12.8	0	0	7	0	15	0	0	0	10
31-Aug-24	116	0.0	0	0	6	0	15	0	0	0	0
1-Sep-24	117	0.0	45	0	6	0	0	0	35	0	0
2-Sep-24	118	0.7	0	0	6	0	0	0	0	0	0
3-Sep-24	119	0.1	30	0	10	0	12	0	0	0	0
4-Sep-24	120	0.0	80	0	9	0	0	0	0	0	0
5-Sep-24	121	0.0	0	0	6	0	0	0	0	0	0
6-Sep-24	122	0.0	0	0	8	0	0	0	0	0	0
7-Sep-24	123	0.0	0	0	6	0	0	0	0	0	0
8-Sep-24	124	0.0	0	0	8	0	0	0	0	0	0
9-Sep-24	125	0.2	0	0	5	0	0	0	0	0	0
10-Sep-24	126	0.0	0	0	5	0	0	0	0	0	0
11-Sep-24	127	0.0	0	0	8	0	12	0	16	0	8
12-Sep-24	128	0.0	0	0	6	0	15	0	0	0	0
13-Sep-24	129	0.0	0	0	8	0	0	0	0	0	15
Total (bbl)		Volume of Built	Dump/Contaminate	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
		776.8	557	107	417	3	240	0	163	0	130.0

Note: Prepared 61.4 bbl Cacl2 Brine (DMR#129) that is consider in completion fluid section.

Table 10-20: AZNS-127 6-1/8" Interval Loss Distribution

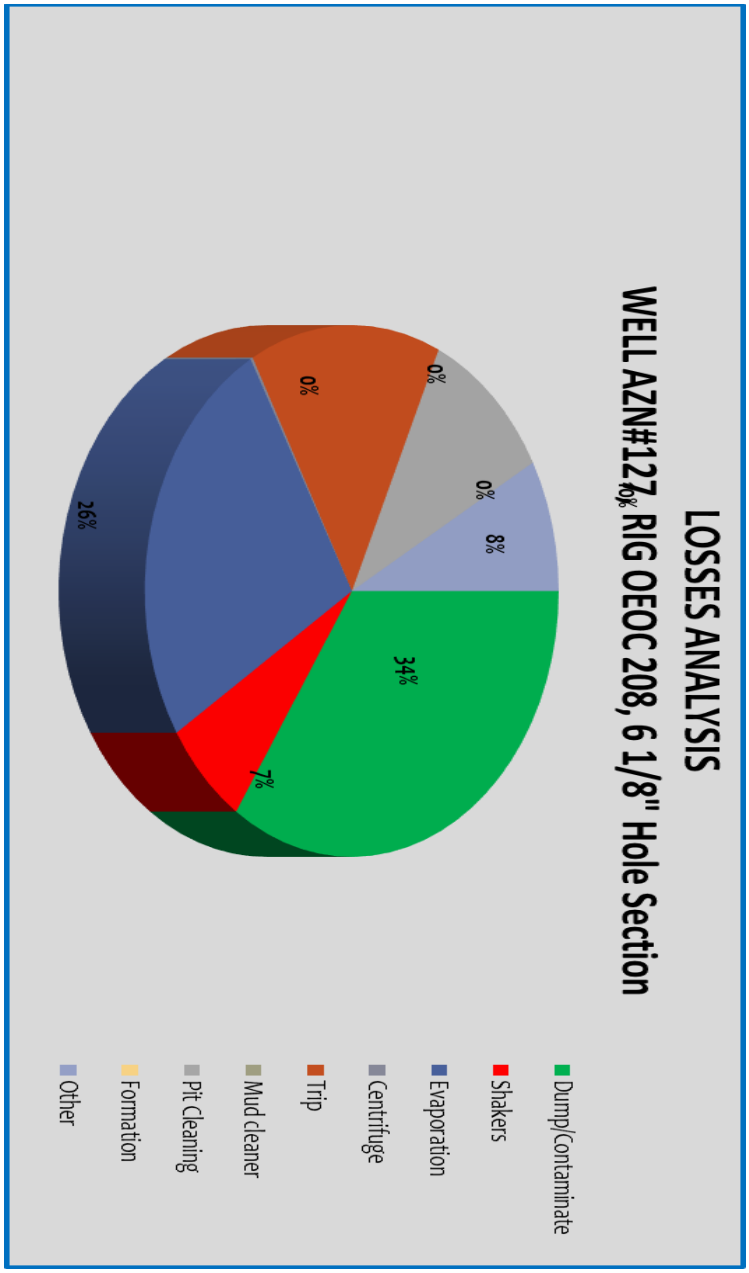


Figure 10-26: AZNS-127 6-1/8” Losses Analysis

10.22 6-1/8” Interval Fluid Properties Recap

Date	DEPTH	Density	VIS	PV	YP	Water Loss	pH	Chloride	KCL (%Wt)	Calcium	Total Hardness	Mud Type
2-Aug-24	3891	88	40	18	10	6	10.7	170000	5	1960	2000	Salt/ Poly Mud
3-Aug-24	3891	88	40	18	11	7	10.7	175000	5	2000	3000	Salt/ Poly Mud
4-Aug-24	3891	88	40	17	10	7	10.6	175000	5	2000	3000	Salt/ Poly Mud
5-Aug-24	3891	88	40	18	11	6	10.6	175000	5	2000	3000	KCL Poly Mud
6-Aug-24	3891	88	40	18	11	6	10.5	175000	5	2000	3000	KCL Poly Mud
7-Aug-24	3891	88	40	18	11	6	10.6	175000	5	2000	3000	KCL Poly Mud
8-Aug-24	3891	88	41	17	11	6	10.8	175000	5	2080	3040	KCL Poly Mud
9-Aug-24	3893	89	42	19	11	6	10.3	180,000	5	1920	2640	KCL Poly Mud
10-Aug-24	3893	100	47	31	18	3	9.4	180,000	5	1400	2080	KCL Poly Mud
11-Aug-24	3938	102	53	37	16	4	9.6	180,000	5	1440	1960	KCL Poly Mud
12-Aug-24	4003	104	55	47	15	3.2	9.6	183,000	5	1440	1960	KCL Poly Mud
13-Aug-24	4035	104	53	38	15	2	9.5	183,000	5	1480	2040	KCL Poly Mud
14-Aug-24	4115	104	55	39	16	2.2	9.7	186,000	5	1400	1960	KCL Poly Mud
15-Aug-24	4140	104	56	42	16	2.4	9.7	180,000	5	1360	1920	KCL Poly Mud
16-Aug-24	4140	104	53	34	17	2.4	9.8	180,000	5	1440	2000	KCL Poly Mud
17-Aug-24	4140	104	54	34	17	2.4	9.8	180,000	5	1440	2000	KCL Poly Mud
18-Aug-24	4140	104	54	34	17	2.4	9.8	180,000	5	1440	2000	KCL Poly Mud
19-Aug-24	4140	104	50	30	10	4	10.9	180,000	5	1800	2400	KCL Poly Mud
20-Aug-24	4140	104	50	30	10	4	10.9	180,000	5	1800	2400	KCL Poly Mud
21-Aug-24	4140	104	50	30	10	4	10.9	180,000	5	1800	2400	KCL Poly Mud
22-Aug-24	4140	104	48	30	8	7	10.9	180,000	5	1800	2400	KCL Poly Mud
23-Aug-24	4140	104	50	30	10	7	10.9	180,000	5	1800	2400	KCL Poly Mud
24-Aug-24	4140	104	50	30	10	7	10.8	180,000	5	1800	2400	KCL Poly Mud
25-Aug-24	4140	104	50	30	10	7	10.8	180,000	5	1800	2400	KCL Poly Mud
26-Aug-24	4140	104	47	38	7	7	10.8	180,000	5	1800	2400	KCL Poly Mud
27-Aug-24	4140	104	47	34	8	8	10.8	180,000	5	1800	2400	KCL Poly Mud
28-Aug-24	4140	104	47	34	8	8	10.8	180,000	5	1800	2400	KCL Poly Mud
29-Aug-24	4140	104	50	31	10	8	10.7	180,000	5	1800	2400	KCL Poly Mud
30-Aug-24	4140	104	50	31	10	8	10.7	180,000	5	1800	2400	KCL Poly Mud
31-Aug-24	4140	104	50	31	10	8	10.7	180,000	5	1800	2400	KCL Poly Mud
1-Sep-24	4140	104	50	31	10	8	10.7	180,000	5	1800	2400	KCL Poly Mud
2-Sep-24	4140	104	49	30	11	8	10.7	180,000	5	1800	2400	KCL Poly Mud
3-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
4-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
5-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
6-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
7-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
8-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
9-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
10-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
11-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
12-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
13-Sep-24	4140	104	49	29	11	6	10.5	180,000	5	1800	2400	KCL Poly Mud
Drilling Start												
Drilling End												

Table 10-21: AZNS-127 6-1/8” Interval Fluid Properties

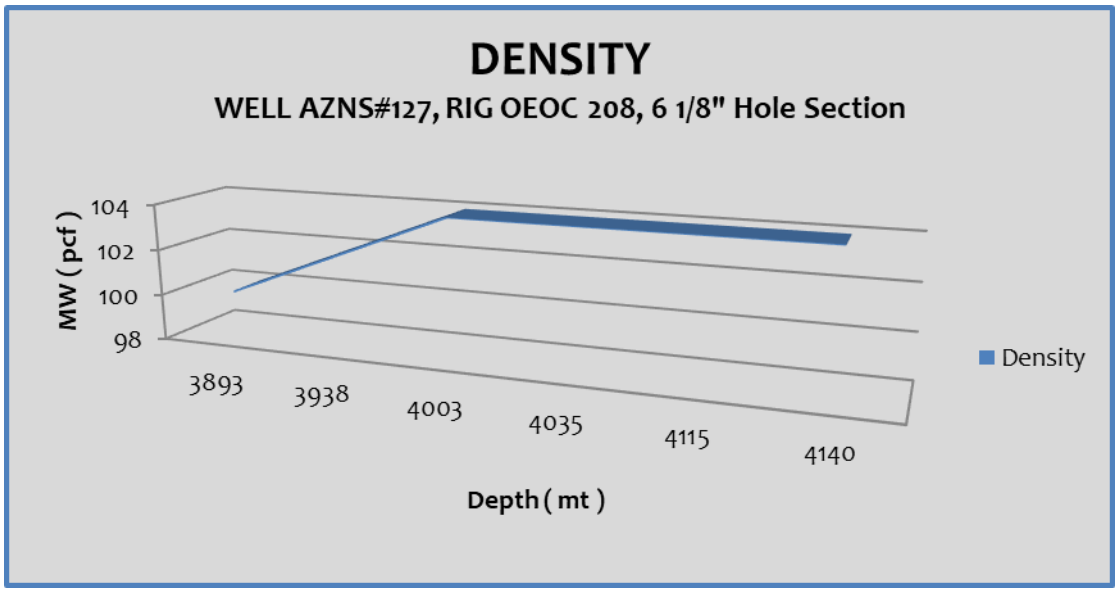


Figure 10-27: AZNS-127 6-1/8" Density Graph

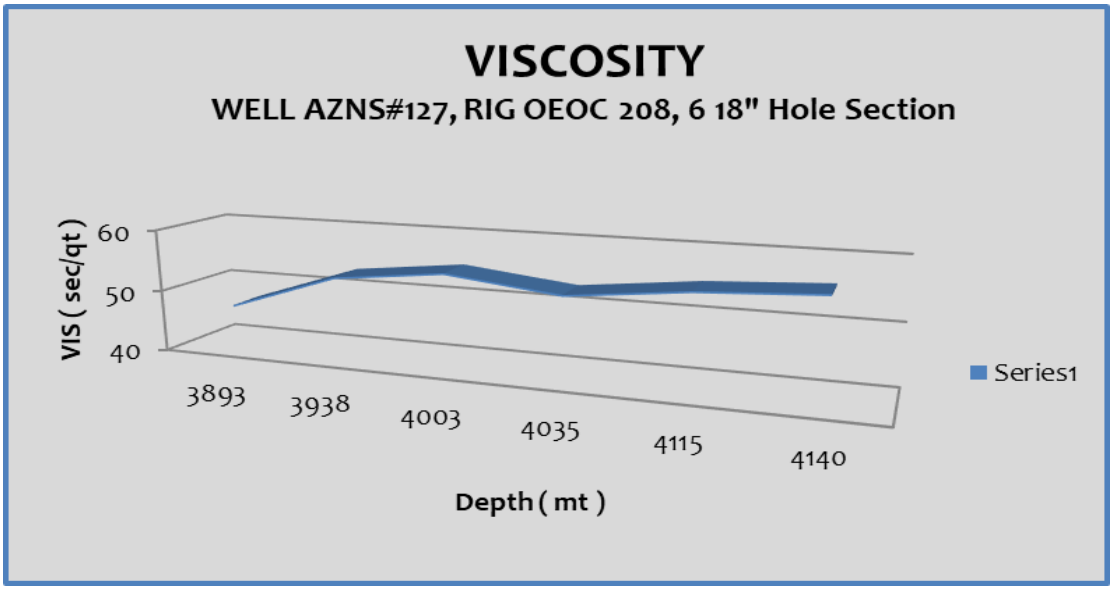


Figure 10-28: AZNS-127 6-1/8" Viscosity Graph

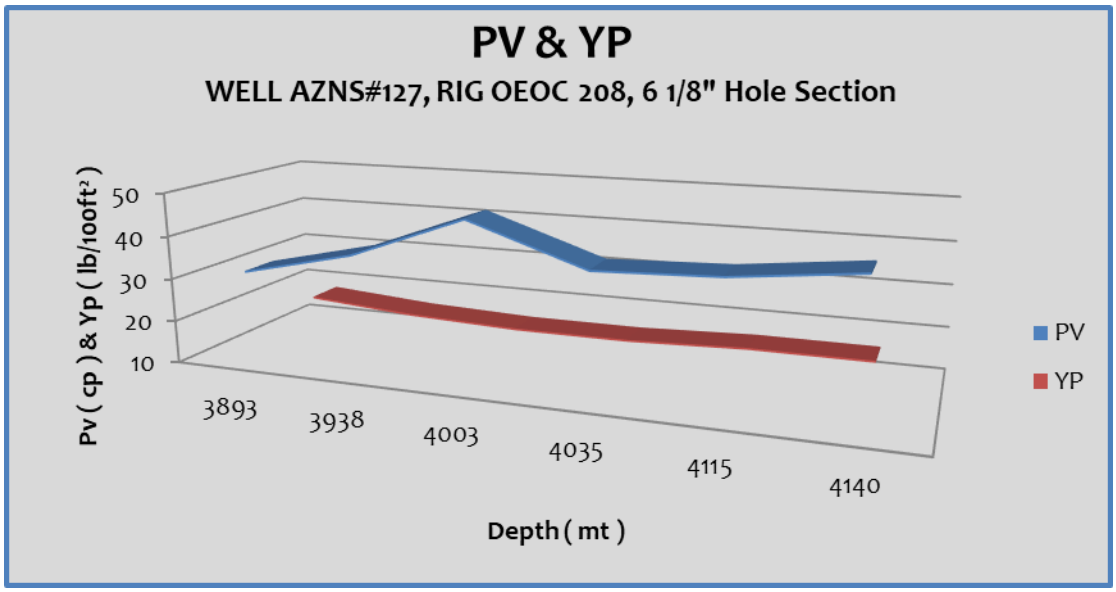


Figure 10-29: AZNS-127 6-1/8" PV & YP Graph

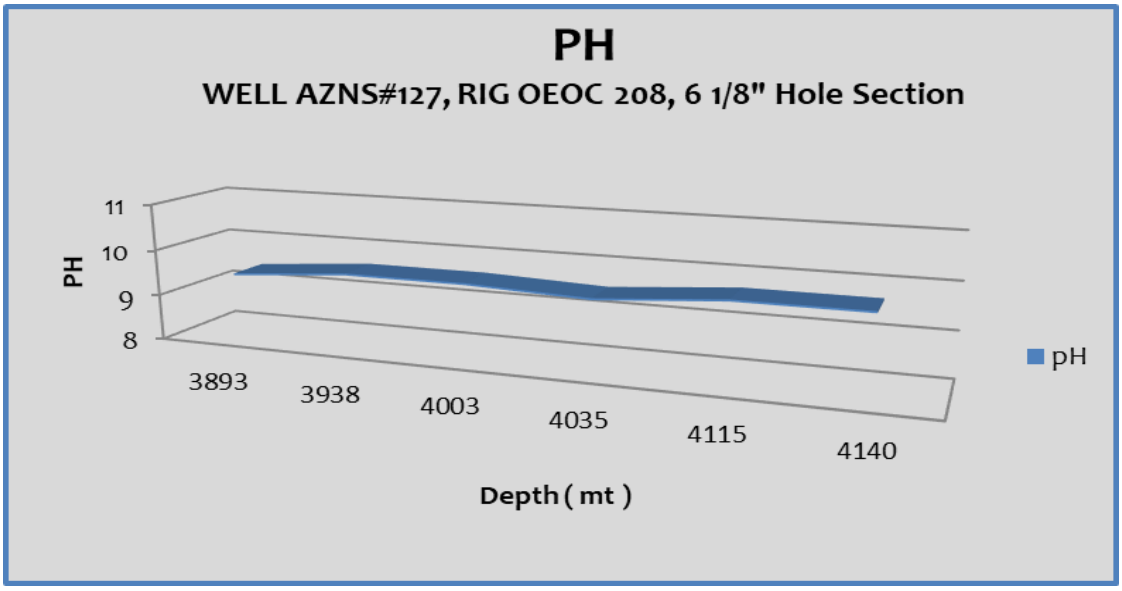


Figure 10-30: AZNS-127 6-1/8" PH Graph

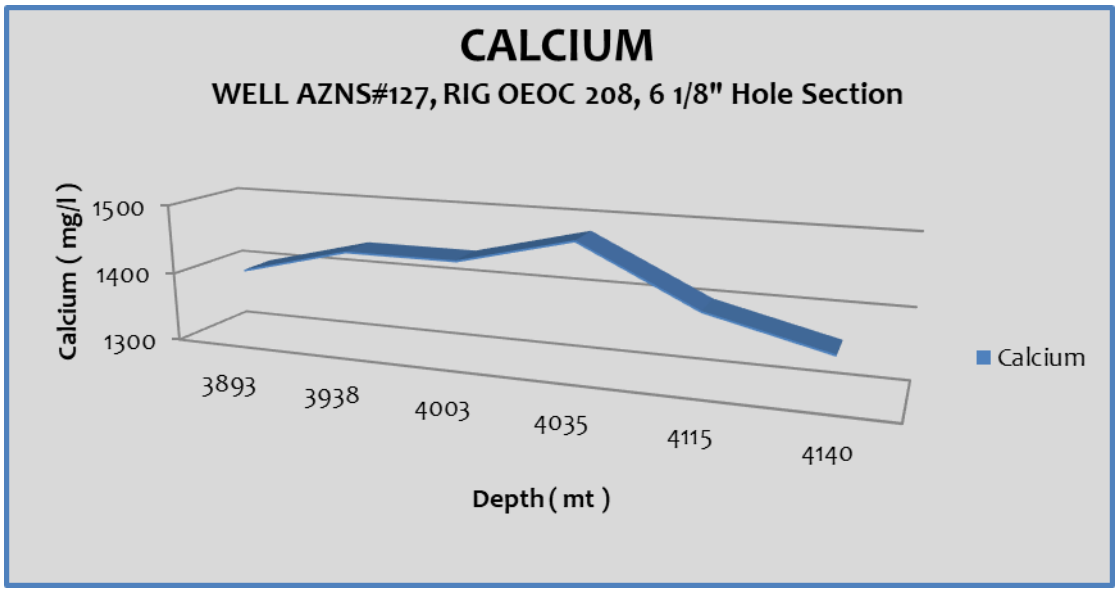


Figure 10-31: AZNS-127 6-1/8" Calcium Graph

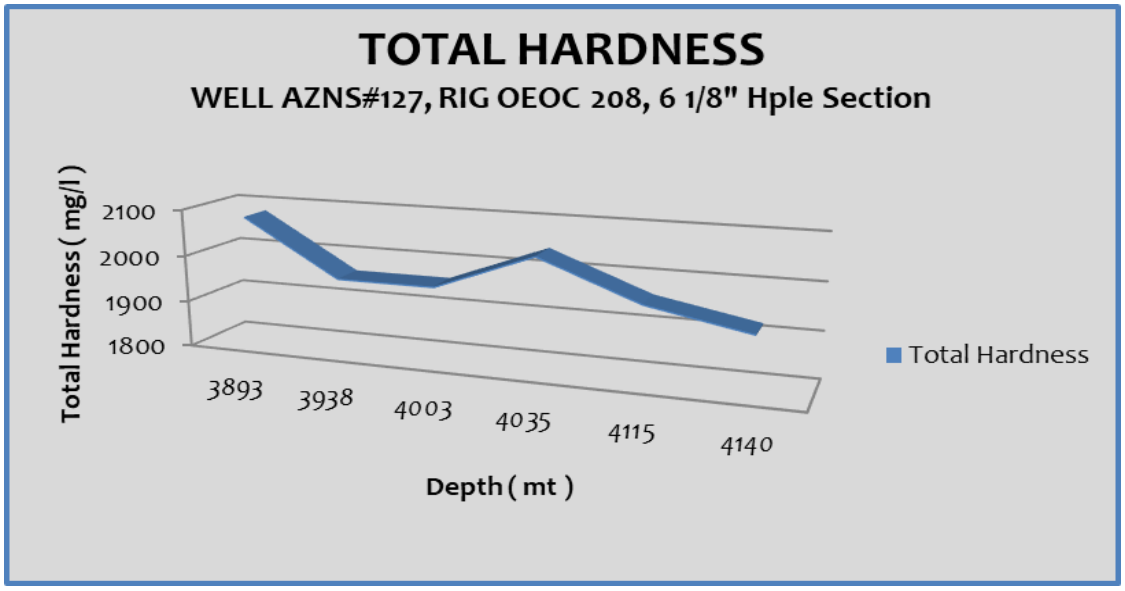


Figure 10-32: AZNS-127 6-1/8" Total Hardness Graph

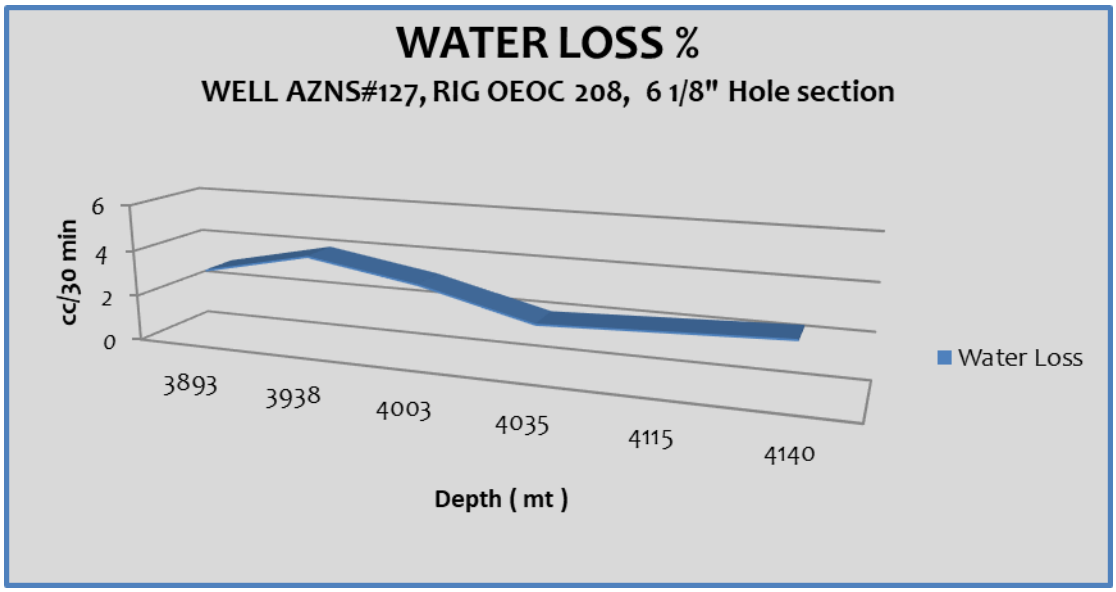


Figure 10-33: AZNS-127 6-1/8" Water Loss% Graph

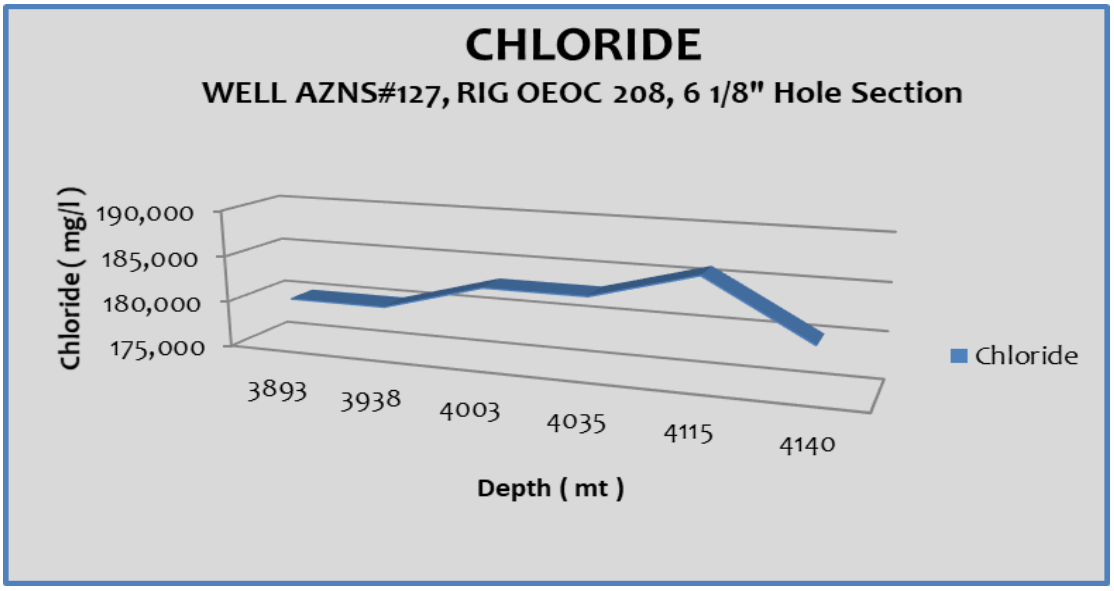


Figure 10-34: AZNS-127 6-1/8" Chloride Graph

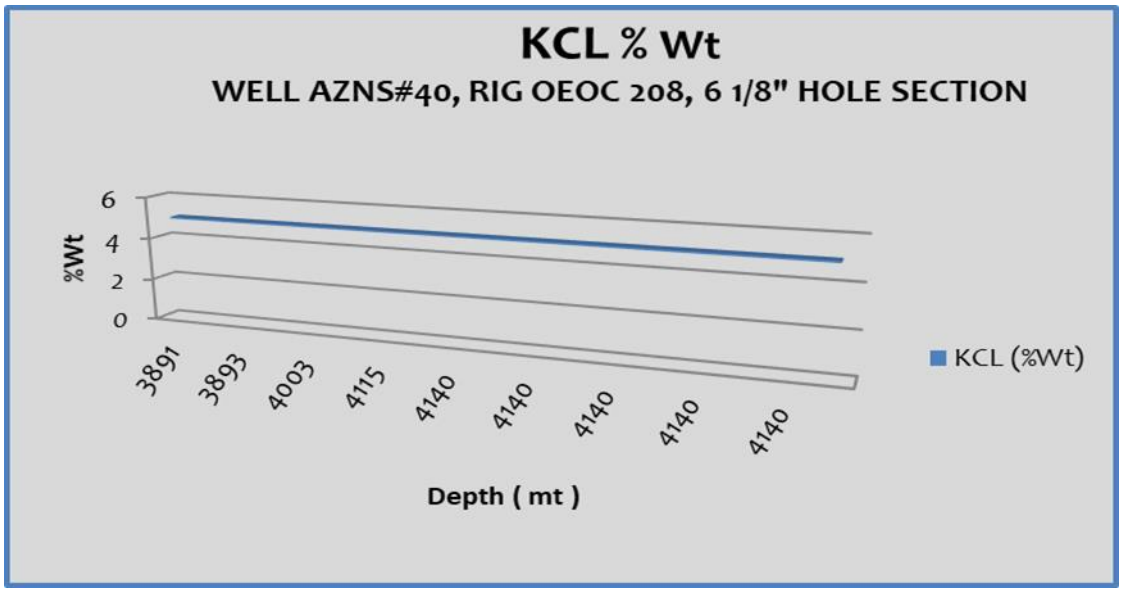


Figure 10-35: AZNS-127 6-1/8" KCl % Wt Graph

10.23 6-1/8” Interval Solid Control Disscusion

Rig OEOC 208 has 3 shale shakers were dressed with:

Shaker#1: 170 – 200 Mesh Screen

Shaker#2: 170 – 200 Mesh Screen

Shaker#3: 170 – 200 Mesh Screen

WELL AZN#85, RIG OEOC 201	
Solid Control Equipment Working Hr.s	
Shaker # 01	38
Shaker # 02	235
Shaker # 03	268

Table 10-22: AZNS-127 6-1/8” Solid Control Desscusion

10.24 Interval Completion Section

Completion						
Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
978 bbls Synthetic Completion fluid has been made						
Synthetic Comp. Fluid	978	BBL	40391.4	gals	41.30	gpb
$CaCl_2$	7	1 MT	1540	lbs	175.57	ppb
Corrision Inhibitor		55 Gal		gals	0.00	gpb
O_2 Scavenger		55 Gal		gals	0.00	gpb
H_2S Scavenger		55 Gal		gals	0.00	gpb
Cacl2 Brine			61.4			
MW :			104.5			*pcf*
Initial Volume :			0.0			*bbl*
Total Volume Built (Completion Fluid) :			978.0			*bbl*
Note:Prepared 60 bbls Hi-Vis Cacl2 Brine pumping 20 bbls as spacer & 40 bbls dead Vol. in the pit.						

Table 10-23: AZNS-127 Completion Section

11. Waste Management Report

11.1 Well & Hole Section

Country / Area	**Iran, Khuzestan.**
Client	MSA
Field	Azadegan
Well	AZNS-127
Location	PAD-54 (Area: II)
Well head co-ordinate [UTM39N, WGS84]	E 196,167m
	N 3,448,561 m
Type	IIF Vertical Well
Objective	De-complete the well, Re-complete the well, Due to abnormal annulus , find leakage point and cure it, Check and smooth collapse interval from 3889-3911.

Hole Section	**De-Completion**
Rig	208 OEOC
Start date	May. 24th 2024
Ended date	July. 13th 2024
Stardet depth (m)	4208 m
Ended depth (m)	3390 m
Duration	51 days

11.2 De-Completion Operation's Recap

Fixation									
Report No.	Date	Depth	Drilled	Total Cuttings		Fixed Cuttings		Sent to Disposal	
				Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
1	24-May-24	4,208	0	20	2,043	0	0	0	0
2	25-May-24	4,208	0	0	2,043	0	0	0	0
3	26-May-24	4,208	0	0	2,043	0	0	0	0
4	27-May-24	4,208	0	168	2,211	0	0	0	0
5	28-May-24	4,208	0	30	2,241	0	0	0	0
6	29-May-24	4,208	0	18	2,259	0	0	0	0
7	30-May-24	4,208	0	18	2,277	0	0	0	0
8	31-May-24	4,208	0	26	2,303	0	0	0	0
9	1-Jun-24	4,208	0	0	2,303	0	0	0	0
10	2-Jun-24	4,208	0	170	2,473	0	0	0	0
11	3-Jun-24	4,208	0	0	2,473	700	700	0	0
12	4-Jun-24	4,208	0	65	2,538	200	900	0	0
13	5-Jun-24	4,208	0	25	2,563	0	900	0	0
14	6-Jun-24	4,208	0	125	2,688	0	900	0	0
15	7-Jun-24	4,208	0	40	2,728	265	1,165	0	0
16	8-Jun-24	4,208	0	68	2,796	0	1,165	0	0
17	9-Jun-24	4,208	0	30	2,826	0	1,165	0	0
18	10-Jun-24	4,208	0	24	2,850	0	1,165	0	0
19	11-Jun-24	4,208	0	24	2,874	0	1,165	0	0
20	12-Jun-24	4,208	0	34	2,908	0	1,165	0	0
21	13-Jun-24	4,208	0	48	2,956	0	1,165	0	0
22	14-Jun-24	4,208	0	0	2,956	0	1,165	0	0
23	15-Jun-24	4,208	0	25	2,981	0	1,165	0	0
24	16-Jun-24	4,208	0	55	3,036	0	1,165	0	0
25	17-Jun-24	4,208	0	13	3,049	0	1,165	0	0
26	18-Jun-24	4,208	0	25	3,074	0	1,165	0	0
27	19-Jun-24	4,208	0	39	3,113	0	1,165	0	0
28	20-Jun-24	4,208	0	12	3,125	0	1,165	0	0
29	21-Jun-24	4,208	0	11	3,136	0	1,165	0	0
30	22-Jun-24	4,208	0	23	3,159	0	1,165	0	0
31	23-Jun-24	4,208	0	0	3,159	0	1,165	0	0
32	24-Jun-24	4,208	0	25	3,184	0	1,165	0	0
33	25-Jun-24	4,208	0	5	3,189	0	1,165	0	0
34	26-Jun-24	4,208	0	20	3,209	0	1,165	0	0
35	27-Jun-24	4,208	0	13	3,222	520	1,685	0	0
36	28-Jun-24	4,208	0	18	3,240	0	1,685	350	350
37	29-Jun-24	4,208	0	15	3,255	0	1,685	300	650
38	30-Jun-24	4,208	0	20	3,275	0	1,685	0	650

39	1-Jul-24	4,208	0	407	3,682	70	1,755	600	1,250
40	2-Jul-24	4,208	0	0	3,682	0	1,755	450	1,700
41	3-Jul-24	4,208	0	25	3,707	0	1,755	0	1,700
42	4-Jul-24	4,208	0	174	3,881	0	1,755	0	1,700
43	5-Jul-24	4,208	0	30	3,911	0	1,755	0	1,700
44	6-Jul-24	4,208	0	250	4,161	0	1,755	0	1,700
45	7-Jul-24	4,208	0	36	4,197	0	1,755	0	1,700
46	8-Jul-24	4,208	0	0	4,197	515	2,270	0	1,700
47	9-Jul-24	4,208	0	482	4,679	0	2,270	0	1,700
48	10-Jul-24	4,208	0	160	4,839	1,046	3,316	0	1,700
49	11-Jul-24	4,208	0	5	4,844	0	3,316	0	1,700
50	12-Jul-24	4,208	0	6	4,850	0	3,316	0	1,700
51	13-Jul-24	3,390	0	51	4,901	0	3,316	0	1,700

Table 11-1: AZNS-127 Fixation Table

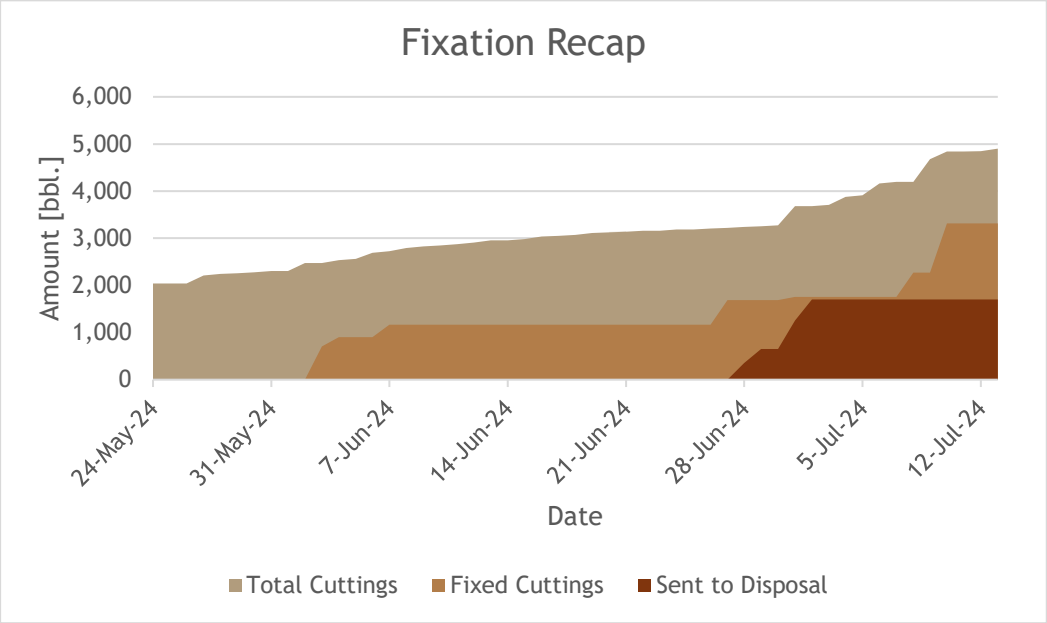


Figure 11-1: AZNS-127 Fixation Graph

Fixation analysis								
Report No.	Date	Depth	Drilled	Can test	Cement/Waste		SodSil/Waste	
				Daily	Daily	Average	Daily	Average
				Pass/NA	kg/M3	kg/M3	kg/M3	kg/M3
1	24-May-24	4,208	0	Pass	0.000	0.000	0.000	0.000
2	25-May-24	4,208	0	NA	0.000	0.000	0.000	0.000
3	26-May-24	4,208	0	NA	0.000	0.000	0.000	0.000
4	27-May-24	4,208	0	Pass	0.000	0.000	0.000	0.000
5	28-May-24	4,208	0	Pass	0.000	0.000	0.000	0.000
6	29-May-24	4,208	0	Pass	0.000	0.000	0.000	0.000
7	30-May-24	4,208	0	Pass	0.000	0.000	0.000	0.000
8	31-May-24	4,208	0	Pass	0.000	0.000	0.000	0.000
9	1-Jun-24	4,208	0	NA	0.000	0.000	0.000	0.000
10	2-Jun-24	4,208	0	Pass	0.000	0.000	0.000	0.000
11	3-Jun-24	4,208	0	NA	2.143	2.143	0.000	0.000
12	4-Jun-24	4,208	0	Pass	5.000	3.571	0.308	0.308
13	5-Jun-24	4,208	0	Pass	0.000	3.571	0.000	0.308
14	6-Jun-24	4,208	0	Pass	0.000	3.571	0.000	0.308
15	7-Jun-24	4,208	0	Pass	3.774	3.639	0.500	0.404
16	8-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
17	9-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
18	10-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
19	11-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
20	12-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
21	13-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
22	14-Jun-24	4,208	0	NA	0.000	3.639	0.000	0.404
23	15-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
24	16-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
25	17-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
26	18-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
27	19-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
28	20-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
29	21-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
30	22-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
31	23-Jun-24	4,208	0	NA	0.000	3.639	0.000	0.404
32	24-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
33	25-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
34	26-Jun-24	4,208	0	Pass	0.000	3.639	0.000	0.404
35	27-Jun-24	4,208	0	Pass	0.962	2.969	2.308	1.038
36	28-Jun-24	4,208	0	Pass	0.000	2.969	0.000	1.038
37	29-Jun-24	4,208	0	Pass	0.000	2.969	0.000	1.038
38	30-Jun-24	4,208	0	Pass	0.000	2.969	0.000	1.038
39	1-Jul-24	4,208	0	Pass	7.143	3.804	0.025	0.785
40	2-Jul-24	4,208	0	NA	0.000	3.804	0.000	0.785
41	3-Jul-24	4,208	0	Pass	0.000	3.804	0.000	0.785
42	4-Jul-24	4,208	0	Pass	0.000	3.804	0.000	0.785
43	5-Jul-24	4,208	0	Pass	0.000	3.804	0.000	0.785
44	6-Jul-24	4,208	0	Pass	0.000	3.804	0.000	0.785
45	7-Jul-24	4,208	0	Pass	0.000	3.804	0.000	0.785
46	8-Jul-24	4,208	0	NA	8.738	4.626	0.000	0.785
47	9-Jul-24	4,208	0	Pass	0.000	4.626	0.000	0.785
48	10-Jul-24	4,208	0	Pass	3.824	4.512	0.500	0.728
49	11-Jul-24	4,208	0	Pass	0.000	4.512	0.000	0.728
50	12-Jul-24	4,208	0	Pass	0.000	4.512	0.000	0.728
51	13-Jul-24	3,390	0	Pass	0.000	4.512	0.000	0.728

Table 11-2: AZNS-127 Fixation Analysis

Dewatering

Report No.	Date	Depth	Drilled	Total waste-water		Total Dewater		Recycled		Disposed and Lost	
				Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
1	24-May-24	4,208	0	0	2,277	0	0	0	0	0	0
2	25-May-24	4,208	0	350	2,627	0	0	0	0	0	0
3	26-May-24	4,208	0	350	2,977	350	350	0	0	0	0
4	27-May-24	4,208	0	200	3,177	0	350	0	0	0	0
5	28-May-24	4,208	0	0	3,177	0	350	0	0	0	0
6	29-May-24	4,208	0	0	3,177	0	350	0	0	0	0
7	30-May-24	4,208	0	0	3,177	0	350	0	0	0	0
8	31-May-24	4,208	0	0	3,177	0	350	0	0	0	0
9	1-Jun-24	4,208	0	0	3,177	0	350	0	0	0	0
10	2-Jun-24	4,208	0	0	3,177	0	350	0	0	0	0
11	3-Jun-24	4,208	0	0	3,177	400	750	300	300	50	50
12	4-Jun-24	4,208	0	0	3,177	0	750	100	400	0	50
13	5-Jun-24	4,208	0	0	3,177	400	1,150	280	680	20	70
14	6-Jun-24	4,208	0	0	3,177	0	1,150	0	680	0	70
15	7-Jun-24	4,208	0	0	3,177	0	1,150	370	1,050	30	100
16	8-Jun-24	4,208	0	0	3,177	400	1,550	0	1,050	0	100
17	9-Jun-24	4,208	0	0	3,177	0	1,550	0	1,050	0	100
18	10-Jun-24	4,208	0	0	3,177	0	1,550	0	1,050	0	100
19	11-Jun-24	4,208	0	0	3,177	200	1,750	200	1,250	0	100
20	12-Jun-24	4,208	0	50	3,227	200	1,950	200	1,450	0	100
21	13-Jun-24	4,208	0	0	3,227	0	1,950	0	1,450	0	100
22	14-Jun-24	4,208	0	0	3,227	400	2,350	200	1,650	0	100
23	15-Jun-24	4,208	0	0	3,227	0	2,350	200	1,850	0	100
24	16-Jun-24	4,208	0	0	3,227	200	2,550	200	2,050	0	100
25	17-Jun-24	4,208	0	0	3,227	400	2,950	200	2,250	0	100
26	18-Jun-24	4,208	0	0	3,227	400	3,350	350	2,600	50	150
27	19-Jun-24	4,208	0	0	3,227	400	3,750	350	2,950	50	200
28	20-Jun-24	4,208	0	0	3,227	400	4,150	350	3,300	50	250
29	21-Jun-24	4,208	0	0	3,227	400	4,550	350	3,650	50	300
30	22-Jun-24	4,208	0	0	3,227	0	4,550	350	4,000	50	350
31	23-Jun-24	4,208	0	0	3,227	0	4,550	350	4,350	50	400
32	24-Jun-24	4,208	0	0	3,227	400	4,950	200	4,550	50	450
33	25-Jun-24	4,208	0	0	3,227	0	4,950	400	4,950	0	450
34	26-Jun-24	4,208	0	300	3,527	400	5,350	0	4,950	0	450
35	27-Jun-24	4,208	0	200	3,727	0	5,350	150	5,100	0	450
36	28-Jun-24	4,208	0	300	4,027	0	5,350	0	5,100	0	450
37	29-Jun-24	4,208	0	100	4,127	0	5,350	300	5,400	50	500
38	30-Jun-24	4,208	0	400	4,527	0	5,350	0	5,400	0	500
39	1-Jul-24	4,208	0	0	4,527	200	5,550	200	5,600	0	500
40	2-Jul-24	4,208	0	100	4,627	0	5,550	250	5,850	0	500
41	3-Jul-24	4,208	0	0	4,627	400	5,950	300	6,150	50	550
42	4-Jul-24	4,208	0	400	5,027	0	5,950	370	6,520	30	580
43	5-Jul-24	4,208	0	400	5,427	0	5,950	0	6,520	0	580
44	6-Jul-24	4,208	0	400	9,720	0	7,500	100	6,920	0	580
45	7-Jul-24	4,208	0	0	9,720	0	7,500	0	6,920	0	580
46	8-Jul-24	4,208	0	50	9,770	0	7,500	0	6,920	0	580

47	9-Jul-24	4,208	0	50	9,820	0	7,500	150	7,070	0	580
48	10-Jul-24	4,208	0	200	10,020	0	7,500	130	7,200	30	610
49	11-Jul-24	4,208	0	30	10,050	390	7,890	0	7,200	0	610
50	12-Jul-24	4,208	0	0	10,050	0	7,890	380	7,580	10	620
51	13-Jul-24	3,390	0	0	10,050	400	8,290	0	7,580	0	620

Table 11-3: AZNS-127 De-watering recap

Dewatering Analysis											
Report No.	Date	Depth	Drilled	pH	Hardn.	Turb.	TSS	Nitrate	Al	Sulfate	Ca
				7-9	<800	<50	< 80	< 50	< 3	< 50	< 600
-	dd-M-yy	m	m	-	mg/L	NTU	mg/L	mg/L	mg/L	mg/L	mg/L
1	24-May-24	4,208	0	0	0	0	0	0	0	0	0
2	25-May-24	4,208	0	0	0	0	0	0	0	0	0
3	26-May-24	4,208	0	7	800	7	28	40	1	30	560
4	27-May-24	4,208	0	0	0	0	0	0	0	0	0
5	28-May-24	4,208	0	0	0	0	0	0	0	0	0
6	29-May-24	4,208	0	0	0	0	0	0	0	0	0
7	30-May-24	4,208	0	0	0	0	0	0	0	0	0
8	31-May-24	4,208	0	0	0	0	0	0	0	0	0
9	1-Jun-24	4,208	0	0	0	0	0	0	0	0	0
10	2-Jun-24	4,208	0	0	0	0	0	0	0	0	0
11	3-Jun-24	4,208	0	7	800	7	28	40	1	30	560
12	4-Jun-24	4,208	0	7	800	7	28	40	1	30	560
13	5-Jun-24	4,208	0	7	800	7	28	40	1	30	560
14	6-Jun-24	4,208	0	7	800	7	28	40	1	30	560
15	7-Jun-24	4,208	0	7	800	7	28	40	1	30	560
16	8-Jun-24	4,208	0	8	800	7	28	40	1	30	560
17	9-Jun-24	4,208	0	8	800	7	28	40	1	30	560
18	10-Jun-24	4,208	0	0	0	0	0	0	0	0	0
19	11-Jun-24	4,208	0	0	0	0	0	0	0	0	0
20	12-Jun-24	4,208	0	0	0	0	0	0	0	0	0
21	13-Jun-24	4,208	0	0	0	0	0	0	0	0	0
22	14-Jun-24	4,208	0	0	0	0	0	0	0	0	0
23	15-Jun-24	4,208	0	0	0	0	0	0	0	0	0
24	16-Jun-24	4,208	0	0	0	0	0	0	0	0	0
25	17-Jun-24	4,208	0	0	0	0	0	0	0	0	0
26	18-Jun-24	4,208	0	0	0	0	0	0	0	0	0
27	19-Jun-24	4,208	0	0	0	0	0	0	0	0	0
28	20-Jun-24	4,208	0	0	0	0	0	0	0	0	0
29	21-Jun-24	4,208	0	0	0	0	0	0	0	0	0
30	22-Jun-24	4,208	0	0	0	0	0	0	0	0	0
31	23-Jun-24	4,208	0	0	0	0	0	0	0	0	0
32	24-Jun-24	4,208	0	0	0	0	0	0	0	0	0
33	25-Jun-24	4,208	0	0	0	0	0	0	0	0	0
34	26-Jun-24	4,208	0	8	730	42	40	30	1	30	500
35	27-Jun-24	4,208	0	0	0	0	0	0	0	0	0
36	28-Jun-24	4,208	0	8	730	38	40	30	1	30	500
37	29-Jun-24	4,208	0	8	730	38	40	30	1	30	500
38	30-Jun-24	4,208	0	8	730	38	40	30	1	30	500
39	1-Jul-24	4,208	0	8	730	38	40	30	1	30	500
40	2-Jul-24	4,208	0	8	730	38	40	30	1	30	500

41	3-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
42	4-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
43	5-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
44	6-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
45	7-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
46	8-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
47	9-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
48	10-Jul-24	**4,208**	**0**	8	730	38	40	30	1	30	500
49	11-Jul-24	**4,208**	**0**	8	800	40	40	30	1	30	500
50	12-Jul-24	**4,208**	**0**	8	800	40	40	30	1	30	500
51	13-Jul-24	**3,390**	**0**	0	0	0	0	0	0	0	0

Table 11-4: AZNS-127 De-watering Analysis

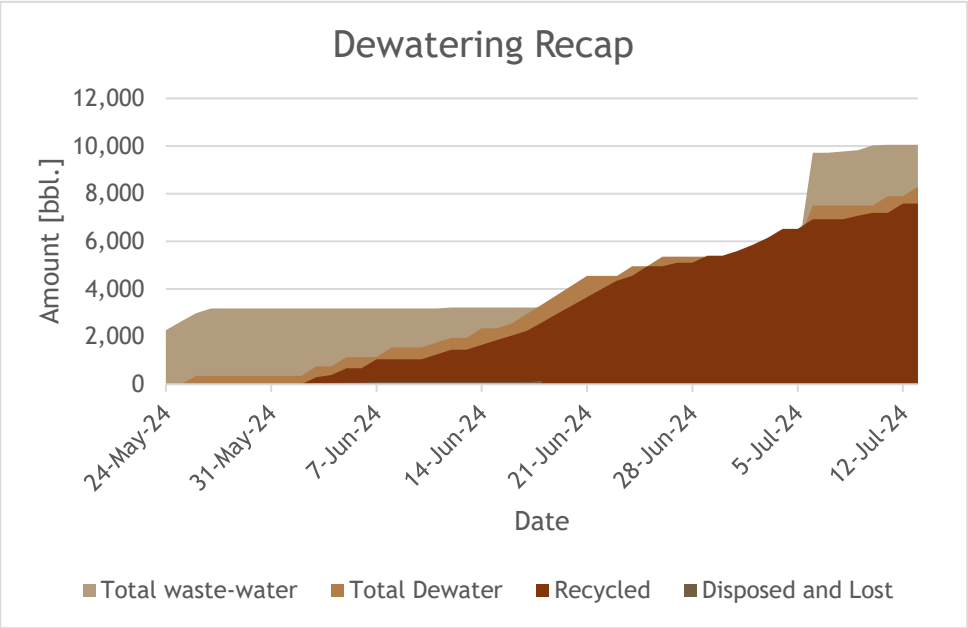


Figure 11-2: AZNS-127 De-watering Graph

Chemical																	
No.	Date	Cement		Na_2SiO_3		Coagulant		Polymer		Lime		Soda ash		Citric acid		Caustic soda	
		Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	Kg		Lit.		Kg		Kg		Kg		Kg		Kg		Kg	
1	24-May-24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	25-May-24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	26-May-24	0	0	0	0	0	0	0	0	50	50	50	50	0	0	0	0
4	27-May-24	0	0	0	0	0	0	0	0	0	50	0	50	0	0	0	0
5	28-May-24	0	0	0	0	0	0	0	0	0	50	0	50	0	0	0	0
6	29-May-24	0	0	0	0	0	0	0	0	0	50	0	50	0	0	0	0
7	30-May-24	0	0	0	0	0	0	0	0	0	50	0	50	0	0	0	0
8	31-May-24	0	0	0	0	0	0	0	0	0	50	0	50	0	0	0	0
9	1-Jun-24	0	0	0	0	0	0	0	0	0	50	0	50	0	0	0	0
10	2-Jun-24	0	0	0	0	0	0	0	0	0	50	0	50	0	0	0	0
11	3-Jun-24	1,500	1,500	30	30	0	0	0	0	50	100	50	100	0	0	0	0
12	4-Jun-24	1,000	2,500	20	50	0	0	0	0	0	100	0	100	0	0	0	0
13	5-Jun-24	0	2,500	0	50	0	0	0	0	50	150	50	150	0	0	0	0
14	6-Jun-24	0	2,500	0	50	0	0	0	0	0	150	0	150	0	0	0	0
15	7-Jun-24	1,000	3,500	20	70	0	0	0	0	0	150	0	150	0	0	0	0
16	8-Jun-24	0	3,500	0	70	0	0	0	0	50	200	50	200	0	0	0	0
17	9-Jun-24	0	3,500	0	70	0	0	0	0	0	200	0	200	0	0	0	0
18	10-Jun-24	0	3,500	0	70	0	0	0	0	0	200	0	200	0	0	0	0
19	11-Jun-24	0	3,500	0	70	0	0	0	0	50	250	50	250	0	0	0	0
20	12-Jun-24	0	3,500	0	70	0	0	0	0	50	300	25	275	0	0	0	0
21	13-Jun-24	0	3,500	0	70	0	0	0	0	50	350	25	300	0	0	0	0
22	14-Jun-24	0	3,500	0	70	0	0	0	0	50	400	50	350	0	0	0	0
23	15-Jun-24	0	3,500	0	70	0	0	0	0	0	400	0	350	0	0	0	0
24	16-Jun-24	0	3,500	0	70	0	0	0	0	50	450	25	375	0	0	0	0
25	17-Jun-24	0	3,500	0	70	0	0	0	0	50	500	50	425	0	0	0	0
26	18-Jun-24	0	3,500	0	70	0	0	0	0	0	500	0	425	0	0	0	0
27	19-Jun-24	0	3,500	0	70	0	0	0	0	0	500	0	425	0	0	0	0
28	20-Jun-24	0	3,500	0	70	0	0	0	0	50	550	25	450	0	0	0	0
29	21-Jun-24	0	3,500	0	70	0	0	0	0	50	600	25	475	0	0	0	0
30	22-Jun-24	0	3,500	0	70	0	0	0	0	50	650	25	500	0	0	0	0
31	23-Jun-24	0	3,500	0	70	0	0	0	0	0	650	0	500	0	0	0	0
32	24-Jun-24	0	3,500	0	70	0	0	0	0	50	700	50	550	0	0	0	0
33	25-Jun-24	0	3,500	0	70	0	0	0	0	0	700	0	550	0	0	0	0
34	26-Jun-24	0	3,500	0	70	0	0	0	0	50	750	50	600	0	0	0	0
35	27-Jun-24	500	4,000	30	100	0	0	0	0	0	750	0	600	0	0	0	0
36	28-Jun-24	0	4,000	0	100	0	0	0	0	25	775	25	625	0	0	0	0
37	29-Jun-24	0	4,000	0	100	0	0	0	0	50	825	50	675	0	0	0	0
38	30-Jun-24	0	4,000	0	100	0	0	0	0	0	825	0	675	0	0	0	0
39	1-Jul-24	500	4,500	10	110	0	0	0	0	25	850	25	700	0	0	0	0
40	2-Jul-24	0	4,500	0	110	0	0	0	0	0	850	0	700	0	0	0	0
41	3-Jul-24	0	4,500	0	110	0	0	0	0	50	900	50	750	0	0	0	0
42	4-Jul-24	0	4,500	0	110	0	0	0	0	0	900	0	750	0	0	0	0
43	5-Jul-24	0	4,500	0	110	0	0	0	0	50	950	50	800	0	0	0	0
44	6-Jul-24	0	4,500	0	110	0	0	0	0	0	950	0	800	0	0	0	0
45	7-Jul-24	0	4,500	0	110	0	0	0	0	0	950	0	800	0	0	0	0
46	8-Jul-24	4,500	9,000	80	190	0	0	0	0	0	950	0	800	0	0	0	0
47	9-Jul-24	0	9,000	0	190	0	0	0	0	0	950	0	800	0	0	0	0
48	10-Jul-24	4,000	13,000	80	270	0	0	0	0	0	950	0	800	0	0	0	0
49	11-Jul-24	0	13,000	0	270	0	0	0	0	50	1,000	50	850	0	0	0	0
50	12-Jul-24	0	13,000	0	270	0	0	0	0	50	1,050	50	900	0	0	0	0
51	13-Jul-24	0	13,000	0	270	0	0	0	0	50	1,100	50	950	0	0	0	0

Table 11-5: AZNS-127 Chemical

11.3 8 – 1/2” Hole Section

Hole Section	8.5”
Rig	208 OEOC
Start date	July. 14th 2024
Ended date	Aug. 11th 2024
Stardet depth (m)	3390 m
Ended depth (m)	3936 m
Duration	29 days

11.4 8 – 1/2" Operation's Recap

Fixation									
Report No.	Date	Depth	Drilled	Total Cuttings		Fixed Cuttings		Sent to Disposal	
				Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
52	14-Jul-24	**3,390**	**0**	87	4,988	0	3,316	0	1,700
53	15-Jul-24	**3,407**	**17**	43	5,031	0	3,316	0	1,700
54	16-Jul-24	**3,407**	**0**	103	5,134	0	3,316	0	1,700
55	17-Jul-24	**3,453**	**46**	33	5,167	0	3,316	0	1,700
56	18-Jul-24	**3,519**	**66**	84	5,251	0	3,316	0	1,700
57	19-Jul-24	**3,625**	**53**	86	5,337	0	3,316	250	1,950
58	20-Jul-24	**3,638**	**14**	77	5,414	0	3,316	250	2,200
59	21-Jul-24	**3,638**	**0**	142	5,556	411	3,727	250	2,450
60	22-Jul-24	**3,638**	**0**	40	5,596	0	3,727	0	2,450
61	23-Jul-24	**3,714**	**76**	70	5,666	0	3,727	250	2,700
62	24-Jul-24	**3,761**	**47**	85	5,750	0	3,727	300	3,000
63	25-Jul-24	**3,790**	**29**	83	5,833	1,500	5,227	250	3,250
64	26-Jul-24	**3,827**	**37**	127	5,960	0	5,227	200	3,450
65	27-Jul-24	**3,864**	**37**	109	6,068	0	5,227	350	3,800
66	28-Jul-24	**3,872**	**8**	52	6,120	0	5,227	300	4,100
67	29-Jul-24	**3,891**	**19**	63	6,183	0	5,227	300	4,400
68	30-Jul-24	**3,891**	**0**	40	6,223	0	5,227	0	4,400
69	31-Jul-24	**3,891**	**0**	40	6,263	0	5,227	427	4,827
70	1-Aug-24	**3,891**	**0**	110	6,373	600	5,827	300	5,127
71	2-Aug-24	**3,891**	**0**	0	6,373	0	5,827	100	5,227
72	3-Aug-24	**3,891**	**0**	43	6,416	0	5,827	0	5,227
73	4-Aug-24	**3,891**	**0**	16	6,432	0	5,827	0	5,227
74	5-Aug-24	**3,891**	**0**	4	6,436	0	5,827	0	5,227
75	6-Aug-24	**3,891**	**0**	100	6,536	0	5,827	0	5,227
76	7-Aug-24	**3,891**	**0**	15	6,551	724	6,551	0	5,227
77	8-Aug-24	**3,891**	**0**	17	6,568	0	6,551	0	5,227
78	9-Aug-24	**3,891**	**0**	23	6,591	0	6,551	0	5,227
79	10-Aug-24	**3,891**	**0**	15	6,606	0	6,551	0	5,227
80	11-Aug-24	**3,936**	**45**	26	6,633	0	6,551	24	5,251

Table 11-6: AZNS-127 Fixation Table

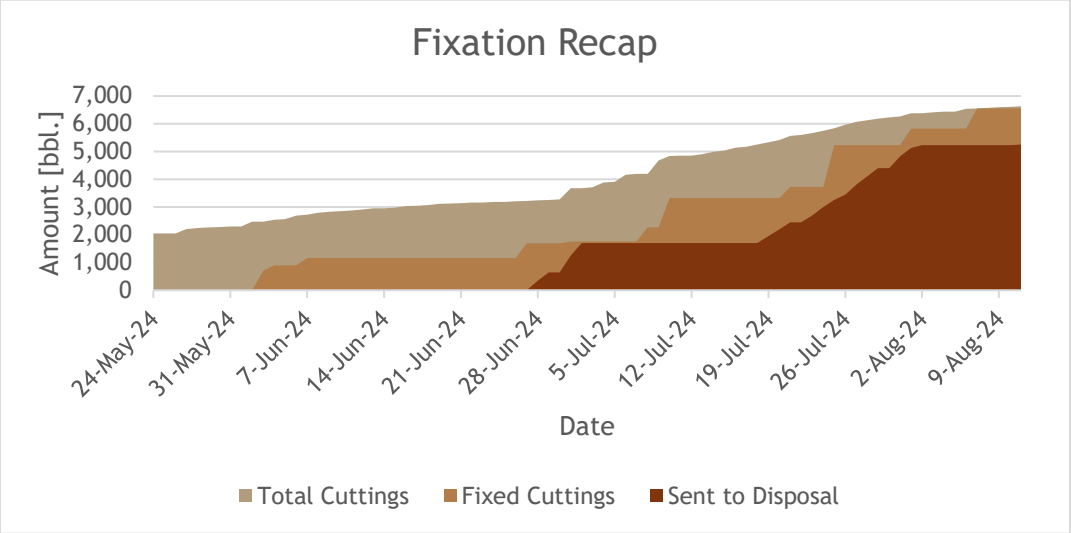


Figure 11-3: AZNS-127 Fixation Graph

Fixation analysis								
Report No.	Date	Depth	Drilled	Can test	Cement/Waste		SodSil/Waste	
				Daily	Daily	Average	Daily	Average
-	dd-M-yy	m	m	Pass/NA	kg/M3	kg/M3	kg/M3	kg/M3
52	14-Jul-24	**3,390**	**0**	Pass	0.000	4.512	0.000	0.728
53	15-Jul-24	**3,407**	**17**	Pass	0.000	4.512	0.000	0.728
54	16-Jul-24	**3,407**	**0**	Pass	0.000	4.512	0.000	0.728
55	17-Jul-24	**3,453**	**46**	Pass	0.000	4.512	0.000	0.728
56	18-Jul-24	**3,519**	**66**	Pass	0.000	4.512	0.000	0.728
57	19-Jul-24	**3,625**	**53**	Pass	0.000	4.512	0.000	0.728
58	20-Jul-24	**3,638**	**14**	Pass	0.000	4.512	0.000	0.728
59	21-Jul-24	**3,638**	**0**	Pass	3.650	4.404	0.106	0.624
60	22-Jul-24	**3,638**	**0**	Pass	0.000	4.404	0.000	0.624
61	23-Jul-24	**3,714**	**76**	Pass	0.000	4.404	0.000	0.624
62	24-Jul-24	**3,761**	**47**	Pass	0.000	4.404	0.000	0.624
63	25-Jul-24	**3,790**	**29**	Pass	1.333	4.063	0.484	0.604
64	26-Jul-24	**3,827**	**37**	Pass	0.000	4.063	0.000	0.604
65	27-Jul-24	**3,864**	**37**	Pass	0.000	4.063	0.000	0.604
66	28-Jul-24	**3,872**	**8**	Pass	0.000	4.063	0.000	0.604
67	29-Jul-24	**3,891**	**19**	Pass	0.000	4.063	0.000	0.604
68	30-Jul-24	**3,891**	**0**	Pass	0.000	4.063	0.000	0.604
69	31-Jul-24	**3,891**	**0**	Pass	0.000	4.063	0.000	0.604
70	1-Aug-24	**3,891**	**0**	Pass	0.833	3.740	0.364	0.574
71	2-Aug-24	**3,891**	**0**	NA	0.000	3.740	0.000	0.574
72	3-Aug-24	**3,891**	**0**	Pass	0.000	3.740	0.000	0.574
73	4-Aug-24	**3,891**	**0**	Pass	0.000	3.740	0.000	0.574
74	5-Aug-24	**3,891**	**0**	Pass	0.000	3.740	0.000	0.574
75	6-Aug-24	**3,891**	**0**	Pass	0.000	3.740	0.000	0.574
76	7-Aug-24	**3,891**	**0**	Pass	7.597	4.091	4.000	0.955
77	8-Aug-24	**3,891**	**0**	Pass	0.000	4.091	0.000	0.955
78	9-Aug-24	**3,891**	**0**	Pass	0.000	4.091	0.000	0.955
79	10-Aug-24	**3,891**	**0**	Pass	0.000	4.091	0.000	0.955
80	11-Aug-24	**3,936**	**45**	Pass	0.000	4.091	0.000	0.955

Table 11-7: AZNS-127 Fixation Analysis

Dewatering											
Report No.	Date	Depth	Drilled	Total waste-water		Total Dewater		Recycled		Disposed and Lost	
				Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
52	14-Jul-24	**3,390**	**0**	65	10,115	0	8,290	70	7,650	0	620
53	15-Jul-24	**3,407**	**17**	50	10,165	0	8,290	100	7,750	0	620
54	16-Jul-24	**3,407**	**0**	2,000	12,165	400	8,690	400	8,150	30	650
55	17-Jul-24	**3,453**	**46**	0	12,165	0	8,690	0	8,150	0	650
56	18-Jul-24	**3,519**	**66**	0	12,165	0	8,690	0	8,150	0	650
57	19-Jul-24	**3,625**	**53**	0	12,165	0	9,200	0	8,150	0	650
58	20-Jul-24	**3,638**	**14**	0	12,165	350	9,550	150	8,300	200	850
59	21-Jul-24	**3,638**	**0**	0	12,165	200	9,750	250	8,550	0	850
60	22-Jul-24	**3,638**	**0**	0	12,165	300	10,050	300	8,850	0	850
61	23-Jul-24	**3,714**	**76**	0	12,165	300	10,350	250	9,100	0	850
62	24-Jul-24	**3,761**	**47**	0	12,165	0	10,350	150	9,250	0	850
63	25-Jul-24	**3,790**	**29**	0	12,165	0	10,350	150	9,400	0	850
64	26-Jul-24	**3,827**	**37**	0	12,165	0	10,350	100	9,500	0	850
65	27-Jul-24	**3,864**	**37**	0	12,165	0	10,350	0	9,500	0	850
66	28-Jul-24	**3,872**	**8**	0	12,165	300	10,650	100	9,600	0	850
67	29-Jul-24	**3,891**	**19**	0	12,165	0	10,650	0	9,600	0	850
68	30-Jul-24	**3,891**	**0**	0	12,165	0	10,650	200	9,800	0	850
69	31-Jul-24	**3,891**	**0**	0	12,165	400	11,050	0	9,800	0	850
70	1-Aug-24	**3,891**	**0**	300	12,465	0	11,050	0	9,800	0	850
71	2-Aug-24	**3,891**	**0**	0	12,465	0	11,050	100	9,900	0	850
72	3-Aug-24	**3,891**	**0**	0	12,465	200	11,250	100	10,000	50	900
73	4-Aug-24	**3,891**	**0**	0	12,465	0	11,250	0	10,000	0	900
74	5-Aug-24	**3,891**	**0**	0	12,465	0	11,250	120	10,120	0	900
75	6-Aug-24	**3,891**	**0**	0	12,465	0	11,250	100	10,220	0	900
76	7-Aug-24	**3,891**	**0**	0	12,465	0	11,250	100	10,320	0	900
77	8-Aug-24	**3,891**	**0**	0	12,465	400	11,650	60	10,380	20	920
78	9-Aug-24	**3,891**	**0**	0	12,465	0	11,650	60	10,440	0	920
79	10-Aug-24	**3,891**	**0**	0	12,465	400	12,050	310	10,750	30	950
80	11-Aug-24	**3,936**	**45**	0	12,465	0	12,050	150	10,900	0	950

Table 11-8: AZNS-127 De-watering Recap

	Dewatering Analysis											
	Report No.	Date	Depth	Drilled	pH	Hardn.	Turb.	TSS	Nitrate	Al	Sulfate	Ca
					7-9	<800	<50	< 80	< 50	< 3	< 50	< 600
	-	dd-M-yy	m	m	-	mg/L	NTU	mg/L	mg/L	mg/L	mg/L	mg/L
52	14-Jul-24	**3,390**	**0**	0	0	0	0	0	0	0	0	
53	15-Jul-24	**3,407**	**17**	7	800	34	40	30	1	30	500	
54	16-Jul-24	**3,407**	**0**	7	800	38	35	30	1	33	500	
55	17-Jul-24	**3,453**	**46**	0	0	0	0	0	0	0	0	
56	18-Jul-24	**3,519**	**66**	0	0	0	0	0	0	0	0	
57	19-Jul-24	**3,625**	**53**	0	0	0	0	0	0	0	0	
58	20-Jul-24	**3,638**	**14**	9	1,240	45	65	14	1	40	800	
59	21-Jul-24	**3,638**	**0**	8	1,280	46	60	15	1	40	800	
60	22-Jul-24	**3,638**	**0**	8	1,240	40	60	15	1	42	880	
61	23-Jul-24	**3,714**	**76**	9	1,200	40	50	15	1	40	800	
62	24-Jul-24	**3,761**	**47**	9	1,200	40	50	15	1	40	800	
63	25-Jul-24	**3,790**	**29**	8	1,200	42	53	15	1	40	800	
64	26-Jul-24	**3,827**	**37**	8	1,200	42	53	15	1	40	800	
65	27-Jul-24	**3,864**	**37**	0	0	0	0	0	0	0	0	
66	28-Jul-24	**3,872**	**8**	9	1,240	32	40	12	1	42	1,000	
67	29-Jul-24	**3,891**	**19**	0	0	0	0	0	0	0	0	
68	30-Jul-24	**3,891**	**0**	8	1,240	32	40	12	1	42	1,000	
69	31-Jul-24	**3,891**	**0**	8	1,240	32	40	12	1	42	1,000	
70	1-Aug-24	**3,891**	**0**	7	1,240	32	40	12	1	42	1,000	
71	2-Aug-24	**3,891**	**0**	7	1,240	32	40	12	1	42	1,000	
72	3-Aug-24	**3,891**	**0**	7	1,240	32	40	12	1	42	1,000	
73	4-Aug-24	**3,891**	**0**	7	1,240	35	40	12	1	42	1,000	
74	5-Aug-24	**3,891**	**0**	7	1,200	35	40	12	1	42	1,000	
75	6-Aug-24	**3,891**	**0**	7	1,200	35	40	12	1	42	1,000	
76	7-Aug-24	**3,891**	**0**	7	1,200	35	40	12	1	42	1,000	
77	8-Aug-24	**3,891**	**0**	8	1,200	35	40	12	1	42	1,000	
78	9-Aug-24	**3,891**	**0**	8	1,200	35	40	12	1	42	1,000	
79	10-Aug-24	**3,891**	**0**	9	1,200	36	42	12	1	42	1,000	
80	11-Aug-24	**3,936**	**45**	8	1,200	36	42	12	1	42	1,000	

Table 11-9: AZNS-127 De-watering Analysis

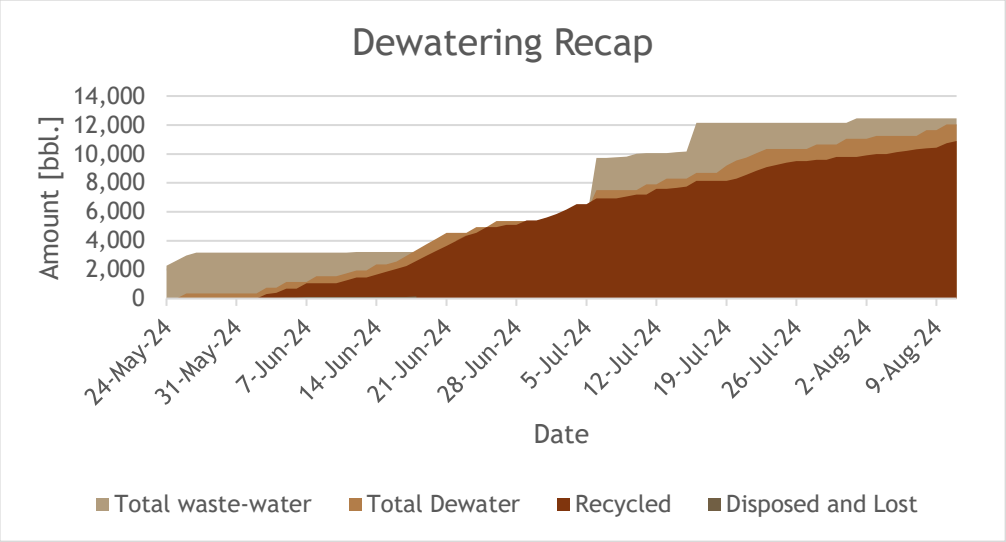


Figure 11-4: AZNS-127 De-watering Graph

Chemical																	
No.	Date	Cement		Na_2SiO_3		Coagulant		Polymer		Lime		Soda ash		Citric acid		Caustic soda	
		Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	Kg		Lit.		Kg		Kg		Kg		Kg		Kg		Kg	
52	14-Jul-24	0	13,000	0	270	0	0	0	0	0	1,100	0	950	0	0	0	0
53	15-Jul-24	0	13,000	0	270	0	0	0	0	0	1,100	0	950	0	0	0	0
54	16-Jul-24	0	13,000	0	270	0	0	0	0	0	1,100	0	950	0	0	0	0
55	17-Jul-24	0	13,000	0	270	0	0	0	0	0	1,100	0	950	0	0	0	0
56	18-Jul-24	0	13,000	0	270	0	0	0	0	0	1,100	0	950	0	0	0	0
57	19-Jul-24	0	13,000	0	270	0	0	0	0	0	1,100	0	950	0	0	0	0
58	20-Jul-24	0	13,000	0	270	25	25	1	1	100	1,200	50	1,000	0	0	0	0
59	21-Jul-24	1,500	14,500	15	285	0	25	0	1	25	1,225	25	1,025	0	0	0	0
60	22-Jul-24	0	14,500	0	285	25	50	1	2	0	1,225	50	1,075	0	0	25	25
61	23-Jul-24	0	14,500	0	285	25	75	0	2	50	1,275	50	1,125	0	0	0	25
62	24-Jul-24	0	14,500	0	285	0	75	0	2	0	1,275	0	1,125	0	0	0	25
63	25-Jul-24	2,000	16,500	40	325	0	75	0	2	0	1,275	0	1,125	0	0	0	25
64	26-Jul-24	0	16,500	0	325	0	75	0	2	0	1,275	0	1,125	0	0	0	25
65	27-Jul-24	0	16,500	0	325	0	75	0	2	0	1,275	0	1,125	0	0	0	25
66	28-Jul-24	0	16,500	0	325	25	100	2	4	25	1,300	50	1,175	0	0	12	37
67	29-Jul-24	0	16,500	0	325	0	100	0	4	0	1,300	0	1,175	0	0	0	37
68	30-Jul-24	0	16,500	0	325	0	125	0	6	0	1,325	0	1,225	0	0	0	49
69	31-Jul-24	0	16,500	0	325	25	150	0	6	25	1,350	50	1,275	0	0	12	61
70	1-Aug-24	500	17,000	40	365	0	150	0	6	0	1,350	0	1,275	0	0	0	61
71	2-Aug-24	0	17,000	0	365	0	150	0	6	0	1,350	0	1,275	0	0	0	61
72	3-Aug-24	0	17,000	0	365	35	185	5	11	0	1,350	75	1,350	0	0	25	86
73	4-Aug-24	0	17,000	0	365	0	185	0	11	0	1,350	0	1,350	0	0	0	86
74	5-Aug-24	0	17,000	0	365	0	185	0	11	0	1,350	0	1,350	0	0	0	86
75	6-Aug-24	0	17,000	0	365	0	185	0	11	0	1,350	0	1,350	0	0	0	86
76	7-Aug-24	5,500	22,500	60	425	0	185	0	11	0	1,350	0	1,350	0	0	0	86
77	8-Aug-24	0	22,500	0	425	25	210	5	16	0	1,350	100	1,450	0	0	14	100
78	9-Aug-24	0	22,500	0	425	0	210	0	16	0	1,350	0	1,450	0	0	0	100
79	10-Aug-24	0	22,500	0	425	25	235	5	21	0	1,350	50	1,500	0	0	12	112
80	11-Aug-24	0	22,500	0	425	0	235	0	21	0	1,350	0	1,500	0	0	0	112

Table 11-10: AZNS-127 Chemical Table

12. List of Contractors

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Haffaran Arvand
4	Electrical Logging and Perforation	PANIZ
5	Mud Logging	MAPSA KISH
6	Directional Drilling (MWD)	VIRA
7	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
8	Casing and Tubing Running	OEOC
9	Cementing	SPAD ENERGY & Mehran
10	Dry Test, DST and RTTS	OEOC
11	Slick Line	OEOC
12	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
13	Surface Well Testing	OEOC
14	Liner Hanger	PARS POOLAD
15	Well Head	OEOC

Table 12-1: AZNS-127 Contractors Involved

13. Appendix

13.1 DDR Summary

Item	Date	Remark
1	08 – May	Install 8” poor boy degasser flare, perform rig function test, L/U to XMT & C/M.
2	09 – May	Monitor the well f/ annulus & TBG, RIH w/ 3.4” gauge cutter to TBG HGR, try to pass this depth, No succ. Retrieve 4” BPV. While perform related P/T & prepare 800^{bbl} 106^{pcf} mud.
3	10 – May	Make & R/B 35 stds to derrick, fix CL port exit block, run 3.4” gauge cutter, RIH w/ 3.437” protection sleeve, stuck tool & work on same.
4	11 – May	Work on fish, retrieve 2.45^{m} slick line wire & 10$^{ft.}$ stem, make & R/B 27 stds to derrick, R/U CTU.
5	12 – May	Perform P/T, RIH w/ 3.2” JDC & CT to work on fish.
6	13 – May	Work on fish & perform injectivity test.
7	14 – May	Killing the well, R/D slick line equipment.
8	15 – May	Pump 146^{pcf} mud, retrieve fish, prepare mud, work on C/M, monitor pressure, pump 137^{pcf} mud.
9	16 – May	Pump mud, bleed off pressure to zero, WH & Ann. Pressure zero, install TWCV & test same, N/D XMT.
10	17 – May	N/D XMT, N/U 13-5/8”, 10k BOP stack, perform pressure test as NIOC policy.
11	18 – May	Screw L. joint on TBG HGR, F/U string, monitor the well Ok, meanwhile monitor pressure zero, R/U golpa flange & BOP & lubricator of slick line, run spear to 61^{m} (3SV) depth to catch possible cable in the well, POOH to surface, Obs. no cable.

12	19 – May	Run protection sleeve & set inside 3SV, run 2.230" gauge cutter to 61^{m}, took weight, POOH to surface, extend protection sleeve, run new protection sleeve & set inside 3SV, RIH 2.230" gauge ring to 3788^{m} (1st CIV depth), POOH to surface.
13	20 – May	R/D slick line equipment, R/U wireline equipment, pump 120^{bbl} 137^{pcf} mud in string, try to unlatch anchor seal No success. RIH with CCL & slinker bar to 3850^{m}, POOH to surface & L/D same, wait on daylight.
14	21 – May	Continue wait on daylight. RIH to 3845^{m}. perform colliding. POOH to surface. Pump totally 800^{bbl} mud, get 88^{pcf} mud in return while applying 700 – 1000^{psi} back pressure. After pumping totally 1800^{bbl} mud, decrease back pressure to zero. Stop pumping & observe the well, static. Due to OEOC refuse to follow the instruction, wait on OEOC while pumping 106^{pcf} mud, NL-NG. also static flow check again, static.
15	22 – May	Continue wait on OEOC while pumping 106^{pcf} mud. Open THS tie down screws. Perform pressure test screws in open position, Ok. R/D TBG L. joint. P/U DP L. joint. Try to release anchor seal several times, No success. B/O & R/D DP L. joint.
16	23 – May	R/U wireline. RIH 2-1/8" CCL & sinker bar to 3786^{m}. Took weight. Try to pass this depth, No success. POOH to surface. Change CCL & RIH 1-7/8" CCL & sinker bar to 3786^{m}, took weight. Try to pass, No success. POOH to surface & R/D wireline. Change L. joint, try to release anchor seal, No success. Establish circulation. Observe gas cut mud in return. Continue circulating while apply 400 – 1400^{psi} back pressure.
17	24 – May	Continue circulation, get 106^{pcf} mud in return. Work on collid point, string get free. R/D wireline. R/U CSG running equipment. POOH completion string while L/D totally 35 jts 4-1/2" TBG. Meanwhile circulate across wellhead. NL-NG.
18	25 – May	Continue POOH completion string while L/D totally 200 jts 4-1/2" TBG. Meanwhile circulate across wellhead. NL-NG.
19	26 – May	Continue POOH completion string while L/D totally 294 jts

		4-1/2” TBG. RIH w/ 7-7/8” overshot & fishing BHAto 220^{m}. NL-NG.
20	27 – May	RIH 7-7/8” overshot & fishing BHA to 2731^{m}. try to catch fish. No success.
21	28 – May	Continue work on fish & try to catch several times. no success. POOH 7-7/8” overshot & fishing BHA f/ 2731^{m} to surface. Run 3.8” magnet to 2731^{m}. POOH to surface. 1^{lb} junk recover. RIH OEDP by 12 jts 6-1/2” DC, 3 jts 5” HWDP, 46 jts 5” DP to 564^{m}.
22	29 – May	Continue RIH Jt by Jt to 1250^{m}. R/B 20 stds 5” DP S-135, 20 stds 5” DP G-105, 1 std 5” HWDP, 4 stds 6-1/2 DC, 4 stds 3-1/2” DP. RIH w/ 8-3/8” milling BHA include 2 No. 6-1/2” junk sub to 200^{m}.
23	30 – May	Cont. RIH w/ 8-3/8" M.M J.Mill, Milling BHA & Followed by 5" DPs f/ 200 to to 2729.5m, Milling obstruction f/ 2729.5 to 2738m.
24	31 – May	Cont. Milling to 2739m, CBU, POOH 8-3/8" M.M J.Mill, Milling BHA to surface, RIH w/ 7-7/8" Overshot to 2739m, Try to catch the fish.
25	01 – June	Work on fish, POOH w/ 7-7/8" Overshot & Fishing BHA to surface, RIH w/ 7-5/8" Overshot & Fishing BHA to 2739, Catch & Engage w/ fish. Work on fish.
26	02 – June	Work on fish.
27	03 – June	Work on fish; H/L 245klbs (25klbs excess on M/D compared with Overshot & Fishing BHA). CBU, Static flow check; Ok, POOH 7-5/8" Overshot & fishing BHA to 950m
28	04 – June	Cont. POOH fishing BHA f/ 950m to surface, L/D Totally 37 Jts (339m) 4-1/2" TBG, RIH w/ 8-3/8" M.M J.Mill, Milling BHA to 1770 m, Circ. thru C/M to uniform MW.
29	05 – June	RIH w/ 8-3/8" M.M J.Mill, Milling BHA f/ 1770 to 3078 m, CBU, Milling f/ 3078.2 to 3079.3 m, Pump & Circ. Out 85 bbl

		Hi-Vis Pill, POOH w/ 8-3/8" M.M J.Mill to surface, RIH w/ 7-5/8" Overshot to 85m.
30	06 – June	Cont. RIH w/ 7-5/8" Overshot & Fishing BHA to 3079.3m, CBU, Catch fish. Work on fish w/ max 390 klbs Wt on M/D. H/L Dropped to 245klbs, POOH to surface, L/D 8Jts 4-1/2" TBG, Perform BOP test.
31	07 – June	RIH 7-5/8" O.Shot to 3152m, CBU, Work on fish w/ different parameter.
32	08 – June	CBU, POOH w/ 7-5/8" Overshot to surface, 1 Jt of 4-1/2" TBG Retrieved, RIH w/ 8-3/8" M.M J.Mill, Milling BHA to 600, Slip & Cut Drilling line, Cont. RIH to 3161.2m, Milling to 3161.4m.
33	09 – June	Cont. Milling f/ 3161.4 to 3162.6m, Pump & Circ. Out 50 bbl Hi-Vis Pill, Work on Junk sub, POOH 8-3/8" M.M J.Mill & Milling BHA f/ 3162.6m to surface, RIH w/ 7-5/8" Overshot to 3162m. CBU.
34	10 – June	Catch fish, R/U W/L Equip, RIH w/ CCL & Wt. bar as dummy run to 3170m, Took Wt, POOH to surf, R/D W/L Equip, Wait on Rec. CT Service, R/U Injector head, wait on Rec. XOS.
35	11 – June	RIH 1-1/2"C.TBG to 3610m, Took Weight, Try to Pass this Point in Several Times, No Succ., Pump 100 bbl Hi-Vis Pill, POOH 1-1/2"C.TBG to 2100m, w/ N.L-N.G, Till M.N.
36	12 – June	Cont. POOH CTU to surf. R/D same. R/U wireline and RIH w/ 2-1/8" Dummy run to 3222m. Try to pass, no success. RIH w/ 1-11/16" Dummy run to 3612m, try to pass, no success. Assemble and RIH w/1-3/8" FPIT to 3600m. POOH same in progress.
37	13 – June	Cont. POOH FPIT Tools f/ 2000m to Surf., RIH Wireline w/ 1-7/8"Colliding Tools to 3612m, Obs. Drop Tension, POOH to 3594m, Apply 180Klb Overpull on String, Shoot the Colliding, Work on String, No Succ., POOH, Run Back Wireline w/ 1-7/8"Colliding Tools as Run#02 to 3398m, Shoot the Colliding, POOH to 3275m, Stuck Tools, Work on

		it, Got Free, POOH to Surf., Obs. Shoot the Colliding, Work on String Upward & at Neutral Point in Several Times, No Succ., w/ N.L-N.G, Till M.N.
38	14 – June	RIH Wireline w/ 2-1/8"CCL as Dummy Run to 3272m, Try to Pass this Point, No Succ., POOH, RIH Wireline w/ 2-1/8"Colliding & CCL Tools as Run#03 to 3272m, Fire Colliding Shot, POOH, Again RIH Wireline w/ New 2-1/8"Colliding & CCL Tools as Run#04 to 3185m, Fire Colliding Shot, POOH, R/D Wireline Equip., Work on String in Several Times, No Succ., w/ N.L-N.G, Till M.N.
39	15 – June	Cont. Work on String w/ Apply Different Parameters in Several Times, String Got Free, Circ. Btm's up, POOH to Surf., Obs. 7-5/8"Over Shot & Retreive 21.1m of 4-1/2"TBG f/ Collided Shot Point, L/D Same (Top of Fish is 3183.1m), RIH New 8-3/8"M.M J.Mill to 2300m, w/ N.L-N.G, Till M.N.
40	16 – June	Cont. RIH New 8-3/8"M.M J.Mill f/ 2300 to 3182.5m, Tag Top of Fish, Start Milling to 3185.8m, Work on J.Sub, Pump Hi-Vis Pill, C.B.U, POOH to 1400m, w/ N.L-N.G, Till M.N.
41	17 – June	Cont. POOH 8-3/8"J.Mill f/ 1400m to Surf., L/D J.Mill & 2No J.Sub, Retreive 10lb Junk, RIH 5-7/8"Overshot to 3185.8m, Tag Top of Fish, Engae w/ Fish, Work on it in Several Times, Got Free, POOH String for 13m While Obs. 30 klbs Overpull, Cont. POOH to 2700m, w/ N.L-N.G, Till M.N.
42	18 – June	Cont. POOH 7-5/8"Overshot to SURFACE,Obs. 7-5/8" Over Shot W/4-1/2" TBG Fish Neck,2.383 + 8 Jts 4-1/2" TBG +,73.318m + 9.268m,RIH W/8-3/8"M.M J.Mill to 2200m w/ N.L-N.G, Till M.N.
43	19 – June	Cont. RIH W/8-3/8"M.M J.Mill f/ 2200 to 3269.5m, Tag Top of Fish, Start Milling to 3274m, Work on J.Sub, Pump Hi-Vis Pill, C.B.U, POOH to 450m, w/ N.L-N.G, Till M.N.
44	20 – June	Cont. POOH 8-3/8"J.Mill f/ 450m to Surf., L/D J.Mill & 2No J.Sub, Retreive 18lb Junk, RIH 5-7/8"Overshot to 3274m, Tag Top of Fish, Engae w/ Fish, Work on it in Several Times,

		Cont. POOH to 2100m, w/ N.L-N.G, Till M.N.
45	21 – June	Cont. POOH w/ 7-5/8" Fishing assmbly f/ 2100m to Surface,RIH W/8-3/8"M.M J.Mill to 3297.5m, Tag Top of Fish, Start Milling to 3298m.
46	22 – June	Cont. milling the fish To 3298.8m,pooh milling bha to surface,P/U & M/U 7-5/8" Overshot & RIH To 1700m.
47	23 – June	Cont. RIH 5-7/8"Overshot to 3298m, Tag Top of Fish, Engae w/ Fish, Work on it in Several Times,Rec. Wire Line,R/u wire line (sealand), assemble & RIh wire line To 3286m,Pooh wire line to surf.
48	24 – June	R/U & Run Wire Line W/ 2-1/8" Colliding & 1-11/16" CCL to 3385M,Cont. POOH w/ 7-5/8" Fishing assmbly to Surfac & L/D Fish
49	25 – June	Cont. POOH w/ 7-5/8" Fishing BHA, Ret. 10 Jts 4-1/2" TBG, RIH w/ 8-3/8" M.M J.Mill to 3390m, Milling the fish to 3390.7m.
50	26 – June	Cont. Milling the fish f/ 3390.6 to 3391m, Pump & Circ. Out Hi-Vis Pill, Work on J.Sub, POOH w/ 8-3/8" M.M J.Mill & Milling BHA to surface, RIH w/ 7-5/8" Overshot BHA to 3391m, Try to catch & engage with fish.
51	27 – June	Cont. Try to catch & engage with fish, POOH w/ 7-5/8" Overshot & Fishing BHA f/ 3391m to surface, RIH w/ 8-3/8" M.M J.Mill & Milling BHA (Include 2 No. J.Sub) to 3390.7m, Milling f/ 3390.7 to 3390.8m.
52	28 – June	Cont. Milling f/ 3390.8 to 3391.9m. Pump and Circ out Hi-vis pill. POOH 8-3/8" Milling BHA to 80m.
53	29 – June	Cont. POOH 8-3/8" milling BHA f/ 80m to Surface. Recover 23lbs junk. Perform BOP test as NIOC policy. RIH w/ 7-5/8" Overshot to 3391.9m, Try to catch fish several times; No-Succ. POOH w/ 7-5/8" Overshot to 2980m.
54	30 – June	Cont. POOH w/ 7-5/8" Overshot & Fishing BHA f/ 2980m to

		surface. R/B 10 Stds 3-1/2" in derrick. RIH w/ 8-3/8" M.M J.Mill & Milling BHA (Include 2 No. J.Sub) to 2400m.
55	01 – July	Cont. RIH w/ 8-3/8" M.M J.Mill & Milling BHA f/ 2400 to 3391.9m, Milling fish to 3392.7m, CBU, POOH w/ 8-3/8" M.M J.Mill to surface. Recovered 39 lbs Junk
56	02 – July	RIH w/ 8-3/8" M.M J.Mill, 9-5/8" Scraper to 3350m, While Slip & Cut, CBU, POOH to surface, R/U WL Equip.
57	03 – July	Run CAST Log f/ 3350 to 35m, RIH w/ OEDP to 3390m.
58	04 – July	CBU, Perform CMT Plug Job as per program, POOH to 3050m, CBU, Apply Press & Monitor same, Open the well, Flow check, POOH to surface, RIH w/ 8-1/2" TCI Bit & D/O BHA to 800m.
59	05 – July	Cont. RIH w/ 8-1/2" TCI Bit &D/O BHA to 3130m, Wash & Ream Dn f/ 3130m to 3170m & Wash Dn To 3347m, D/O Hard CMT to 3350m while CBU @ 3310, 3349 & 3350m. Perform Positive Press test. POOH to 250m.
60	06 – July	POOH w/ 8-1/2" TCI Bit & D/O BHA f/ 250m to surface, RIH w/ 8-1/2" TCI Bit & 9-5/8" Scraper to 3350, scrap 3130 to 3350m, Circ. Out 50 bbl Hi-Vis Pill, Flush all surfce lines, Displace the well w/ 85 pcf mud, static flow check.
61	07 – July	Circ. Out 70 bbl Hi-Vis Pill, Spot 70 bbl Hi-Vis Pill, Static flow check, POOH w/ 8-1/2" TCI Bit & 9-5/8" Scraper to surface, Run whipstock Assy to 240m.
62	08 – July	Cont. Run 9-5/8" Hydraulic set whipstock f/ 240 to 3340m, RIH w/ Gyro tools to 3317m.
63	09 – July	Orient Whipface on Desirable Azi. (90 deg.), POOH Gyro Tools to Surface, Set Whipstock PKR @ 3339.08m, Work on TDS, Commence Milling as Side Track#01 f/ 3333.3 to 3337.2m, w/ N.L-N.G, While Add 4% Kcl to Mud System, Till M.N.
64	10 – July	Cont. milling as side track #1 w/ control parameters f/ 3337.2

		to 3341.3^{m}. smooth window interval in progress.
65	11 – July	Cont. Work on Smooth Window Interval, POOH w/ 8-1/2"Tri-Mill f/ 3341.3m to Surf., RIH 8-1/2"PDC Bit & 6-3/4"D.H.M (Ako: 1.5 Deg) & MWD Tool to 300, w/ N.L-N.G, Till M.N.
66	12 – July	Cont. RIH 8-1/2"PDC Bit & 6-3/4"D.H.M (Ako: 1.5 Deg) & MWD Tool to 3341.3m (BTM). Commence Drilling Formation to 3383.5m. POOH to 3220m , w/ N.L-N.G, Till M.N.
67	13 – July	Cont. POOH f/ 3220m to Surf., While Test MWD Tools, No Obs. Signal & L/D it, RIH 8-1/2"B.Bit & Last 6-3/4"D.H.M & New 6-3/4"MWD Tools & 6-3/4"Gyro Sub to 304m, Slip & Cut DRLG Line, Cont. RIH to 1900m, w/ N.L-N.G, Till M.N.
68	14 – July	Cont. RIH Directional BHA (6-3/4"D.H.M & MWD Tools) w/ Gyro Sub & Followed by 5"DPs f / 1900 to 3383.5m (BTM), POOH tp 3377.5m, R/U Logging Equip., RIH Gyro Tools to 3358m (Gyro Sub Depth), Adjust Tool Face of D.H.M (105.8 Deg), POOH & R/D Logging Equip., RIH & Resume D.DRLG as Slide Mode f/ 3383.5 to 3390.7m, w/ N.L-N.G, Till M.N.
69	15 – July	Cont. D.DRLG as Slide & Rotation Mode f/ 3390.7 to 3407.5m, w/ N.L-N.G, Till M.N.
70	16 – July	Cont. D.DRLG as Slide & Rotation Mode f/ 3407.5 to 3415.5m, Circ. Btm's up. POOH to Surf. Change 8-1/2" Bit and 6-3/4" D.H.M w/ N.L-N.G, Till M.N.
71	17 – July	M/U BHA. RIH w/ D.Drilling Assembly to 3415.5 (Btm). Resume Drlg 8-1/2" as slide & rotation mode H.S. to 3454m. w/ N.L-N.G, Till M.N.
72	18 – July	Cont. Drlg 8-1/2" as slide & rotation mode H.S. f/ 3454 to 3518m. w/ N.L-N.G, Till M.N.
73	19 – July	Cont. D.Drlg 8-1/2" to 3522m. POOH as wiper trip to 3400m,

		RIH to Btm, Resume drilling 8-1/2"H.S. to 3572m. w/ N.L-N.G, Till M.N.
74	20 – July	Cont. D.Drlg 8-1/2" to 3522m. POOH as wiper trip to 3572m, RIH to Btm, Resume drilling 8-1/2"H.S. to 3638m. w/ N.L-N.G, Till M.N.
75	21 – July	Cont. D.Drlg 8-1/2" to 3666.5m. POOH as wiper trip to 3430m, RIH to Btm, w/ good hole condition. Spot 70 bbl inhibited pill @ btm. POOH to 600m till MN. w/ N.L-N.G.
76	22 – July	Cont. POOH to surf. L/D directional BHA. M/U BHA and RIH to 3445m. R/D to 3588m, Work on stuck. Cont hard R/D to 3514m while inc. MW to 87 pcf. till MN. w/ N.L-N.G.
77	23 – July	Cont hard R/D f/ 3514 to 3666.5m (btm). Commenced drlg 8-1/2" H.S. to 3715m till MN. w/ N.L-N.G.
78	24 – July	Cont hard R/D f/ 3514 to 3666.5m (btm). Commenced drlg 8-1/2" H.S. to 3715m till MN. w/ N.L-N.G.
79	25 – July	Cont. Drlg 8-1/2" H.S. to 3791m, While work on TDS and fix same.
80	26 – July	Cont. Drlg 8-1/2" H.S. to 3827m.
81	27 – July	Cont. Drlg 8-1/2" H.S. to 3835m, POOH w/ 8-1/2" Drlg BHA to surface, RIH w/ 8-1/2" New PDC Bit & Previous BHA to 270m.
82	28 – July	Cont. RIH w/ 8-1/2" New PDC Bit & Previous BHA to 3835m, While clear tight spots, Drlg 8-1/2" H.S f/ 3835 to 3875m.
83	29 – July	Cont. Drlg 8-1/2" H.S to 3891m, CBU & take last sample, Perform wiper trip to 3840m, Run Back to BTM, Circ. Out Hi-Vis Pill, Spot Hi-Vis Pill, POOH to surface.
84	30 – July	Perform BOP Test, Run 7" LNR (Totally 61 Jts) to 3200m.
85	31 – July	Cont. RIH w/ 7" LNR on 5" DPs to 3890.5m, Perform CMT

		Job, POOH 17 Stds, CBU, Cont. POOH to surface.
86	01 – August	RIH w/8-1/2" TCI Bit & D/O BHA to 3129m, D/O CMT to 3149.5m, Circ. Out 50 bbl Hi-Vis Pill, POOH w/8-1/2" TCI Bit & D/O BHA to surface, while Slip & Cut Drlg Line.
87	02 – August	RIH w/ 6-1/8" MT Bit & Clean out BHA to 3150.5 m, While stop operation due to bad weather @ 1800m, D/O Hard CMT & POB.
88	03 – August	Cont. RIH w/ 6-1/8" MT Bit & Clean out BHA to 3845m, D/O Hard CMT above Plugs to 3853.5m (LDC), Circ. Out Hi-Vis Pill, POOH to 3098m, Perform 7" LNR CIT, Resume POOH to 600m.
89	04 – August	RIH w/ Tandem Scraper to top of LDC while scrap dry test PKR Setting depth, CBU, POOH to 2375m.
90	05 – August	Cont. POOH w/ 6-1/8" MT Bit & 9-5/8" X 7" Tandem Scraper to surface, Run W/L logging tools (CCL/GR/SBT/VDL), Run 9-5/8" Dry test tools for 7" L.Lap to 300m.
91	06 – August	Cont. Run 9-5/8" Dry test tools for 7" L.Lap to 3120m, Set 9-5/8" Dry test PKR at 3119 m (30.5m above 7" L.Lap). Obs. Hydro Spring Tester Valve Opening Indication on M/D, Monitor Open period for 3Hrs; "No-Bubble". Close DCIP Valve. Record Close Period for 1 Hr is in progress till M.N
92	07 – August	Record Close Period for totally 6 Hrs, Rev. Circ. To uniform MW, Unset the 9-5/8" Dry Test PKR, Change TDS link & CBU, Static flow check, POOH 9-5/8" Dry test Assy to surface.
93	08 – August	Cont. POOH 9-5/8" Dry test Assy to surface. Retreive 2No. Memory gauges. Check sample properteis Between Hydro spring & DCIP. M/U 6" M.M.J. Mill and RIH to 3853m. Commence D/O hard cmt to 3862m till M.N.
94	09 – August	Cont. D/O F.collar, hard cmt, F.shoe & R.Hole to 3891m, Commence drlg formation to 3893.5m, circ. Btm's up. Work on junk sub. POOH to 1450m w/ NL-NG, till M.N.

95	10 – August	Cont. POOH to surf. . M/U 6-1/8" PDC Bit. RIH to 3870m. Displace the well with 100pcf fresh mud in progress w/ NL-NG, till M.N.
96	11 – August	Cont. Displace the well with 100pcf fresh mud in progress, perform FIT. commence drlg formation f/ 3893.5 to 3938m, w/ NL-NG, till M.N.
97	12 – August	Cont. drlg formation f/ 3938 to 4003m, due to obs. Connection gas stop operation, weight up active mud system in progress. w/ NL-NG, till M.N.
98	13 – August	Cont. drlg formation f/4003 to 4035m, POOH with hard B/R to 3870m (Inside 7" Shoe). Circulated and worked on mud parameters. RIH back with W/R twice to 4000m. w/ NL-NG, till M.N.
99	14 – August	Cont. RIH back with W/R twice to 4000 to 4035m(btm). Resume drlg formation f/4035 to 4113m,while worked on mud parameters. w/ NL-NG, till M.N.
100	15 – August	Cont. drlg FM to 4140m as section TD, Circ. Btm's up. Perform condition trip. Spot Inhibited Hi-Vis, Hi-Dense pill @ btm w/ NL-NG, till M.N.
101	16 – August	POOH to surf. w/ good hole condition. R/U Lnr runnig equip. RIH 4-1/2" Lnr as per tally. Cont RIH on 3-1/2" DP to 700m till M.N. w/ NL-NG.
102	17 – August	Cont RIH 4-1/2" liner on 3-1/2" & 5"DP to 4140m (btm) Set hanger. M/U CMT Head on string. L/U CMT Line to CMT Head in progress. till M.N. w/ NL-NG.
103	18 – August	Cont. L/U CMT Line to CMT Head. Perform cmt job as per plan. POOH to surf. M/U 6" MMJ Mill and RIH to 1800m till M.N. w/ NL-NG.
104	19 – August	Cont. RIH w/ 6" MMJ Mill and RIH to 3480m, W/D to 3720m. D/O cmt to 3780.5m. Circ. Btm's up. POOH to 200mwhile L/D excess tubulars till M.N. w/ NL-NG.

105	20 – August	Cont. POOH f/ 200m to Surf. L/D excess tubulars in derrick. RIH w/ 3-1/2" J.Mill on 3-1/8" DC & 2-7/8" DP Jt by Jt to 254m till M.N. w/ NL-NG.
106	21 – August	Cont. RIH w/ 3-1/2" J.Mill on 3-1/8" to 3780.5m (4-1/2" L.Lap). commence D/O med. Cmt f/ 3780.5 to 3784m.till M.N. w/ NL-NG.
107	22 – August	D/O POB, Cont. wash and ream & Clean out inside 4-1/2" LNR to 4030m.
108	23 – August	Cont. wash and ream & Clean out inside 4-1/2" LNR to 4085m. D/O Med to Hard CMT f/ 4085 to 4102m. Tag LDC @ 4102m. Pump & Circ. Out Hi-Vis Pill. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 4102 to 1150m.
109	24 – August	Cont. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 1150m to surface, RIH w/ open end string & 7" Scraper BHA to 917m, Due to change in plan from client side, POOH w/ OEDPs to surface, RIH w/ Clean out BHA to 450m.
110	25 – August	Cont. RIH w/ Clean out BHA to 4100m.
111	26 – August	Cont. Wash & Ream Dn to 4102m, D/O Pump Dn Plug, Wiper Plug & 4-1/2" LNR LDC. Cont. D/O Hard CMT to 4114.5m.
112	27 – August	Cont. D/O Hard CMT to 4134m, Circ. Out Hi-Vis Pill, Spot 20 bbl Non-Solid Brine, POOH to 3100m.
113	28 – August	Cont. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 3100m to surface, Change top rams to 4-1/2" x 7" VBR, Press test same, Run CCL/GR/RBT Wireline logging tools to 4120m, POOH to 1900m.
114	29 – August	Cont. POOH w/ CCL/GR/RBT Wireline logging tools to surface. Run CCL/ 2-7/8" Gun as dummy run to 4120m. RIH w/ 3-1/2" Mill & Clean Out BHA to 2550m, String plug, work on same & Unplug.
115	30 – August	RIH w/ clean out BHA to 4134m, Circ. Out tandem pill, Spot Non-Solid Brine, POOH to 3350m.

116	31 – August	Cont. POOH w/ 3-1/2" M.M. J.Mill & Clean out BHA f/ 3350m to surface. Run CCL & 2-7/8" Blank Gun as dummy run to 4134m. RIH w/ 2-7/8" WCP HSD Gun Assy as Run #1 to 4124.7m. Fire the gun. POOH to 100m. Wait on daylight.
117	01 – Sept	Perform 2nd run of perforation f/ 4120.7 - 4117.7m to & f/ 4092 to 4088.2 m.
118	02 – Sept	Wait on daylight, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 4 to 3587m, Try to pass this depth, RIH w/ Clean out BHA to 1600m, Meanwhile Slip & Cut Drlg line @ 450m.
119	03 – Sept	Cont. RIH w/ Clean out BHA f/ 1600 to 4128m, CBU, Spot Non-Solid Brine, POOH to 1750m.
120	04 – Sept	Cont. POOH w/ 3-1/2" M.M J.Mill f/ 1750m to surface, Run dummy run to 4080.2m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #4 to 4080.2m. Fire the gun. POOH to 100m. Wait on daylight.
121	05 – Sept	Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4075m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #5 & #6 as per plan. Fire the gun. POOH to Run #6 to 100m. Wait on daylight.
122	06 – Sept	Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #7 & #8 as per plan. Fire the gun. POOH to Run #8 to 100m. Wait on daylight.
123	07 – Sept	Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 9 & # 10 as per plan. Fire the gun. POOH to Run # 10 to 100m. Wait on daylight.
124	08 – Sept	Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 11 & # 12 as per plan. Fire the gun. POOH to Run # 12 to 100m. Wait on daylight.

125	09 – Sept	Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, Run dummy run to 3989.65m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 13. Fire the gun. POOH to surface. R/D WL unit. Pressure test BOP in progress.
126	10 – Sept	Cont. Press. Test BOP. P/U and M/U 3-1/2" J.Mill and RIH to 4050m till MN. NL-NG.
127	11 – Sept	Cont. RIH w/ 3-1/2" M.M.J.Mill and RIH to 4134m (btm) Circ. Btm's up. Spud 25 bbl Comp. fluid @ btm. POOH to 2300m till MN. NL-NG.
128	12 – Sept	Cont. POOH f/2300m to surface while L/D exceess tubulars. Perform CCL logging. RIH & L/D 6 Jts 3-1/2" DP & 15 Jts 4-3/4" DC till MN, NL-NG.
129	13 – Sept	RIH w/ 7" Scrapper to 3286m. Scrap PKR interval 3286 to 3325m. Cont RIH to 3777m. Pump Hi-Vis. Circ. Btm's up in progress. NL-NG.
130	14 – Sept	Pump Hi-Vis. Circ. Btm's up. Pump 60 bbl Hi-Vis pill (104pcf, Ferobar free) + 20 bbl Hi-Vis pill (82 pcf brine) followed by Synthetic Completion Fluid and displace the well, Obs. The well, static. POOH and L/D DPs jt by jt to 2700m till MN, NL-NG.
131	15 – Sept	Cont. POOH w/ 7" scrapper while L/D Jt by Jt f/ 2700 to 770m till, NL-NG.
132	16 – Sept	Cont. POOH w/ 7" f/ 770m to surface, R/U slick line & TBG running equip. Start RIH 3-1/2" X 4-1/2" comp string to 73m till, NL-NG.
133	17 – Sept	Cont. RIH 3-1/2" X 4-1/2" comp string to 406m. Run 2.8" gauge ring. Work on rectify lakages of pump truck lines till MN, NL-NG.
134	18 – Sept	Cont. RIH 3-1/2" X 4-1/2" comp string to 1470m. While press. Test string @ 406 & 1224m; OK, till MN, NL-NG.

135	19 – Sept	Cont. RIH w/ Totally 284 Jts 4-1/2" TBG (18.9#, L-80, HSM-3) & 22 Jts 3-1/2" TBG (10.2#, L-80, HSM-3) to 2932m. While Press test completion string @ 2010 & 2932m; "OK".
136	20 – Sept	Run completion string to 3257m, M/U TR3SV Assy & Perform P/T TR3SV, RIH to 3274m, Perform P/T Completion string & inflow test of flapper, Cont. Run to 3325m, Perform P/T completion string in progress till M.N.
137	21 – Sept	Change L/J, P/T Completion string; OK. Set Protection Sleeve. Ret. check valve f/ SSD. Set Plug in No-Go. Due to Client Request, Decided to repeat string Press test. Ret. Plug f/ No-Go. Set Check valve in SSD. P/T Completion string; OK. Ret. check valve f/ SSD. RIH w/ Plug to top of SSD. In order to record upward Wt.; Obs. Suddenly drop tension. POOH to surface; Obs. About 830m Slick line wire, rope socket, 12 ft stem, 6 ft Mech. Jar, Running tools & 2.75" Plug left in hole. Due to lack of stem; wait on Rec. stem.
138	22 – Sept	Run LIB, Run 2.8" Wire grab w/o no catch wire, Ret. P.Sleeve, Run 3.2" Wire grab; Recover 2 ft wire, Run 3.2" 3 leg wire grab; Recover fish w/o plug, Set P.Sleeve, Run 2.8" Wire grab.
139	23 – Sept	Ret. Plug, Run Gauge ring to 3330m, Set Plug & Prong in No-Go, Run Shifting tool to SSD, Set PKR, Perform Pull test, Land TBG HGR, P/T String to 5000 psi.
140	24 – Sept	P/T tie downs, Ann, Ret. P.Sleeve, Back off & L/D L.J, Install TWCV & P/T same against SBR, N/D BOP Stack, N/U BMV & Related P/T.
141	25 – Sept	N/U XMT, P/T same, Ret. TWCV, Set P.Sleeve, Ret. Prong & Plug, Ret. P.Sleeve, Install BPV, Close All XMT Valve except BMV & Secure the well. Released the rig.

Table 13-1: AZNS-127 Daily Drilling Report Summary

14. Comments & Revision

Drilling Engineering							
1	RTE.به 13.45 اصلاح گردد	12	CM	• اصلاح شد			AP
2	به صورت زیر اصلاح گردد:Targetمختصات E:196,163.4 N:3448538.7	12	CM	• اصلاح شد			AP
3	عمق لوله جداری 20" به 146 متر اصلاح گردد	12	CM	• اصلاح شد			AP
4	هر دو نوع لوله جداری 9 5/8" استفاده شده در جدول مشخص گردد: 9 5/8” 53.5#-T95-TSH BLUE 9 5/8” 53.5#-T95-TPCQ	12	CM	• اصلاح شد			AP
5	در صفحه ذکر شده بخش تست خشک لبه آستری 7"، افزایش فشار مواجه PSIذکر گردد نتیجه تست با 213 بوده است.	29	CM	• اصلاح شد			AP
6	به شرح EMW عمق و FITدر صفحه 29، بخش اخذ زیر ذکر گردد: FIT @ 3871 M , EMW:108 PCF	29	CM	• اصلاح شد			AP

7	در شماتیک، عمق لوله جداری 20" به 146 متر ، عمق لوله جداری 13 3/8 " به 1441 متر، عمق نصب پکر به 3308 متر اصلاح گردند.	33	CM	• اصلاح شد			AP
8	تعداد روز واقعی اعلام شده در جدول شکست زمانی با PTگزارش روزانه مغایرت دارد. تعداد روز عملیات 73.85 می باشد.	40	CM	• اصلاح شد • اصلاح شد.	اصلاحی صورت نگرفته است. واقعی ,با توجه به گزارشات مشاور تعداد روز NPT، 73.85 روز و با احتساب NPTبدون به صورت کلی می بایست در جدول زمانی. تعداد روز واقعی 139.6 روز وارد گردد.		CM
9	اعلام شده با گزارش مشاور مغایرت دارد.NPT صحیح 1577.5 ساعت معادل 65.72 روز می NPT باشد.	43-41	CM	• اصلاح شد.	اعلام شده با گزارش مشاور مغایرت NPT دارد. صحیح 1577.5 ساعت معادل 65.72 NPT روز می باشد.		CM
10	5 با گزارش روزانه مغایرت دارد به BHAمجموع طول 115.12 متر اصلاح گردد.	47	CM	• اصلاح شد			AP
11	در جدول مشخصات لوله جداری، وزن لوله آستری 7" به 29Ib/ft و ID.آن نیز اصلاح گردد	74	CM	• اصلاح شد • اصلاح شد.	7" به 6.184 اصلاح شود.ID	74	CM

12	لوله آستری gradeدر جدول مشخصات لوله جداری، اصلاح و با صفحه 73 یکسان سازی 4 1/2ss" به 110 شود.	80	CM	• اصلاح شد			AP
13	عمق آستری 4 1/2 به 4140 اصلاح گردد.	80, 135	CM	• اصلاح شد			AP
14	لبه آستری 7" از 3150.2 به 3147 متر اصلاح گردد	135	CM	• اصلاح شد			AP
15	مربوط به کارفرما حذف گردد.NPT	39-197-30	CM	• NPT مربوط به کارفرما لحاظ نگردیده است.			
16	مدارک مربوط به تست کارگاهی ابزار رشته تکمیلی (با امضا صورت جلسه توسط نماینده کارفرما) می بایست ارائه گردد.		CM	• فایل Handover Booklet مورد تأیید کارفرما در فایل جداگانه ارائه می‌گردد.			
Fluid Engineering							
1	میزان مصرفی افزایه های سیال تکمیلی ذکر نشده.	184	CM	• در جدول صفحه 185 لحاظ گردیده است.			
2	برنامه پسماند جهت حفره های 6 1/8"و8.5" اعمال نشده.		CM	• گزارش مربوط به پسماند اضافه گردید.			

3	در سیال تکمیلی از سیال ایرانی(سنتزی) جایگزین استفاده شده.		CM	• اصلاح شد.			